ASTA

The ASTA team

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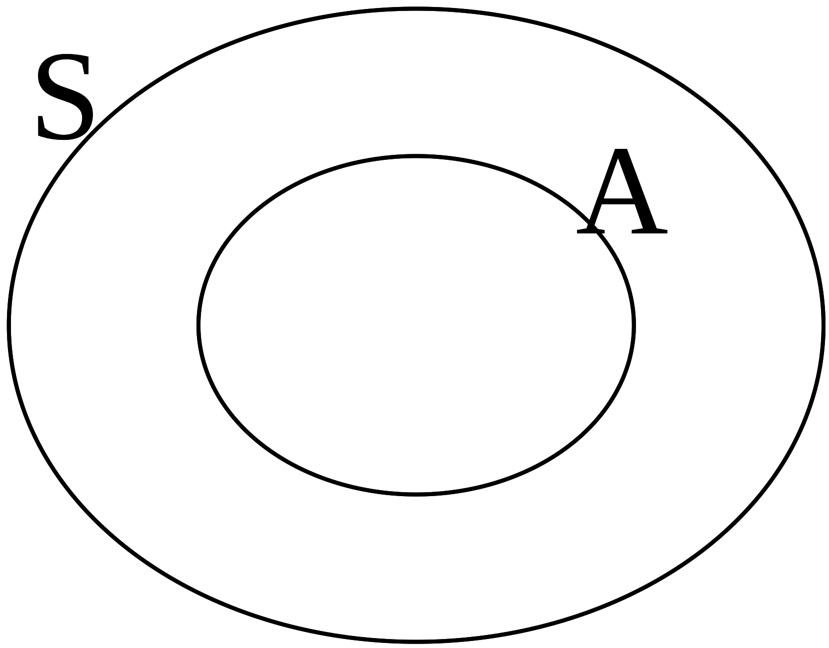
1 Introduction to probability

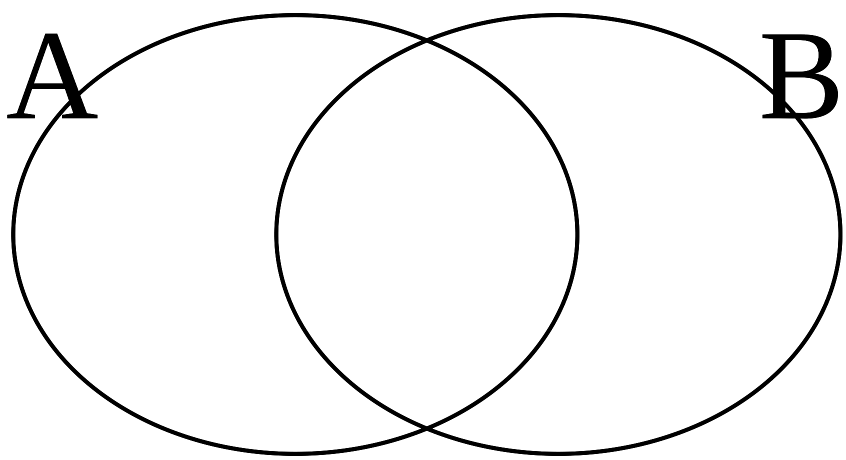
1.1 Events

- Consider an experiment.
- The **sample space** S is the set of all possible outcomes.
  - **Example:** We roll a die. The possible outcomes are $S = \{1, 2, 3, 4, 5, 6\}$.
  - **Example:** We measure wind speed (in m/s). The sample space is $[0, \infty)$.
- An **event** is a subset $A \subseteq S$ of the sample space.
  - **Example:** Rolling a die and getting an even number is the event $A = \{2, 4, 6\}$.
  - **Example:** Measuring a wind speed of at least 5m/s is the event $[5, \infty)$.

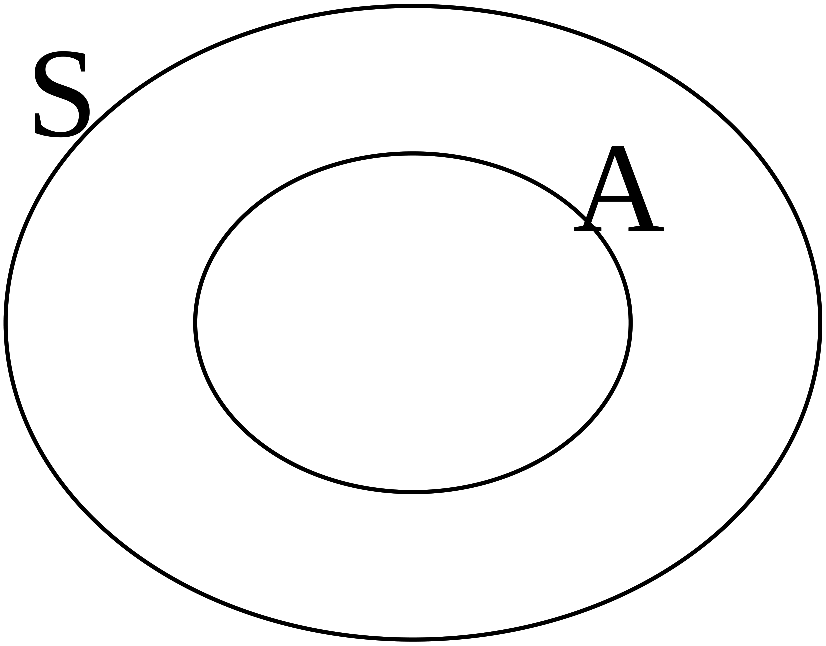


1.2 Combining events

- Consider two events A and B.
  - The **union** $A \cup B$ is the event that either A or B occurs.
  - The **intersection** $A \cap B$ of is the event that both A and B occurs.



- The **complement** A^c of A of is the event that A does not occur.



- **Example:** We roll a die and consider the events $A = \{2, 4, 6\}$ that we get an even number and $B = \{4, 5, 6\}$ that we get at least 4. Then
  - $A \cup B = \{2, 4, 5, 6\}$
  - $A \cap B = \{4, 6\}$
  - $A^c = \{1, 3, 5\}$

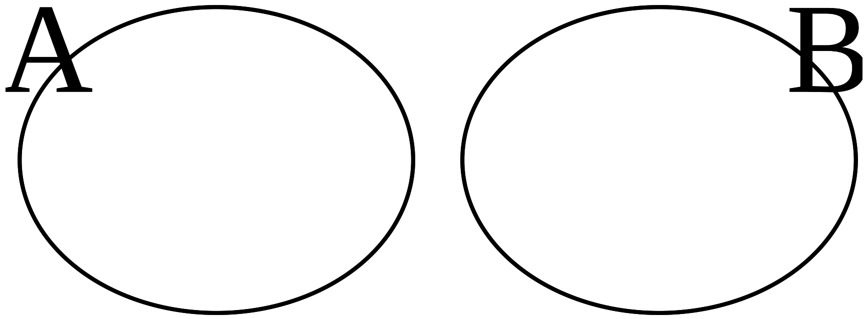
1.3 Probability of event

- The **probability** of an event is the proportion of times the event A would occur when the experiment is repeated many times.
- The probability of the event A is denoted $P(A)$.
  - **Example:** We throw a coin and consider the outcome $A = \{Head\}$. We expect to see the outcome $\{Head\}$ half of the time, so $P(Head) = \frac{1}{2}$.
  - **Example:** We throw a die and consider the outcome $A = \{4\}$. Then $P(4) = \frac{1}{6}$.
- Properties:
  1. $P(S) = 1$
  2. $P(\emptyset) = 0$
  3. $0 \leq P(A) \leq 1$ for all events A

1.4 Probability of mutually exclusive events

- Consider two events A and B.
- If A and B are **mutually exclusive** (never occur at the same time, i.e. $A \cap B = \emptyset$), then

$$P(A \cup B) = P(A) + P(B).$$



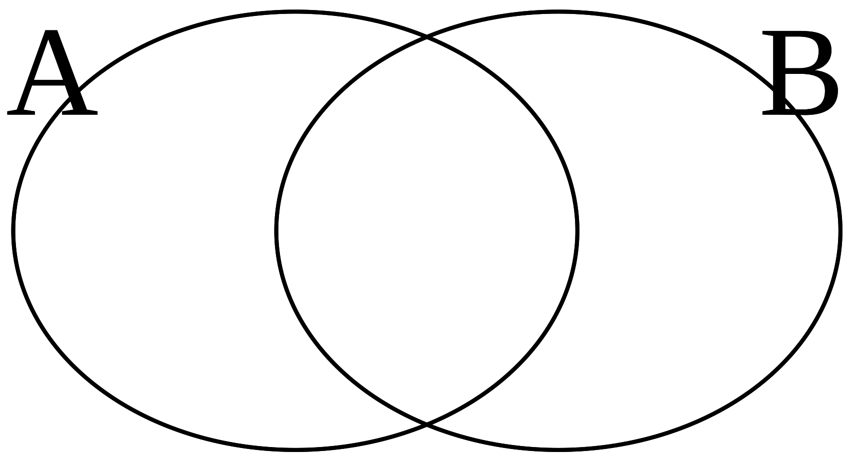
- **Example:** We roll a die and consider the events $A = \{1\}$ and $B = \{2\}$. Then

$$P(A \cup B) = P(A) + P(B) = \tfrac{1}{6} + \tfrac{1}{6} = \tfrac{1}{3}.$$

1.5 Probability of union

- For general events A an B,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$



- **Example:** We roll a die and consider the events $A = \{1, 2\}$ and $B = \{2, 3\}$. Then $A \cap B = \{2\}$, so

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = \tfrac{1}{3} + \tfrac{1}{3} - \tfrac{1}{6} = \tfrac{1}{2}.$$

1.6 Probability of complement

- Since A and A^c are mutually exclusive with $A \cup A^c = S$, we get

$$1 = P(S) = P(A \cup A^c) = P(A) + P(A^c),$$

so

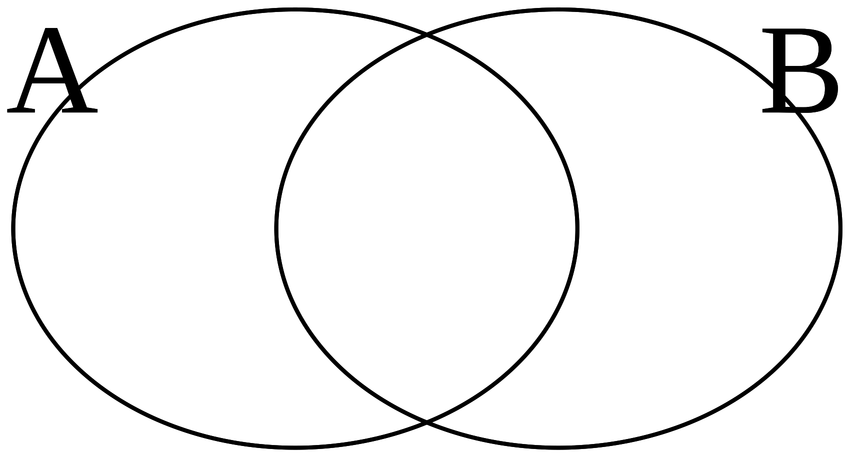
$$P(A^c) = 1 - P(A).$$

1.7 Conditional probability

- Consider events A and B.
- The **conditional probability** of A given B is defined by

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

if $P(B) > 0$.



- **Example:** We toss a coin two times. The possible outcomes are $S = \{HH, HT, TH, TT\}$. Each outcome has probability $\frac{1}{4}$. What is the probability of at least one head if we know there was at least one tail?
  - Let $A = \{\text{at least one H}\}$ and $B = \{\text{at least one T}\}$. Then

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{2/4}{3/4} = \frac{2}{3}.$$

1.8 Independent events

- Two events A and B are said to be **independent** if

$$P(A|B) = P(A).$$

  - **Example:** Consider again a coin tossed two times with possible outcomes HH, HT, TH, TT.
    - Let $A = \{\text{at least one H}\}$ and $B = \{\text{at least one T}\}$.
    - We found that $P(A|B) = \frac{2}{3}$ while $P(A) = \frac{3}{4}$, so A and B are not independent.

1.9 Independent events - equivalent definition

- Two events A and B are **independent** if and only if

$$P(A \cap B) = P(A)P(B).$$

- Proof: A and B are independent if and only if

$$P(A) = P(A|B) = \frac{P(A \cap B)}{P(B)}.$$

Multiplying by $P(B)$ we get $P(A)P(B) = P(A \cap B)$.

- **Example:** Roll a die and let $A = \{2, 4, 6\}$ be the event that we get an even number and $B = \{1, 2\}$ the event that we get at most 2. Then,
  - $P(A \cap B) = P(2) = \frac{1}{6}$
  - $P(A)P(B) = \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{6}$.
  - So A and B are independent.

2 Stochastic variables

2.1 Definition of stochastic variables

- A **stochastic variable** is a function that assigns a real number to every element of the sample space.
  - **Example:** Throw a coin three times. The possible outcomes are

$$S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}.$$

    - The random variable X assigns to each outcome the number of heads, e.g.

$$X(HHH) = 3, \quad X(HTT) = 1.$$

  - **Example:** Consider the question whether a certain machine is defect. Define
    - $X = 0$ if the machine is not defect,
    - $X = 1$ if the machine is defect.
  - **Example:** X is the temperature in the lecture room.

2.2 Discrete or continuous stochastic variables

- A stochastic variable X may be
- **Discrete:** X can take a finite or infinite list of values.
  - **Examples:**
    - Number of heads in 3 coin tosses (can take values $0, 1, 2, 3$)
    - Number of machines that break down over a year (can take values $0, 1, 2, 3, \ldots$)
- **Continuous:** X takes values on a continuous scale.
  - **Examples:**
    - Temperature, speed, voltage,...

3 Discrete random variables

3.1 Discrete random variables

- Let X be a discrete stochastic variable which can take the values $x_1, x_2, \ldots$
- The distribution of X is given by the **probability function**, which is given by

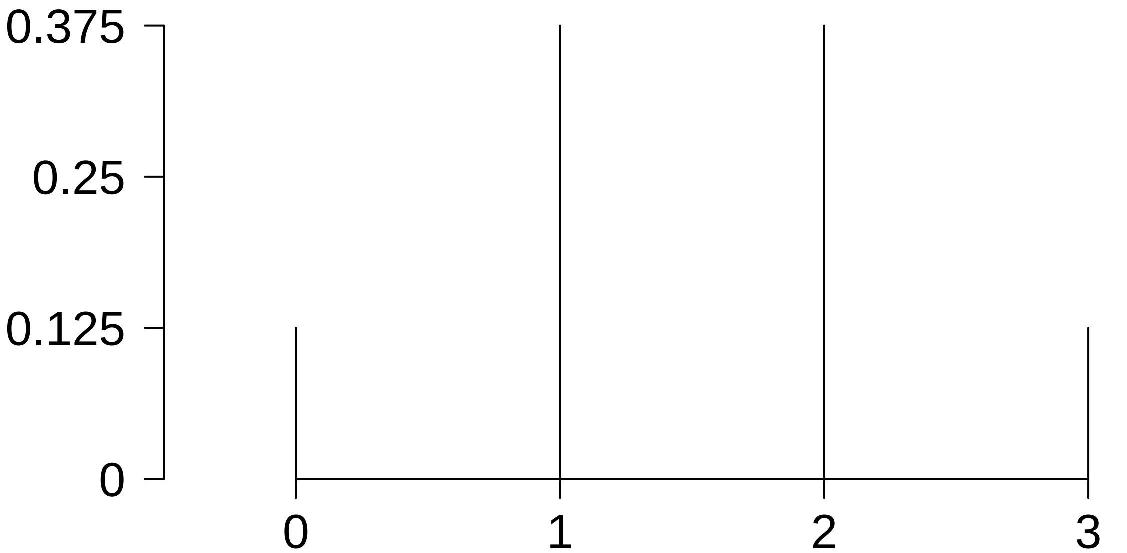
$$f(x_i) = P(X = x_i), \quad i = 1, 2, \ldots$$

– **Example:** We throw a coin three times and let X be the number of heads. The possible outcomes are

$$S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}.$$

The probability function is

* $f(0) = P(X = 0) = \frac{1}{8}$
* $f(1) = P(X = 1) = \frac{3}{8}$
* $f(2) = P(X = 2) = \frac{3}{8}$
* $f(3) = P(X = 3) = \frac{1}{8}$



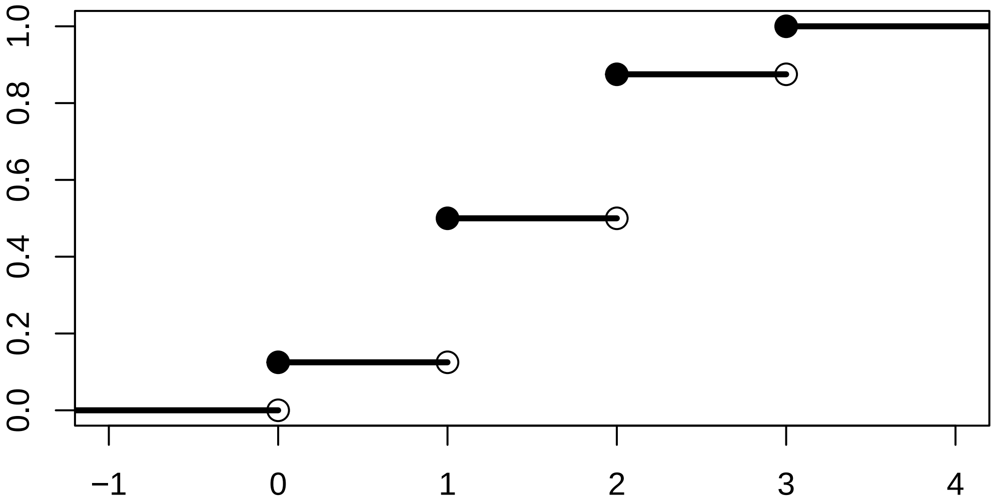
3.2 The distribution function

- Let X be a discrete random variable with probability function f. The **distribution function** of X is given by

$$F(x) = P(X \leq x) = \sum_{x_i \leq x} f(x_i), \quad x \in \mathbb{R}.$$

– **Example:** For the three coin tosses, we have

* $F(0) = P(X \leq 0) = \frac{1}{8}$
* $F(1) = P(X \leq 1) = P(X = 0) + P(X = 1) = \frac{1}{2}$
* $F(2) = P(X \leq 2) = P(X = 0) + P(X = 1) + P(X = 2) = \frac{7}{8}$
* $F(3) = P(X \leq 3) = 1$



- For a discrete variable, the result is an increasing step function.

3.3 Mean of a discrete variable

- The **mean** or **expected value** of a discrete random variable X with values $x_1, x_2, \ldots$ and probability function $f(x_i)$ is
$$\mu = E(X) = \sum_i x_i P(X = x_i) = \sum_i x_i f(x_i).$$
- Interpretation: A weighted average of the possible values of X, where each value is weighted by its probability. A sort of "center" value for the distribution.
  - **Example:** Toss a coin 3 times. What are the expected number of heads?

    $E(X) = 0 \cdot P(X = 0) + 1 \cdot P(X = 1) + 2 \cdot P(X = 2) + 3 \cdot P(X = 3) = 0 \cdot \frac{1}{8} + 1 \cdot \frac{3}{8} + 2 \cdot \frac{3}{8} + 3 \cdot \frac{1}{8} = 1.5.$

3.4 Variance of a discrete variable

- The **variance** is the mean squared distance between the values of the variable and the mean value. More precisely,
$$\sigma^2 = \sum_i (x_i - \mu)^2 P(X = x_i) = \sum_i (x_i - \mu)^2 f(x_i).$$
- A high variance indicates that the values of X have a high probability of being far from the mean values.
- The **standard deviation** is the square root of the variance
$$\sigma = \sqrt{\sigma^2}.$$
- The advantage of the standard deviation over the variance is that it is measured in the same units as X.
  - **Example** Let X be the number of heads in 3 coin tosses. What is the variance and standard deviation?
    - Solution: The mean was found to be 1.5. Thus,

      $\sigma^2 = (0-1.5)^2 \cdot f(0) + (1-1.5)^2 \cdot f(1) + (2-1.5)^2 \cdot f(2) + (3-1.5)^2 \cdot f(3) = (0-1.5)^2 \cdot \frac{1}{8} + (1-1.5)^2 \cdot \frac{3}{8} + (2-1.5)^2 \cdot \frac{3}{8} + (3-$

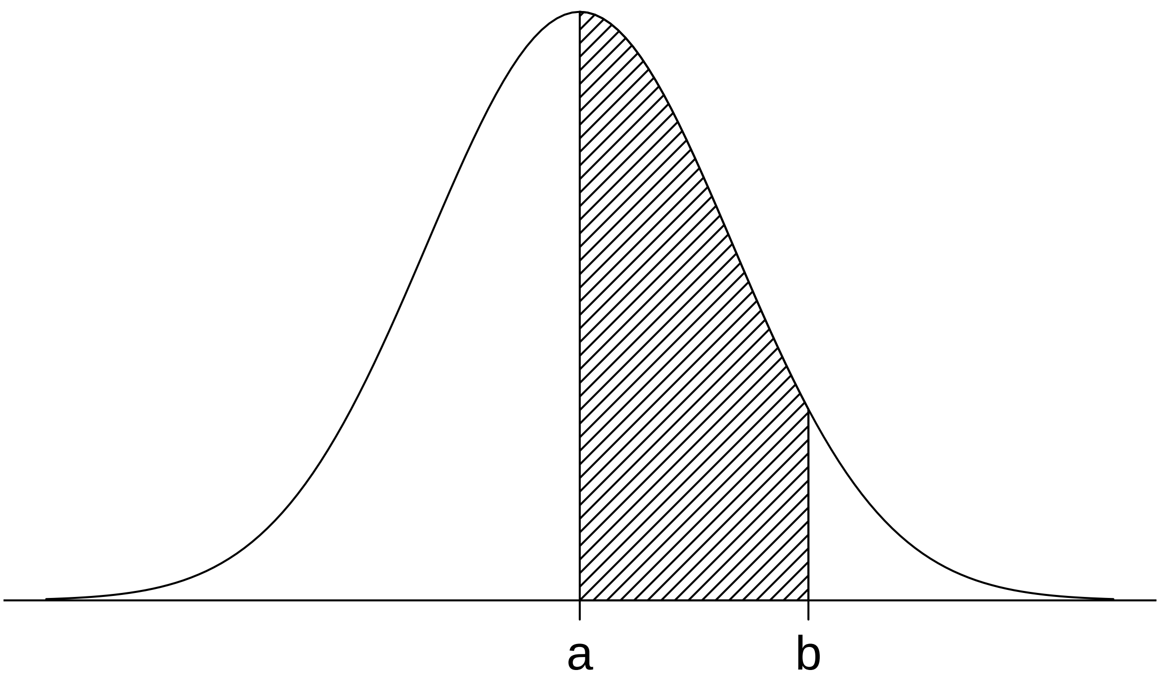
      The standard deviation is $\sigma = \sqrt{0.75} \approx 0.866$.

4 Continuous random variables

4.1 Distribution of continuous random variables

- The distribution of a continuous random variable X is given by a **probability density function** f, which is a function satisfying
  1. $f(x)$ is defined for all x in $\mathbb{R}$,
  2. $f(x) \geq 0$ for all x in $\mathbb{R}$,
  3. $\int_{-\infty}^{\infty} f(x)dx = 1$.
- The probability that X lies between the values a and b is given by

$$P(a < X < b) = \int_a^b f(x)dx.$$

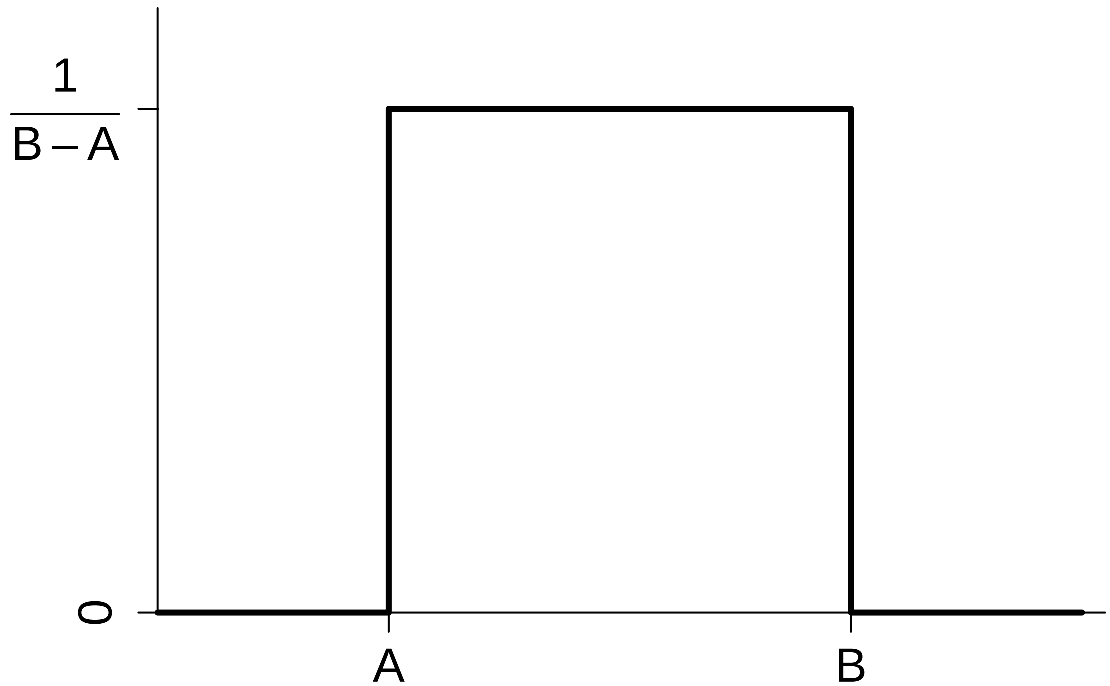


- Notes:
  - Condition 3. ensures that $P(-\infty < X < \infty) = 1$.
  - The probability of X assuming a specific value a is zero, i.e. $P(X = a) = 0$.

4.2 Example: The uniform distribution

- The **uniform distribution** on the interval (A, B) has density

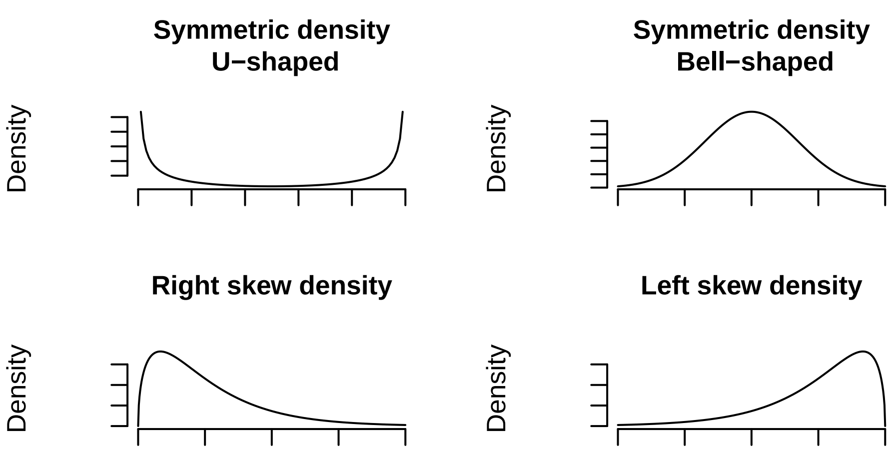
$$f(x) = \begin{cases} \frac{1}{B-A} & A \leq x \leq B \\ 0 & \text{otherwise} \end{cases}$$



- **Example:** If X has a uniform distribution on $(0,1)$, find $P(\frac{1}{3} < X \leq \frac{2}{3})$.
  - Solution:

$$P\left(\tfrac{1}{3} < X \leq \tfrac{2}{3}\right) = P\left(\tfrac{1}{3} < X < \tfrac{2}{3}\right) + P\left(X = \tfrac{2}{3}\right) = \int_{1/3}^{2/3} f(x)dx + 0 = \int_{1/3}^{2/3} 1dx = \frac{1}{3}.$$

4.3 Density shapes



4.4 Distribution function of continuous variable

- Let X be a continuous random variable with probability density f. The **distribution function** of X is given by

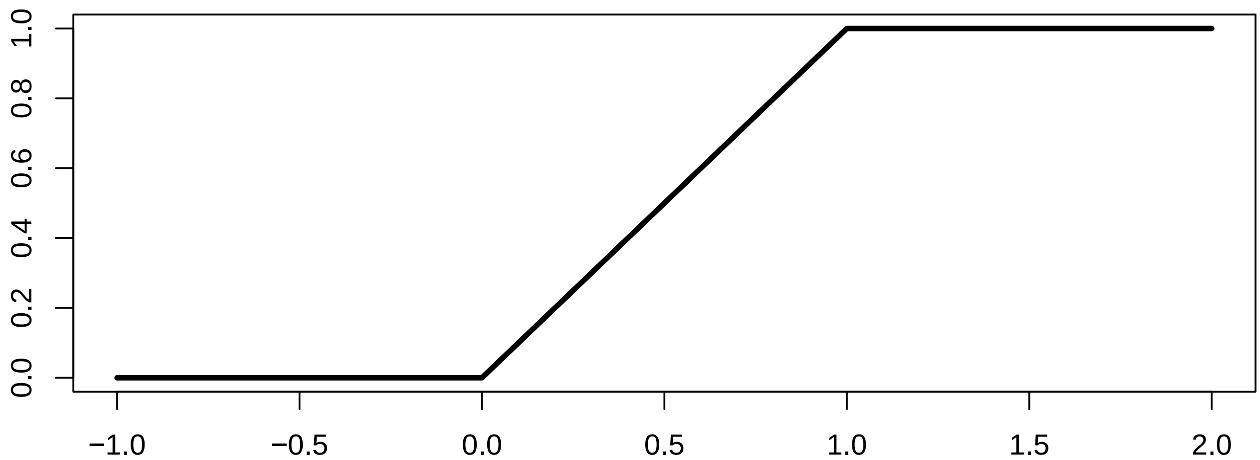
$$F(x) = P(X \leq x) = \int_{-\infty}^{x} f(y)dy, \quad x \in \mathbb{R}.$$

– **Example:** For the uniform distribution on $[0, 1]$, the density was

$$f(x) = \begin{cases} 1, & 0 \le x \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

Hence,

$$F(x) = P(X \le x) = \int_{-\infty}^{x} f(y)dy = \int_{0}^{x} 1dy = x, \quad x \in [0, 1].$$



4.5 Mean and variance of a continuous variable

- The **mean** or **expected value** of a continuous random variable X is

$$\mu = E(X) = \int_{-\infty}^{\infty} xf(x)dx.$$

- The **variance** is given by

$$\sigma^2 = \int_{-\infty}^{\infty} (x - \mu)^2 f(x)dx.$$

  – In calculations, it is often more convenient to use the formula

$$\sigma^2 = E(X^2) - E(X)^2 = \int_{-\infty}^{\infty} x^2 f(x)dx - \mu^2.$$

4.5.1 Example: Mean and variance in the uniform distribution

- Consider again the uniform distribution on the interval $(0, 1)$ with density

$$f(x) = \begin{cases} 1 & 0 \le x \le 1 \\ 0 & \text{otherwise} \end{cases}$$

  Find the mean and variance.

- **Solution:** The mean is

$$\mu = E(X) = \int_{-\infty}^{\infty} xf(x)dx = \int_{0}^{1} x \cdot 1dx = \left[\tfrac{1}{2}x^2\right]_0^1 = \tfrac{1}{2},$$

  and the variance is computed using the formula

$$\sigma^2 = E(X^2) - E(X)^2 = \int_{-\infty}^{\infty} x^2 f(x)dx - \mu^2 = \int_{0}^{1} x^2 dx - \mu^2 = \left[\tfrac{1}{3}x^3\right]_0^1 - \left(\tfrac{1}{2}\right)^2 = \tfrac{1}{3} - \tfrac{1}{4} = \tfrac{1}{12}.$$

4.6 Rules for computing mean and variance

- Let X be a random variable and a, b be constants. Then,
  1. $E(aX+b) = aE(X)+b$.
  2. $\text{Var}(aX+b) = a^2\text{Var}(X)$.
  - **Example:** If X has mean μ and variance σ^2, then
    * $E\left(\frac{X-\mu}{\sigma}\right) = \frac{1}{\sigma}E(X-\mu) = \frac{1}{\sigma}(E(X)-\mu) = 0,$
    * $\text{Var}\left(\frac{X-\mu}{\sigma}\right) = \frac{1}{\sigma^2}\text{Var}(X-\mu) = \frac{1}{\sigma^2}\text{Var}(X) = \frac{1}{\sigma^2}\sigma^2 = 1.$
    * So $\frac{X-\mu}{\sigma}$ is a standardization of X that has mean 0 and variance 1.

5 Two random variables

5.1 Joint distribution of two discrete variables

- Let X and Y be two discrete random variables. The **joint distribution** of X and Y is given by their **joint probability function**

$$f(x,y) = P(X=x, Y=y).$$

- We find the probability of $(X,Y) \in A$ by summing probabilities:

$$P((X,Y) \in A) = \sum_{(x,y)\in A} f(x,y).$$

- **Example:** We roll two dies and let X be the outcome of die 1 and Y be the outcome of die 2. Since all 36 combinations are equally likely,

$$f(x,y) = P(X=x, Y=y) = \frac{1}{36}, \qquad x, y = 1, 2, \ldots, 6.$$

  We can now compute:

$$P(X+Y=4) = f(1,3)+f(2,2)+f(3,1) = \frac{1}{36}+\frac{1}{36}+\frac{1}{36} = \frac{1}{12}.$$

5.2 Marginal distributions and independence

- Let (X,Y) be a pair of discrete variables with joint probability function $f(x,y)$. The **marginal probability function** for X is found by

$$f(x) = P(X=x) = \sum_y P(X=x, Y=y) = \sum_y f(x,y).$$

- Similarly, the **marginal probability function** for Y is

$$g(y) = \sum_x f(x,y).$$

- We say that X and Y are **independent** if

$$f(x,y) = f(x)g(y).$$

- Note: Recalling the definition of the probability function, the independence condition says that

$$f(x,y) = P(X = x, Y = y) = P(X = x) \cdot P(Y = y) = f(x)g(y),$$

which corresponds to independence of the events $\{X = x\}$ and $\{Y = y\}$.

- **Example:** We roll two dice and let X and Y be the outcome of die 1 and die 2, respectively. We found earlier that $f(x,y) = \frac{1}{36}$ for $x, y = 1, 2, \ldots, 6$. From this we can find the marginal distribution of X

$$f(x) = \sum_{y=1}^{6} f(x,y) = \frac{1}{36} + \frac{1}{36} + \frac{1}{36} + \frac{1}{36} + \frac{1}{36} + \frac{1}{36} = \frac{1}{6}, \quad x = 1, 2, \ldots, 6,$$

as we would expect. Similarly, the marginal distribution of Y is $g(y) = \frac{1}{6}, \quad y = 1, 2, \ldots, 6$. We can now check that the two dice are statistically independent:

$$f(x,y) = \frac{1}{36} = \frac{1}{6} \cdot \frac{1}{6} = f(x)g(y).$$

5.3 Joint distribution of two continuous variables

- Let X and Y be two continuous random variables. The **joint distribution** of X and Y is given by their **joint density function** $f(x,y)$.
- To find the probability of $(X,Y) \in A$ we integrate over A:

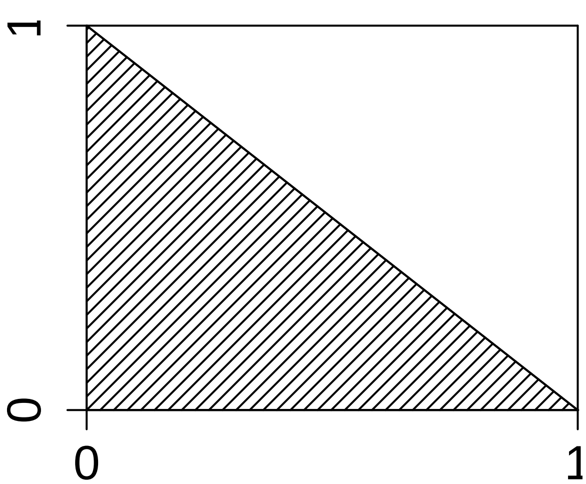
$$P((X,Y) \in A) = \iint_A f(x,y)dxdy.$$

- **Example:** Suppose that (X,Y) have the joint density

$$f(x,y) = \begin{cases} 1, & 0 \le x, y \le 1 \\ 0, & \text{otherwise.} \end{cases}$$

Suppose we want to find the probability $P(X + Y \le 1)$. This means (X,Y) should belong to the set $A = \{(x,y) : x + y \le 1\}$. Thus,

$$P(X+Y \le 1) = \iint_A f(x,y)dxdy = \int_0^1 \int_0^{1-x} 1dydx = \int_0^1 [y]_0^{1-x} = \int_0^1 (1-x)dx = [-\tfrac{1}{2}(1-x)^2]_0^1 = \tfrac{1}{2}.$$



5.4 Marginal distributions and independence

- Let (X, Y) be a pair of continuous variables with joint density function $f(x, y)$. Then the **marginal density functions** for X and Y are found by the formula

$$f(x) = \int_{-\infty}^{\infty} f(x, y)dy, \qquad g(y) = \int_{-\infty}^{\infty} f(x, y)dx.$$

- We say that X and Y are **independent** if

$$f(x, y) = f(x)g(y).$$

5.5 Covariance

- For two random variables, the dependence between them can be measured by the **covariance** between them. This is given by

$$\sigma_{XY} = E((X-\mu_X)(Y-\mu_Y)) = \sum_{(x,y)} (x-\mu_X)(y-\mu_Y)f(x, y), \sigma_{XY} = E((X-\mu_X)(Y-\mu_Y)) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} (x-\mu_X)(y-\mu_Y)f(x$$

  in the discrete and continuous case, respectively.

- Properties:
  - $\sigma_{XY} > 0$ indicates that the values of X tend to be large when Y is large and X tends to be small when Y is small.
  - $\sigma_{XY} < 0$ indicates that the values of X tend to be large when Y is small and small when Y is large.
  - If X and Y are statistically independent, then $\sigma_{XY} = 0$.
  - If $\sigma_{XY} = 0$ it is not guaranteed that X and Y are independent!
- Apart from this, the values of σ_{XY} are hard to interpret since they depend on the units that X and Y are measured in.

5.6 Correlation

- To obtain a unit free version of the covariance, we define the **correlation coefficient**

$$\rho_{XY} = \frac{\sigma_{XY}}{\sigma_X \sigma_Y}.$$

  This can be thought of as the covariance when X and Y are measured in standard deviation units.

- Properties:
  - $-1 \le \rho_{XY} \le 1$.
  - $\rho_{XY} = 1$ means one of the variables is linearly determined by the other, say $Y = a + bX$, where the slope $b > 0$.
  - $\rho_{XY} = -1$ means one of the variables is linearly determined by the other, say $Y = a + bX$, where the slope $b < 0$.
  - If X and Y are independent, then $\rho_{XY} = 0$. Again, one cannot conclude that X and Y are independent if $\rho_{XY} = 0$.
- More on correlation in Module 3.

6 Important distributions

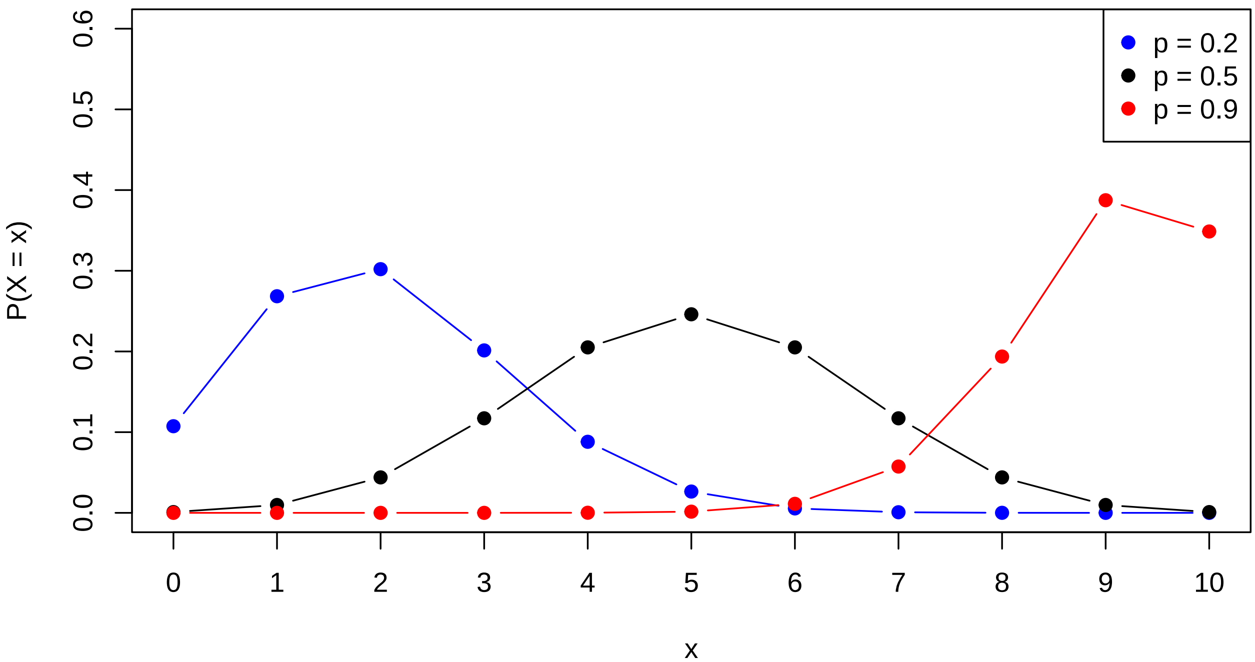
6.1 The binomial distribution

- The **binomial distribution:** An experiment with two possible outcomes (success/failure) is repeated n times, each independent of each other and with probability p of success.
- Let X be the number of successes. Then X can take the values $0, 1, \dots, n$.
- X follows a binomial distribution, denoted $X \sim \text{binom}(n, p)$ (or $\text{Bin}(n, p)$).
  - **Example:** Flip a coin n times. In each flip, the probability of head is $p = \frac{1}{2}$. Let X be the number of heads. Then $X \sim \text{binom}(n, 1/2)$
  - **Example:** We buy n items of the same type. Each has probability p of being defect. Let X be the number of defect items. Then $X \sim \text{binom}(n, p)$

Probability function for binomial distribution, $\binom{n}{x}$ is the binomial coefficient:

$$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, \dots, n$$

Graph of probability functions for binomial distributions with $n = 10$:



7 The normal distribution

7.1 Definition of the normal distribution

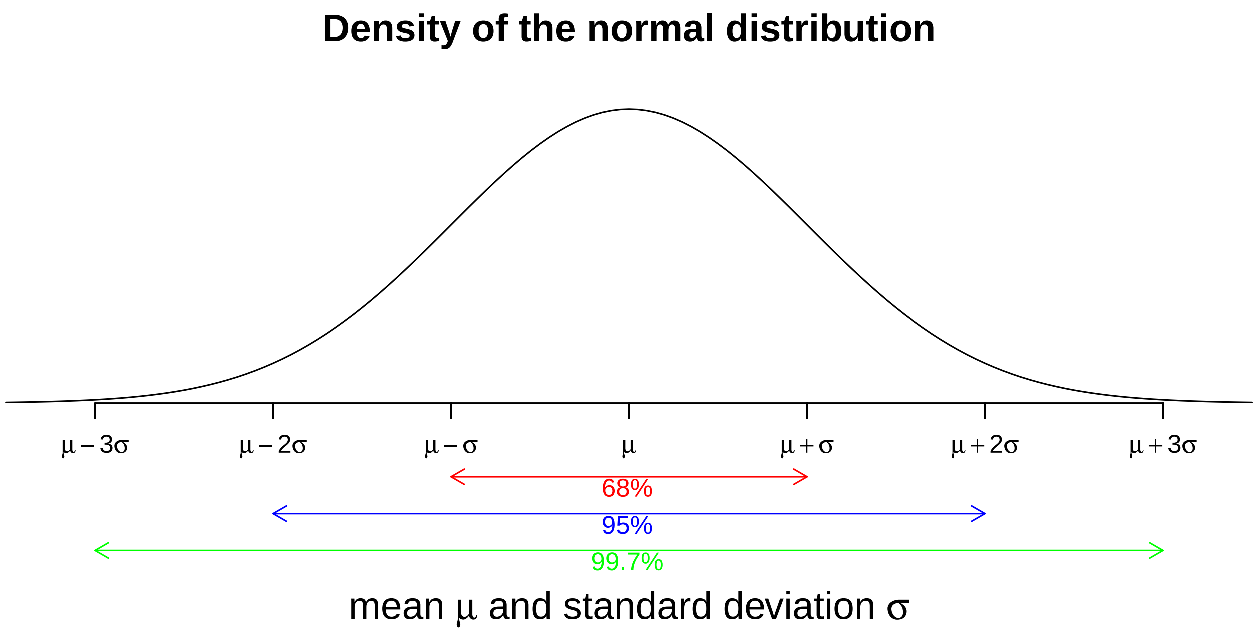
- The **normal distribution** is a continuous distribution with probability density function

$$n(x; \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}.$$

- It depends on two parameters:
  - The mean μ
  - The standard deviation σ
- When a random variable Y follows a normal distribution with mean μ and standard deviation σ, we write $Y \sim \texttt{norm}(\mu, \sigma)$.

7.2 The normal distribution - interpretation of parameters

- The probability density function of a normal distribution is a symmetric bell-shaped curve centered around μ.



- Interpretation of standard deviation:
  - $\approx$ 68% of the population is within 1 standard deviation of the mean.
  - $\approx$ 95% of the population is within 2 standard deviations of the mean.
  - $\approx$ 99.7% of the population is within 3 standard deviations of the mean.

7.3 Normal z-scores

- The normal distribution with mean $\mu = 0$ and standard deviation $\sigma = 1$ is called the **standard normal distribution**.
- If $Y \sim \texttt{norm}(\mu, \sigma)$ then the corresponding **z-score** is

$$Z = \frac{Y - \mu}{\sigma}$$

- Interpretation: Z is the number of standard deviations that Y is away from the mean, where a negative value tells that we are below the mean.
- We have that $Z \sim \texttt{norm}(0, 1)$, i.e. Z follows a standard normal distribution.
- This implies that

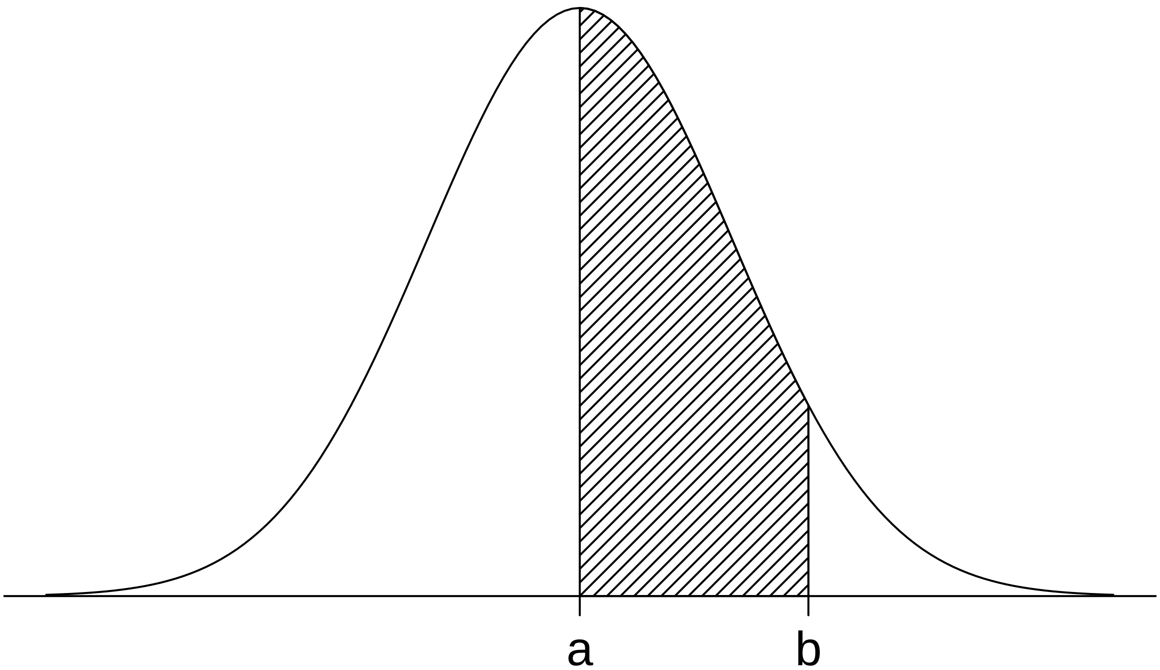
  - Z lies between -1 and 1 with probability 68%
  - Z lies between -2 and 2 with probability 95%
  - Z lies between -3 and 3 with probability 99.7%

- It also implies that:

  - The probability of Y being between $\mu - z\sigma$ and $\mu + z\sigma$ is equal to the probability of Z being between $-z$ and z.

7.4 Probabilities in a normal distribution

- To find the probabilities $P(a < X < b)$ in a normal distribution, we need to find the area under the density curve:



- This is given by

$$P(a < X < b) = \int_a^b \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx.$$

- This integral cannot be computed by hand!

7.5 Getting started with R

- To calculate normal probabilities in R we use the `mosaic` package.
- The first time you use the `mosaic` package, you need to install it first. This is done via the command:

```
install.packages("mosaic")
```

- At the beginning of each new R session you need to load it through the `library` command:

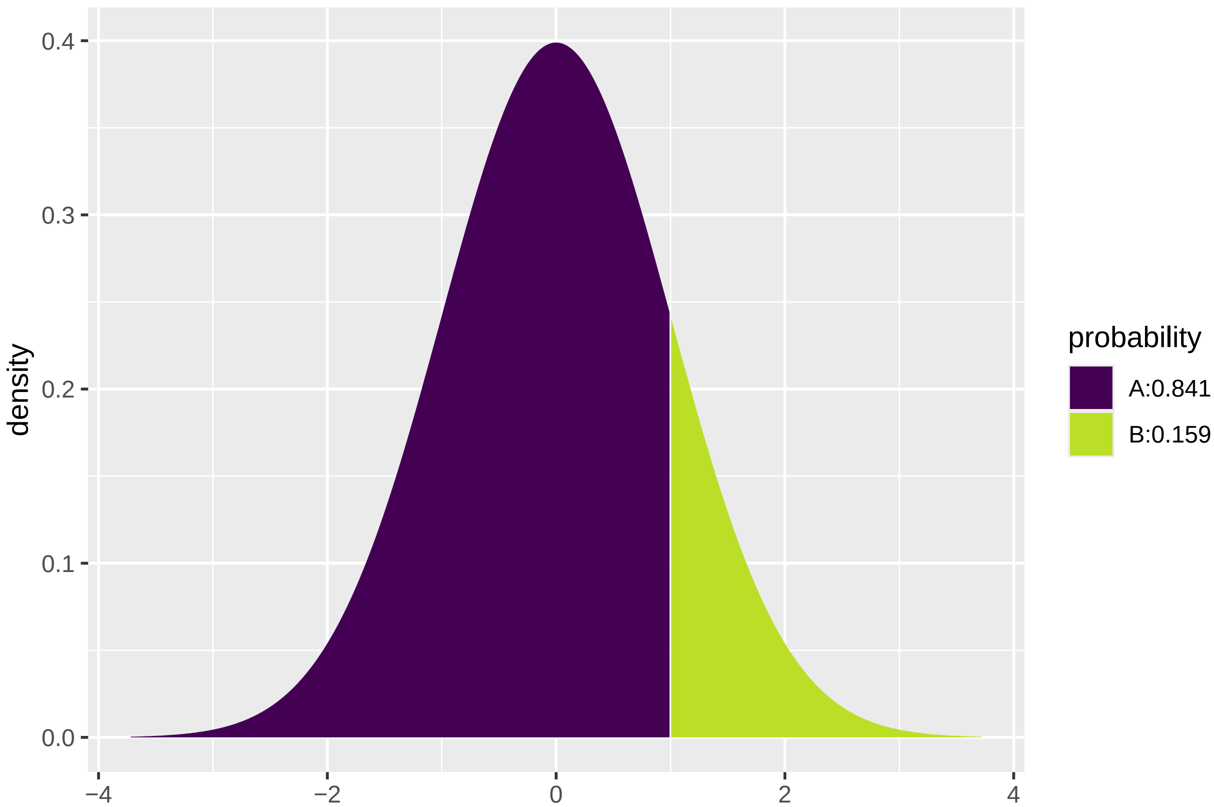
```
library(mosaic)
```

- This loads the `mosaic` package which has a lot of convenient functions for this course (we will get back to that later). It also prints a lot of info about functions that have been changed by the `mosaic` package, but you can safely ignore that.

7.6 Computing probabilities in a normal distribution

- To find the probability $P(X \leq q)$ when $X \sim \texttt{norm}(\mu, \sigma)$, we use the `pdist` function in **R**.
- For instance with $q = 1$, $\mu = 0$ and $\sigma = 1$, we type

```
# For a standard normal distribution the probability of getting a value less than 1 is:
pdist("norm", q = 1, mean = 0, sd = 1)
```




```
## [1] 0.84
```

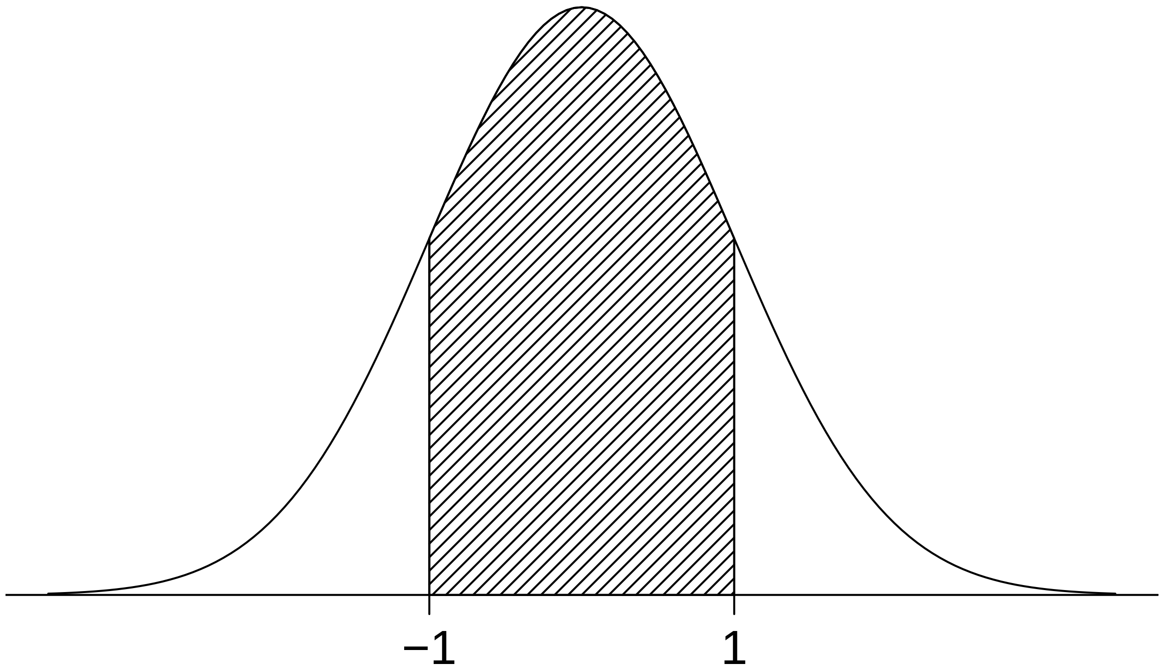
- The output is always the probability of being *to the left* of q, which is marked as the purple area.
- To get the probability of being *to the right* of q, we compute

$$P(X > q) = 1 - P(X \leq q) = 1 - 0.8413447 = 0.1586553.$$

- We can also get the probability of an observation lying between -1 and 1 by

$$P(-1 \leq X \leq 1) = 1 - P(X > 1) - P(X < -1) = 1 - 2 \cdot 0.1587 = 0.683,$$

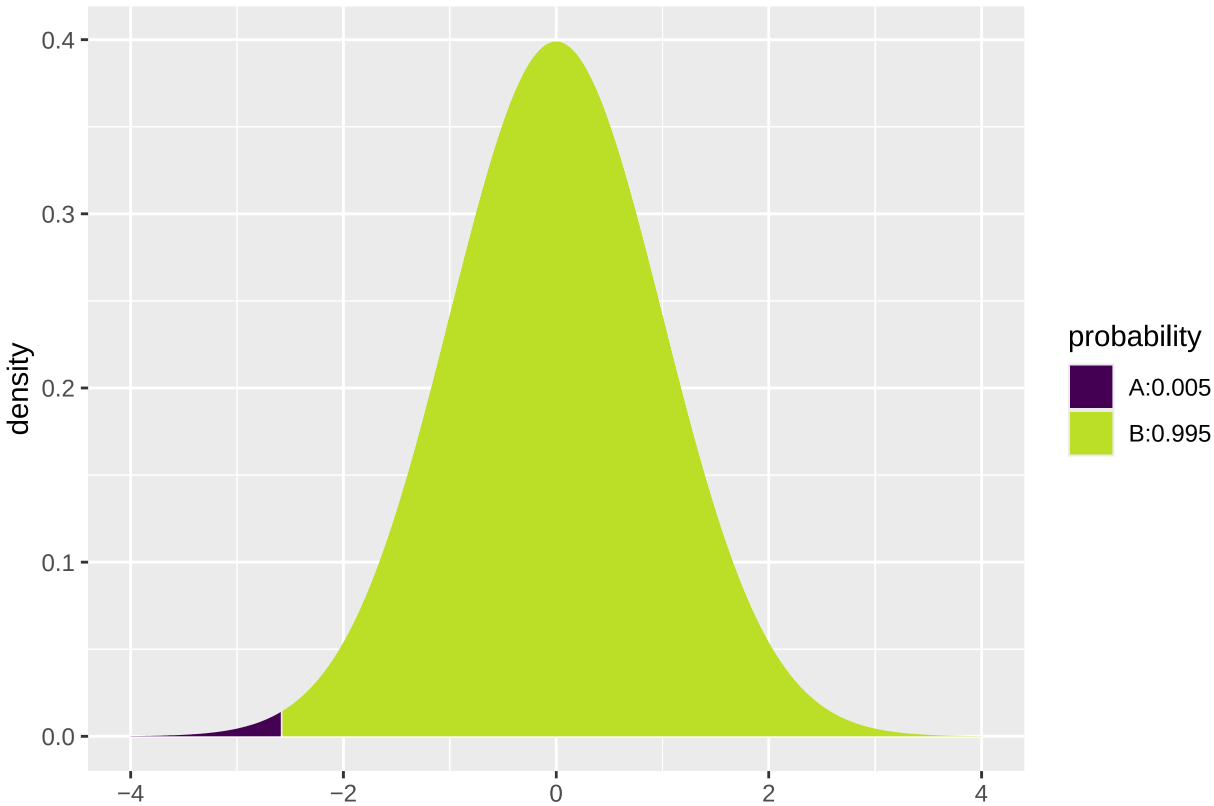
  where we used that by symmetry of the normal curve, $P(X > 1) = P(X < -1)$.



7.7 Calculating z-values in the standard normal distribution

- We can also go in the other direction using `qdist`: Given a probability p, find the value z such that $P(X \leq z) = p$ when $X \sim \texttt{norm}(\mu, \sigma)$.
- For instance with $p = 0.005$, $\mu = 0$ and $\sigma = 1$:

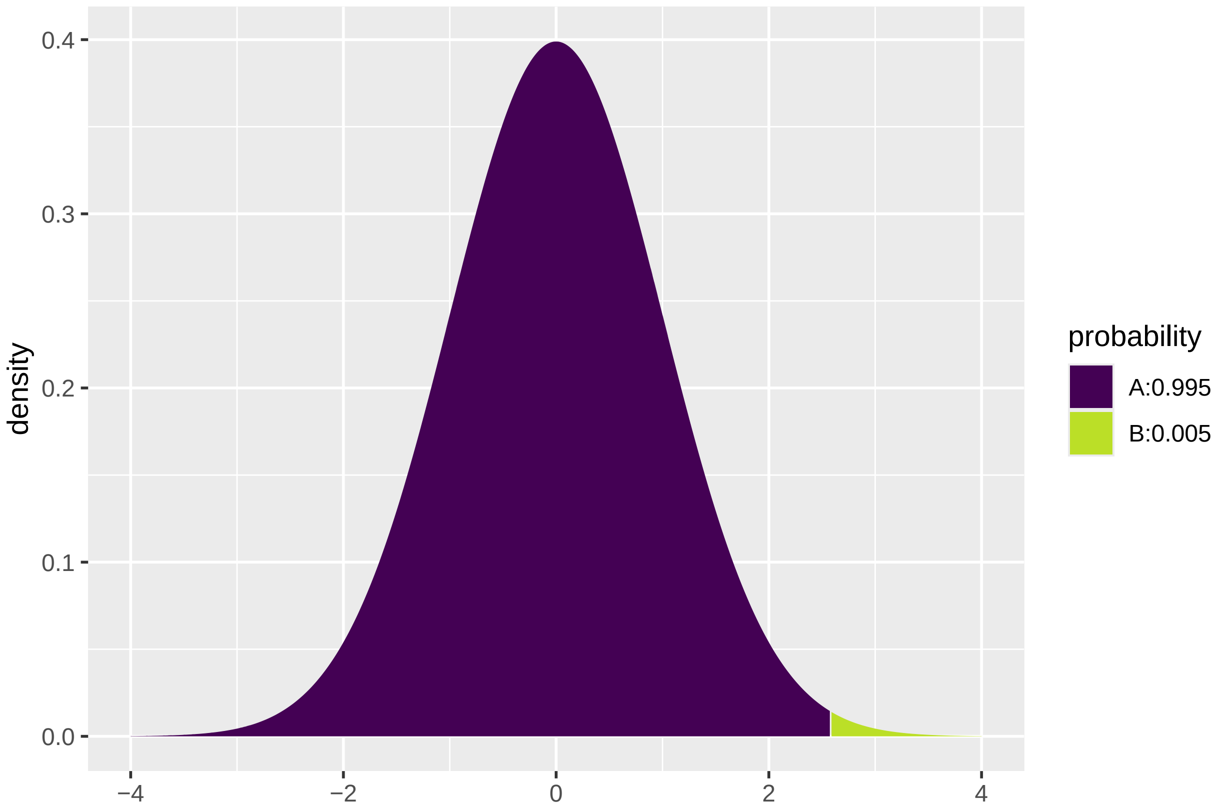
```
qdist("norm", p = 0.005, mean = 0, sd = 1, xlim = c(-4, 4))
```




```
## [1] -2.6
```

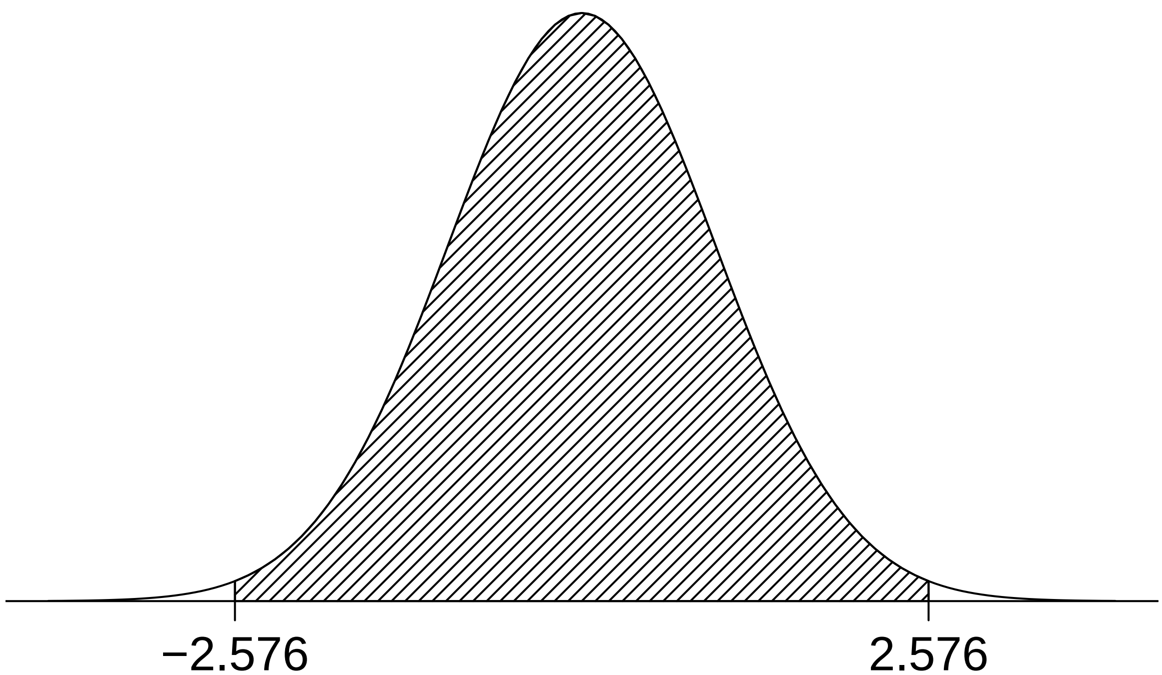
- Sometimes we want to find z such that $P(X > z) = p$. Since this is the same as $P(X \leq z) = 1 - p$, we may do as follows:

```
qdist("norm", p = 1-0.005, mean = 0, sd = 1, xlim = c(-4, 4))
```




```
## [1] 2.6
```

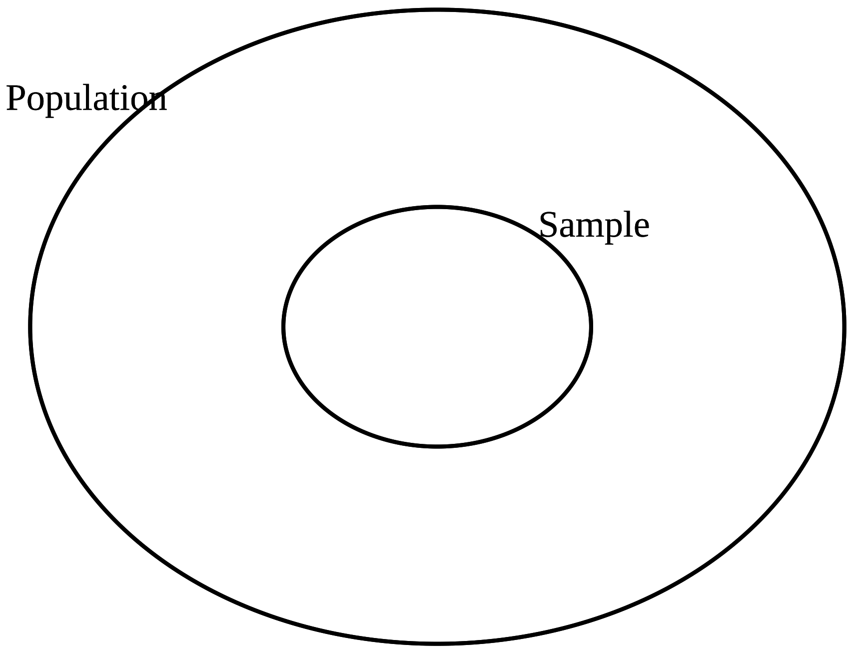
- Thus, the probability of an observation between -2.576 and 2.576 equals $1 - 2 \cdot 0.005 = 99\%$.



8 Sampling

8.1 Population and sample

- In statistics, the word **population** refers to the collection of all the objects we are interested in.
- **Examples:**
  - The Danish population
  - All possible outcomes of a lab experiment
- A **sample** consists of finitely many elements selected randomly and independently of each other from the population.
- **Examples:**
  - People selected for an opinion poll
  - The experiments we actually carried out



8.2 Sampling principles

- If we draw a random element from the population, the result will be a random variable X with a certain distribution.
- When we sample, we draw n elements from the population *independently* of each other. This results in n independent random variables $X_1, \ldots, X_n$, each having the *same distribution* as X.
- Sampling principles:
  - Independence: If you make experiments in the lab, reusing parts of an experiment for the next one might cause dependence between outcomes. If you measure the same quantity at a sequence of time points, measurements close in time are typically not independent.
  - Same distribution as the population: If we only go out and make weather measurements when the weather is good, our sample does not have the same distribution as measurements from any randomly selected day.

- Note: We use capital letters $X_1, \dots, X_n$ to indicate that the elements of the sample are random and small letters $x_1, \dots, x_n$ to denote the values that are actually observed in the experiment. These values are called **observations**.

8.3 Statistical inference

- **Statistical inference** means drawing conclusions about the population based on the sample.
- Typically, we want to draw conclusions about some parameters of the population, e.g. mean μ and standard deviation σ.
- Note: The number of elements n in the sample is called the **sample size**. In general: the larger n, the more precise conclusions we can draw about the population.

8.4 Sample proportion

- Consider an experiment with two possible outcomes, e.g. flipping a coin or testing whether a component is defect or not.
- Call the two outcomes 0 and 1. We are interested in the probability p of getting the outcome 1.
- Given a sample $X_1, \dots, X_n$, we estimate p by

$$\hat{P} = \frac{\text{number of 1's among } X_1, \dots, X_n}{n} = \frac{\sum_{i=1}^n X_i}{n}.$$

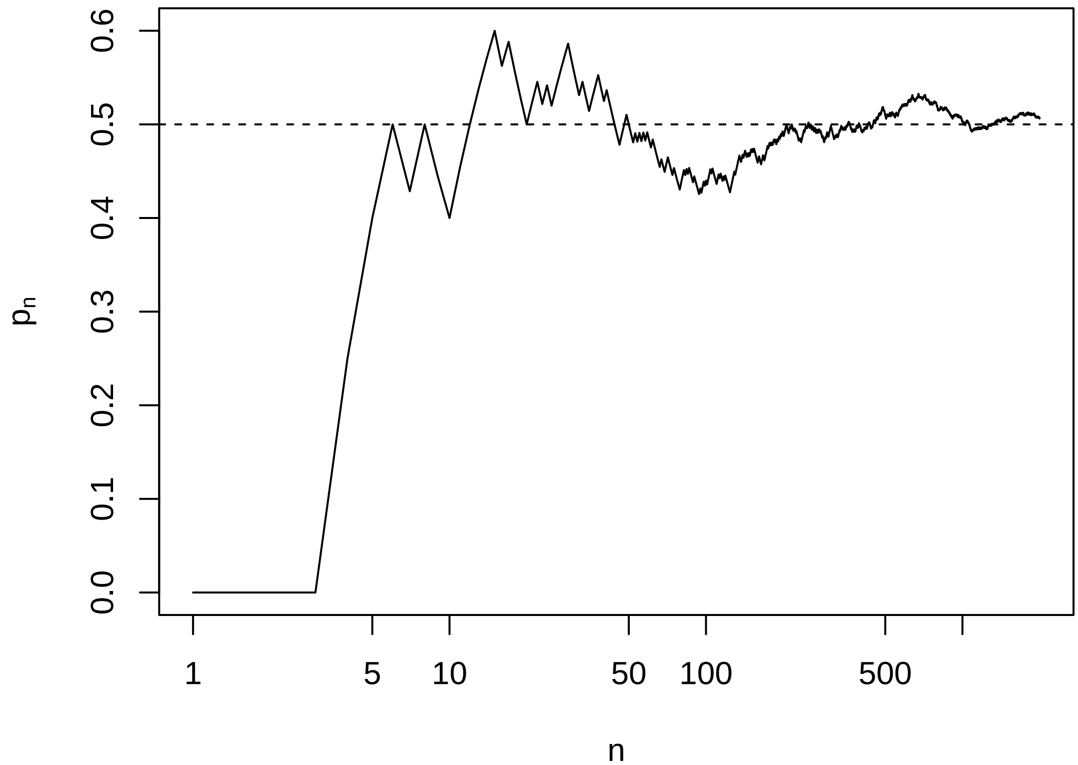
- $\hat{P}$ is a so-called **summary statistics**, i.e. a function of the sample.
- Since $\hat{P}$ is a function of the random sample $X_1, \dots, X_n$, $\hat{P}$ is itself a random variable. Different samples may lead to different values of $\hat{P}$.
- $E(\hat{P}) = p$.
- $\lim_{n\to\infty} \hat{P} = p$.

8.5 A real experiment

- John Kerrich, a South African mathematician, was visiting Copenhagen when World War II broke out. Two days before he was scheduled to fly to England, the Germans invaded Denmark. Kerrich spent the rest of the war interned at a camp in Hald Ege near Viborg, Jutland. To pass the time he carried out a series of experiments in probability theory. In one, he tossed a coin 10,000 times.
- The first 25 observations were (0 = tail, 1 = head):

$$0, 0, 0, 1, 1, 1, 0, 1, 0, 0, 1, 1, 1, 1, 1, 0, 1, 0, 0, 0, 1, 1, 0, 1, 0, \dots$$

- Plot of the empirical probability $\hat{p}$ of getting a head against the number of tosses n:



(The horizontal axis is on a log scale).

8.6 Sample mean

- Suppose we are interested in the mean value μ of a population and we have drawn a random sample $X_1, \ldots, X_n$.
- Based on the sample we estimate μ by the **sample mean**, which is the average of all the elements

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

- Properties:
  - $\bar{X}$ is random, as it depends on the random sample $X_1, \ldots, X_n$. Different samples might result in different values of $\bar{X}$.
  - $E(\bar{X}) = \mu$.
  - $\bar{X}$ has standard deviation $\frac{\sigma}{\sqrt{n}}$, where σ is the population standard deviation. Note that increasing n decreases $\frac{\sigma}{\sqrt{n}}$.
  - To distinguish between the standard deviation of the population and the standard deviation of $\bar{X}$, we call the standard deviation of $\bar{X}$ the **standard error**.
  - $\lim_{n\to\infty} \bar{X} = \mu$.

8.7 Central limit theorem

- When the population distribution is a normal distribution $\texttt{norm}(\mu, \sigma)$, then

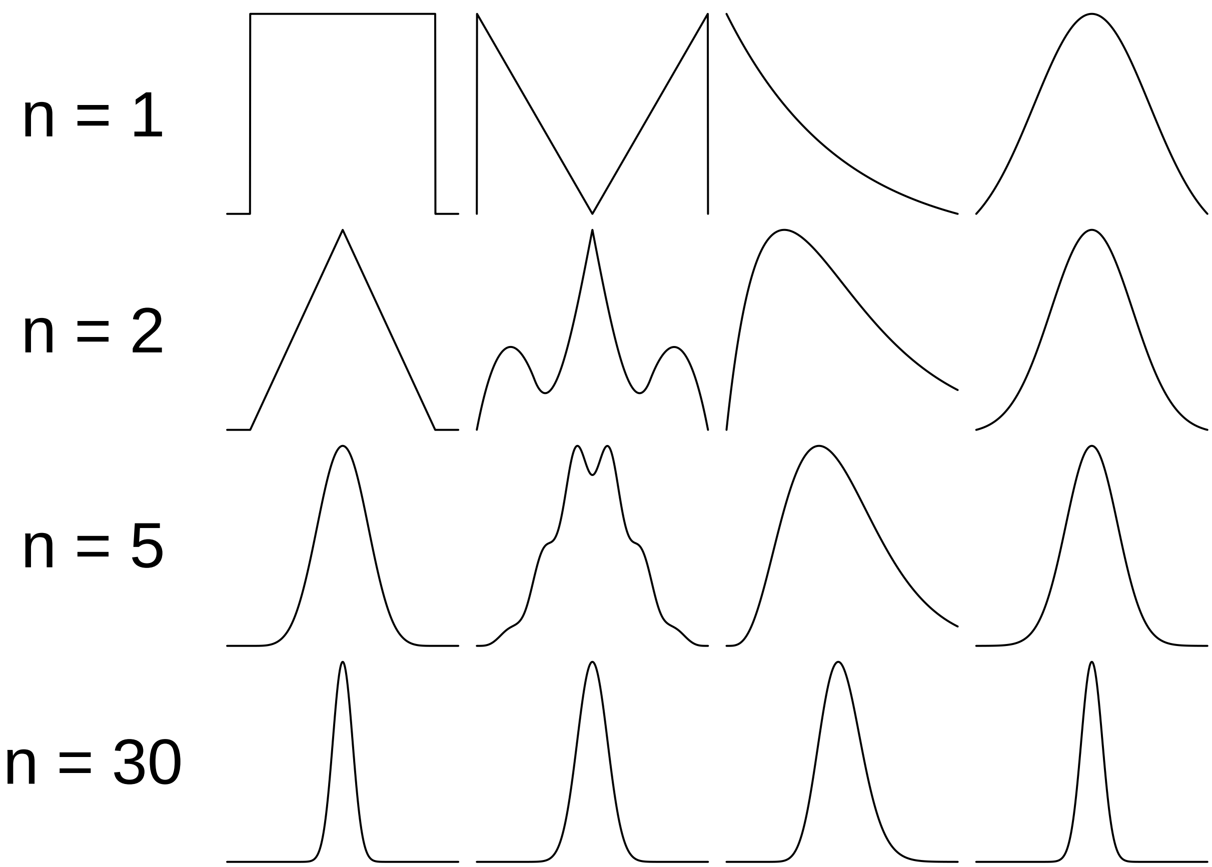
$$\bar{X} \sim \texttt{norm}\left(\mu, \frac{\sigma}{\sqrt{n}}\right).$$

- For any population distribution, the **central limit theorem** states:
  - When n goes to ∞, the distribution of $\bar{X}$ approaches a normal distribution with mean μ and standard deviation $\frac{\sigma}{\sqrt{n}}$. Thus, for large n,

$$\bar{X} \approx \texttt{norm}\left(\mu, \frac{\sigma}{\sqrt{n}}\right).$$

  - The corresponding z-score $Z = \frac{\bar{X}-\mu}{\sigma/\sqrt{n}}$ follows a standard normal distribution $\texttt{norm}(0, 1)$.
- As a rule of thumb, n is large enough when $n \geq 30$.

8.8 Illustration of CLT



- The top row shows 4 different population distributions. The plots below show the distribution of $\bar{X}$ when $n = 2$, 5, and 30.

8.8.1 Example: use of CLT

- A company produces cylindrical components for automobiles. It is important that the mean component diameter is $\mu = 5$mm. The standard deviation is $\sigma = 0.1$mm.
- An engineer takes a random sample of $n = 100$ components. These have an average diameter of $\bar{x} = 5.027$. Is it reasonable to think $\mu = 5$?
- If the population of components has the correct mean, then

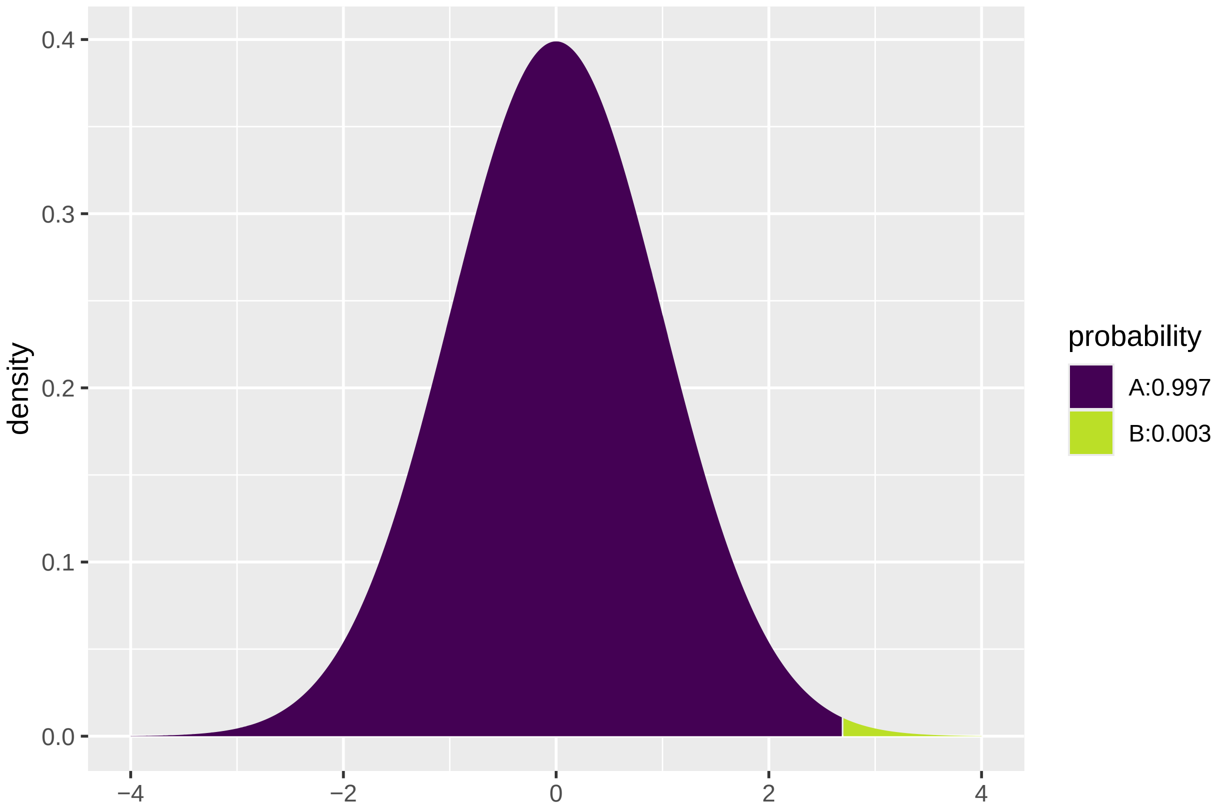
$$\bar{X} \approx \texttt{norm}\left(\mu, \frac{\sigma}{\sqrt{n}}\right) = \texttt{norm}\left(5, \frac{0.1}{\sqrt{100}}\right) = \texttt{norm}(5, 0.01).$$

- For the actual sample this gives the observed z-score

$$z_{obs} = \frac{\bar{x} - 5}{0.01} = 2.7$$

which should come from an approximate standard normal distribution.
- The probability of getting a higher z-score is:

```
1 - pdist("norm", mean = 0, sd = 1, q = 2.7, xlim = c(-4, 4))
```




```
## [1] 0.0035
```

- Thus, it is highly unlikely that a random sample has such a high z-score. A better explanation might be that the produced components have a population mean larger than 5mm.

8.9 Sample variance and standard deviation

- Suppose we are interested in the variance σ^2 of a population and we have drawn a random sample $X_1, \ldots, X_n$.

- Based on the sample we estimate the population variance σ^2 by the **sample variance**, which is

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2.$$

- We estimate the population standard deviation σ by the **sample standard deviation**

$$S = \sqrt{S^2}.$$

- Properties:
  - S^2 is again a random variable.
  - $E(S^2) = \sigma^2$.

8.10 z-scores for the sample mean

- According to the central limit theorem $\bar{X} \approx \texttt{norm}(\mu, \frac{\sigma}{\sqrt{n}})$ when the population follows a normal distribution or n is large.
- The corresponding z-score $Z = \frac{\bar{X}-\mu}{\sigma/\sqrt{n}}$ follows a standard normal distribution $\texttt{norm}(0, 1)$.
- Problem: We don't know σ.
- We may insert the sample standard deviation to get the t**-score**

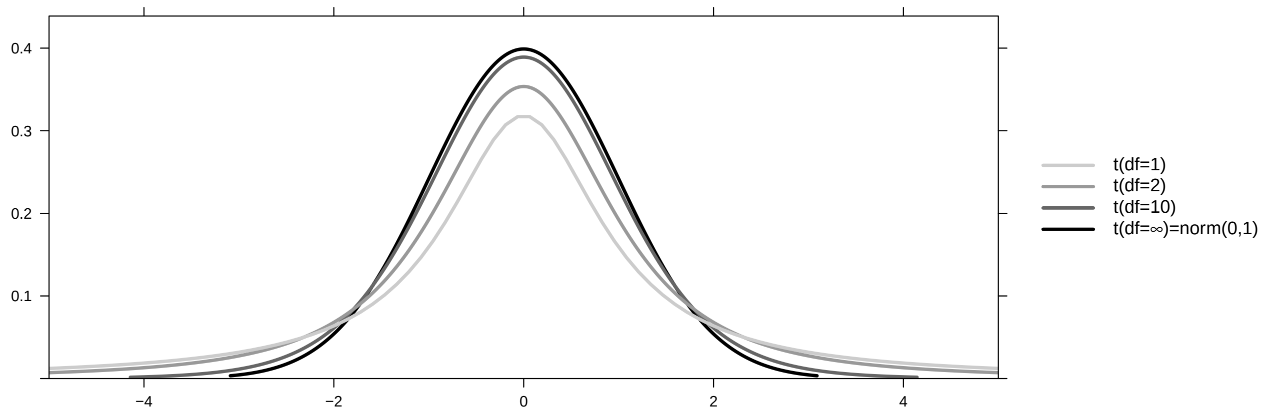
$$T = \frac{\bar{X}-\mu}{S/\sqrt{n}}.$$

- Since S is random with a certain variance, this causes T to vary more than Z.
- As a consequence, T no longer follows a normal distribution, but a t**-distribution** with $n-1$ degrees of freedom.

8.11 t-distribution and t-score

- The t**-distribution** is very similar to the standard normal distribution:
  - it is symmetric around zero and bell shaped, but
  - it has "heavier" tails and thereby
  - a slightly larger standard deviation than the standard normal distribution.
  - Further, the t-distribution's standard deviation decays as a function of its **degrees of freedom**, which we denote df,
  - and when df grows, the t-distribution approaches the standard normal distribution.

The expression of the density function is of slightly complicated form and will not be stated here, instead the t-distribution is plotted below for $df = 1, 2, 10$ and ∞.



9 Introduction to R

9.1 Rstudio

- Make a folder on your computer where you want to keep files to use in **Rstudio**. **Do NOT use special characters like æ, ø, å** in the folder name (or anywhere in the path to the folder).
- Set the working directory to this folder: `Session -> Set Working Directory -> Choose Directory` (shortcut: Ctrl+Shift+H).
- Make the change permanent by setting the default directory in: `Tools -> Global Options -> Choose Directory`.

9.2 R basics

- Ordinary calculations:

```
4.6 * (2 + 3)^4
```

```
## [1] 2875
```

- Make a (scalar) object and print it:

```
a <- 4
a
```

```
## [1] 4
```

- Make a (vector) object and print it:

```
b <- c(2, 5, 7)
b
```

```
## [1] 2 5 7
```

- Make a sequence of numbers and print it:

```
s <- 1:4
s
```

```
## [1] 1 2 3 4
```

- Note: A more flexible command for sequences:

```
s <- seq(1, 4, by = 1)
```

- **R** does elementwise calculations:

```
a * b
```

```
## [1]  8 20 28
```

```
a + b
```

```
## [1]  6  9 11
```

```
b ^ 2
```

```
## [1]  4 25 49
```

- Sum and product of elements:

```
sum(b)
```

```
## [1] 14
```

```
prod(b)
```

```
## [1] 70
```

9.3 R markdown

- The slides and all exercises in R (including the exam questions) are made in the special Rmarkdown format.
- This allows you to combine text and R code.
- You can write formulas using standard LaTeX commands.

9.4 R extensions

- The functionality of **R** can be extended through libraries or packages (much like plugins in browsers etc.). Some are installed by default in **R** and you just need to load them.
- To install a new package in **Rstudio** use the menu: `Tools -> Install Packages`
- You need to know the name of the package you want to install. You can also do it through a command:

```
install.packages("mosaic")
```

- When it is installed you can load it through the `library` command:

```
library(mosaic)
```

- This loads the `mosaic` package which has a lot of convenient functions for this course (we will get back to that later). It also prints a lot of info about functions that have been changed by the `mosaic` package, but you can safely ignore that.

9.5 R help

- You get help via `?<command>`:

```
?sum
```

- Use `tab` to make **Rstudio** guess what you have started typing.
- Search for help:

```
help.search("plot")
```

- You can find a cheat sheet with the **R** functions we use for this course here.
- Save your commands in a file for later usage:
  - Select history tab in top right pane in **Rstudio** .
  - Mark the commands you want to save.
  - Press `To Source` button.

10 Data in R

10.1 Data example

- Chile dataset in R is a data frame with 2700 rows and 8 columns from the 1988 plebiscite in Chile for or against Pinochet to continue for another eight years as leader. The sample consists of voting intentions for voters from the Chilean population. There are missing values in the dataset.

- The data set contains the variables:

  - `region`: The region in Chile where the voter lives
  - `population`: Population of the region.
  - `sex`: The gender of the voter.
  - `age`: The age of the voter.
  - `education`: Education level of the voter (primary, secondary, post-secondary).
  - `income`: Monthly income of the voter.
  - `statusquo`: To which degree the voter supports the status quo (numbers ranging from about -2 to 2).
  - `vote`: Should Pinochet continue? `Y` = yes, `N`= no, `U`=undecided, `A`= will abstain from voting.

Note: The referendum was held in Chile on 5 October 1988. The "No" side won with 56% of the vote. Democratic elections were held in 1989, leading to the establishment of a new government in 1990.

- More information about the data set may be found here.

```
Chile <- read.delim("https://asta.math.aau.dk/datasets?file=Chile.txt")
head(Chile)
```

```
##   region population sex age education income statusquo vote
## 1      N     175000   M  65         P  35000      1.01    Y
## 2      N     175000   M  29        PS   7500     -1.30    N
## 3      N     175000   F  38         P  15000      1.23    Y
## 4      N     175000   F  49         P  35000     -1.03    N
## 5      N     175000   F  23         S  35000     -1.10    N
## 6      N     175000   F  28         P   7500     -1.05    N
```

10.2 Data types

10.2.1 Quantitative variables

- The measurements have numerical values.
- Quantative data often comes about in one of the following ways:
  - **Continuous variables**: measurements of e.g. speed, temperature, etc.

- **Discrete variables**: counts of e.g. number of household members, hits on a webpage, cars passing on a road in one hour, etc.
- Measurements like this have a well-defined scale and in **R** they are stored as the type **numeric**.
- It is important to be able to distinguish between discrete count variables and continuous variables, since this often determines how we describe the uncertainty of a measurement.

10.2.2 Categorical/qualitative variables

- The measurement is one of a set of given categories, e.g. sex (male/female), education level, satisfaction score (low/medium/high), etc.
- Factors have two so-called scales:
  - **Nominal scale**: There is no natural ordering of the factor levels, e.g. sex and hair color.
  - **Ordinal scale**: There is a natural ordering of the factor levels, e.g. education level and satisfaction score.
- The measurement is usually stored (which is also recommended) as a **factor** in **R**. The possible categories are called **levels**. Example: the levels of the factor "sex" is male/female. A factor in **R** can have a so-called **attribute** assigned, which tells if it is ordinal.

10.3 Variables in the data set

```
head(Chile)
```

```
##   region population sex age education income statusquo vote
## 1      N     175000   M  65         P  35000      1.01    Y
## 2      N     175000   M  29        PS   7500     -1.30    N
## 3      N     175000   F  38         P  15000      1.23    Y
## 4      N     175000   F  49         P  35000     -1.03    N
## 5      N     175000   F  23         S  35000     -1.10    N
## 6      N     175000   F  28         P   7500     -1.05    N
```

- Quantitative variables in the `Chile` data set:
  - `population`, `age`, `income`, `statusquo`
- Categorical variables:
  - `region`, `sex`, `education`, `vote`
- All the categorical variables are nominal except `education`, which has three ordered categories (primary, secondary, post-secondary).

11 Descriptive statistics of categorical data

11.1 Tables

- To summarize the the variable `vote` we can use the function `tally` from the `mosaic` package (remember the package **must be loaded** via `library(mosaic)` if you did not do so yet):

```
tally( ~ vote, data = Chile)
```

```
## vote
##    A    N    U    Y <NA>
##  187  889  588  868  168
```

- In percent:

```
tally( ~ vote, data = Chile, format = "percent")
```

```
## vote
##     A     N     U     Y  <NA>
##  6.93 32.93 21.78 32.15  6.22
```

- Here we use an **R** `formula` (characterized by the "tilde" sign `~`) to indicate that we want this variable from the dataset `Chile` (without the tilde it would look for a global variable called `vote` and use that rather than the one in the dataset).

11.2 2 factors: Cross tabulation

- To get an overview over the relation between two categorical variables, we can make a cross tabulation.
- To make a table of all combinations of the two factors `vote` and `sex`, we use `tally` again:

```
tally( ~ vote + sex, data = Chile)
```

```
##       sex
## vote     F   M
##   A    104  83
##   N    363 526
##   U    362 226
##   Y    480 388
##   <NA>  70  98
```

- We can also get the relative frequencies (in percent) columnwise:

```
tally( ~ vote | sex, data = Chile, format = "percent")
```

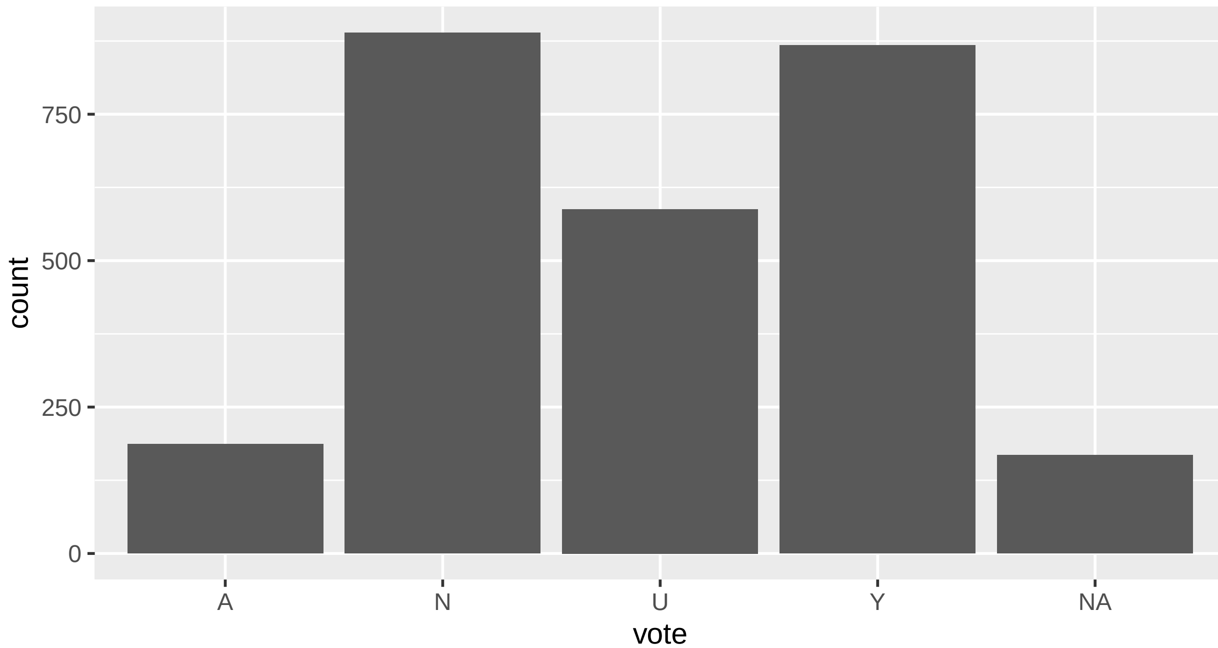
```
##       sex
## vote       F     M
##   A     7.54  6.28
##   N    26.32 39.82
##   U    26.25 17.11
##   Y    34.81 29.37
##   <NA>  5.08  7.42
```

- For instance we see that 34.8% of the women said they would vote yes, while this holds for only 29.4% of the men.

11.3 Visualizing categorical data: Bar graph

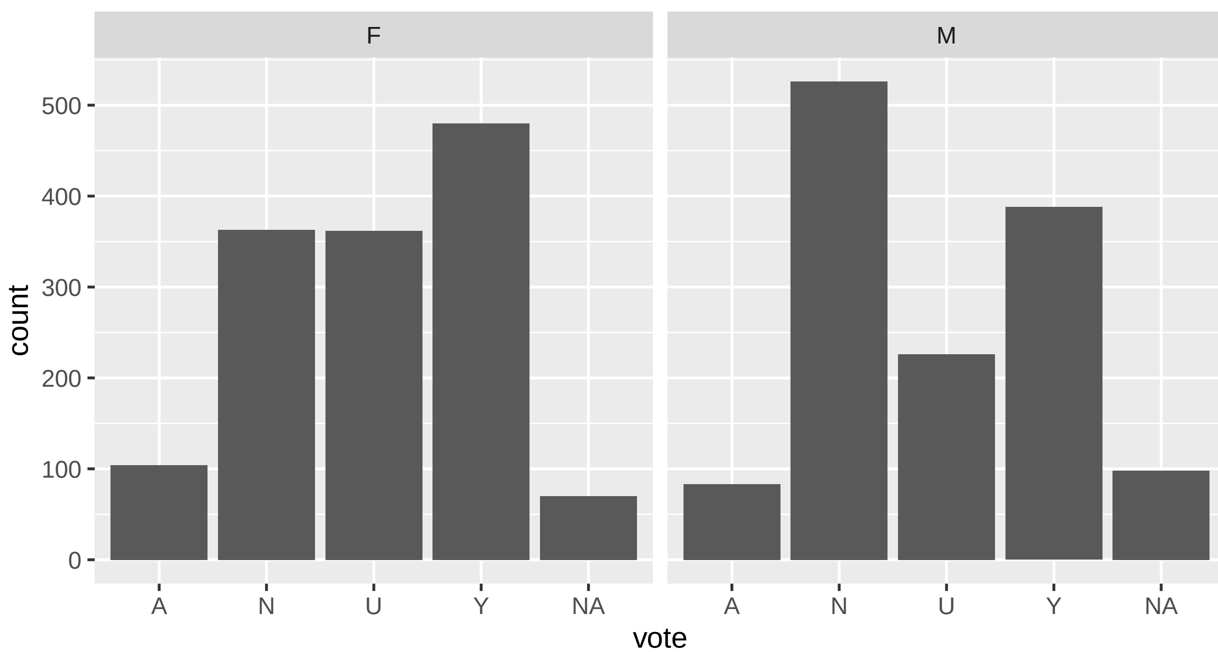
- To create a bar graph plot of table data we use the function `gf_bar` from `mosaic`. For each level of the factor, a box is drawn with the height proportional to the frequency (count) of the level.

```
gf_bar( ~ vote, data = Chile)
```



- The bar graph can also be split by gender:

```
gf_bar( ~ vote | sex, data = Chile)
```



12 Descriptive statistics of quantitative variables

12.1 Data example: Fuel consumption of cars

- The data was extracted from the 1974 *Motor Trend* US magazine, and comprises fuel consumption and 10 aspects of automobile design and performance for 32 automobiles (1973-74 models).

- The data set is built into **R** under the name `mtcars`, so it does not need to be loaded before use.
- A description of the variables can be found here: https://rstudio-pubs-static.s3.amazonaws.com/61800_faea93548c6b49cc91cd0c5ef5059894.html
- In particular: vs: cylinder configuration a V-shape (vs=0) or Straight Line (vs=1). am: automatic (am=0) or manual (am=1) transmission

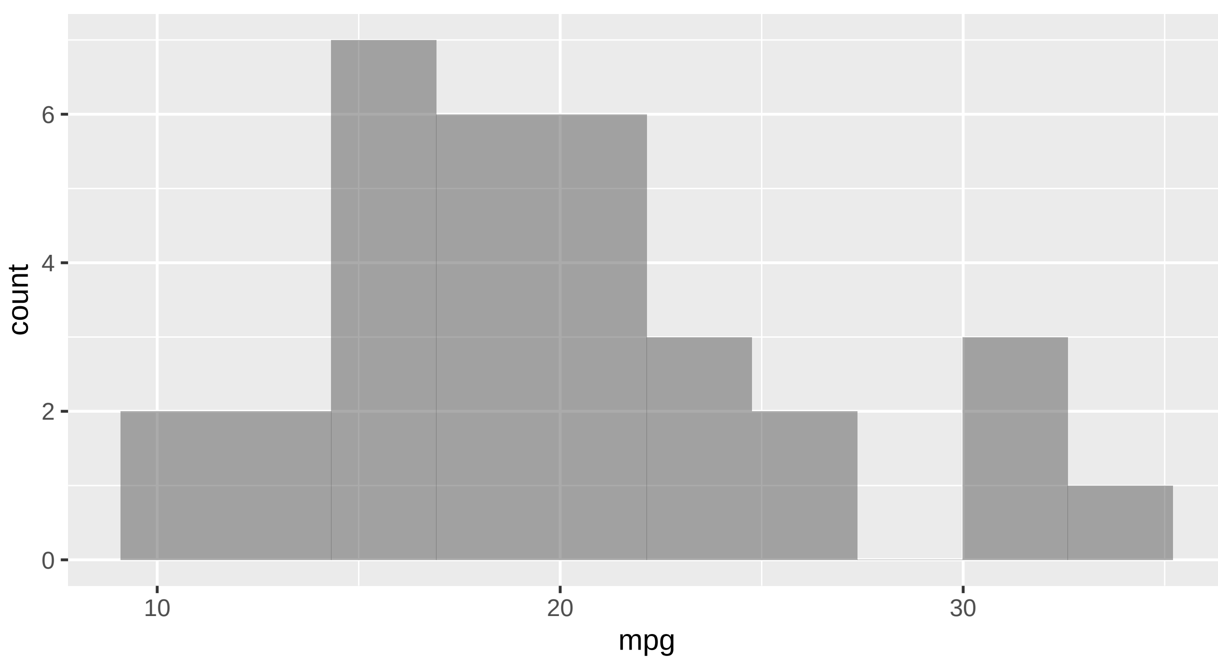
```
head(mtcars)
```

```
##                    mpg cyl disp  hp drat    wt  qsec vs am gear carb
## Mazda RX4         21.0   6  160 110 3.90 2.62 16.5  0  1    4    4
## Mazda RX4 Wag     21.0   6  160 110 3.90 2.88 17.0  0  1    4    4
## Datsun 710        22.8   4  108  93 3.85 2.32 18.6  1  1    4    1
## Hornet 4 Drive    21.4   6  258 110 3.08 3.21 19.4  1  0    3    1
## Hornet Sportabout 18.7   8  360 175 3.15 3.44 17.0  0  0    3    2
## Valiant           18.1   6  225 105 2.76 3.46 20.2  1  0    3    1
```

12.2 Visualizing quantitative data: Histogram

- The way to get a first impression of a quantitative variable is to draw a histogram.
- The histogram of a variable `x` is made as follows:
  - Divide the interval from the minimum value of `x` to the maximum value of `x` in an appropriate number of equal sized sub-intervals.
  - Draw a box over each sub-interval with the height being proportional to the number of observations in the sub-interval.
- Histogram of `mpg` for the `mtcars` data. The `bins` option sets the number of subintervals to 10.

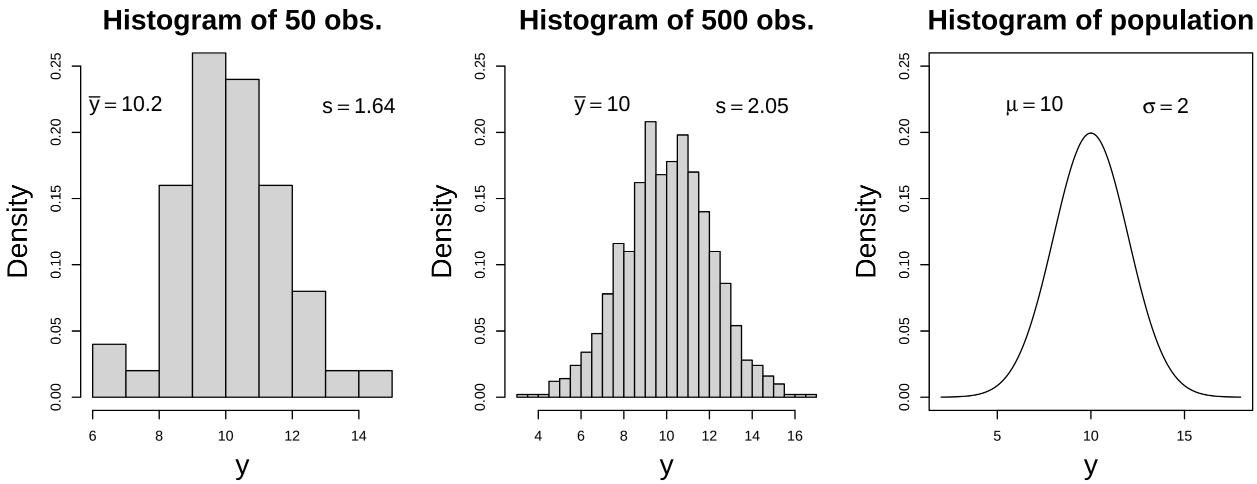
```
gf_histogram( ~ mpg, data=mtcars, bins=10)
```



12.3 Relation between histogram and denity function

- Suppose a sample comes from a population having a continuous distribution with density function f.

- Draw a histogram where the y-axis is scaled such that the total area of the bars is 1.
- When the number of observations (the sample size) increases we can make a finer interval division and get a more smooth histogram.
- When the number of observations tends to infinity, we obtain a nice smooth curve, where the area below the curve is 1. This curve is exactly the probability density function f.



- If the histogram looks bell-shaped this may suggest a normal distribution.

12.4 Summary statistics for quantitative data

- We return to the `mtcars` example. A summary of the fuel consumption `mpg` can be retrieved using the `favstats` function:

```
favstats( ~ mpg, data = mtcars)
```

```
##   min   Q1 median   Q3  max mean   sd  n missing
##  10.4 15.4   19.2 22.8 33.9 20.1 6.03 32       0
```

- The output contains the following information
  - **min** The minimal value in the sample is 10.4.
  - **max** The maximal value in the sample is 33.9.
  - **n** The sample size (number of observations) is 32.
  - **mean** The sample mean is 20.1. Recall that this was the average of all observations $x_1, \dots, x_n$, i.e.

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

  - **sd** The sample standard deviation is 6.03. Recall that this was given by

$$s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}.$$

  - **missing** There are no missing values.
  - **median** The median (or 50-percentile) is the value such that half of the sample has lower values than the median and half the sample has larger values.
  - **Q1** and **Q3** will be introduced on later slides.
- Both the mean and the median can be considered the center of a distribution. In a symmetric distribution (such as the normal distribution) they are equal, while in a skewed distribution, they tend to be different.

12.5 Calculation of mean, median and standard deviation using R

- The mean, median and standard deviation are just some of the summaries that can be read of the `favstats` output (shown on previous page). They may also be calculated separately in the following way:
- Sample size of `mpg`:

```
length(mtcars$mpg)
```

```
## [1] 32
```

- Mean of `mpg`:

```
mean( ~ mpg, data = mtcars)
```

```
## [1] 20.1
```

- Median of `mpg`:

```
median( ~ mpg, data = mtcars)
```

```
## [1] 19.2
```

- Standard deviation for `mpg`:

```
sd( ~ mpg, data = mtcars)
```

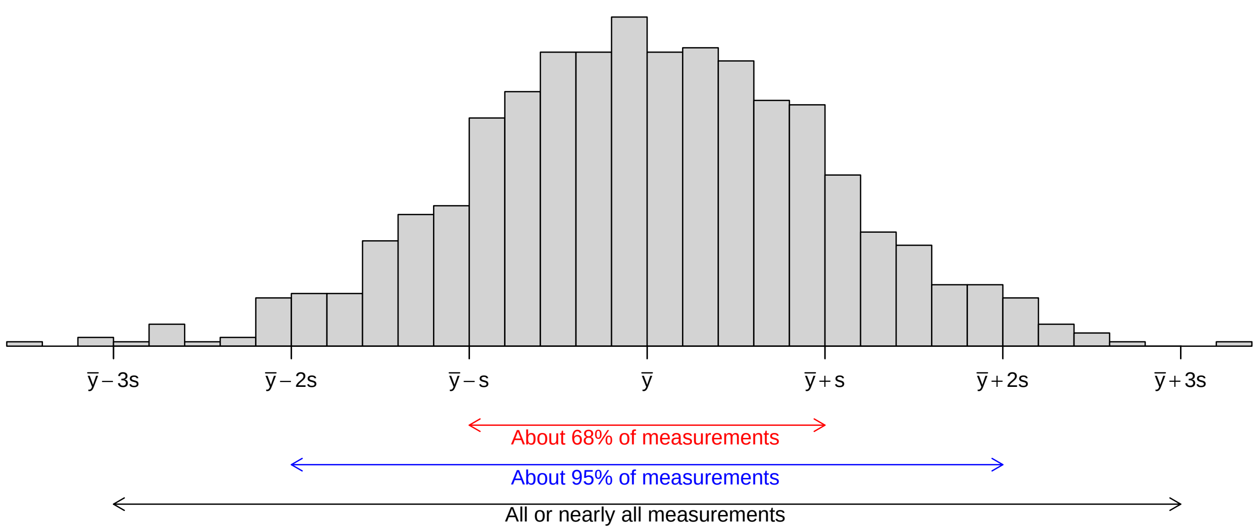
```
## [1] 6.03
```

- We may also calculate the summaries within groups. For instance, for each engine type (variable `vs`) the sample mean is:

```
mean( ~ mpg | factor(vs), data = mtcars)
```

```
##    0    1
## 16.6 24.6
```

12.6 Interpretation of summary statistics: The empirical rule



12.7 Very practical rules of thumb

If the histogram of the sample is unimodal approximately bell shaped, then

- The mean and median are approximately equal.

```
mean(mtcars$mpg)
```

```
## [1] 20.1
```

```
median(mtcars$mpg)
```

```
## [1] 19.2
```

And the median is easy to find: Sort data and locate the middle observation.

- about 95% of the observations lie between $\bar{y} - 2s$ and $\bar{y} + 2s$.

If we say that 95% of the observations are "all observations" then we get the very practical rule of thumb: The range of all or nearly all observations is approximately $4s$. That is a very useful interpretation of s.

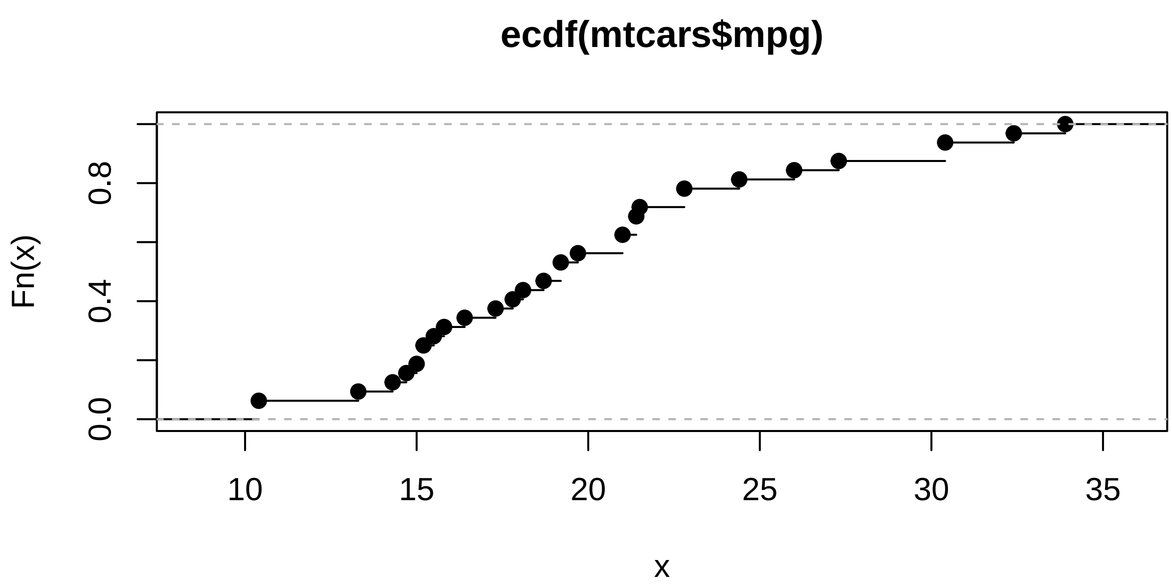
```
4*sd(mtcars$mpg)
```

```
## [1] 24.1
```

```
range(mtcars$mpg)[2]-range(mtcars$mpg)[1]
```

```
## [1] 23.5
```

```
plot(ecdf(mtcars$mpg))
```



- Natural to approximate F at $x_{(i)}$ by empirical distribution $\hat{F}()$ so

$$\hat{F}(x_{(i)}) = \frac{i}{n}.$$

- Hence: $x_{(i)}$ is approximately the $\frac{i}{n}$-quantile.
- Note: some authors use slightly different quantiles, e.g. $\frac{i-0.5}{n}$-quantile.

12.8 Normal quantile-quantile plots

- Above we needed to specify distribution exactly, i.e. $U(10, 34)$.
- For the normal distribution things are much easier. Want to investigate whether the sample comes from a normal distribution `norm`(μ, σ) and we need not know μ or σ.
- Recall

$$i/n = \hat{F}(x_{(i)}) = P(X \leq x_{(i)})$$

- Recall this: If Z has a standard normal distribution `norm`$(0, 1)$ then $Y = \mu + \sigma Z$ has a `norm`(μ, σ)–distribution.
- Let q_i be the $\frac{i}{n}$-quantile of a standard normal distribution, i.e. $P(Z \leq q_i) = \frac{i}{n}$.
- Then

$$\begin{aligned} \frac{i}{n} &= P(Z \leq q_i) = P(\mu + \sigma Z \leq \mu + \sigma q_i) \\ &= P(Y \leq \mu + \sigma q_i) \end{aligned}$$

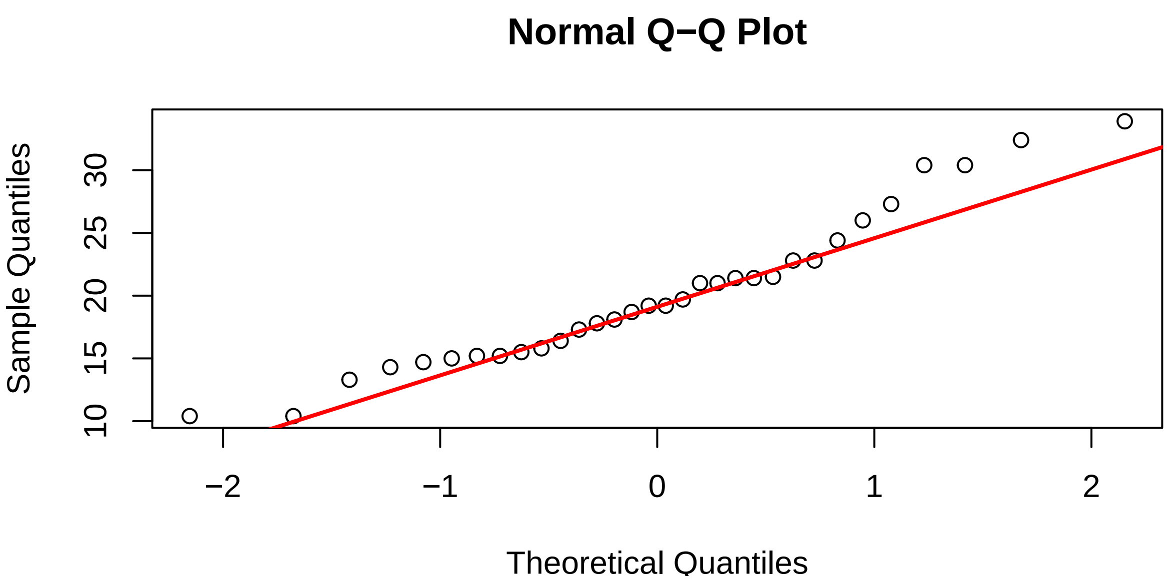
So $\mu + \sigma q_i$ is the corresponding $\frac{i}{n}$-quantile of a `norm`(μ, σ) distribution.

Hence if the sample comes from a `norm`(μ, σ) distribution, then the sample quantiles $x_{(i)}$ should be approximately equal to the population quantiles $\mu + \sigma q_i$:

$$x_{(i)} \approx \mu + \sigma q_i$$

So if we plot $(x_{(i)}, \mu + \sigma q_i)$ and if the sample comes from a `norm`(μ, σ) distribution the points should be on a straight line with intercept μ and slope σ. Looks mostly like a straight line, so `mpg` could be described - approximately - by a normal distribution:

```r
qqnorm(mtcars$mpg)
qqline(mtcars$mpg, col="red", lwd=2)
```



13 Point and interval estimates

13.1 Point and interval estimates

- Suppose we study a population and we are interested in certain parameters of the population distribution, e.g. the mean μ and the standard deviation σ.
- Based on a sample we can make a **point estimate** of the parameter. We have already seen the following examples:
  - $\bar{x}$ is a point estimate of μ
  - s is a point estimate of σ
- We often supplement the point estimate with an **interval estimate** (also called a **confidence interval**). This is an interval around the point estimate, in which we are confident (to a certain degree) that the population parameter is located.

13.2 Point estimators: Bias

- If we want to estimate the population mean μ we have several possibilities e.g.
  - the sample mean $\bar{X}$
  - the average X_T of the sample upper and lower quartiles
- Advantage of X_T: Very large/small observations have little effect, i.e. it has practically no effect if there are a few errors in the data set.
- Disadvantage of X_T: If the distribution of the population is skewed, i.e. asymmetrical, then X_T is **biased**, i.e. $E(X_T) \neq \mu$. This means that in the long run this estimator systematically over or under estimates the value of μ.
- Generally we prefer that an estimator is **unbiased**, i.e. its expected value equals the true parameter value.
- Recall that for a sample from a population with mean μ, the sample mean $\bar{X}$ also has mean μ. That is, $\bar{X}$ is an unbiased estimate of the population mean μ.

13.3 Point estimators: Consistency

- From previous lectures we know that the standard error of $\bar{X}$ is $\frac{\sigma}{\sqrt{n}}$, so
  - the standard error decreases when the sample size increases.
- In general an estimator with this property is called **consistent**.
- X_T is also a consistent estimator, but has a variance that is greater than $\bar{X}$.

13.4 Point estimators: Efficiency

- Since the variance of X_T is greater than the variance of $\bar{X}$, we prefer $\bar{X}$.
- In general, we prefer the estimator with the smallest possible variance. This estimator is said to be **efficient**.
- $\bar{X}$ is an efficient estimator.

13.5 Notation

- The symbol ˆ above a parameter denotes a (point) estimate of the parameter. We have looked at an
  - estimate of the population mean μ, namely $\hat{\mu} = \bar{x}$.

- estimate of the population standard deviation σ, namely $\hat{\sigma} = s$
- estimate of the population proportion p, namely the sample proportion $\hat{p}$.

14 Confidence intervals

14.1 Confidence Interval

- A **confidence interval** for a parameter is constructed as an interval, where we expect the population parameter to be.
- The probability that this construction yields an interval which includes the population parameter is called the **confidence level**.
  - We write the confidence level as $100(1-\alpha)\%$.
  - The confidence level is typically chosen to be 95%.
  - α is called the **error probability**.
  - For a 95% confidence level $\alpha = 1 - 0.95 = 0.05$.
- In practice the interval is often constructed as a symmetric interval around a point estimate:
  - **point estimate±margin of error**
  - Rule of thumb: With a margin of error of 2 times the standard error you get a confidence interval, where the confidence level is approximately 95%.
  - I.e: **point estimate ± 2 x standard error** has confidence level of approximately 95%.
- Interpretation: We say that we are $100(1-\alpha)\%$ confident that the population parameter lies in the confidence interval.

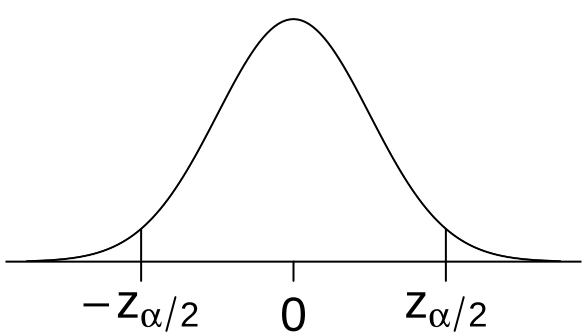
14.2 Confidence interval for the mean (known standard deviation)

- Consider a population with population mean μ and standard deviation σ. We would like to make a $100(1-\alpha)\%$ confidence interval for μ.
- Suppose we draw a random sample $X_1, \ldots, X_n$. As a point estimate for μ we use $\bar{X}$.
- If the population follows a normal distribution or if $n \geq 30$, we may assume $\bar{X} \sim \texttt{norm}(\mu, \frac{\sigma}{\sqrt{n}})$.
- The z-score of $\bar{X}$ follows a standard normal distribution:

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim \texttt{norm(0,1)}.$$

- We determine the **critical z-value** $z_{\alpha/2}$ such that $P(Z > z_{\alpha/2}) = \alpha/2$. This implies by symmetry that

$$P(-z_{\alpha/2} \leq Z \leq z_{\alpha/2}) = 1 - \alpha.$$



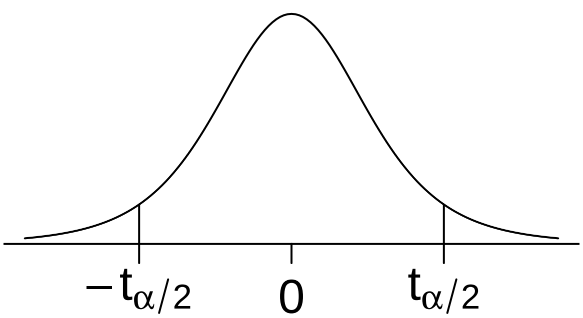
- Inserting what Z is, we get
$$P\left(-z_{\alpha/2} \leq \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \leq z_{\alpha/2}\right) = 1-\alpha.$$
- Isolating μ in both in inequalities, we get
$$P\left(\bar{X} - z_{\alpha/2}\tfrac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + z_{\alpha/2}\tfrac{\sigma}{\sqrt{n}}\right) = 1-\alpha.$$
- That is, for $100(1-\alpha)\%$ of all samples, the population mean μ lies in the interval
$$\left[\bar{X} - z_{\alpha/2}\tfrac{\sigma}{\sqrt{n}}; \bar{X} + z_{\alpha/2}\tfrac{\sigma}{\sqrt{n}}\right].$$

14.3 Unknown standard deviation

- The confidence interval was derived using the z-score $Z = \frac{\bar{X}-\mu}{\sigma/\sqrt{n}}$.
- Problem: In practice σ is typically unknown.
- If we replace σ by the sample standard deviation S, we get the **t-score**
$$T = \tfrac{\bar{X}-\mu}{S/\sqrt{n}}.$$
- Since S is random with a certain variance, this causes T to vary more than Z.
- As a consequence, T no longer follows a normal distribution, but a **t-distribution** with $n-1$ degrees of freedom.

14.4 Confidence interval (unknown standard deviation)

- In the situation where σ is unknown, we use that
$$T = \frac{\bar{X}-\mu}{S/\sqrt{n}} \sim \mathtt{t}(n-1).$$
- We determine the **critical t-value** $t_{\alpha/2}$ such that $P(T > t_{\alpha/2}) = \alpha/2$. This implies by symmetry that
$$P(-t_{\alpha/2} \leq T \leq t_{\alpha/2}) = 1-\alpha.$$

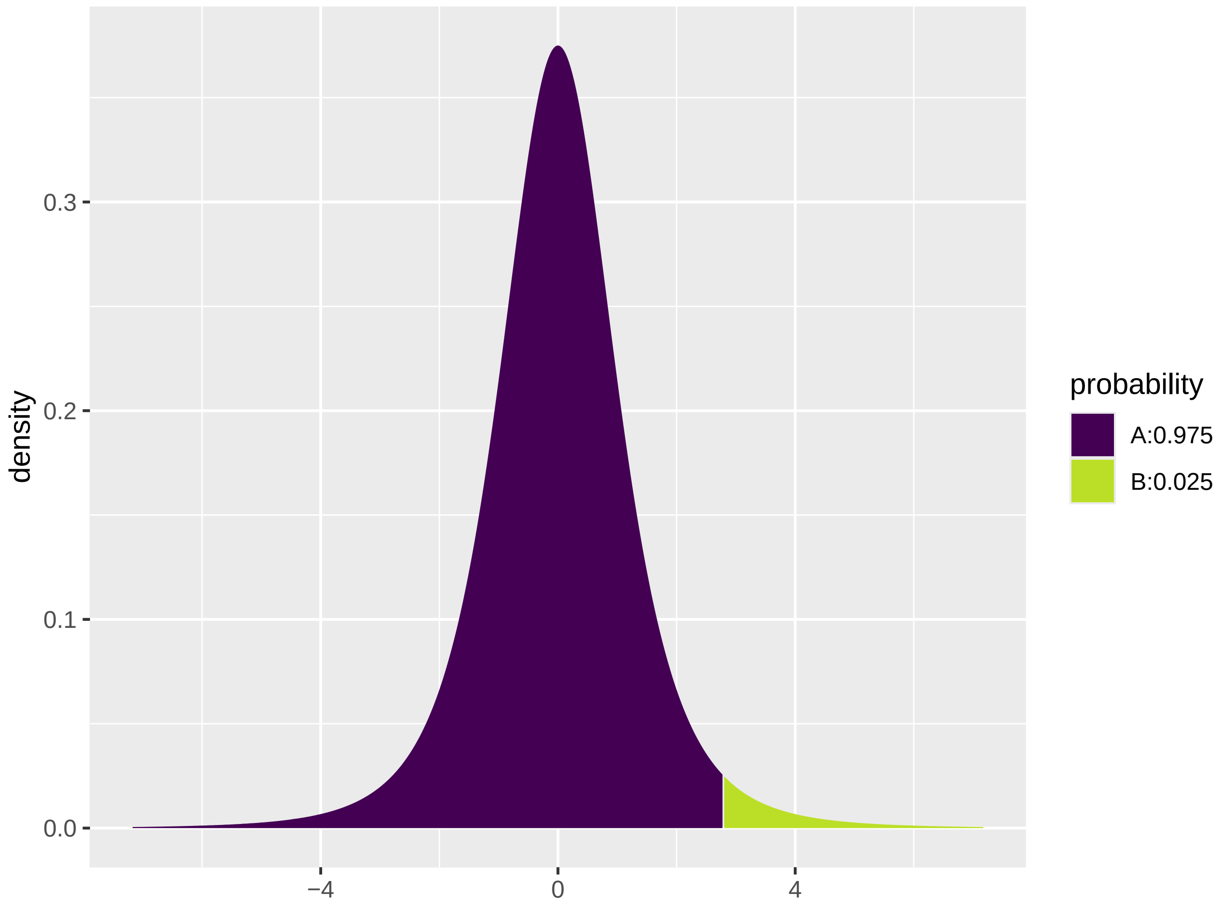


- By exactly the same computations as before, we find that for $100(1-\alpha)\%$ of all samples, μ lies in
$$\left[\bar{X} - t_{\alpha/2}\tfrac{S}{\sqrt{n}}; \bar{X} + t_{\alpha/2}\tfrac{S}{\sqrt{n}}\right].$$
- This interval is what we call the $100(1-\alpha)\%$ **confidence interval** for μ.

14.5 Calculation of critical t-value in R

- To apply the formula, we need to be able to compute the critical t-value $t_{\alpha/2} = P(T > \alpha/2)$.
- This can be done in R via the function `qdist`.
- Note that we need the point with **right tail** probability $\alpha/2$ while R gives the **left tail** probabilities. The right tail probability $\alpha/2$ corresponds to the left tail probability $1 - \alpha/2$.
- So to find $t_{\alpha/2}$ with $\alpha = 0.05$ (corresponding to a 95% confidence level) in a t-distribution with 4 degrees of freedom, we type:

```
qdist("t", p = 1 - 0.025, df = 4)
```




```
## [1] 2.78
```

14.5.1 Example: Confidence interval for mean

- We return to the dataset `mtcars`. We want to construct a 95% confidence interval for the population mean μ of the fuel consumption.

```
stats <- favstats( ~ mpg, data = mtcars)
stats
```

```
##   min   Q1 median   Q3  max mean   sd  n missing
##  10.4 15.4   19.2 22.8 33.9 20.1 6.03 32       0
```

```
qdist("t", 1 - 0.025, df = 32 - 1, plot = FALSE)
```

```
## [1] 2.04
```

- I.e. we have
  - $\bar{x} = 20.1$
  - $s = 6$
  - $n = 32$
  - $df = n - 1 = 31$
  - $t_{crit} = 2.04$.
- The confidence interval is $\bar{x} \pm t_{crit}\frac{s}{\sqrt{n}} = [17.9, 22.3]$
- All these calculations can be done automatically by R:

```
t.test( ~ mpg, data = mtcars, conf.level = 0.95)
```

```
##
##  One Sample t-test
##
## data:  mpg
## t = 19, df = 31, p-value <2e-16
## alternative hypothesis: true mean is not equal to 0
## 95 percent confidence interval:
##  17.9 22.3
## sample estimates:
## mean of x
##      20.1
```

14.5.2 Example: Plotting several confidence intervals in R

- We shall look at a built-in **R** dataset `chickwts`.
- `?chickwts` yields a page with the following information

  > An experiment was conducted to measure and compare the effectiveness of various feed supplements on the growth rate of chickens. Newly hatched chicks were randomly allocated into six groups, and each group was given a different feed supplement. Their weights in grams after six weeks are given along with feed types.

- `chickwts` is a data frame with 71 observations on 2 variables:
  - `weight`: a numeric variable giving the chick weight.
  - `feed`: a factor giving the feed type.
- Calculate a confidence interval for the mean weight for each feed separately; the confidence interval is from `lower` to `upper` given by `mean`±`tscore * se`:

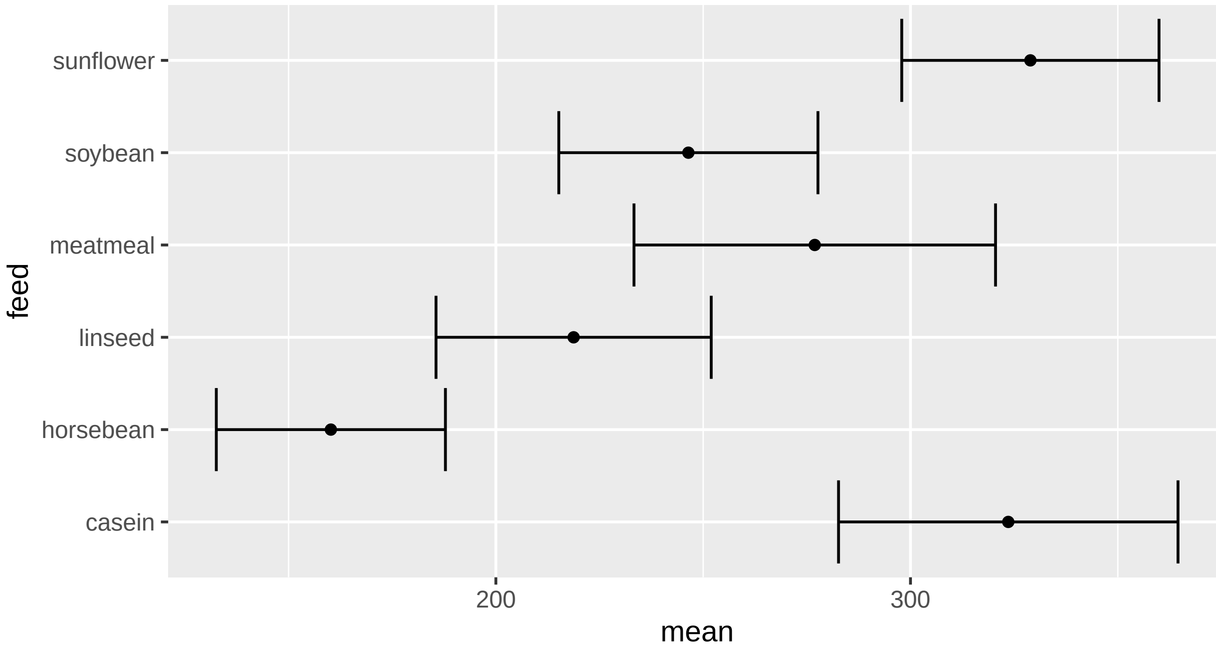
```
cwei <- favstats( weight ~ feed, data = chickwts)
se <- cwei$sd / sqrt(cwei$n) # Standard errors
tscore <- qt(p = .975, df = cwei$n - 1) # t-scores for 2.5% right tail probability (qdist cannot take m
cwei$lower <- cwei$mean - tscore * se
cwei$upper <- cwei$mean + tscore * se
cwei[, c("feed", "mean", "lower", "upper")]
```

```
##        feed mean lower upper
## 1    casein  324   283   365
## 2 horsebean  160   133   188
## 3   linseed  219   186   252
```

```
## 4  meatmeal  277   233   321
## 5   soybean  246   215   278
## 6 sunflower  329   298   360
```

- We can plot the confidence intervals as horizontal line segments using `gf_errorbar`:

```
gf_errorbar(feed ~ lower + upper, data = cwei) %>%
  gf_point(feed ~ mean)
```



14.6 Confidence interval for proportion

- Consider a population with a distribution that can only take the values 0 and 1. The interesting parameter of this distribution is the proportion p of the population that has the value 1.

- Given a random sample $X_1, \ldots, X_n$, recall that we estimate p by

$$\hat{P} = \frac{\sum_{i=1}^{n} X_i}{n} = \bar{X}.$$

- Based on the central limit theorem we have:

$$\hat{P} \approx N\left(p, \frac{\sigma}{\sqrt{n}}\right)$$

  if both $n\hat{p}$ *and* $n(1-\hat{p})$ *are at least* 15.
- It can be shown that a random variable that takes the value 1 with probability p and 0 with probability $(1-p)$ has standard deviation

$$\sigma = \sqrt{p(1-p)}.$$

  That is, the standard deviation is not a "free" parameter for a 0/1 variable as it is determined by the probability p.
- With a sample size of n, the standard error of $\hat{P}$ will be:

$$\frac{\sigma}{\sqrt{n}} = \sqrt{\frac{p(1-p)}{n}}.$$

- We do not know p but we insert the estimate $\hat{P}$ and get the **estimated standard error** of $\hat{P}$:

$$se = \sqrt{\frac{\hat{P}(1-\hat{P})}{n}}.$$

- By similar calculations as in the case of confidence intervals for the mean, we find the limits of the confidence interval to be

$$\hat{P} \pm z_{\alpha/2}\sqrt{\frac{\hat{P}(1-\hat{P})}{n}}.$$

- Here $z_{\alpha/2}$ (or z_{crit}) is the critical value for which the upper tail probability in the standard normal distribution is $\alpha/2$. (E.g. we have $z = 1.96$ when $\alpha = 5\%$.)

14.6.1 Example: Point and interval estimate for proportion

- We consider again the data set concerning votes in Chile.
- We are interested in the unknown proportion p of females in the population of Chile.
- The gender distribution in the sample is:

```
library(mosaic)
tally( ~ sex, data = Chile)
```

```
## sex
##    F    M
## 1379 1321
```

```
tally( ~ sex, data = Chile, format = "prop")
```

```
## sex
##     F     M
## 0.511 0.489
```

- Estimate of p (sample proportion): $\hat{p} = \frac{1379}{1379+1321} = 0.5107$
- An approximate 95% confidence interval for p is:

$$\hat{p} \pm z_{crit} \times se = 0.5107 \pm 1.96\sqrt{\frac{0.5107(1-0.5107)}{1379+1321}} = (0.49, 0.53)$$

- Interpretation: We are 95% confident that there is between 49% and 53% females in Chile.

14.6.2 Example: Confidence intervals for proportion in R

- R automatically calculates the confidence interval for the proportion of females when we do a so-called hypothesis test (we will get back to that later):

```
prop.test( ~ sex, data = Chile, correct = FALSE)
```

```
##
##  1-sample proportions test without continuity correction
##
## data:  Chile$sex  [with success = F]
## X-squared = 1, df = 1, p-value = 0.3
```

```
## alternative hypothesis: true p is not equal to 0.5
## 95 percent confidence interval:
##  0.492 0.530
## sample estimates:
##     p
## 0.511
```

- The argument `correct = FALSE` is needed to make R use the "normal" formulas as on the slides and in the book. When `correct = TRUE` (the default) a mathematical correction which you have not learned about is applied and slightly different results are obtained.

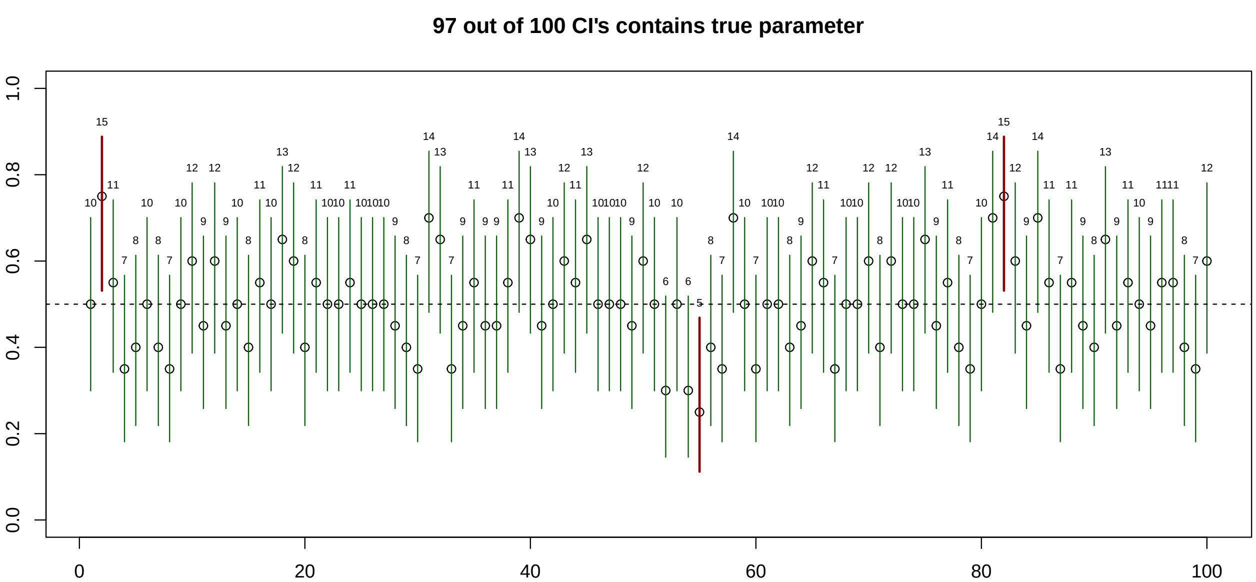
14.6.3 Example: Chile data

We could also have computed a 99% confidence interval for the proportion of females in Chile:

- For a 99%-confidence level we have $\alpha = 1\%$ and
  1) z_{crit}=`qdist("norm", 1 - 0.01/2)`=2.576.
  2) We still have $\hat{p} = 0.5107$ and $n = 2700$, so $se = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = 0.0096$.
  3) Thereby, a 99%-confidence interval is: $\hat{p} \pm z_{crit} \times se = [0.49, 0.54]$.
- Note that the confidence interval becomes wider, when we want to be more confident that the interval contains the population parameter.

15 Confidence interval interpretation

- Assume a population parameter $p = 0.5$ (e.g. a fair coin)
- Draw 100 samples of size $n = 20$ and for each estimate $\hat{p}$ and 95% confidence interval



- Long run interpretation (repeating the experiment many, many times)
  - Each sample gives new CI limits
  - Before collecting data, the procedure for calculating a CI will provide a CI that with 95% probability will contain the true parameter value
  - Once data collected: Contains the true value or not
- CI contains values of the parameter that are compatible with the observed data
  - Either the CI contains the true parameter, or something "extraordinary" has happened (not impossible, just improbable/surprising)

- We are 95% confident that there is between 49% and 53% females in Chile

16 Confidence interval for the variance

16.1 The sample variance

- Suppose we are interested in the variance σ^2 of a population.
- We draw a random sample $X_1, \ldots, X_n$ and use the sample variance S^2 as a point estimate or σ^2.
- When the population distribution is normal, or $n \geq 30$,

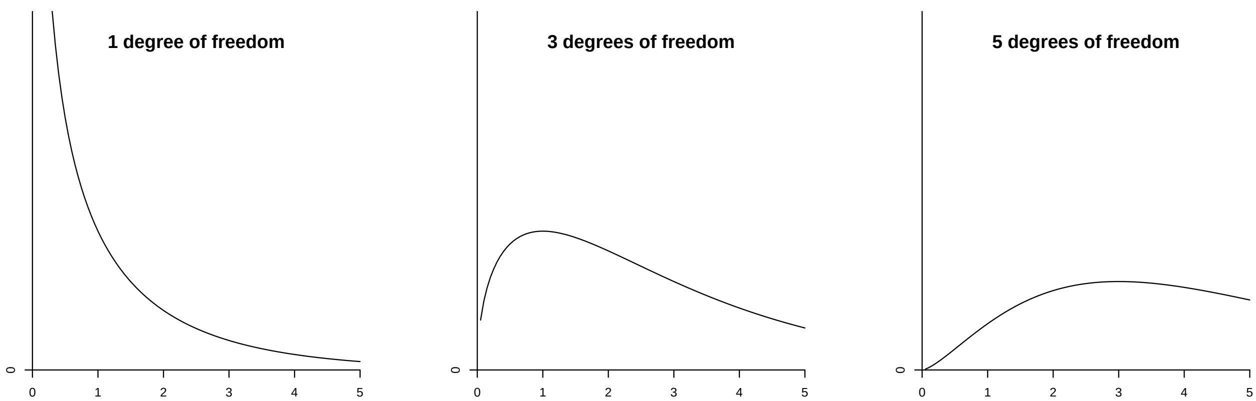
$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1),$$

  where $\chi^2(n-1)$ is the **chi-square distribution** with $(n-1)$ degrees of freedom (se next slide).

  - As a rule of thumb, the degrees of freedom are found as the number of observations (n) minus the number of unknown parameters describing the mean (one, namely μ).

16.2 The χ^2-distribution

- The distribution $\chi^2(k)$ is called the **chi-square** distribution.
  - It is a continuous distribution on $(0, \infty)$.
  - It depends on a the parameter k called the **degrees of freedom**.
  - The degrees of freedom determine the shape of the distribution.
  - The mean value is k.

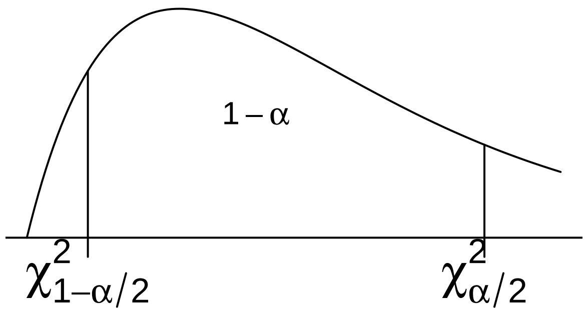


16.3 Confidence interval for variance

- To make a confidence interval for σ^2, we draw a random sample $X_1, \ldots, X_n$, compute S^2 and recall that

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1).$$

- Let $\chi^2_{\alpha/2}$ and $\chi^2_{1-\alpha/2}$ be the critical values in a χ^2-distribution with $(n-1)$ degrees of freedom such the right tail probabilities are $\alpha/2$ and $1-\alpha/2$, respectively.



- Then

$$P\left(\chi^2_{1-\alpha/2} \leq \frac{(n-1)S^2}{\sigma^2} \leq \chi^2_{\alpha/2}\right) = 1-\alpha.$$

- Isolating σ^2, this is equivalent to

$$P\left(\frac{(n-1)S^2}{\chi^2_{\alpha/2}} \leq \sigma^2 \leq \frac{(n-1)S^2}{\chi^2_{1-\alpha/2}}\right) = 1-\alpha.$$

- So we get the confidence interval for σ^2:

$$\left[\frac{(n-1)S^2}{\chi^2_{\alpha/2}}; \frac{(n-1)S^2}{\chi^2_{1-\alpha/2}}\right].$$

- A confidence interval for σ can be found by taking square roots:

$$\left[\sqrt{\frac{(n-1)S^2}{\chi^2_{\alpha/2}}}; \sqrt{\frac{(n-1)S^2}{\chi^2_{1-\alpha/2}}}\right].$$

- Note that these confidence intervals are not symmetric around the point estimate.

16.3.1 Example: confidence interval for a variance

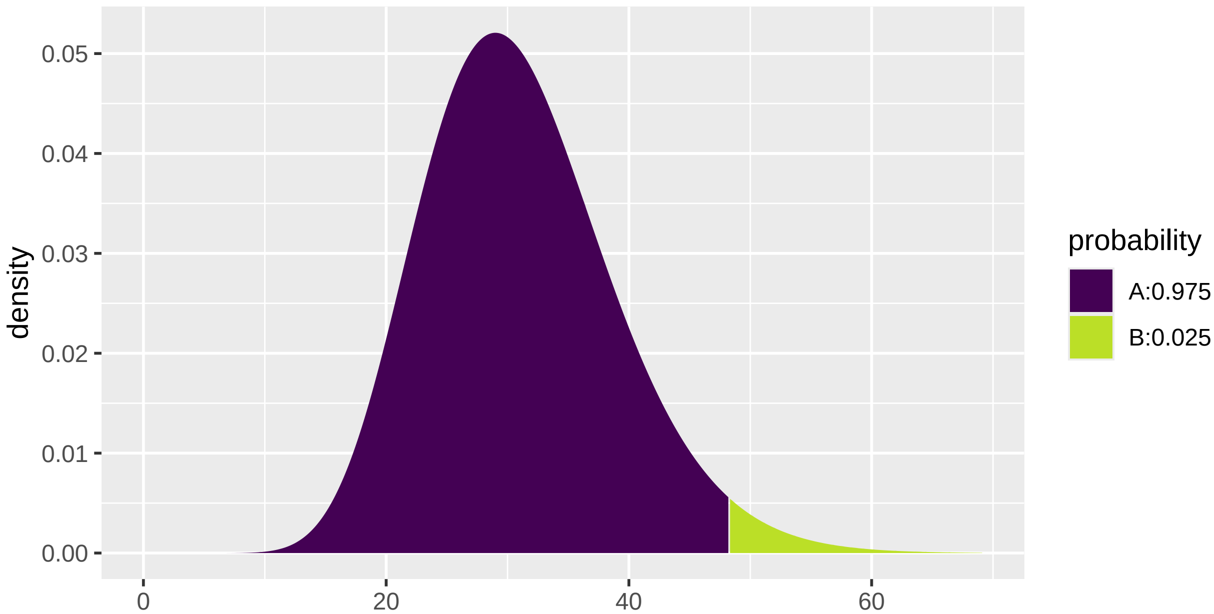
- We return to the dataset `mtcars` and construct a 95% confidence interval for the population variance σ^2 of the fuel consumption. We find the sample variance to be $6.026948^2 \approx 36.3$ using ´favstats´.

```
stats <- favstats( ~ mpg, data = mtcars)
stats
```

```
##   min   Q1 median   Q3  max mean   sd  n missing
##  10.4 15.4   19.2 22.8 33.9 20.1 6.03 32       0
```

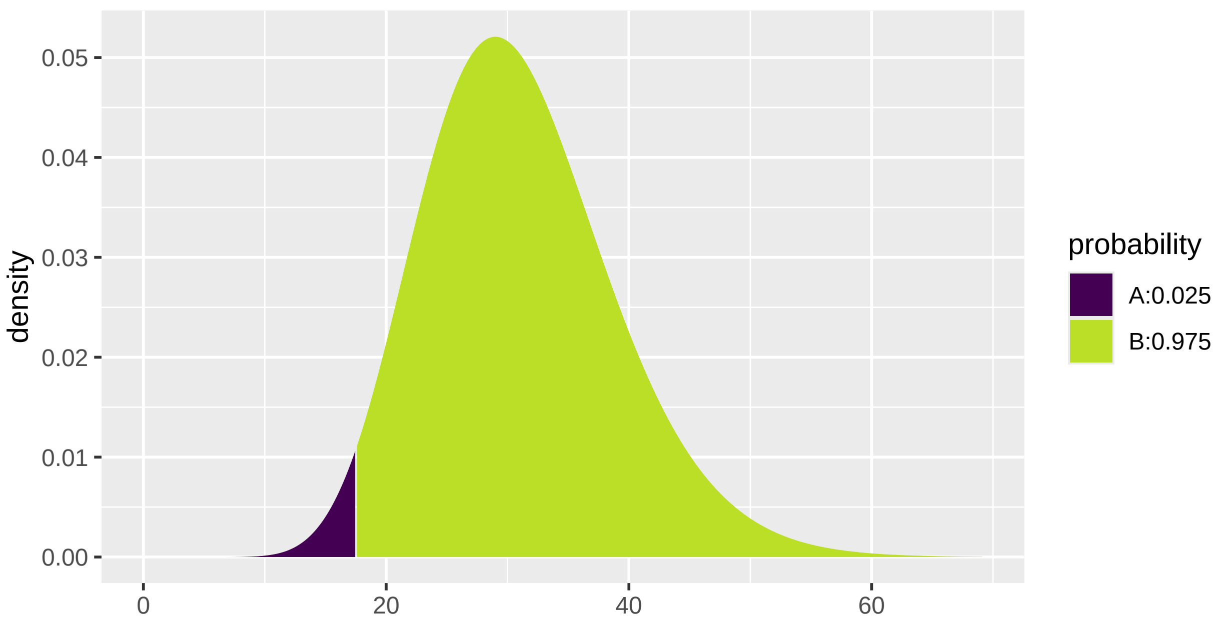
- The critical χ^2-values are found using `qdist`. The degrees of freedom are $n-1$. The χ^2-distribution is not symmetric, so we need to find both $\chi^2_{\alpha/2}$ and $\chi^2_{1-\alpha/2}$.

```
qdist("chisq", 1 - 0.025, df = 32 - 1 )
```




```
## [1] 48.2
```

```r
qdist("chisq",  0.025, df = 32 -1)
```




```
## [1] 17.5
```

- I.e. we have
  - $\chi^2_{\alpha/2} = 48.23189$
  - $\chi^2_{1-\alpha/2} = 17.53874$
- So we get the confidence interval for σ^2:

$$\left[\frac{(n-1)s^2}{\chi^2_{\alpha/2}}; \frac{(n-1)s^2}{\chi^2_{1-\alpha/2}}\right] = \left[\frac{31 \cdot 6.026948^2}{48.23189}; \frac{(31 \cdot 6.026948^2}{17.53874}\right] = [23.3, 64.2].$$

- A confidence interval for σ is $[\sqrt{23.3}, \sqrt{64.2}] = [4.83, 8.01]$.

17 Determining sample size

17.1 Determining sample size

- When planning an experiment, one has to decide how large the sample size should be.
  - If the sample size is too small, the parameter estimates will have high variance and hence confidence intervals will be large.
  - A sample size that is too large is costly in terms of time, money, etc.

17.2 Sample size for proportion

- The confidence interval is of the form point estimate±estimated margin of error.
- Imagine that we want to plan an experiment, where we **want to achieve a certain margin of error** M (and thus a specific width of the associated confidence interval).
- When we estimate a proportion the margin of error is

$$M = z_{crit}\sqrt{\frac{p(1-p)}{n}},$$

where the critical z-score, z_{crit}, is determined by the specified confidence level.
- If we solve the equation above we see that if we choose sample size

$$n = p(1-p)\left(\frac{z_{crit}}{M}\right)^2,$$

then we obtain an estimate of π with margin of error M.
- If we do not have a good guess for the value of p we can use the worst case value $p = 50\%$. The corresponding sample size $n = \left(\frac{z_{crit}}{2M}\right)^2$ ensures that we obtain an estimate with a margin of error, which is at the *most* M.

17.2.1 Example

- We want to make a survey to determine the proportion of the Danish population that will a certain party at the next election. How many voters should we ask to get a margin of error, which equals 1%?
- We set the confidence level to be 95%, which means that $z_{crit} = 1.96$.
- Worst case is $p = 0.5$, yielding:

$$n = p(1-p)\left(\frac{z_{crit}}{M}\right)^2 = \frac{1}{4}\left(\frac{1.96}{0.01}\right)^2 = 9604.$$

- If we are interested in the proportion of voters that vote for "socialdemokratiet' ' a good guess is $p = 0.23$, yielding

$$n = p(1-p)\left(\frac{z_{crit}}{M}\right)^2 = 0.23(1-0.23)\left(\frac{1.96}{0.01}\right)^2 = 6804.$$

- If we instead are interested in "liberal alliance' ' a good guess is $p = 0.05$, yielding

$$n = p(1-p)\left(\frac{z_{crit}}{M}\right)^2 = 0.05(1-0.05)\left(\frac{1.96}{0.01}\right)^2 = 1825.$$

17.3 Sample size for mean

- The confidence interval is of the form point estimate±estimated margin of error.
- Imagine that we want to plan an experiment, where we **want to achieve a certain margin of error** M.
- When we estimate a mean the margin of error is

$$M = z_{crit} \frac{\sigma}{\sqrt{n}},$$

  where the critical z-score, z_{crit}, is determined by the specified confidence level.
- If we solve the equation above we see:
  - If we choose sample size $n = \left(\frac{z_{crit}\sigma}{M}\right)^2$, then we obtain an estimate with margin of error M.
- Problem: We usually do not know σ. Possible solutions:
  - Based on similar studies conducted previously, we make a qualified guess at σ.
  - Based on a pilot study a value of σ is estimated.

ASTA

The ASTA team

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1 Statistical inference: Hypothesis and test

1.1 Concept of hypothesis

- A **hypothesis** is a statement about a given population. Usually it is stated as a population parameter having a given value or being in a certain interval.
- Examples:
  - Quality control of products: The hypothesis is that the products e.g. have a certain weight, a given power consumption or a minimal durability.
  - Scientific hypothesis: There is no dependence between a company's age and level of return.

1.2 Significance test

- A significance test is used to investigate, whether data is contradicting the hypothesis or not.
- If the hypothesis says that a parameter has a certain value, then the test should tell whether the sample estimate is "far" away from this value.
- For example:
  - Waiting times in a queue. We sample n customers and count how many that have been waiting more than 5 minutes. The company policy is that at most 10% of the customers should wait more than 5 minutes. In a sample of size $n = 32$ we observe 4 with waiting time above 5 minutes, i.e. the estimated proportion is $\hat{\pi} = \frac{4}{32} = 12.5\%$. Is this "much more" than (i.e. significantly different from) 10%?
  - The blood alcohol level of a student is measured 4 times with the values $0.504, 0.500, 0.512, 0.524$, i.e. the estimated mean value is $\bar{y} = 0.51$. Is this "much different" than a limit of 0.5?

1.3 Null and alternative hypothesis

- **The null hypothesis** - denoted H_0 - usually specifies that a population parameter has some given value. E.g. if μ is the mean blood alcohol level we can state the null hypothesis
  - $H_0 : \mu = 0.5$.
- The **alternative hypothesis** - denoted H_a - usually specifies that the population parameter is contained in a given set of values different than the null hypothesis. E.g. if μ again is the population mean of a blood alcohol level measurement, then
  - the null hypothesis is $H_0 : \mu = 0.5$
  - the alternative hypothesis is $H_a : \mu \neq 0.5$.

1.4 Test statistic

- We consider a population parameter μ and write the null hypothesis

$$H_0 : \mu = \mu_0,$$

  where μ_0 is a known number, e.g. $\mu_0 = 0.5$.
- Based on a sample we have an estimate $\hat{\mu}$.
- A **test statistic** T will typically depend on $\hat{\mu}$ and μ_0 (we may write this as $T(\hat{\mu}, \mu_0)$) and measures "how far from μ_0 is $\hat{\mu}$?"
- Often we use $T(\hat{\mu}, \mu_0) =$ "the number of standard deviations from $\hat{\mu}$ to μ_0".
- For example it would be very unlikely to be more than 3 standard deviations from μ_0, i.e. in that case μ_0 is probably not the correct value of the population parameter.

1.5 P-value

- We consider
  - H_0: a null hypothesis.
  - H_a: an alternative hypothesis.
  - T: a test statistic, where the value calculated based on the current sample is denoted t_{obs}.

- To investigate the plausibility of H_0, we measure the evidence against H_0 by the so-called p-value:
  - The p-value is the probability of observing a more extreme value of T (if we were to repeat the experiment) than t_{obs} *under the assumption that* H_0 *is true.*
  - "Extremity" is measured relative to the alternative hypothesis; a value is considered extreme if it is "far from" H_0 and "closer to" H_a.
  - If the p-value is small then there is a small probability of observing t_{obs} if H_0 is true, and thus H_0 is not very probable for our sample and we put more support in H_a, so:

The smaller the p-value, the less we trust H_0.

- What is a small p-value? If it is below 5% we say it is **significant** at the 5% level.

1.6 Significance level

- We consider
  - H_0: a null hypothesis.
  - H_a: an alternative hypothesis.
  - T: a test statistic, where the value calculated based on the current sample is denoted t_{obs} and the corresponding p-value is p_{obs}.
- Small values of p_{obs} are critical for H_0.
- In practice it can be necessary to decide whether or not we are going to reject H_0.
- The decision can be made if we previously have decided on a so-called **α-level**, where
  - α is a given percentage
  - we reject H_0, if p_{obs} is less than or equal to α
  - α is called the **significance level** of the test
  - typical choices of α are 5% or 1%.

1.7 Significance test for mean

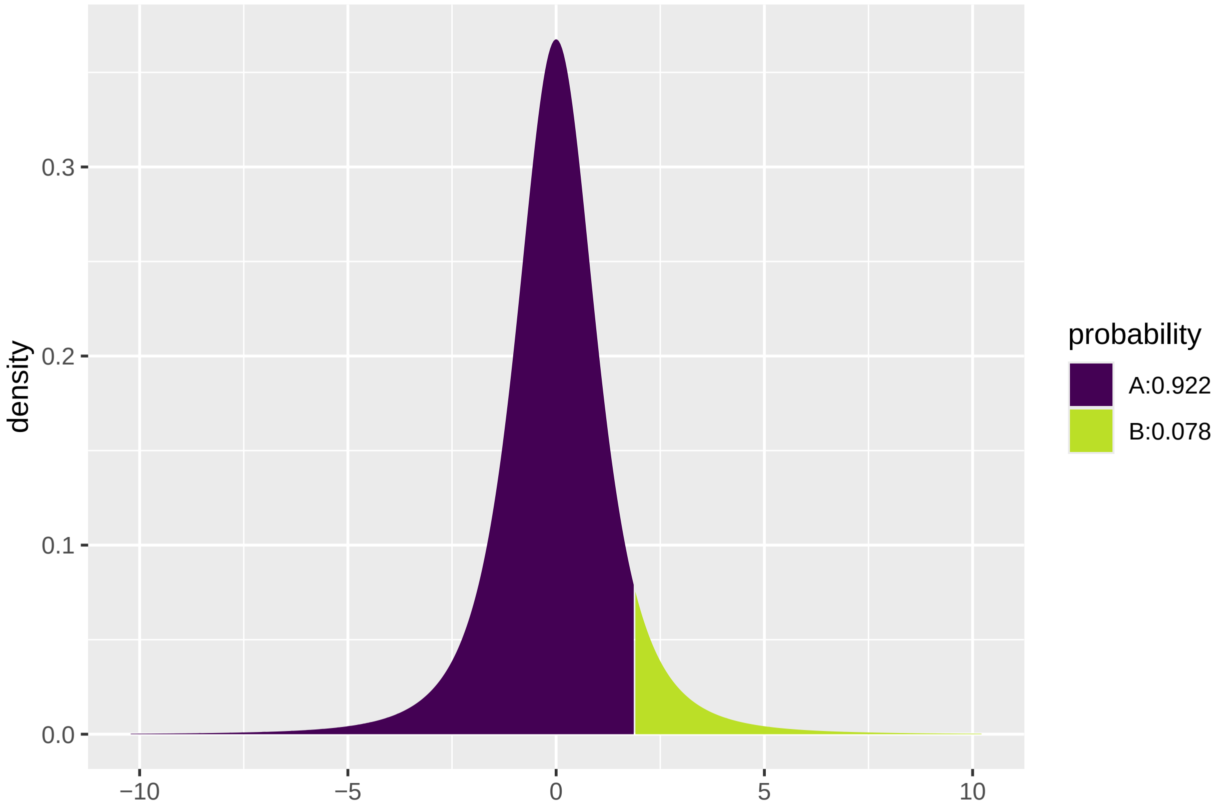
1.7.1 Two-sided t-test for mean:

- We assume that data is a sample from $\texttt{norm}(\mu, \sigma)$.
- The estimates of the population parameters are $\hat{\mu} = \bar{y}$ and $\hat{\sigma} = s$ based on n observations.
- Null hypothesis: $H_0: \ \mu = \mu_0$, where μ_0 is a known value.
- **Two-sided alternative hypothesis**: $H_a: \ \mu \neq \mu_0$.
- Observed test statistic: $t_{obs} = \frac{\bar{y}-\mu_0}{se}$, where $se = \frac{s}{\sqrt{n}}$.
- I.e. t_{obs} measures, how many standard deviations (with $\pm$ sign) the empirical mean lies away from μ_0.
- If H_0 is true, then t_{obs} is an observation from the t-distribution with $df = n - 1$.
- P-value: Values bigger than $|t_{obs}|$ or less than $-|t_{obs}|$ puts more support in H_a than H_0.
- The p-value = 2 x "upper tail probability of $|t_{obs}|$". The probability is calculated in the t-distribution with df degrees of freedom.

1.7.2 Example: Two-sided t-test

- Blood alcohol level measurements: $0.504, 0.500, 0.512, 0.524$.
- These are assumed to be a sample from a normal distribution.
- We calculate
  - $\bar{y} = 0.51$ and $s = 0.0106$
  - $se = \frac{s}{\sqrt{n}} = \frac{0.0106}{\sqrt{4}} = 0.0053$.
  - $H_0: \mu = 0.5$, i.e. $\mu_0 = 0.5$.
  - $t_{obs} = \frac{\bar{y}-\mu_0}{se} = \frac{0.51-0.5}{0.0053} = 1.89$.
- So we are almost 2 standard deviations from 0.5. Is this extreme in a t-distribution with 3 degrees of freedom?

```r
library(mosaic)
1 - pdist("t", q = 1.89, df = 3)
```




```
## [1] 0.07757725
```

- The p-value is $2 \cdot 0.078,$ i.e. more than 15%. On the basis of this we do not reject H_0.

1.8 One-sided t-test for mean

The book also discusses one-sided t-tests for the mean, but we will not use those in the course.

1.9 Agresti: Overview of t-test

TABLE 6.3: The Five Parts of Significance Tests for Population Means

1. **Assumptions**
   Quantitative variable
   Randomization
   Normal population (robust, especially for two-sided H_a, large n)
2. **Hypotheses**
   H_0: $\mu = \mu_0$
   H_a: $\mu \neq \mu_0$ (or H_a: $\mu > \mu_0$ or H_a: $\mu < \mu_0$)
3. **Test statistic**
   $t = \dfrac{\bar{y} - \mu_0}{se}$ where $se = \dfrac{s}{\sqrt{n}}$
4. ***P*-value**
   In t curve, use
   P = Two-tail probability for H_a: $\mu \neq \mu_0$
   P = Probability to right of observed t-value for H_a: $\mu > \mu_0$
   P = Probability to left of observed t-value for H_a: $\mu < \mu_0$
5. **Conclusion**
   Report P-value. Smaller P provides stronger evidence against H_0 and supporting H_a. Can reject H_0 if $P \leq \alpha$-level.

1.10 Significance test for proportion

- Consider a sample of size n, where we observe whether a given property is present or not.
- The relative frequency of the property in the population is π, which is estimated by $\hat{\pi}$.
- Null hypothesis: $H_0 : \pi = \pi_0$, where π_0 is a known number.
- **Two-sided alternative** hypothesis: $H_a : \pi \neq \pi_0$.
- *If* H_0 *is true* the standard error for $\hat{\pi}$ is given by $se_0 = \sqrt{\frac{\pi_0(1-\pi_0)}{n}}$.
- Observed test statistic: $z_{obs} = \frac{\hat{\pi}-\pi_0}{se_0}$
- I.e. z_{obs} measures, how many standard deviations (with $\pm$ sign) there is from $\hat{\pi}$ to π_0.

1.10.1 Approximate test

- If both $n\hat{\pi}$ and $n(1-\hat{\pi})$ are larger than 15 we know from previously that $\hat{\pi}$ follows a normal distribution (approximately), i.e.
  - If H_0 is true, then z_{obs} is an observation from the standard normal distribution.
- P-value for **two-sided** test: Values greater than $|z_{obs}|$ or less than $-|z_{obs}|$ point more towards H_a than H_0.
- The p-value=2 x "upper tail probability for $|z_{obs}|$". The probability is calculated in the standard normal distribution.

1.10.2 Example: Approximate test

- We consider a study from Florida Poll 2006:
  - In connection with problems financing public service a random sample of 1200 individuals were asked whether they preferred less service or tax increases.
  - 52% preferred tax increases. Is this enough to say that the proportion is significantly different from fifty-fifty?
- Sample with $n = 1200$ observations and estimated proportion $\hat{\pi} = 0.52$.
- Null hypothesis $H_0 : \pi = 0.5$.
- Alternative hypothesis $H_a : \pi \neq 0.5$.
- Standard error $se_0 = \sqrt{\frac{\pi_0(1-\pi_0)}{n}} = \sqrt{\frac{0.5\times 0.5}{1200}} = 0.0144$
- Observed test statistic $z_{obs} = \frac{\hat{\pi}-\pi_0}{se_0} = \frac{0.52-0.5}{0.0144} = 1.39$
- "upper tail probability for 1.39" in the standard normal distribution is 0.0823, i.e. we have a p-value of $2\cdot$ 0.0823$\approx$ 16%.
- Conclusion: There is not sufficient evidence to reject H_0, i.e. we do not reject that the preference in the population is fifty-fifty.
- Note, the above calculations can also be performed automatically in **R** by (a little different results due to rounding errors in the manual calculation):

```
count <- 1200 * 0.52 # number of individuals preferring tax increase
prop.test(x = count, n = 1200, correct = F)
```

```
##
##  1-sample proportions test without continuity correction
##
## data:  count out of 1200
## X-squared = 1.92, df = 1, p-value = 0.1659
## alternative hypothesis: true p is not equal to 0.5
## 95 percent confidence interval:
##  0.4917142 0.5481581
## sample estimates:
##    p
## 0.52
```

1.10.3 Binomial (exact) test

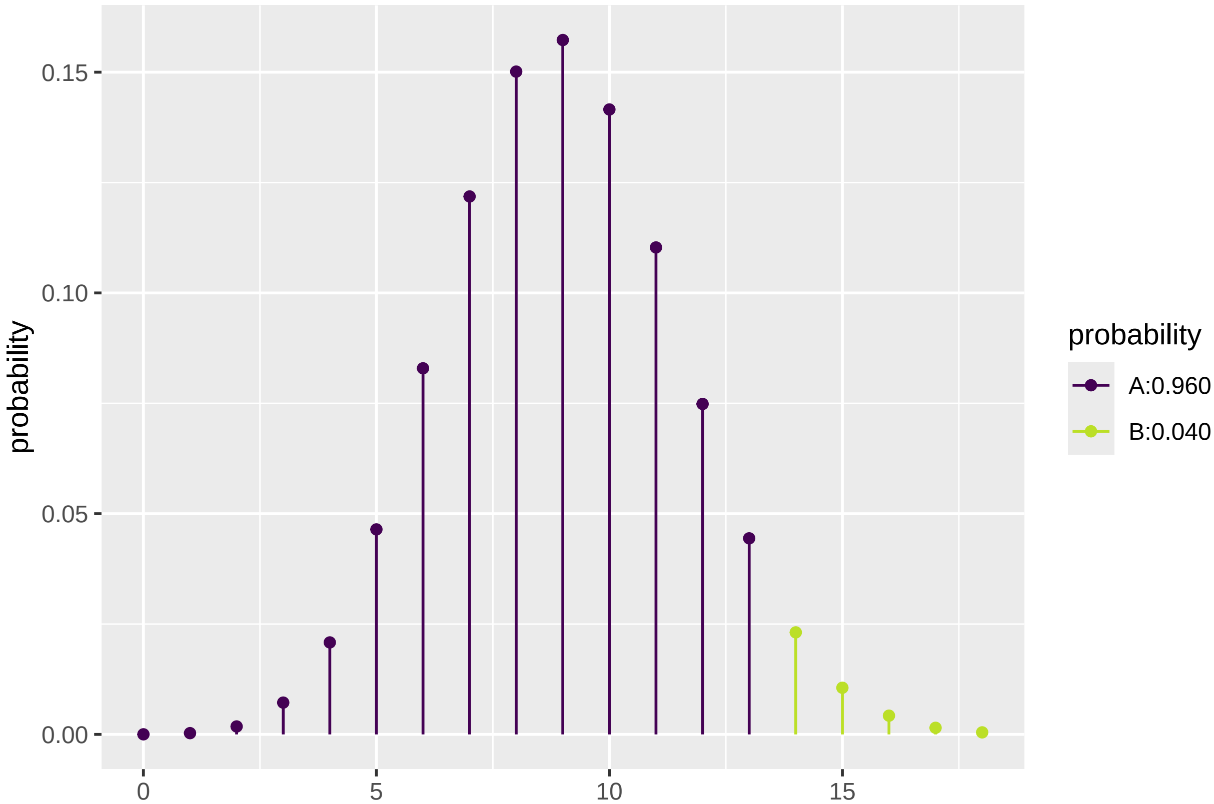
- Consider again a sample of size n, where we observe whether a given property is present or not.
- The relative frequency of the property in the population is π, which is estimated by $\hat{\pi}$.
- Let $y_+ = n\hat{\pi}$ be the frequency (total count) of the property in the sample.
- It can be shown that y_+ follows the **binomial distribution** with size parameter n and success probability π. We use $Bin(n, \pi)$ to denote this distribution.
- Null hypothesis: $H_0 : \pi = \pi_0$, where π_0 is a known number.
- Alternative hypothesis: $H_a : \pi \neq \pi_0$, where π_0 is a known number.
- P-value for **two-sided** binomial test:
  - If $y_+ \geq n\pi_0$: 2 x "upper tail probability for y_+" in the $Bin(n, \pi_0)$ distribution.
  - If $y_+ < n\pi_0$: 2 x "lower tail probability for y_+" in the $Bin(n, \pi_0)$ distribution.

1.10.4 Example: Binomial test

- Experiment with $n = 30$, where we have $y_+ = 14$ successes.
- We want to test $H_0 : \pi = 0.3$ vs. $H_a : \pi \neq 0.3$.

- Since $y_+ > n\pi_0 = 9$ we use the upper tail probability corresponding to the sum of the height of the red lines to the right of 14 in the graph below. (Notice, the graph continues on the right hand side to $n = 30$, but it has been cut off for illustrative purposes.)
- The upper tail probability from 14 and up (i.e. greater than 13) is:

```
lower_tail <- pdist("binom", q = 13, size = 30, prob = 0.3)
```




```
1 - lower_tail
```

```
## [1] 0.04005255
```

- The two-sided p-value is then 2 x 0.04 = 0.08.

1.10.5 Binomial test in R

- We return to the Chile data, where we again look at the variable `sex`.
- Let us test whether the proportion of females is different from 50 %, i.e., we look at $H_0: \ \pi = 0.5$ and $H_a: \ \pi \neq 0.5$, where π is the unknown population proportion of females.

```
Chile <- read.delim("https://asta.math.aau.dk/datasets?file=Chile.txt")
binom.test( ~ sex, data = Chile, p = 0.5, conf.level = 0.95)
```

```
##
##
##
## data:  Chile$sex  [with success = F]
## number of successes = 1379, number of trials = 2700, p-value = 0.2727
## alternative hypothesis: true probability of success is not equal to 0.5
## 95 percent confidence interval:
```

```
##  0.4916971 0.5297610
## sample estimates:
## probability of success
##              0.5107407
```

- The p-value for the binomial exact test is 27%, so there is no significant difference between the proportion of males and females.
- The approximate test has a p-value of 26%, which can be calculated by the command

```
prop.test( ~ sex, data = Chile, p = 0.5, conf.level = 0.95, correct = FALSE)
```

(note the additional argument `correct = FALSE`).

1.11 Agresti: Overview of tests for mean and proportion

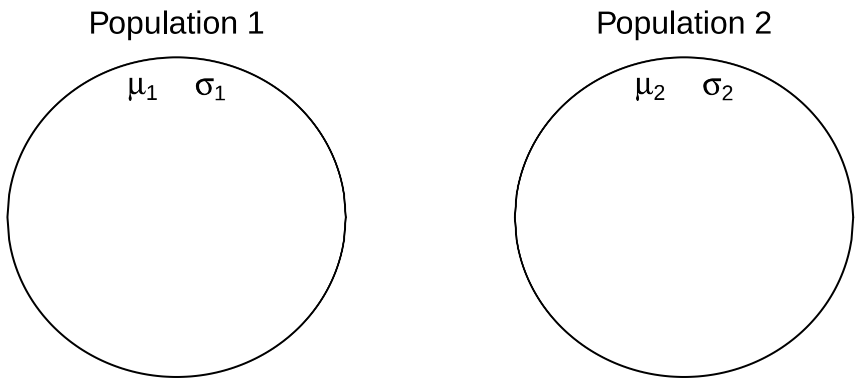
TABLE 6.7: Summary of Significance Tests for Means and Proportions

Parameter	Mean	Proportion
1. Assumptions	Random sample, quantitative variable normal population	Random sample, categorical variable null expected counts at least 10
2. Hypotheses	$H_0: \mu = \mu_0$ $H_a: \mu \neq \mu_0$ $H_a: \mu > \mu_0$ $H_a: \mu < \mu_0$	$H_0: \pi = \pi_0$ $H_a: \pi \neq \pi_0$ $H_a: \pi > \pi_0$ $H_a: \pi < \pi_0$
3. Test statistic	$t = \frac{\bar{y} - \mu_0}{se}$ with $se = \frac{s}{\sqrt{n}}, df = n - 1$	$z = \frac{\hat{\pi} - \pi_0}{se_0}$ with $se_0 = \sqrt{\pi_0(1 - \pi_0)/n}$
4. P-value	Two-tail probability in sampling distribution for two-sided test (H_0: $\mu \neq \mu_0$ or H_a: $\pi \neq \pi_0$); One-tail probability for one-sided test	
5. Conclusion	Reject H_0 if P-value $\leq \alpha$-level such as 0.05	

2 Comparison of two populations

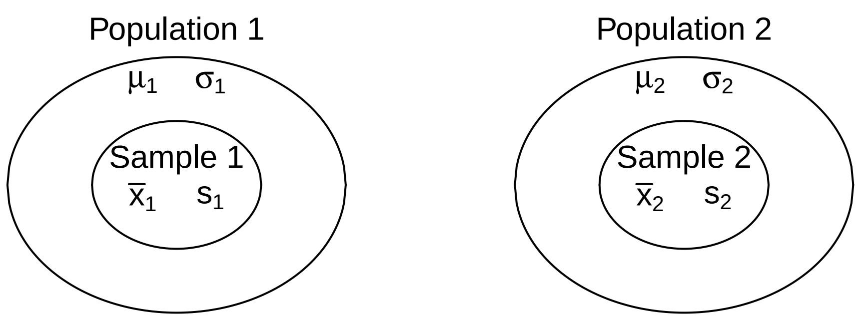
2.1 Two populations

- Consider two populations:
  - Population 1 has mean μ_1 and standard deviation σ_1.
  - Population 2 has mean μ_2 and standard deviation σ_2.
- We want to compare the means by looking at the difference $\mu_1 - \mu_2$.



2.2 Two samples

- We now take a sample from each population.
  - The sample from Population 1 has sample mean $\bar{x}_1$, sample standard deviation s_1 and sample size n_1.
  - The sample from Population 2 has sample mean $\bar{x}_2$, sample standard deviation s_2 and sample size n_2.



2.3 Dependent and independent samples

- We distinguish between two types of samples:
  - The two samples are **independent**.
  - The two samples are **paired**.
- **Example:** Suppose we consider the fuel consumption of cars.
  - If we compare two samples of cars with different engine types, then the two samples are *independent*, since each car can only have one of the two engine types.
  - If we compare the fuel consumption of cars at two different speed levels by testing each car at both speed levels, then then samples are *paired*.

2.4 Comparison of two means (Independent samples)

- We consider the situation, where we have two independent samples of a quantitative variable.
- We estimate the difference $\mu_1 - \mu_2$ by
$$d = \bar{x}_1 - \bar{x}_2.$$
- Assume that we can find the **estimated standard error** se_d of the difference.

- If the samples come from two normal distributions, or if both samples are large ($n_1, n_2 \geq 30$), then one can show

$$T_{obs} = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{se_d} \sim \mathtt{t}(df),$$

where $\mathtt{t}(df)$ is a t-distribution with df degrees of freedom.

- By the usual procedure, we can use this to construct a confidence interval for the unknown population difference of means $\mu_1 - \mu_2$ by

$$(\bar{x}_1 - \bar{x}_2) \pm t_{crit} se_d,$$

where the critical t-score, t_{crit}, is determined by the confidence level and the df.

2.5 Significance test (Independent samples)

- We may be interested the testing the **null-hypothesis** that the population means are the same, which we can formulated as:
  - $H_0 : \mu_1 - \mu_2 = 0$.
  - $H_a : \mu_1 - \mu_2 \neq 0$.
- If the null hypothesis is true, then the **test statistic**:

$$T_{obs} = \frac{(\bar{X}_1 - \bar{X}_2) - 0}{se_d},$$

has a t-distribution with df degrees of freedom.

- The **p-value** is the probability of observing something further away from 0 than t_{obs} in a $\mathtt{t}(df)$ distribution.
- It remains to find the estimated standard error se_d and the degrees of freedom df. We distinguish between two cases:
  - The two populations have equal variances $\sigma_1^2 = \sigma_2^2$.
  - The two populations have different variances $\sigma_1^2 \neq \sigma_2^2$.

2.6 Standard error (Independent samples, equal variances)

- The standard error of $d = \bar{x}_1 - \bar{x}_2$ is given by the formula:

$$\sqrt{\tfrac{\sigma_1^2}{n_1} + \tfrac{\sigma_2^2}{n_2}}.$$

- If the **variances are equal**, $\sigma_1^2 = \sigma_2^2$, then we estimate the common value by the **pooled variance estimate**

$$s_p^2 = \tfrac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}.$$

- Inserting this estimate in the formula for the standard error we obtain the estimated standard error

$$se_d = \sqrt{\tfrac{s_p^2}{n_1} + \tfrac{s_p^2}{n_2}} = s_p\sqrt{\tfrac{1}{n_1} + \tfrac{1}{n_2}}.$$

- In this situation, the degrees of freedom are $df = n_1 + n_2 - 2$.

2.7 Example: Comparing two means (independent samples, equal variances)

We return to the `mtcars` data. We study the association between the variables `vs` and `mpg` (engine type and fuel consumption). So, we will perform a significance test to test the null-hypothesis that there is no difference between the mean of fuel consumption for the two engine types.

- We will test the null-hypothesis assuming equal variances:

```
library(mosaic)
fv <- favstats(mpg ~ vs, data = mtcars)
fv
```

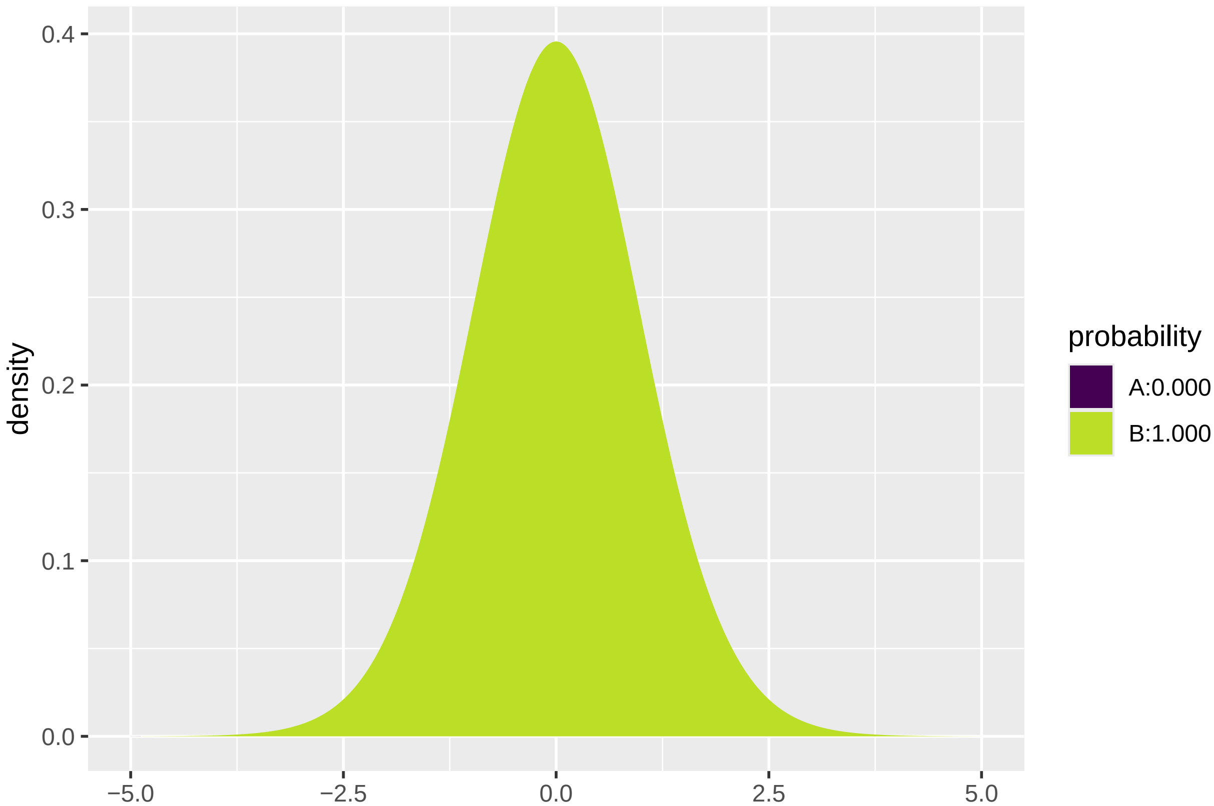
```
##   vs  min   Q1 median   Q3  max mean   sd  n missing
## 1  0 10.4 14.8   15.7 19.1 26.0 16.6 3.86 18       0
## 2  1 17.8 21.4   22.8 29.6 33.9 24.6 5.38 14       0
```

- Difference: $d = 16.6167 - (24.5571) = -7.9405$.
- Sample sizes: $n_1 = 18$ and $n_2 = 14$.
- Estimated standard deviations: $s_1 = 3.8607$ (not v-shaped) and $s_2 = 5.379$ (v-shaped).
- Pooled variance:

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2} = \frac{17 \cdot 3.8607^2 + 13 \cdot 5.379^2}{18 + 14 - 2} = 20.984.$$

- Estimated standard error of difference: $se_d = s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}} = \sqrt{20.984}\sqrt{\frac{1}{18} + \frac{1}{14}} = 1.6324.$
- Observed t-score for $H_0: \ \mu_1 - \mu_2 = 0$ is: $t_{obs} = \frac{d-0}{se_d} = \frac{-7.9405}{1.6324} = -4.864.$
- The degrees of freedom are $df = n_1 + n_2 - 2 = 30$.
- We find the p-value:

```
2*pdist("t", q = -4.864, df=30, xlim = c(-5, 5))
```




```
## [1] 3.419648e-05
```

2.8 Standard error (Independent samples, unequal variances)

- If the **variances are unequal**, then we simply insert the two estimates s_1^2 and s_2^2 for σ_1^2 and σ_2^2 in the formula for the standard error to obtain the estimated standard error

$$se_d = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}.$$

- The degrees of freedom *df* for se_d can be estimated by a complicated formula, which we will not present here (see p.365 in the book).
- Note:
  - If both n_1 and n_2 are above 30, then we may use the standard normal distribution to compute a z-score rather than the t-distribution to compute the t-score. This way we avoid computing *df*.
  - If n_1 or n_2 are below 30, then we let **R** calculate the degrees of freedom and the p-value/confidence interval.

2.9 Example: Comparing two means (independent samples, unequal variances)

We return to the `mtcars` data. We study the association between the variables `vs` and `mpg` (engine type and fuel consumption). So, we will perform a significance test to test the null-hypothesis that there is no difference between the mean of fuel consumption for the two engine types.

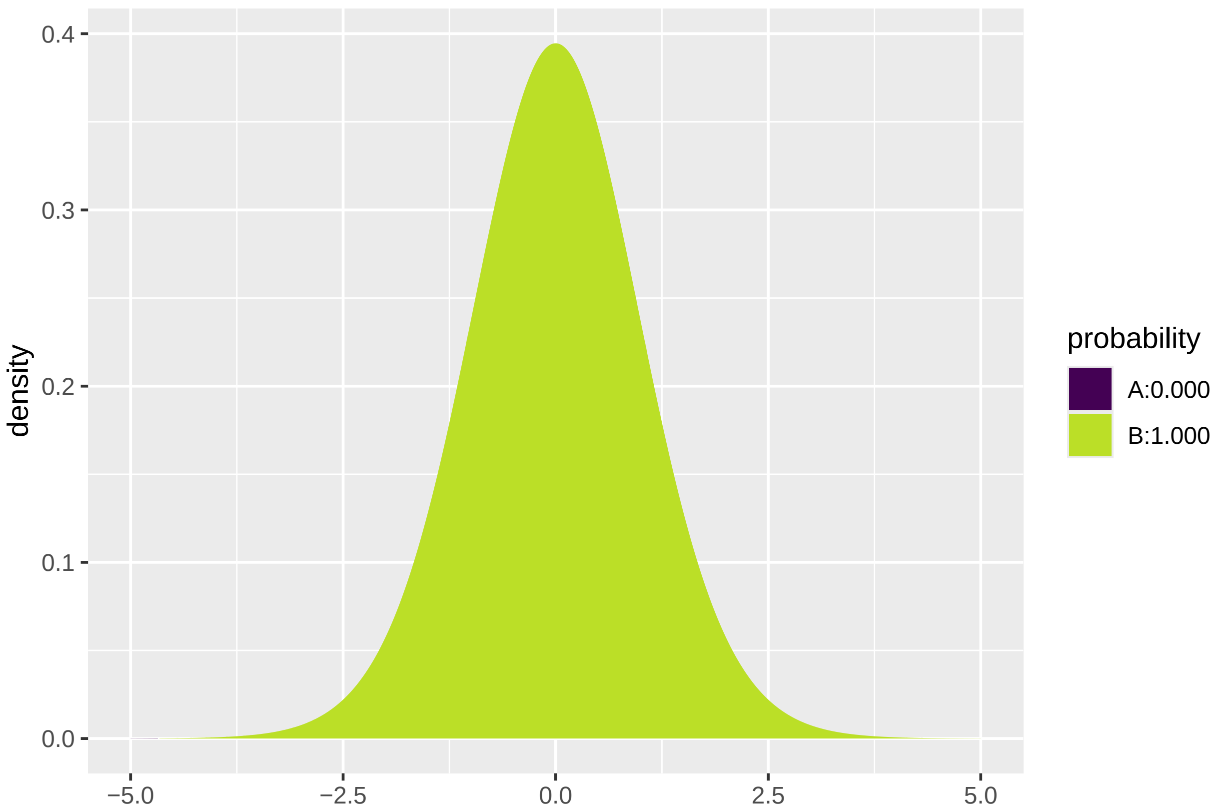
- We now make the analysis without assuming equal variances:

```
library(mosaic)
fv <- favstats(mpg ~ vs, data = mtcars)
fv
```

```
##   vs  min   Q1 median   Q3  max mean   sd  n missing
## 1  0 10.4 14.8   15.7 19.1 26.0 16.6 3.86 18       0
## 2  1 17.8 21.4   22.8 29.6 33.9 24.6 5.38 14       0
```

- Difference: $d = 16.6167 - (24.5571) = -7.9405$.
- Sample sizes: $n_1 = 18$ and $n_2 = 14$.
- Estimated standard deviations: $s_1 = 3.8607$ (not v-shaped) and $s_2 = 5.379$ (v-shaped).
- Estimated standard error of difference: $se_d = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} = \sqrt{\frac{3.8607^2}{18} + \frac{5.379^2}{14}} = 1.7014$.
- Observed t-score for $H_0 : \ \mu_1 - \mu_2 = 0$ is: $\quad t_{obs} = \frac{d-0}{se_d} = \frac{-7.9405}{1.7014} = -4.6671$.
- The degrees of freedom can be found using R (see below) to be $df = 22.716$.
- We find the p-value:

```
2* pdist("t", q = -4.6671, df=22.716, xlim = c(-5, 5))
```




```
## [1] 0.0001098212
```

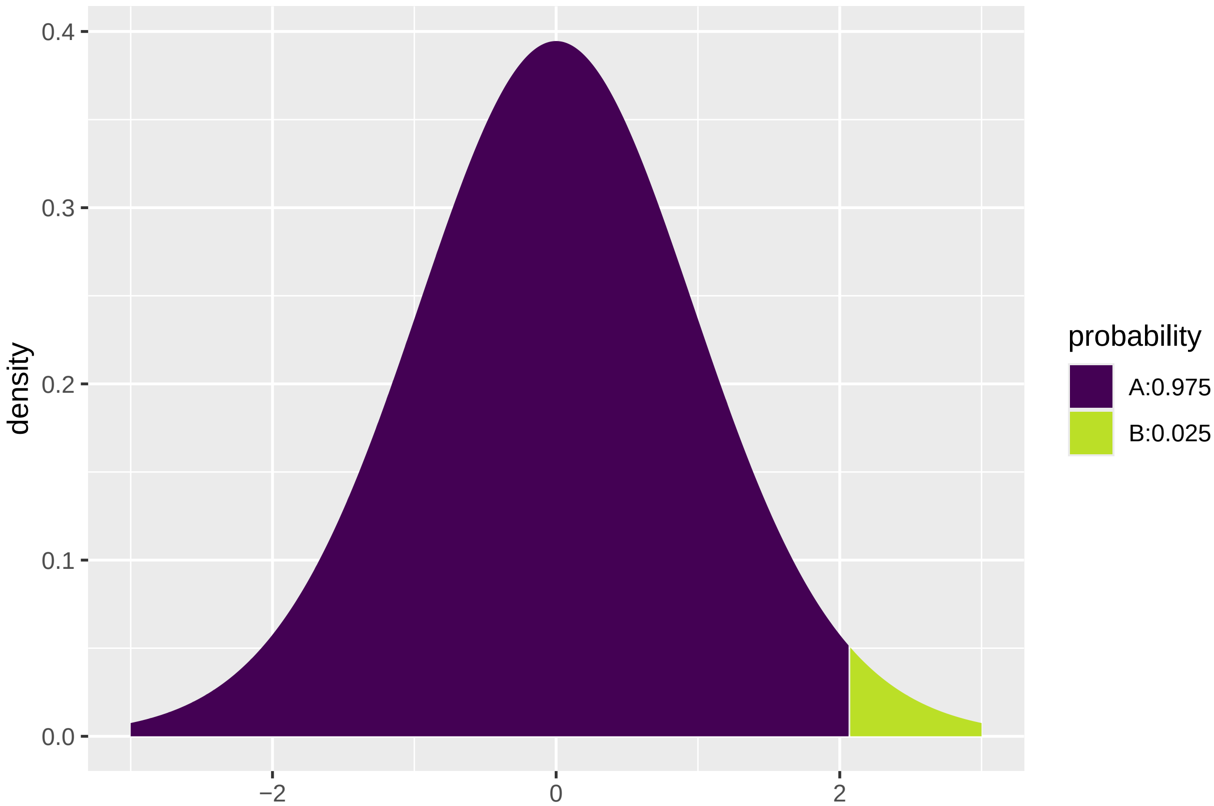
- We reject the null-hypothesis and conclude that the fuel consumption is different for the two engine types.

2.10 Example: Comparing two means (independent samples)

- Now we know there is a difference between the two population means. We can also make a 95% confidence interval for how large the difference $\mu_1 - \mu_2$ actually is by the formula

$$d \pm t_{crit} se_d$$

```r
qdist("t", p = 1-0.05/2, df=22.716, xlim = c(-3, 3))
```




```
## [1] 2.07009
```

- Inserting the values from the previous slide yields

$$[-7.94 - 2.07 * 1.70; -7.94 + 2.07 * 1.70] = [-11.5, -4.4].$$

- We are 95% confident that the difference in fuel consumption is between the two engine types is between -4.4mpg and -11.5mpg.

2.11 T-test in R (Independent samples)

- We can leave all the calculations to **R** by using `t.test`:

```r
t.test(mpg ~ vs, data = mtcars,var.equal = FALSE)
```

```
##
##  Welch Two Sample t-test
##
## data:  mpg by vs
## t = -4.6671, df = 22.716, p-value = 0.0001098
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
## 95 percent confidence interval:
##  -11.462508  -4.418445
## sample estimates:
## mean in group 0 mean in group 1
##        16.61667        24.55714
```

- We recognize the t-score -4.6671, the p-value 0.0001, and the confidence interval $[-11.5; -4.4]$. The estimated degrees of freedom can be found in the output to be $df = 22.716$.

2.12 Test for equal variances (Independent samples)

- In order to decide whether to use the t-test with equal or unequal variance, we may test the hypothesis $H_0 : \sigma_1^2 = \sigma_2^2$.
- As test statistic we use
$$F_{obs} = \frac{s_1^2}{s_2^2}.$$
- If the null-hypothesis is true, we expect F_{obs} to take values close to 1. Small and large values are critical for H_0.
- Under H_0, F_{obs} follows a so-called F-distribution with $df_1 = n_1 - 1$ and $df_2 = n_2 - 1$ degrees of freedom.
  - If $F_{obs} < 1$ we reject the null-hypothesis if two times the probability of getting something smaller than F_{obs} is less than the significance level.
  - If $F_{obs} > 1$ we reject the null-hypothesis if two times the probability of getting something larger than F_{obs} is less than the significance level.

2.12.1 Example: Test for equal variances (Independent samples)

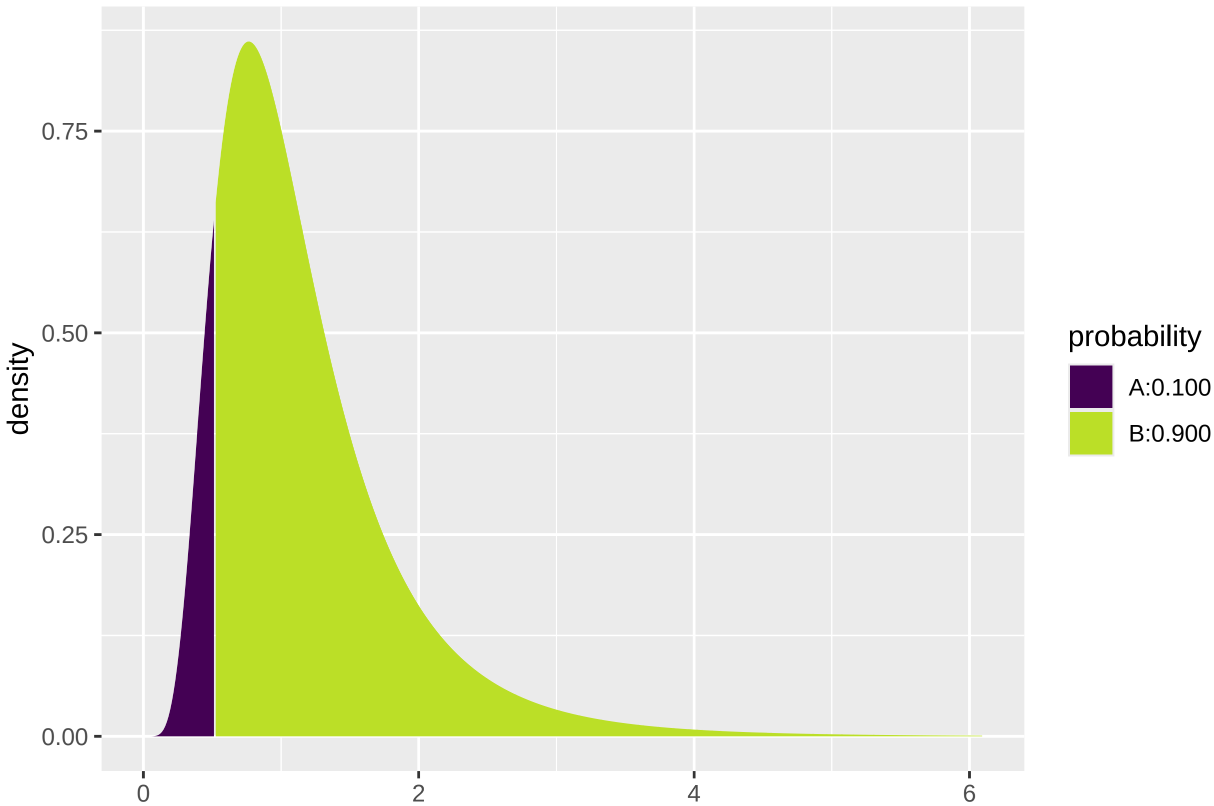
- To test whether the variance is the same for the two engine types in the `mtcars` example, we first compute the sample variances.

```
var(mpg~vs,data=mtcars)
```

```
##        0        1
## 14.90500 28.93341
```

- We compute $F_{obs} = \frac{s_1^2}{s_2^2} = \frac{14.9}{28.9} = 0.516$.
- The probability of observing something smaller than F_{obs} in an F-distribution with $df_1 = n_1 - 1 = 17$ and $df_2 = n_2 - 1 = 13$:

```
pdist("f", 0.516, df1=17, df2=13)
```




```
## [1] 0.1004094
```

- The p-value is $2 * 0.1004 = 0.2008$. Here we multiply by two because the test is two-sided (large values would also have been critical).
- We find no evidence against the null-hypothesis.

2.13 Comparison of two means: paired t-test (dependent samples)

- We now consider the case where we have two samples from two different populations but the observations in the two samples are **paired**.
  - For each pair, we can compute the difference between the two observations.
  - We now have one sample of observed differences.
  - We apply the the one-sample t-test from Lecture 2.1 to test whether the mean difference is zero.
- **Example:** Suppose we make the following experiment:
  - Choose 32 students at random and measure their average reaction time in a driving simulator while they are listening to radio or audio books.
  - Later the same 32 students redo the simulated driving while talking on a cell phone.
  - We are interested in whether or not the fact that you are actively participating in a conversation changes your average reaction time compared to when you are passively listening.
- So we have 2 samples corresponding to with/without phone. In this case we have **paired** samples, since we have 2 measurement for each student.
- We use the following strategy for analysis:
  - For each student calculate **the change** in average reaction time with and without talking on the phone.
  - The changes $d_1, d_2, \ldots, d_{32}$ are now considered as **ONE** sample from a population with mean μ.

- Test the hypothesis $H_0 : \mu = 0$ as usual (using a one-sample t-test).

2.13.1 Reaction time: data example

- Data is organized in a data frame with 3 variables:
  - `student` (integer – a simple id)
  - `reaction_time` (numeric – average reaction time in milliseconds)
  - `phone` (factor – `yes`/`no` indicating whether speaking on the phone)

```
reaction <- read.delim("https://asta.math.aau.dk/datasets?file=reaction.txt")
head(reaction, n = 3)
```

```
##   student reaction_time phone
## 1       1           604    no
## 2       2           556    no
## 3       3           540    no
```

- We first manually find the reaction time difference for each student and do a one sample t-test on this difference:

```
yes <- subset(reaction, phone == "yes")
no  <- subset(reaction, phone == "no")
all(yes$student == no$student)
```

```
## [1] TRUE
```

```
reaction_paired <- data.frame(student = no$student, yes = yes$reaction_time, no = no$reaction_time)
reaction_paired$diff <- reaction_paired$yes - reaction_paired$no
head(reaction_paired)
```

```
##   student yes  no diff
## 1       1 636 604   32
## 2       2 623 556   67
## 3       3 615 540   75
## 4       4 672 522  150
## 5       5 601 459  142
## 6       6 600 544   56
```

```
t.test( ~ diff, data = reaction_paired)
```

```
## 
##  One Sample t-test
## 
## data:  diff
## t = 5.4563, df = 31, p-value = 5.803e-06
## alternative hypothesis: true mean is not equal to 0
## 95 percent confidence interval:
##  31.70186 69.54814
## sample estimates:
## mean of x 
##    50.625
```

- With a p-value of 0.0000058 we reject the null-hypothesis that speaking on the phone has no influence on the reaction time.
- We can avoid the manual calculations and let **R** perform the significance test by using `t.test` with `paired = TRUE`:

```r
t.test(reaction_paired$no, reaction_paired$yes, paired = TRUE)
```

```
##
##  Paired t-test
##
## data:  reaction_paired$no and reaction_paired$yes
## t = -5.4563, df = 31, p-value = 5.803e-06
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
##  -69.54814 -31.70186
## sample estimates:
## mean difference
##         -50.625
```

2.14 Response variable and explanatory variable

- The situation with two populations is an example where we have: * A **response variable** (or outcome, dependent variable).
  - An **explanatory variable** (or independent variable, covariate) that divides data in 2 groups.
- We are interested in the effect of the explanatory variable on the response variable.
  - For instance in the `mtcars` data, `mpg` is the response variable and `vs` is the explanatory variable.
- In this lecture we consider the case with one discrete explanatory variable. Module 3 is concerned with the case of one or more continuous variables.

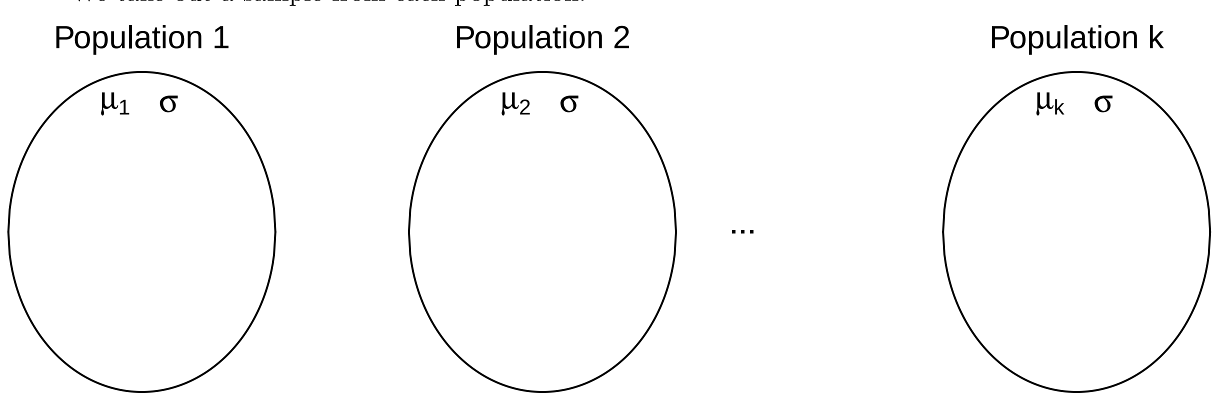
3 More than two groups (Analysis of variance)

3.1 More than two populations

- We are now going to consider a situation where we have k populations with mean values $\mu_1, \dots, \mu_k$.
- We assume that each population follows a normal distribution and that the standard deviation is the same in all populations.
- We are interested in the null-hypothesis that all k populations have the same mean, i.e.

$$H_0: \mu_1 = \dots = \mu_k.$$
$$H_a: \text{ not all } \mu_1, \dots \mu_k \text{ are the same.}$$

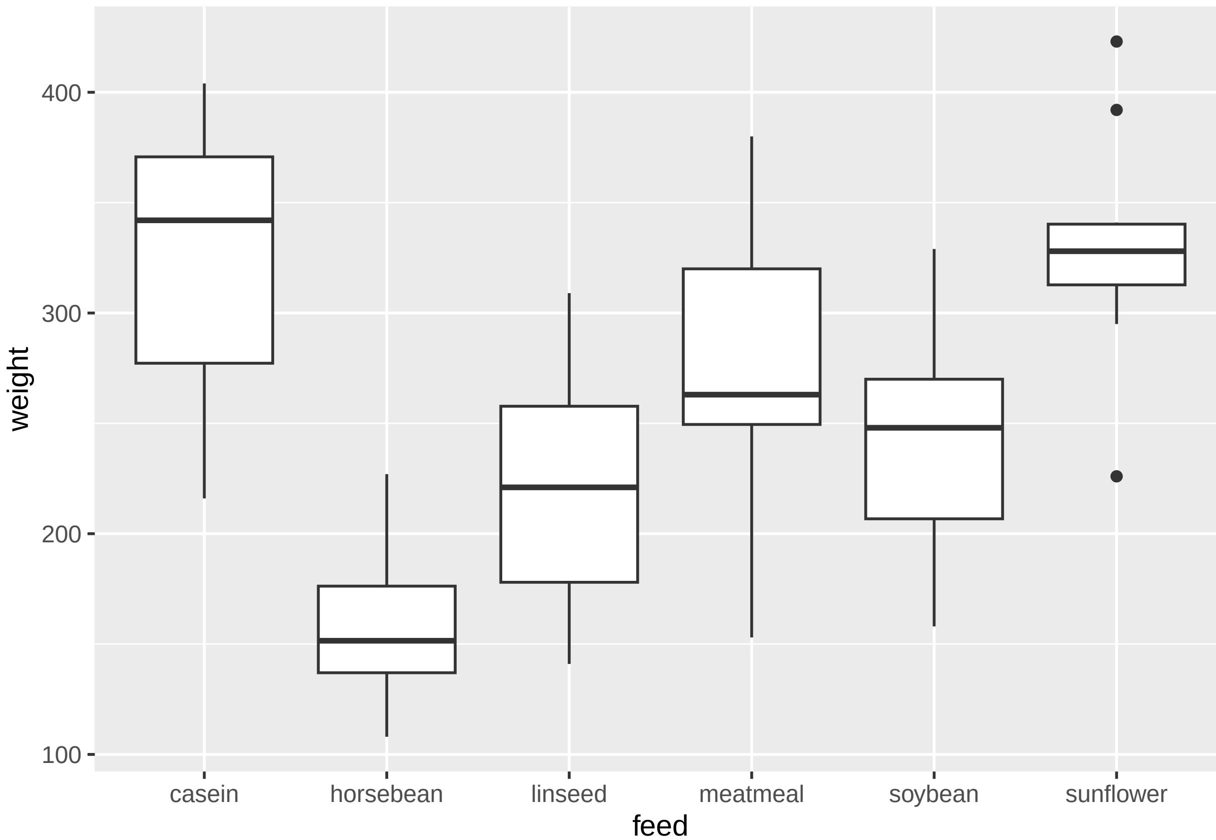
- We take out a sample from each population.



3.1.1 Data example

- The data set `chickwts` is available in `R`, and on the course webpage.
- 71 newly hatched chickens were randomly allocated into six groups, and each group was given a different feed supplement.
- Their weights in grams after six weeks are given along with feed types, i.e. we have a sample with corresponding measurements of 2 variables:
  - `weight`: a numeric variable giving the chicken weight.
  - `feed`: a factor giving the feed type.
- Always start with some graphics:

```
library(mosaic)
gf_boxplot(weight ~ feed, data = chickwts)
```



3.2 Estimation of mean values

- We estimate the mean in each group by the sample mean inside that group, i.e. $\hat{\mu}_i = \bar{x}_i$, $i = 1, \ldots, k$.
- We use `mean` to find the mean, for each group:

```
mean(weight ~ feed, data = chickwts)
```

```
##    casein horsebean   linseed  meatmeal   soybean sunflower
##  323.5833  160.2000  218.7500  276.9091  246.4286  328.9167
```

- We can e.g. see that the sample mean is 323.6, when `feed=casein` but 160.2, when `feed=horsebean`.
- Is this a significant difference ?

3.3 Contrasts

- If we want compare groups, it is convenient to formulate the model using contrasts.
- One group is chosen as the **reference group**, which all other groups are compared to.
    - Sometimes there is a group corresponding to "no treatment" and we are interested in the effect of different treatments. Other times the reference group can be arbitrary.
- If group 1 is the reference group, the mean values in the remaining groups groups can be expressed as

$$\mu_i = \mu_1 + \alpha_i,$$

where $\alpha_i = (\mu_i - \mu_1)$ is the difference between group i and the reference group. The α_i are called **contrasts**.

3.3.1 Example: contrast estimates

```
model <- lm(weight ~ feed, data = chickwts)
summary(model)
```

```
##
## Call:
## lm(formula = weight ~ feed, data = chickwts)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -123.909  -34.413    1.571   38.170  103.091
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)    323.583     15.834  20.436  < 2e-16 ***
## feedhorsebean -163.383     23.485  -6.957 2.07e-09 ***
## feedlinseed   -104.833     22.393  -4.682 1.49e-05 ***
## feedmeatmeal   -46.674     22.896  -2.039 0.045567 *
## feedsoybean    -77.155     21.578  -3.576 0.000665 ***
## feedsunflower    5.333     22.393   0.238 0.812495
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 54.85 on 65 degrees of freedom
## Multiple R-squared:  0.5417, Adjusted R-squared:  0.5064
## F-statistic: 15.36 on 5 and 65 DF,  p-value: 5.936e-10
```

- In the example the groups are different `feeds`. R chooses the lexicographically smallest, which is `casein`, to be the reference group.
- We get information about contrasts and their significance:
- `Intercept` is the estimated mean $\hat{\mu}_{casein} = 323.583$ in the reference group.
    - In the same line, there is also a test of the null-hypothesis $H_0 : \mu_1 = 0$ that the weight after 6 weeks is 0 ($p < 2 \times 10^{-16}$) (of course, chickens grow a lot over 6 weeks).
- The line `feedhorsebean` estimates the contrast $\alpha_{horsebean}$ between the `casein` and `horsebean` group to be $\hat{\alpha}_{horsebean} = -163.383$.
    - The null-hypothesis that there is no difference between casein and horsebean ($H_0 : \alpha_{horsebean} = 0$) is rejected with p=$2 \times 10^{-9}$.

3.4 Overall test for effect

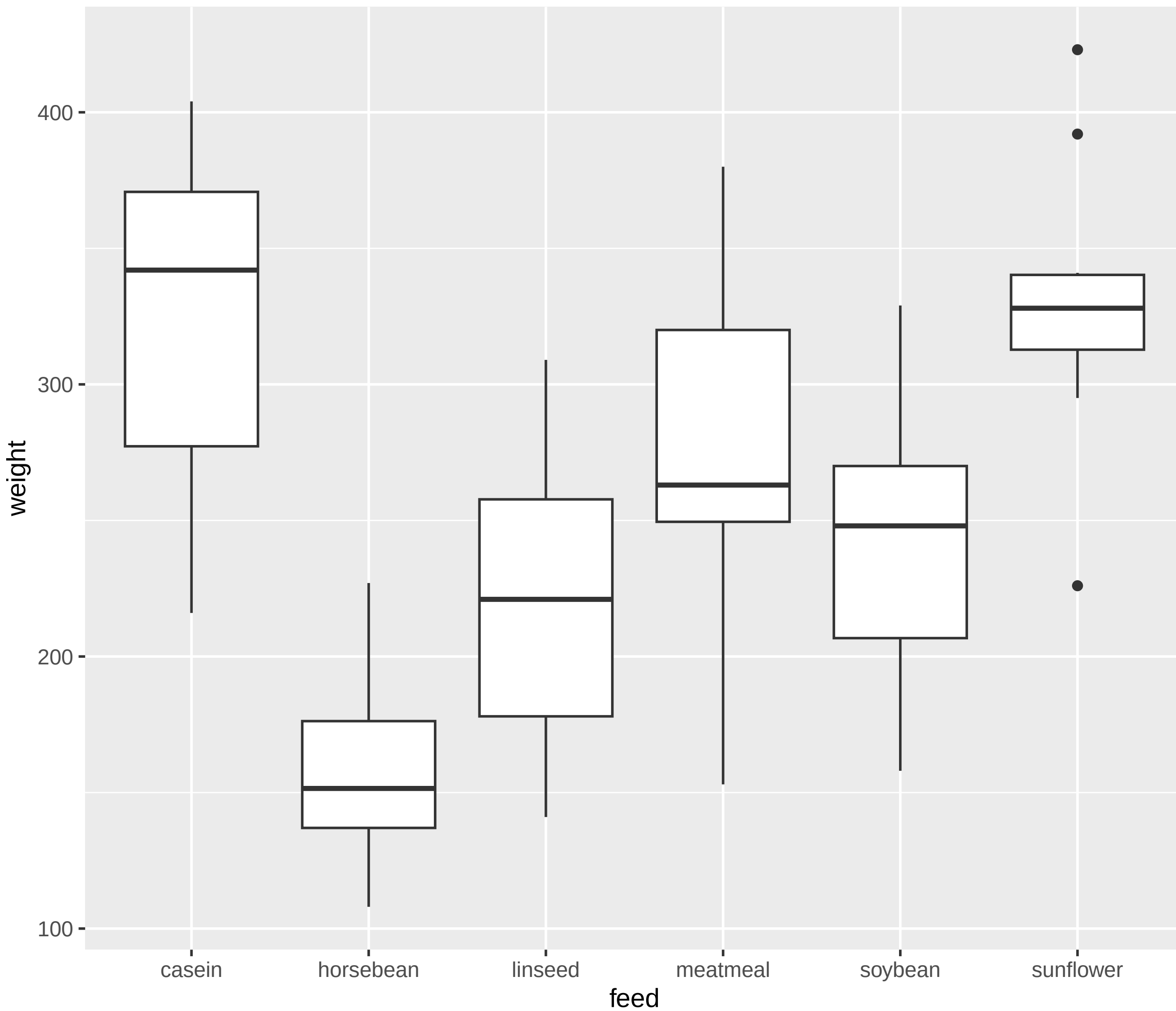
- We are now interested in testing the null-hypothesis

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_k \quad \text{against} \quad H_a : \text{ Not all of the population means are the same}$$

- Alternatively

$$H_0 : \alpha_2 = \alpha_3 = \cdots = \alpha_k = 0, \quad H_a : \text{ At least one contrast is non-zero.}$$

- Idea: Compare variation within groups and variation between groups.



3.5 Test statistic

- We use the test statistic

$$F_{obs} = \frac{(TSS - SSE)/(k-1)}{SSE/(n-k)}.$$

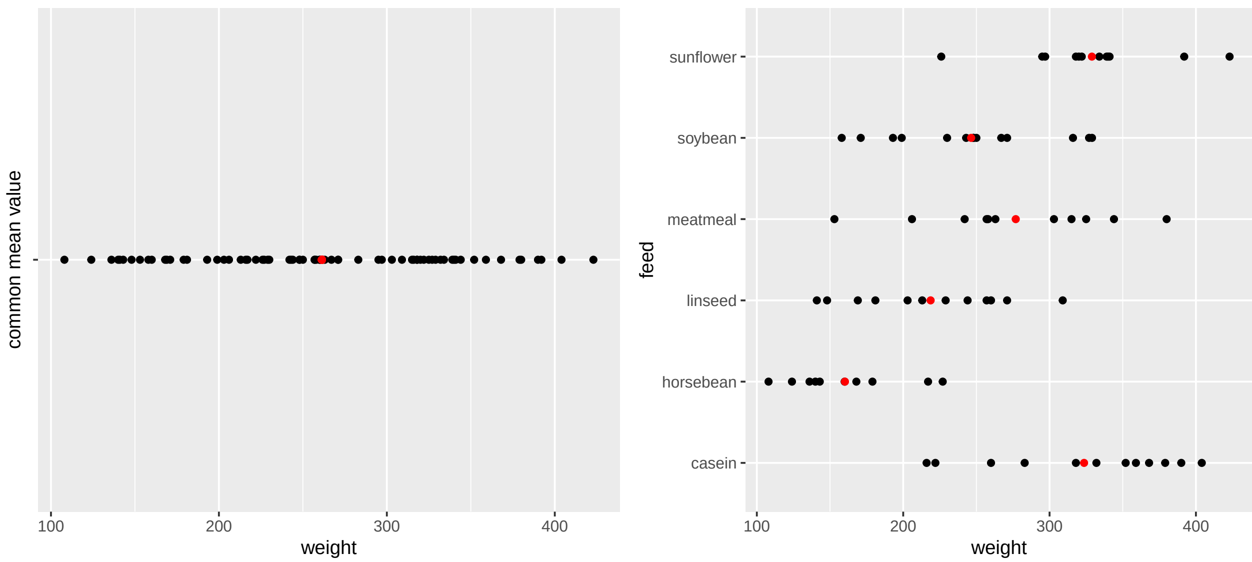
- If observations from group i are called x_{ij}, $j = 1, \ldots, k$, we have:
  - $TSS = \sum_i \sum_j (x_{ij} - \bar{x})^2$, where $\bar{x}$ is the average of all observations from all groups.

- $SSE = \sum_i \sum_j (x_{ij} - \bar{x}_i)^2$.

- Interpretation:
  - TSS: error sum of squares if common mean.
  - SSE: error sum of squares if different means.
  - TSS-SSE: how much does error sum of squares increase if means are restricted to be equal.
- One can show that TSS-SSE measures the variance of group means around common mean.
- Interpretation is thus

$$F_{obs} = \frac{\text{variance between groups}}{\text{variance within groups}}.$$

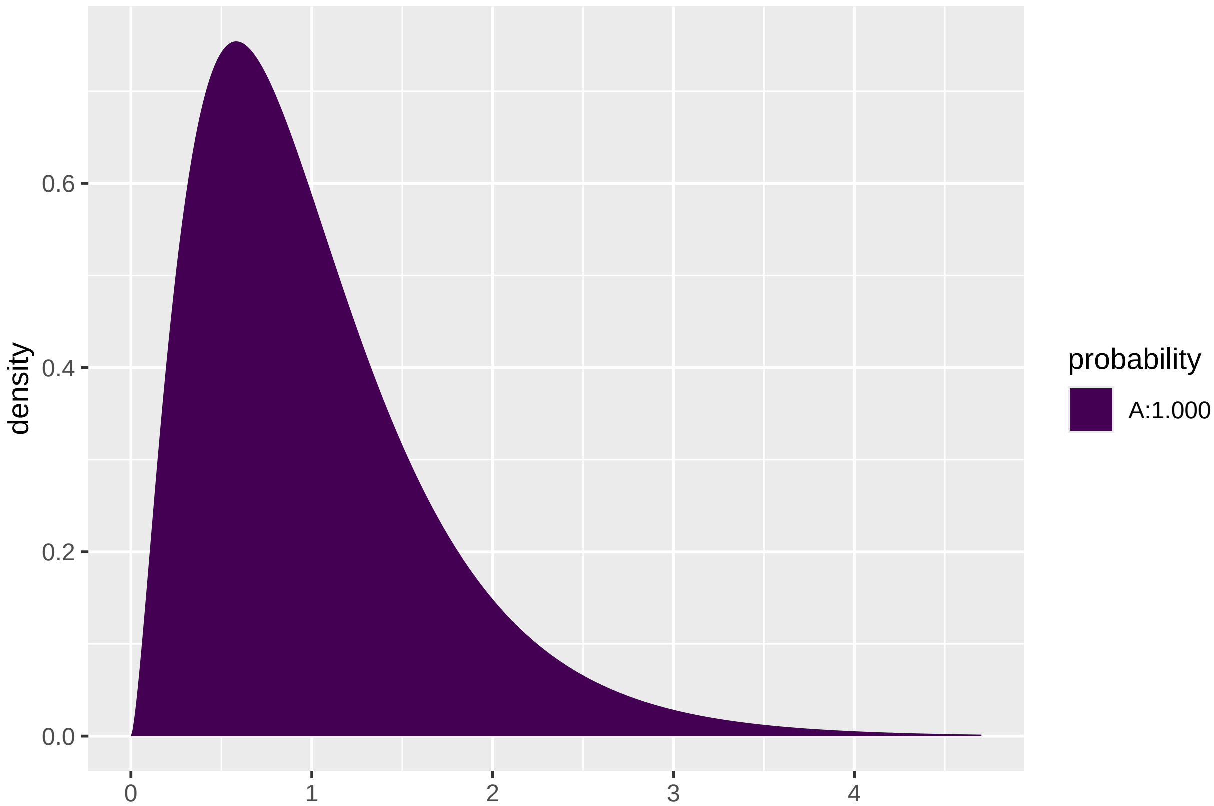
```
## Warning in geom_point(aes(x = red_dot), color = "red"): All aesthetics have length 1, but the data ha
## i Please consider using `annotate()` or provide this layer with data containing a single row.
```



3.6 The F-test

- A large variation between groups compared to the variation within groups points against H_0.
- Thus, large values are critical for the null-hypothesis.
- Under the null-hypothesis, F_{obs} follows an F-distribution with $df_1 = k - 1$ and $df_2 = n - k$ degrees of freedom.
- A p-value for the null-hypothesis is the probability of observing something larger than F_{obs} in an F-distribution with df_1 and df_2 degrees of freedom.
- For instance if $F_{obs} = 15.36$ with $df_1 = 5$ and $df_2 = 65$ degrees of freedom:

```
1 - pdist("f", 15.36, df1=5, df2=65)
```




```
## [1] 5.967948e-10
```

3.7 Example

```r
model <- lm(weight ~ feed, data = chickwts)
summary(model)
```

```
##
## Call:
## lm(formula = weight ~ feed, data = chickwts)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -123.909  -34.413    1.571   38.170  103.091
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)    323.583     15.834  20.436  < 2e-16 ***
## feedhorsebean -163.383     23.485  -6.957 2.07e-09 ***
## feedlinseed   -104.833     22.393  -4.682 1.49e-05 ***
## feedmeatmeal   -46.674     22.896  -2.039 0.045567 *
## feedsoybean    -77.155     21.578  -3.576 0.000665 ***
## feedsunflower    5.333     22.393   0.238 0.812495
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 54.85 on 65 degrees of freedom
## Multiple R-squared:  0.5417, Adjusted R-squared:  0.5064
```

```
## F-statistic: 15.36 on 5 and 65 DF,  p-value: 5.936e-10
```

- The last line gives us the value of $F_{obs} = 15.36$ and the corresponding p-value (5.9×10^{-10}). Clearly there is a significant difference between the types of `feed`.

ASTA

The ASTA team

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1 The regression problem

1.1 We want to predict

- We will study the dataset `trees`, which is on the course website (and actually also already available in R).

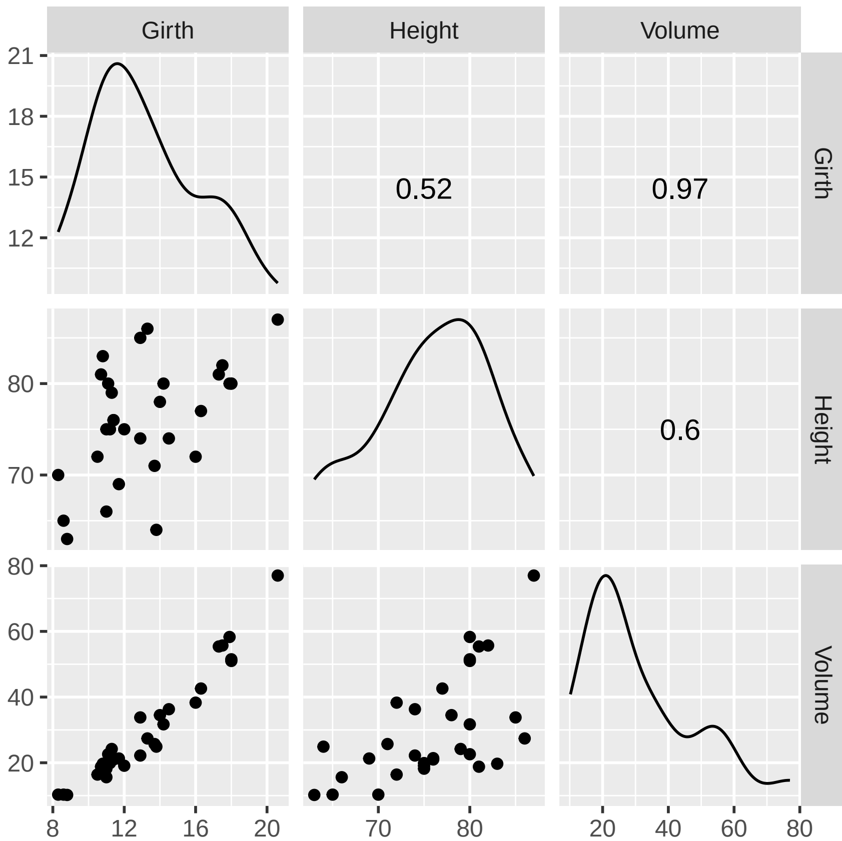
```
trees <- read.delim("https://asta.math.aau.dk/datasets?file=trees.txt")
```

- In this experiment we have measurements of 3 variables for 31 randomly chosen trees:
- `Girth` numeric. Tree diameter in inches.
- `Height` numeric. Height in ft.
- `Volume` numeric. Volume of timber in cubic ft.
- We want to predict the tree volume, if we measure the tree height and/or the tree girth (diameter).
- This type of problem is called **regression**.
- Relevant terminology:
  - We measure a quantitative **response** y, e.g. `Volume`.
  - In connection with the response value y we also measure one (later we will consider several) potential **explanatory** variable x. Another name for the explanatory variable is **predictor**.

1.2 Initial graphics

- Any analysis starts with relevant graphics.

```
library(mosaic)
library(GGally)
ggscatmat(trees) # Scatter plot matrix from GGally package
```



- For each combination of the variables we plot the (x, y) values.
- It looks like `Girth` is a good predictor for `Volume`.
- If we only are interested in the association between two (and not three or more) variables we use the usual `gf_point` function.

1.3 Simple linear regression

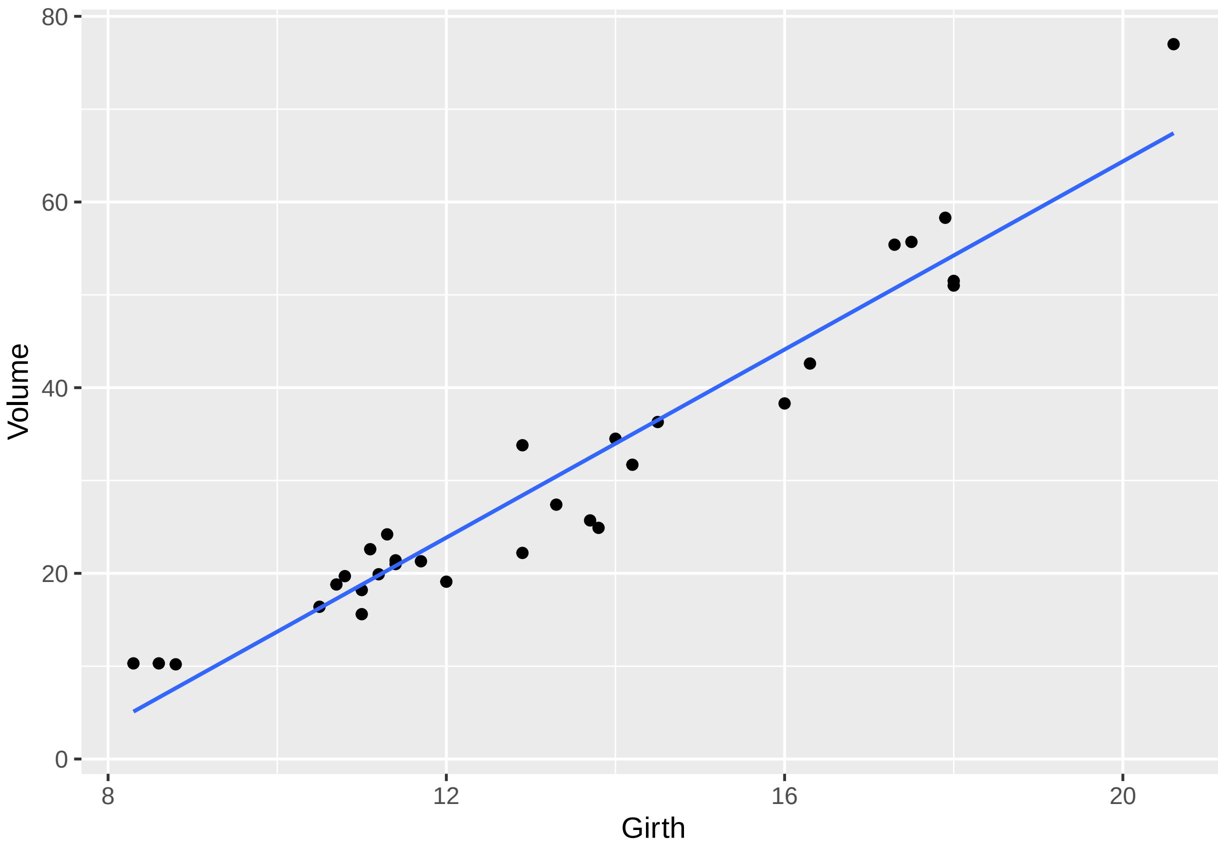
- We choose to use `x=Girth` as predictor for `y=Volume`. When we only use one predictor we are doing **simple regression**.
- The simplest **model** to describe an association between **response** y and a **predictor** x is **simple linear regression**.
- I.e. ideally we see the picture

$$y(x) = \alpha + \beta x$$

  where
  - α is called the `Intercept` - the line's intercept with the y-axis, corresponding to the response for $x = 0$.
  - β is called `Slope` - the line's slope, corresponding to the change in response, when we increase the predictor by one unit.

```
gf_point(Volume ~ Girth, data = trees) %>% gf_lm()
```

```
## Warning: Using the `size` aesthetic with geom_line was deprecated in ggplot2 3.4.0.
## i Please use the `linewidth` aesthetic instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was generated.
```



1.4 Model for linear regression

- Assume we have a sample with joint measurements (x, y) of predictor and response.
- Ideally the model states that

$$y(x) = \alpha + \beta x,$$

  but due to random variation there are deviations from the line.
- What we observe can then be described by

$$y = \alpha + \beta x + \varepsilon,$$

  where ε is a **random error**, which causes deviations from the line.
- We will continue under the following **fundamental assumption**:
  - The errors ε are normally distributed with mean zero and standard deviation σ.
- We call σ the **conditional standard deviation** given x, since it describes the variation in y around the regression line, when we know x.

1.5 Least squares

- In summary, we have a model with 3 parameters:
  - (α, β) which determine the line
  - σ which is the standard deviation of the deviations from the line.
- How are these estimated, when we have a sample $(x_1, y_1), \ldots, (x_n, y_n)$ of pairs of (x, y) values?
- To do this we focus on the errors

$$\varepsilon_i = y_i - \alpha - \beta x_i$$

  which should be as close to 0 as possible in order to fit the data best possible.

- We will choose the line, which minimizes the sum of squares of the errors:

$$\sum_{i=1}^{n} \varepsilon_i^2 = \sum_{i=1}^{n} (y_i - \alpha - \beta x_i)^2.$$

- If we set the partial derivatives to zero we obtain two linear equations for the unknowns (α, β), where the solution (a, b) is given by:

$$b = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2} \quad \text{and} \quad a = \bar{y} - b\bar{x}$$

1.6 The prediction equation and residuals

- The equation given by the estimates $(\hat{\alpha}, \hat{\beta}) = (a, b)$,

$$\hat{y} = a + bx$$

  is called the **regression equation** or **the prediction equation**, since it can be used to predict y for any value of x.
- Note: The prediction equation is determined by the current sample. I.e. there is an uncertainty attached to it. A new sample would without any doubt give a different prediction equation.
- Our best estimate of the errors is

$$e_i = y_i - \hat{y} = y_i - a - bx_i,$$

  i.e. the vertical deviations from the prediction line.
- These quantities are called **residuals**.
- We have that
  - The prediction line passes through the point $(\bar{x}, \bar{y})$.
  - The sum of the residuals is zero.

1.7 Estimation of conditional standard deviation

- To estimate σ we need the **Sum of Squared Errors**

$$SSE = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2,$$

  which is the squared distance between the model and data.

- We then estimate σ by the quantity

$$s = \sqrt{\frac{SSE}{n-2}}$$

- Instead of n we divide SSE with the **degrees of freedom** $df = n - 2$. Theory shows, that this is reasonable.

- The degrees of freedom df are determined as the sample size minus the number of parameters in the regression equation.

- In the current setup we have 2 parameters: (α, β).

1.8 Example in R

```
model <- lm(Volume ~ Girth, data = trees)
summary(model)
```

```
##
## Call:
## lm(formula = Volume ~ Girth, data = trees)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -8.065 -3.107  0.152  3.495  9.587
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -36.9435     3.3651  -10.98 7.62e-12 ***
## Girth         5.0659     0.2474   20.48  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.252 on 29 degrees of freedom
## Multiple R-squared:  0.9353, Adjusted R-squared:  0.9331
## F-statistic: 419.4 on 1 and 29 DF,  p-value: < 2.2e-16
```

- The estimated residuals vary from -8.065 to 9.578 with median 0.152.
- The estimate of `Intercept` is $a = -36.9435$
- The estimate of slope of `Girth` is $b = 5.0659$
- The estimate of the conditional standard deviation (called residual standard error in **R**) is $s = 4.252$ with $31 - 2 = 29$ degrees of freedom.

1.9 Test for independence

- We consider the regression model
$$y = \alpha + \beta x + \varepsilon$$
where we use a sample to obtain estimates (a, b) of (α, β), the estimate s of σ and the degrees of freedom $df = n - 2$.
- We are going to test
$$H_0: \beta = 0 \quad \text{against} \quad H_a: \beta \neq 0$$
- The null hypothesis specifies, that y **doesn't** depend linearly on x.
- Observed values of b far away from zero are critical for the null-hypothesis?
- It can be shown that b has standard error
$$se_b = \frac{s}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}}$$
with $df = n - 2$ degrees of freedom.
- So, we want to use the test statistic
$$t_{\text{obs}} = \frac{b}{se_b}$$
which has to be evaluated in a t-distribution with df degrees of freedom.

1.10 Example

- Recall the summary of our example:

```
summary(model)
```

```
##
## Call:
## lm(formula = Volume ~ Girth, data = trees)
```

```
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -8.065 -3.107  0.152  3.495  9.587
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -36.9435     3.3651  -10.98 7.62e-12 ***
## Girth         5.0659     0.2474   20.48  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.252 on 29 degrees of freedom
## Multiple R-squared:  0.9353, Adjusted R-squared:  0.9331
## F-statistic: 419.4 on 1 and 29 DF,  p-value: < 2.2e-16
```

- As we noted previously $b = 5.0659$ and $s = 4.252$ with $df = 29$ degrees of freedom.
- In the second column(`Std. Error`) of the `Coefficients` table we find $se_b = 0.2474$.
- The observed t-score (test statistic) is then

$$t_{\text{obs}} = \frac{b}{se_b} = \frac{5.0659}{0.2474} = 20.48$$

  which also can be found in the third column(`t value`).
- The corresponding p-value is found in the usual way by using the t-distribution with 29 degrees of freedom.
- In the fourth column(`Pr(>|t|)`) we see that the p-value is less than 2×10^{-16}. This is no surprise since the t-score was way above 3.

1.11 Confidence interval for slope

- When we have both the standard error and the reference distribution, we can construct a confidence interval in the usual way:

$$b \pm t_{crit} se_b,$$

  where the t-score is determined by the confidence level and we find this value using `qdist` in **R**.

- In our example we have 29 degrees of freedom and with a confidence level of 95% we get $t_{crit} =$ `qdist("t", 0.975, df = 29)`$= 2.045$.

- If you are lazy (like most statisticians are):

```
confint(model)
```

```
##                  2.5 %     97.5 %
## (Intercept) -43.825953 -30.060965
## Girth         4.559914   5.571799
```

- i.e. $(4.56, 5.57)$ is a 95% confidence interval for the slope of `Girth`.

1.12 Correlation

- The estimated slope b in a linear regression doesn't say anything about the strength of association between y and x.
- `Girth` was measured in inches, but if we rather measured it in kilometers the slope is much larger: An increase of 1km in `Girth` yield an enormous increase in `Volume`.
- Let s_y and s_x denote the sample standard deviation of y and x, respectively.

- The corresponding t-scores
$$y_t = \frac{y}{s_y} \quad \text{and} \quad x_t = \frac{x}{s_x}$$
are independent of the chosen measurement scale.
- The corresponding prediction equation is then
$$\hat{y}_t = \frac{a}{s_y} + \frac{s_x}{s_y} b x_t$$
- i.e. **the standardized regression coefficient** (slope) is
$$r = \frac{s_x}{s_y} b$$
which also is called **the (sample) correlation** between y and x.

- It can be shown that:
  - $-1 \leq r \leq 1$
  - The absolute value of r measures the (linear) strength of dependence between y and x.
  - When $r = 1$ all the points are on the prediction line, which has positive slope.
  - When $r = -1$ all the points are on the prediction line, which has negative slope.
- To calculate the sample correlation in **R**:

```
cor(trees)
```

```
##            Girth    Height    Volume
## Girth  1.0000000 0.5192801 0.9671194
## Height 0.5192801 1.0000000 0.5982497
## Volume 0.9671194 0.5982497 1.0000000
```

- There is a strong positive correlation between `Volume` and `Girth` (r=0.967).
- Note, calling `cor` on a `data.frame` (like `trees`) only works when all columns are numeric. Otherwise the relevant numeric columns should be extracted like this:

```
cor(trees[,c("Height", "Girth", "Volume")])
```

which produces the same output as above.

- Alternatively, one can calculate the correlation between two variables of interest like:

```
cor(trees$Height, trees$Volume)
```

```
## [1] 0.5982497
```

2 R-squared: Reduction in prediction error

2.1 R-squared: Reduction in prediction error

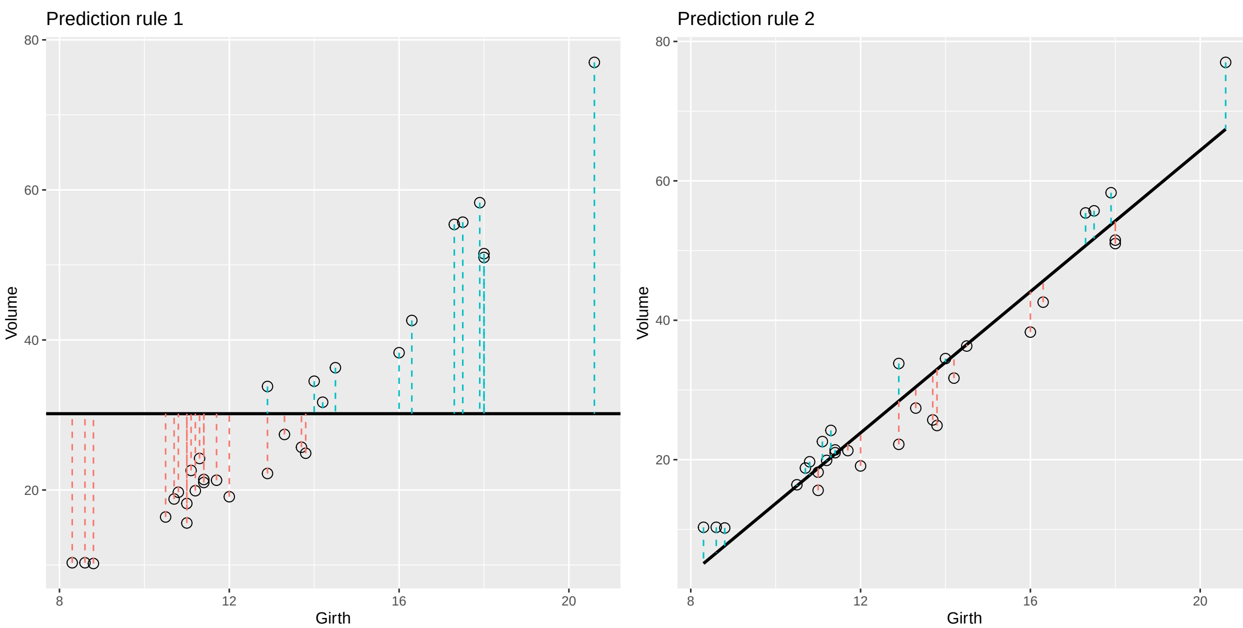
- We want to compare two different models used to predict the response y.
- Model 1: We do not use the knowledge of x, and use $\bar{y}$ to predict any y-measurement. The corresponding prediction error is defined as
$$TSS = \sum_{i=1}^{n} (y_i - \bar{y})^2$$
and is called the **Total Sum of Squares**.
- Model 2: We use the prediction equation $\hat{y} = a + bx$ to predict y_i. The corresponding prediction error is then the Sum of Squared Errors
$$SSE = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2.$$

- We then define
$$r^2 = \frac{TSS - SSE}{TSS}$$
which can be interpreted as the relative reduction in the prediction error, when we include x as explanatory variable.
- This is also called the **fraction of explained variation**, **coefficient of determination** or simply **r-squared**.
- For example if $r^2 = 0.65$, the interpretation is that x explains about 65% of the variation in y, whereas the rest is due to other sources of random variation.

2.2 Graphical illustration of sums of squares

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was generated.

## `geom_smooth()` using formula = 'y ~ x'
```



- Note the data points are the same in both plots. Only the prediction rule changes.
- The error of using Rule 1 is the total sum of squares $E_1 = TSS = \sum_{i=1}^n (y_i - \bar{y})^2$.
- The error of using Rule 2 is the residual sum of squares (sum of squared errors) $E_2 = SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2$.

2.3 r^2: Reduction in prediction error

- For the simple linear regression we have that
$$r^2 = \frac{TSS - SSE}{TSS}$$
is equal to the square of the correlation between y and x, so it makes sense to denote it r^2.
- Towards the bottom of the output below we can read off the value $r^2 = 0.9353 = 93.53\%$, which is a large fraction of explained variation.

```r
summary(model)
```

```
##
## Call:
## lm(formula = Volume ~ Girth, data = trees)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -8.065 -3.107  0.152  3.495  9.587
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -36.9435     3.3651  -10.98 7.62e-12 ***
## Girth         5.0659     0.2474   20.48  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.252 on 29 degrees of freedom
## Multiple R-squared:  0.9353, Adjusted R-squared:  0.9331
## F-statistic: 419.4 on 1 and 29 DF,  p-value: < 2.2e-16
```

3 Introduction to multiple regression model

3.1 Multiple regression model

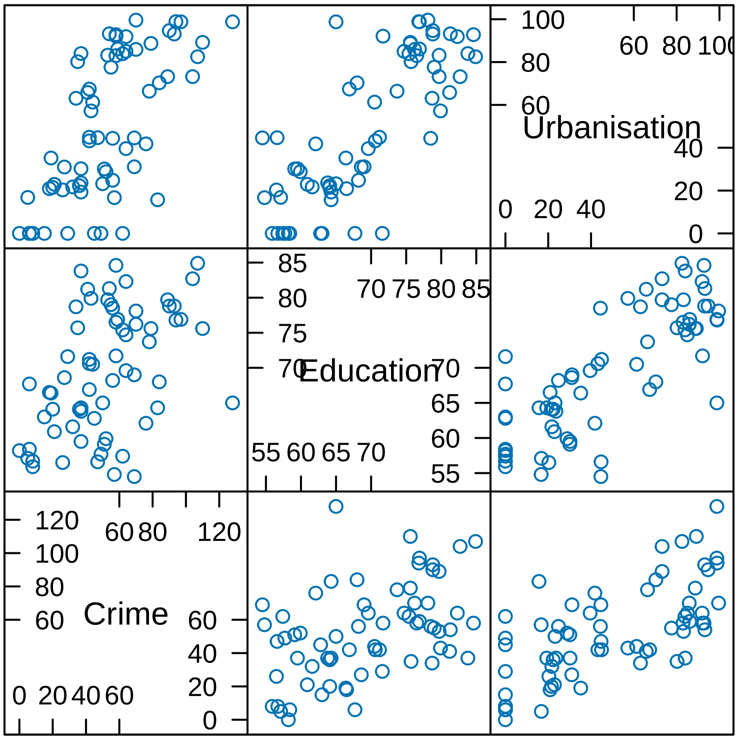
- We look at data set containing measurements from the 67 counties of Florida.
- Our focus is on
  - The response y: `Crime` which is the crime rate
  - The predictor x_1: `Education` which is proportion of the population with high school exam
  - The predictor x_2: `Urbanisation` which is proportion of the population living in urban areas

3.2 Example

```r
FL <- read.delim("https://asta.math.aau.dk/datasets?file=fl-crime.txt")
head(FL, n = 3)
```

```
##   Crime Education Urbanisation
## 1   104      82.7         73.2
## 2    20      64.1         21.5
## 3    64      74.7         85.0
```

```r
library(mosaic)
splom(FL) # Scatter PLOt Matrix
```



Scatter Plot Matrix

3.3 Correlations

- There is significant (p$\approx 7 \times 10^{-5}$) positive correlation (r=0.47) between `Crime` and `Education`
- Then there is also significant positive correlation (r=0.68) between `Crime` and `Urbanisation`

```
cor(FL)
```

```
##                  Crime Education Urbanisation
## Crime        1.0000000 0.4669119    0.6773678
## Education    0.4669119 1.0000000    0.7907190
## Urbanisation 0.6773678 0.7907190    1.0000000
```

```
cor.test(~ Crime + Education, data = FL)
```

```
##
##  Pearson's product-moment correlation
##
## data:  Crime and Education
## t = 4.2569, df = 65, p-value = 6.806e-05
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.2553414 0.6358104
## sample estimates:
##       cor
## 0.4669119
```

3.4 Several predictors

- Both `Education` (x_1) and `Urbanisation` (x_2) are pretty good predictors for `Crime` (y).

- We therefore want to consider the model

$$y = \alpha + \beta_1 x_1 + \beta_2 x_2 + \epsilon$$

- The errors ϵ are random noise with mean zero and standard deviation σ.
- The graph for the mean response is in other words a 2-dimensional plane in a 3-dimensional space.
- We determine estimates (a, b_1, b_2) for $(\alpha, \beta_1, \beta_2)$ via the least squares method, i.e deviations from the plane.

3.5 Example

```
model <- lm(Crime ~ Education + Urbanisation, data = FL)
summary(model)
```

```
##
## Call:
## lm(formula = Crime ~ Education + Urbanisation, data = FL)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -34.693 -15.742  -6.226  15.812  50.678
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   59.1181    28.3653   2.084   0.0411 *
## Education     -0.5834     0.4725  -1.235   0.2214
## Urbanisation   0.6825     0.1232   5.539 6.11e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 20.82 on 64 degrees of freedom
## Multiple R-squared:  0.4714, Adjusted R-squared:  0.4549
## F-statistic: 28.54 on 2 and 64 DF,  p-value: 1.379e-09
```

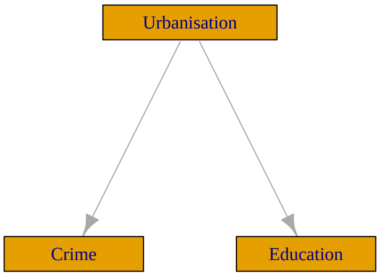
- From the output we find the prediction equation

$$\widehat{y} = 59 - 0.58x_1 + 0.68x_2$$

- Not exactly what we expected based on the correlation.
- Now there appears to be a negative association between y and x_1 (Simpsons Paradox)!
- We can also find the standard error (0.4725) and the corresponding t-score (-1.235) for the the slope of `Education`
- This yields a p-value of 22%, i.e. the slope is not significantly different from zero.

3.6 Simpsons paradox

- The example illustrates **Simpson's paradox**.
- When considered alone `Education` is a good predictor for `Crime` (with positive correlation).
- When we add `Urbanisation`, then `Education` has a negative effect on `Crime` (but not significant).



- A possible explanation is illustrated by the graph above.
  - Urbanisation has positive effect on both Education and Crime.
  - For a given level of urbanisation there is a (non-significant) negative association between Education and Crime.
  - Viewed alone Education is a good predictor for Crime. If Education has a large value, then this indicates a large value of Urbanisation and thereby a large value of Crime.

4 Example from last lecture

4.1 Crime data set

- The data are measurements from the 67 counties of Florida.
- Our focus is on
  - The response y: Crime which is the crime rate
  - The predictor x_1: Education which is proportion of the population with high school exam
  - The predictor x_2: Urbanisation which is proportion of the population living in urban areas

```
FL <- read.delim("https://asta.math.aau.dk/datasets?file=fl-crime.txt")
head(FL, n = 3)
```

```
##   Crime Education Urbanisation
## 1   104      82.7         73.2
## 2    20      64.1         21.5
## 3    64      74.7         85.0
```

4.2 Multiple regression model for crime data

- Both Education (x_1) and Urbanisation (x_2) were correlated with Crime (y).
- We consider the model

$$Y = \alpha + \beta_1 x_1 + \beta_2 x_2 + \epsilon$$

- The errors ϵ are random terms with a norm$(0, \sigma)$ distribution.
- The graph for the mean response is a 2-dimensional plane in a 3-dimensional space.

```
model <- lm(Crime ~ Education + Urbanisation, data = FL)
summary(model)
```

```
##
## Call:
## lm(formula = Crime ~ Education + Urbanisation, data = FL)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -34.693 -15.742  -6.226  15.812  50.678
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept)   59.1181    28.3653   2.084   0.0411 *
## Education     -0.5834     0.4725  -1.235   0.2214
## Urbanisation   0.6825     0.1232   5.539 6.11e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 20.82 on 64 degrees of freedom
## Multiple R-squared:  0.4714, Adjusted R-squared:  0.4549
## F-statistic: 28.54 on 2 and 64 DF,  p-value: 1.379e-09
```

- From the output we find the prediction equation

$$\widehat{y} = 59 - 0.58x_1 + 0.68x_2.$$

5 The general model

5.1 Regression model

- In a multiple regression we have
  - a response variable Y.
  - k predictor variables $x_1, x_2, \dots, x_k$.
- Multiple regression model:
$$Y = \alpha + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + \epsilon.$$
- The **systematic** part of the model says that **the mean response** is a linear function of the predictors:

$$E(Y|x_1, x_2, \dots, x_k) = \alpha + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k.$$

  - Here $E(Y|x_1, x_2, \dots, x_k)$ is used to denote the mean value of the response when we know the value of the predictors $x_1, \dots, x_k$.
- The **error** ϵ is a random variable having a distribution with
  - a normal distribution
  - mean 0
  - standard deviation σ

5.2 Interpretation of parameters

- In the multiple linear regression model

$$E(y|x_1, x_2, \dots, x_k) = \alpha + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k$$

  - The parameter α is the `Intercept`, corresponding to the mean response, when all predictors are equal to zero.
  - The parameters $(\beta_1, \beta_2, \cdots, \beta_k)$ are called **partial regression coefficients**.
- Imagine that all predictors but x_1 are held fixed. Then

$$E(y|x_1, x_2, \dots, x_k) = \tilde{\alpha} + \beta_1 x_1,$$

  where
$$\tilde{\alpha} = \alpha + \beta_2 x_2 + \cdots + \beta_k x_k.$$
- So the mean response depends linearly on x_1 when all other variables are kept fixed.
  - The line has slope β_1, which describes the change in the mean response, when x_1 is changed one unit.
  - The rate of change β_1 does not depend on the value of the remaining predictors. In this case we say that there is **no interaction** between the effects of the predictors on the response.
- The above holds similarly for the other partial regression coefficients.

6 Estimation

6.1 Estimation of model parameters

- Suppose we have a sample of size n.
- Based on the sample, we estimate $(\alpha, \beta_1, \beta_2, \dots, \beta_k)$ by the values $(a, b_1, b_2, \dots, b_k)$.
- Based on this estimate we obtain the **prediction equation** (or **regression equation**) as

$$\widehat{y} = a + b_1 x_1 + b_2 x_2 + \cdots + b_k x_k$$

- The difference between our observations and the predictions made by the prediction equation are called the **residuals** $e_i = y_i - \widehat{y}_i$.
- The estimates $(a, b_1, b_2, \dots, b_k)$ are chosen by the **least squares method**, which seeks to minimize the sum of squared residuals

$$\sum_{i=1}^{n} (y_i - \widehat{y}_i)^2.$$

6.2 Estimation of error variance

- Recall that σ^2 is the variance of the error terms in the model. Using the residuals as approximations of the errors in our sample, we estimate σ^2 by

$$s^2 = \frac{\sum_{i=1}^{n} (y_i - \widehat{y}_i)^2}{n - k - 1}.$$

- Rather than n we divide by **the degrees of freedom** $df = n - k - 1$. Theory shows, that this is reasonable.
  - The degrees of freedom df are determined by the sample size n minus the number of parameters in the regression equation.
  - Currently we have $k+1$ parameters: 1 intercept and k slopes.

7 Multiple R-squared

7.1 Multiple R^2

- We can compare two models to predict the response y.
- Model 1: We do not use the predictors, and use $\bar{y}$ to predict any y-measurement. The corresponding sum of squared prediction errors is
  - $TSS = \sum_{i=1}^{n} (y_i - \bar{y})^2$ and is called the **Total Sum of Squares**.
- Model 2: We use the multiple prediction equation with predictors $x_1, \dots, x_k$ to predict y. The sum of squared prediction errors is now
  - $SSE = \sum_{i=1}^{n} (y_i - \widehat{y}_i)^2$ and is called **Sum of Squared Errors**.
- We then define **the multiple coefficient of determination**

$$R^2 = \frac{TSS - SSE}{TSS}.$$

- Thus, R^2 is the relative reduction in squared prediction errors, when we use $x_1, x_2, \dots, x_k$ as explanatory variables.
  - We say that R^2 is the proportion of the total variation in the data that can be explained by $x_1, \dots, x_k$.
- Properties:
  - $0 \leq R^2 \leq 1$
  - $R^2 = 0$ means $TSS = SSE$, so the model does not improve when we include $x_1, \dots, x_k$ in the model.
  - $R^2 = 1$ means $SSE = 0$, so the observations are predicted perfectly by $x_1, \dots, x_k$.

7.2 Example

```r
summary(model)
```

```
##
## Call:
## lm(formula = Crime ~ Education + Urbanisation, data = FL)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -34.693 -15.742  -6.226  15.812  50.678
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   59.1181    28.3653   2.084   0.0411 *
## Education     -0.5834     0.4725  -1.235   0.2214
## Urbanisation   0.6825     0.1232   5.539 6.11e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 20.82 on 64 degrees of freedom
## Multiple R-squared:  0.4714, Adjusted R-squared:  0.4549
## F-statistic: 28.54 on 2 and 64 DF,  p-value: 1.379e-09
```

- The prediction equation is $\widehat{y} = 59 - 0.58x_1 + 0.68x_2$
- The estimate for σ is $s = 20.82$ (`Residual standard error` in **R**) with $df = 67 - 3 = 64$ degrees of freedom.
- Multiple $R^2 = 0.4714$, i.e. 47% of the variation in the response is explained by including the predictors in the model.
- The output provides a test of the hypothesis $H_0 : \beta_1 = 0$ corresponding to no effect of the predictor x_1.
  - The estimate $b_1 = -0.5834$ has standard error (`Std. Error`) $se = 0.4725$ with corresponding observed t-value (`t value`) $t_{\text{obs}} = \frac{-0.5834}{0.4725} = -1.235$.
  - This means that b_1 isn't significantly different from zero, since the p-value (`Pr(>|t|)`) is 22%. That means that we can exclude `Education` as a predictor.

7.3 Example

- Our final model is then a simple linear regression:

```r
model2 <- lm(Crime ~ Urbanisation, data = FL)
summary(model2)
```

```
##
## Call:
## lm(formula = Crime ~ Urbanisation, data = FL)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -34.766 -16.541  -4.741  16.521  49.632
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  24.54125    4.53930   5.406 9.85e-07 ***
## Urbanisation  0.56220    0.07573   7.424 3.08e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Residual standard error: 20.9 on 65 degrees of freedom
## Multiple R-squared:  0.4588, Adjusted R-squared:  0.4505
## F-statistic: 55.11 on 1 and 65 DF,  p-value: 3.084e-10
```

- The coefficient of determination always decreases, when the model is simpler. Now we have $R^2 = 46\%$, where before we had 47%. But the decrease is not significant as we will see next.

7.4 F-test for comparing two models

- We can compare two models, where one is obtained from the other by setting m parameters to zero, by an F-test.
- We can compare R^2 for the two models via the following F-statistic:

$$F_{obs} = \frac{(R_2^2 - R_1^2)/\mathrm{df}_1}{(1 - R_2^2)/\mathrm{df}_2}$$

  - R_2^2 corresponds to the larger model and R_1^2 corresponds to the smaller model.
  - Large values of F_{obs} means that R_2^2 is large compared to R_1^2, pointing towards the alternative hypothesis.
  - $\mathrm{df}_1 = m$ is the number of parameters set to zero in the null-hypothesis.
  - $\mathrm{df}_2 = n - k - 1$ where n is sample size and $k + 1$ is the number of unknown parameters in the larger model.
- In R the calculations are done using `anova`:

```r
anova(model2, model)
```

```
## Analysis of Variance Table
##
## Model 1: Crime ~ Urbanisation
## Model 2: Crime ~ Education + Urbanisation
##   Res.Df   RSS Df Sum of Sq      F Pr(>F)
## 1     65 28391
## 2     64 27730  1    660.61 1.5247 0.2214
```

8 Overall F-test for effect of predictors

8.1 F-test

- We consider the hypothesis

$$H_0 : \beta_1 = \beta_2 = \ldots = \beta_k = 0$$

  against the alternative that at least one β_i are non-zero.
  - This is the hypothesis that there is no effect of any of the predictors.
- As test statistic we use

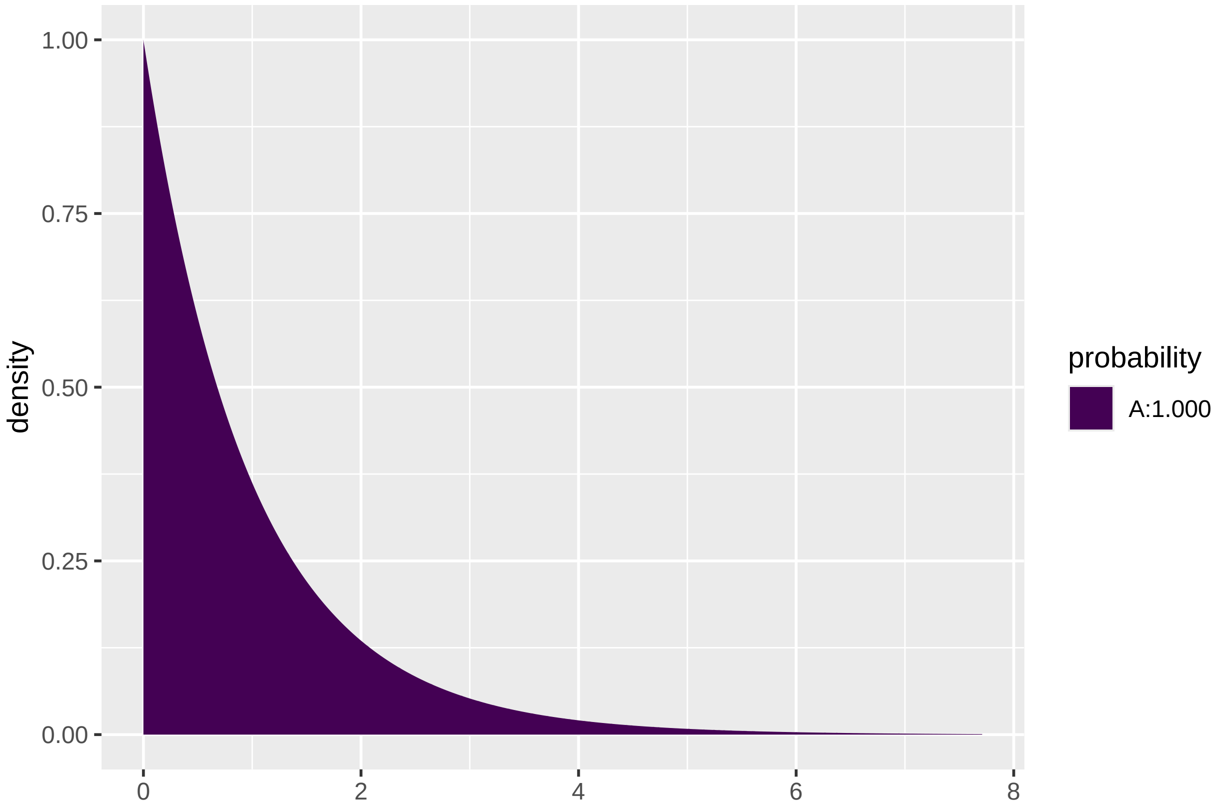
$$F_{obs} = \frac{R^2/k}{(1 - R^2)/(n - k - 1)}$$

- Large values of R^2 implies large values of F_{obs}, which points to the alternative hypothesis.
- Thus, the p-value is the probability of observing something larger than the computed F_{obs}.
- The distribution of F_{obs} under the null-hypothesis is an F-distribution with degrees of freedom
  - $df_1 = k$ (the number of parameters set to zero in the null-hypothesis).
  - $df_2 = n - k - 1$ (number of observations minus number of unknown parameters in the model).

8.2 Example

- We return to `Crime` and the prediction equation $\widehat{y} = 59 - 0.58x_1 + 0.68x_2$, where $n = 67$ and $R^2 = 0.4714$.
- We test the hypothesis $H_0 : \beta_1 = \beta_2 = 0$. We have
  - $df_1 = k = 2$ since 2 parameters are set to zero under H_0.
  - $df_2 = n - k - 1 = 67 - 2 - 1 = 64$.
  - Then we can calculate $F_{obs} = \frac{R^2/k}{(1-R^2)/(n-k-1)} = 28.54$
- To evaluate the value 28.54 in the relevant F-distribution:

```
1 - pdist("f", 28.54, df1=2, df2=64)
```




```
## [1] 1.378612e-09
```

- So p-value=1.38×10^{-9} (notice we don't multiply by 2 since this is a one-sided test; only large values point more towards the alternative than the null hypothesis).
- All this can be found in the summary output we already have:

```
summary(model)
```

```
##
## Call:
## lm(formula = Crime ~ Education + Urbanisation, data = FL)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -34.693 -15.742  -6.226  15.812  50.678
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  59.1181    28.3653   2.084   0.0411 *
```

```
## Education     -0.5834     0.4725  -1.235   0.2214
## Urbanisation   0.6825     0.1232   5.539 6.11e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 20.82 on 64 degrees of freedom
## Multiple R-squared:  0.4714, Adjusted R-squared:  0.4549
## F-statistic: 28.54 on 2 and 64 DF,  p-value: 1.379e-09
```

9 Interaction model

9.1 Interaction between effects of predictors

- Could it be possible that a combination of `Education` and `Urbanisation` is good for prediction? We investigate this using the **interaction model**

$$E(y|x_1, x_2) = \alpha + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2,$$

  where we have extended with a possible effect of the product $x_1 x_2$.
- If we fix x_2 in this model, the mean response is linear in x_1 with intercept $\alpha + \beta_2 x_2$ and slope $\beta_1 + \beta_3 x_2$, since

$$E(y|x_1, x_2) = (\alpha + \beta_2 x_2) + (\beta_1 + \beta_3 x_2)x_1.$$

  - The slope for x_1 now depends on the value of x_2!
  - Interaction means that the effect of x_1 on the response depends on the value of x_2.
  - Interaction **does not** mean that x_1 and x_2 affect each other.

9.2 Example - interaction model

- We fit the model for the `Crime` data set:

```r
model3 <- lm(Crime ~ Education * Urbanisation, data = FL)
summary(model3)
```

```
##
## Call:
## lm(formula = Crime ~ Education * Urbanisation, data = FL)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -35.181 -15.207  -6.457  14.559  49.889
##
## Coefficients:
##                        Estimate Std. Error t value Pr(>|t|)
## (Intercept)            19.31754   49.95871   0.387    0.700
## Education               0.03396    0.79381   0.043    0.966
## Urbanisation            1.51431    0.86809   1.744    0.086 .
## Education:Urbanisation -0.01205    0.01245  -0.968    0.337
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 20.83 on 63 degrees of freedom
## Multiple R-squared:  0.4792, Adjusted R-squared:  0.4544
## F-statistic: 19.32 on 3 and 63 DF,  p-value: 5.371e-09
```

- When we look at the p-values in the table nothing is significant at the 5% level!

- But the F-statistic tells us that the predictors collectively have a significant prediction ability.
- Why has the highly significant effect of x_2 disappeared? Because the predictors x_1 and $x_1 x_2$ are able to explain the same as x_2.
- Previously we only had x_1 as alternative explanation to x_2 - and that wasn't enough.
- The phenomenon is called **multicollinearity**. It happens because the predictors are highly correlated.
- It also illustrates that we can have different models with equally good predictive properties.
  - In the case of an interaction model we always choose the model without interaction because it is simpler.
- However, in general it can be difficult to choose between models. For example, if both height and weight are good predictors of some response, but one of them can be left out, which one do we choose?

10 Multiple linear regression with categorical predictors

10.1 Dummy variables

- Suppose we want to predict the response variable using a categorical predictor variable x with k categories.
- We choose one group, say group k, as the reference category.
- For the remaining groups $1, \dots, k-1$, we define dummy variables

$$z_i = \begin{cases} 0, & \text{if } x \neq i, \\ 1, & \text{if } x = i. \end{cases}$$

  for $i = 1, \dots, k-1$.
  - The dummy variable z_i is 1 if an observation is in group i and 0 otherwise.
  - When all dummy variables $z_i = 0$, $i = 1, \dots, k-1$, it means that the observation belongs to the reference group k.
- We can use the variables $z_1, \dots, z_{k-1}$ in a multiple regression along with other predictor variables.

10.2 Example

- Consider the dataset `mtcars`. We are interested in how engine type `vs` (categorical) and weight of the car `wt` (quantitative, x_1) are associated with fuel consumption `mpg`.
- The variable `vs` is already coded as a dummy variable z in R, taking the value 1 if the engine is v-shaped and 0 otherwise.
- The multiple regression model becomes

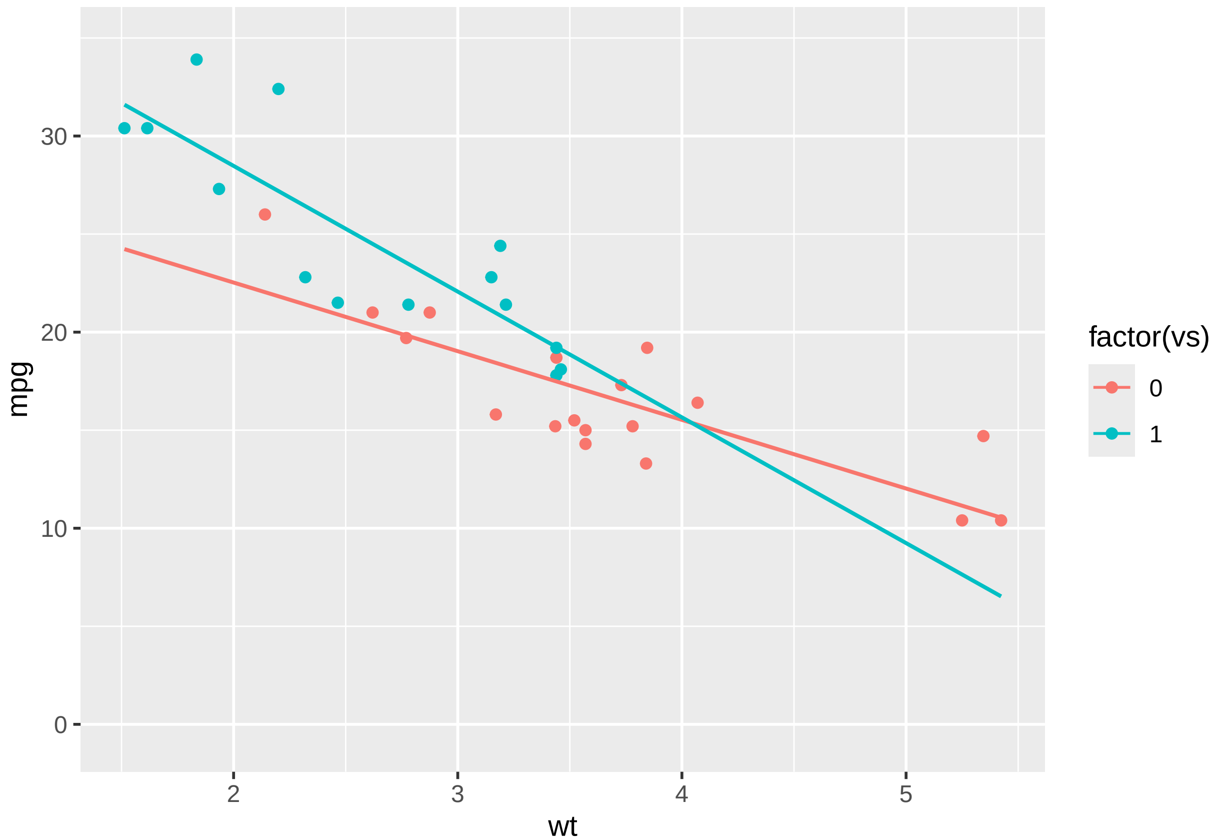
$$E(Y|x_1, z) = \alpha + \beta_1 x_1 + \beta_2 z.$$

  - For $z = 0$:
$$E(Y|x_1, z) = \alpha + \beta_1 x_1.$$
  - For $z = 1$:
$$E(Y|x_1, z) = \alpha + \beta_2 + \beta_1 x_1.$$
- So we get two different regression lines for the two groups.
  - The lines have a common slope β_1 (parallel lines).
  - The lines have different intercepts. The difference in intercepts is β_2.

10.3 Example

- We always start with some graphics (remember the function `gf_point` for plotting points and `gf_lm` for adding a regression line).

```
library(mosaic)
gf_point(mpg ~ wt, color = ~factor(vs), group=~factor(vs), data = mtcars) %>% gf_lm()
```



- An unclear picture, but a tendency to decreasing number of miles per gallon with increasing weight for both engine types.
- The slope of the lines for the two engine types look different. But is the difference significant? Or can the difference be explained by sampling variation?

10.4 Example

- We fit a multiple regression model without interaction in R:

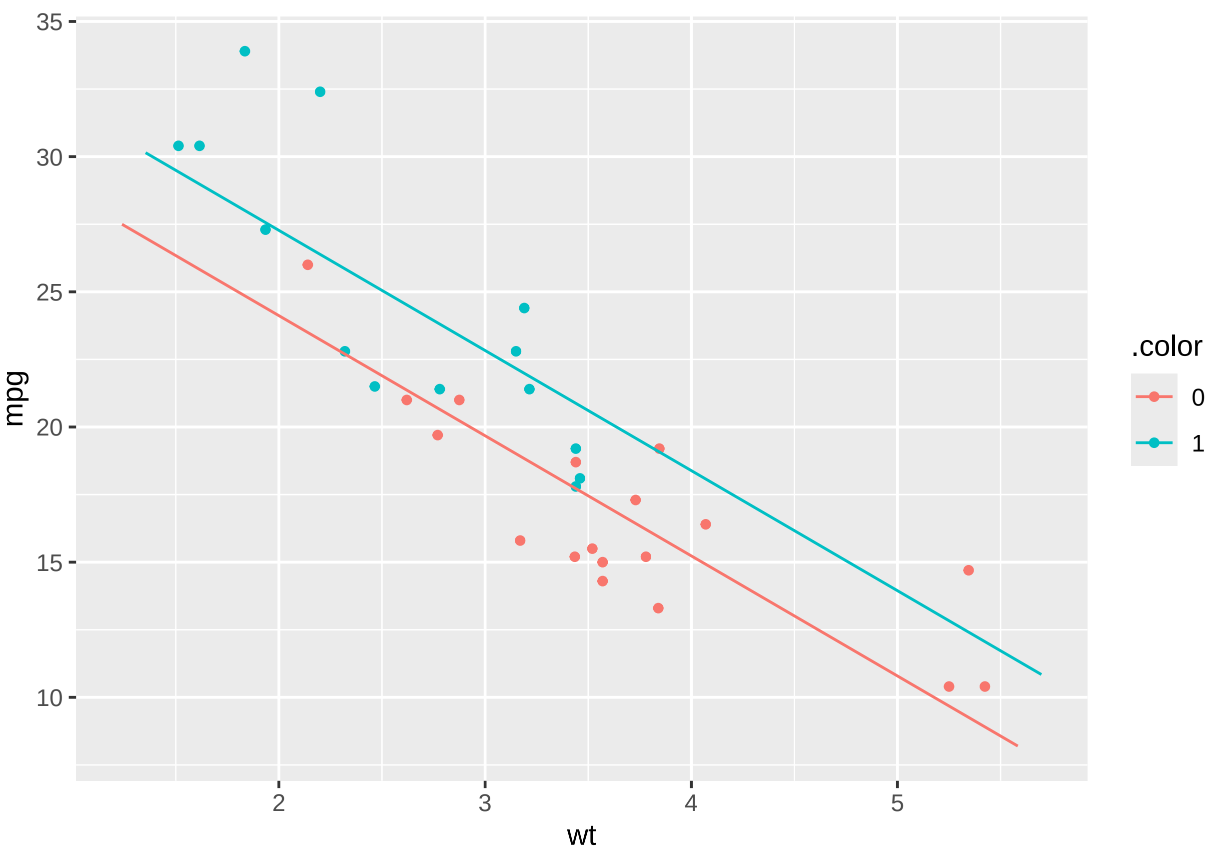
```
model1 <- lm(mpg ~ wt + factor(vs) , data = mtcars)
summary(model1)
```

```
##
## Call:
## lm(formula = mpg ~ wt + factor(vs), data = mtcars)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.7071 -2.4415 -0.3129  1.4319  6.0156
##
## Coefficients:
```

```
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  33.0042     2.3554  14.012 1.92e-14 ***
## wt           -4.4428     0.6134  -7.243 5.63e-08 ***
## factor(vs)1   3.1544     1.1907   2.649   0.0129 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.78 on 29 degrees of freedom
## Multiple R-squared:  0.801,  Adjusted R-squared:  0.7873
## F-statistic: 58.36 on 2 and 29 DF,  p-value: 6.818e-11
```

- The common slope to `wt` is estimated to be $\widehat{\beta}_1 = -4.44$, with corresponding p-value $5.63 \cdot 10^{-8}$, so the effect of `wt` is significantly different from zero.
  - The estimate is negative, so increasing weight decreases the number of miles per gallon.
- The estimate for intercept in the reference group ("not v-shaped") is $\widehat{\alpha} = 33.0$, which is significantly different from zero if we test at level 5% (this test is not really of interest).
- The difference between intercepts for the two engine types is $\widehat{\beta}_1 = 3.15$, which is significant with p-value=1%.
  - This suggests that the regression lines are not the same for the two engine types.
  - The value 3.15 is the vertical distance between the two regression lines.

```r
plotModel(model1)
```



10.5 Example: Prediction equations

```r
summary(model1)
```

```
##
```

```
## Call:
## lm(formula = mpg ~ wt + factor(vs), data = mtcars)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.7071 -2.4415 -0.3129  1.4319  6.0156
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  33.0042     2.3554  14.012 1.92e-14 ***
## wt           -4.4428     0.6134  -7.243 5.63e-08 ***
## factor(vs)1   3.1544     1.1907   2.649   0.0129 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.78 on 29 degrees of freedom
## Multiple R-squared:  0.801,  Adjusted R-squared:  0.7873
## F-statistic: 58.36 on 2 and 29 DF,  p-value: 6.818e-11
```

- Reference/baseline group (not v-shaped):

$$\widehat{y} = 33.0 - 4.44x$$

- V-shaped:

$$\widehat{y} = 33.0 + 3.15 - 4.44x = 36.15 - 4.44x.$$

10.6 Interaction model

- We can expand the regression model by including an interaction between x and z:

$$E(y|x, z) = \alpha + \beta_1 x + \beta_2 z + \beta_3 z \cdot x.$$

- This yields a regression line for engine type:
- Not v-shaped ($z = 0$): $E(y|x, z) = \alpha + \beta_1 x$
- V-shaped ($z = 1$): $E(y|x, z) = \alpha + \beta_2 + (\beta_1 + \beta_3)x$.
- β_2 is *the difference* in `Intercept` between the two groups, while β_3 is *the difference* in `slope` between the two groups.

10.7 Example: Prediction equations

- When we use `*` in the model formula we include interaction between `vs` and `wt`:

```r
model2 <- lm(mpg ~ wt * factor(vs), data = mtcars)
summary(model2)
```

```
##
## Call:
## lm(formula = mpg ~ wt * factor(vs), data = mtcars)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.9950 -1.7881 -0.3423  1.2935  5.2061
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)     29.5314     2.6221  11.263 6.55e-12 ***
## wt              -3.5013     0.6915  -5.063 2.33e-05 ***
```

```
## factor(vs)1     11.7667     3.7638   3.126   0.0041 **
## wt:factor(vs)1  -2.9097     1.2157  -2.393   0.0236 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.578 on 28 degrees of freedom
## Multiple R-squared:  0.8348, Adjusted R-squared:  0.8171
## F-statistic: 47.16 on 3 and 28 DF,  p-value: 4.497e-11
```

- We use the output to write the prediction equations:
  - Reference/baseline group (not v-shaped):

$$\widehat{y} = 29.5 - 3.5x$$

  - V-shaped:

$$\widehat{y} = (29.5 + 11.8) + (-3.5 - 2.9)x = 41.3 - 6.4x.$$

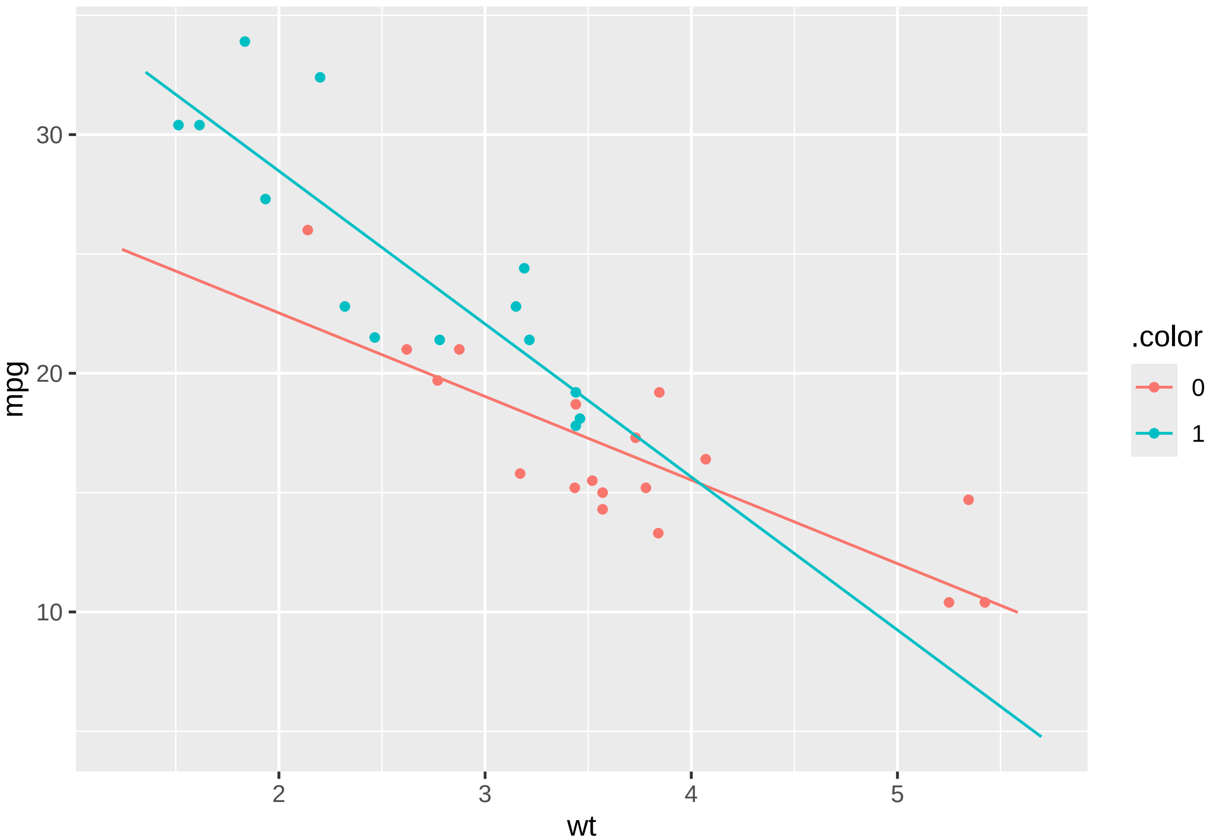
10.8 Example: Individual tests

```r
summary(model2)
```

```
##
## Call:
## lm(formula = mpg ~ wt * factor(vs), data = mtcars)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.9950 -1.7881 -0.3423  1.2935  5.2061
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)     29.5314     2.6221  11.263 6.55e-12 ***
## wt              -3.5013     0.6915  -5.063 2.33e-05 ***
## factor(vs)1     11.7667     3.7638   3.126   0.0041 **
## wt:factor(vs)1  -2.9097     1.2157  -2.393   0.0236 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.578 on 28 degrees of freedom
## Multiple R-squared:  0.8348, Adjusted R-squared:  0.8171
## F-statistic: 47.16 on 3 and 28 DF,  p-value: 4.497e-11
```

- *The difference* in slope between the two engine types is estimated to $\widehat{\beta}_3 = -2.9$ which is significant with p-value=0.0236, so the slopes are significantly different.

```r
plotModel(model2)
```



10.9 Hierarchy of models

- Always test for no interaction ($\beta_3 = 0$) before making tests for main effects ($\beta_1 = 0$ or $\beta_2 = 0$).

$$Model: E(Y|x,z) = \alpha + \beta_1 x + \beta_2 z + \beta_3 xz$$

Interaction

$\downarrow$ H_0: $\beta_3 = 0$

Main effects only

H_0: $\beta_2 = 0$ $\swarrow$ $\searrow$ H_0: $\beta_1 = 0$

Only effect of x Only effect of z

H_0: $\beta_1 = 0$ $\searrow$ $\swarrow$ H_0: $\beta_2 = 0$

No effect of x or z

10.10 F-test

- We can compare the two models in the `mtcars` example, namely the model with and without interaction via

```
anova(model1, model2)
```

```
## Analysis of Variance Table
##
## Model 1: mpg ~ wt + factor(vs)
## Model 2: mpg ~ wt * factor(vs)
##   Res.Df    RSS Df Sum of Sq      F  Pr(>F)
## 1     29 224.09
## 2     28 186.03  1    38.062 5.7287 0.02363 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

ASTA

The ASTA team

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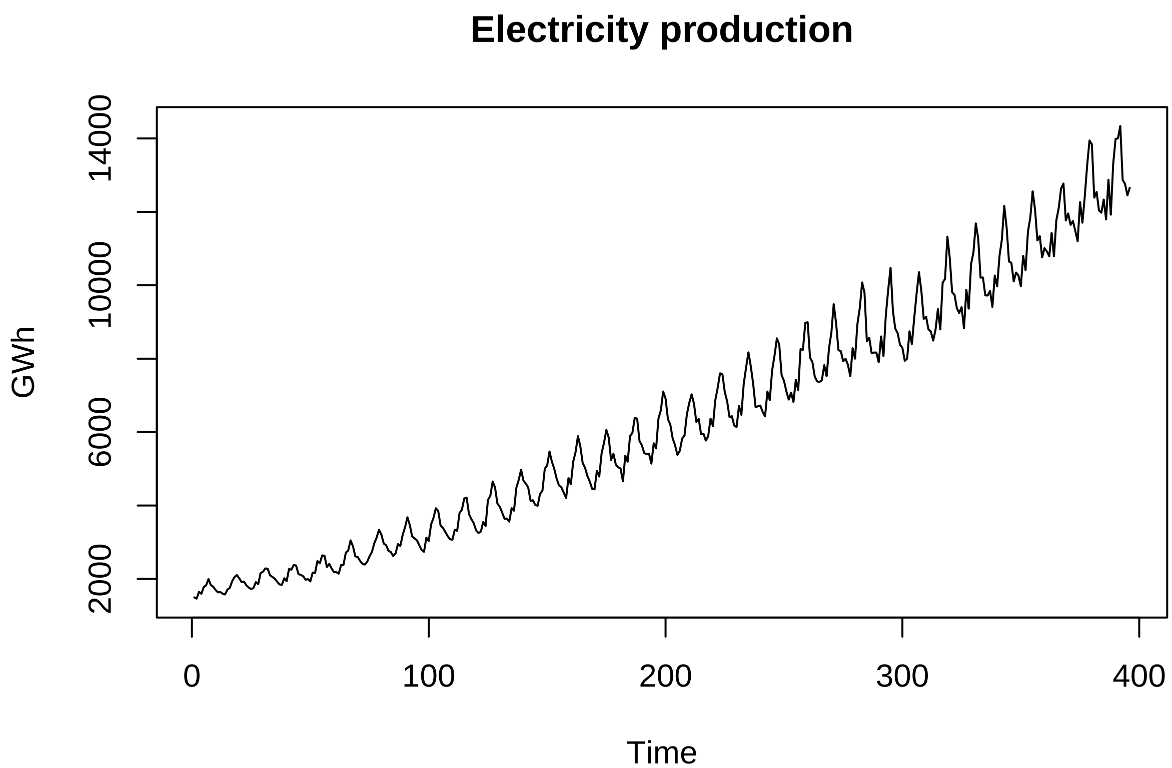
1 Introduction to stochastic processes

1.1 Data examples

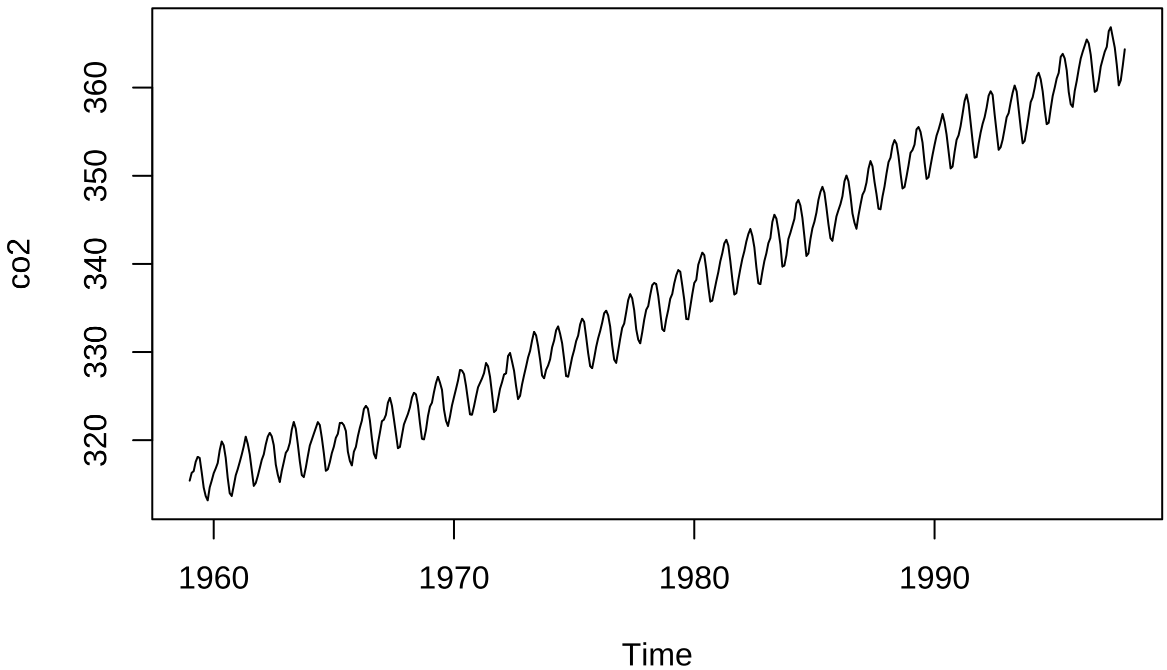
- A special type of data arises when we measure the same variable at different points in time with equal steps between time points.
- This data type is called a (discrete time) **stochastic process** or a **time series**
- One example is the time series of monthly electricity production (GWh) in Australia from Jan. 1958 to Dec. 1990 :

```r
CBEdata <- read.table("https://asta.math.aau.dk/eng/static/datasets?file=cbe.dat", header = TRUE)
CBE <- ts(CBEdata[,3])
plot(CBE, ylab="GWh",main="Electricity production")
```



- Another example is monthly measurements of the atmospheric CO_2 concentration measured at Mauna Loa 1959 - 1997:

```r
dat<-ts(co2)
plot(co2)
```



- Other examples:
  - Hourly wind speed measurements
  - Daily elspot prices
  - An electrical signal measured each millisecond
- Aim: Model, analyse and make predictions for such datasets.

1.2 Stochastic processes

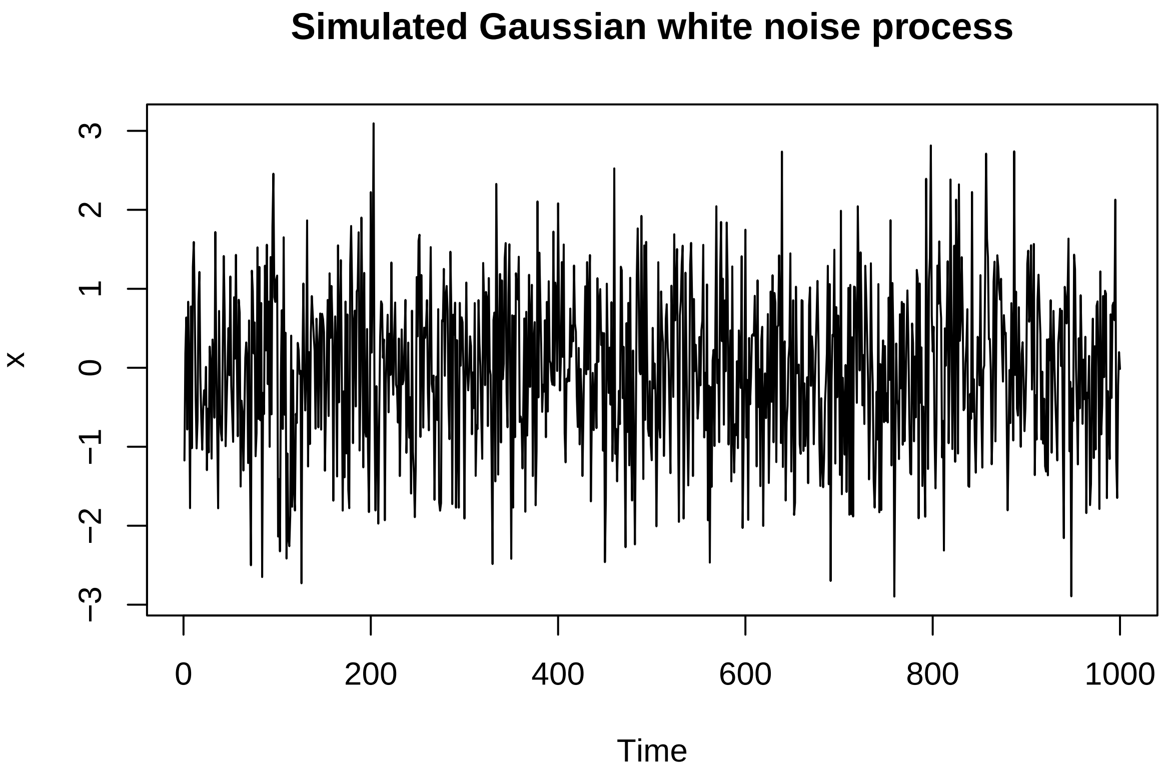
- We denote by X_t the variable at time t. We denote the time points by $t = 1, 2, 3, \ldots, n$.
  - We will always assume the data is observed at equidistant points in time (i.e. time steps between consecutive observations are the same).
- Measurements that are close in time will typically be similar: observations are not statistically independent!
- Measurements that are far apart in time will typically be less correlated.

2 Important stochastic processes

2.1 Example 1: White noise

- A stochastic process is called a **white noise process** if the X_t are
  - statistically independent
  - identically distributed
  - have mean 0 and variance σ^2
- It is called **Gaussian white noise**, if
  - $X_t \sim norm(0, \sigma^2)$

```r
x = rnorm(1000,0,1)
ts.plot(x, main = "Simulated Gaussian white noise process")
```



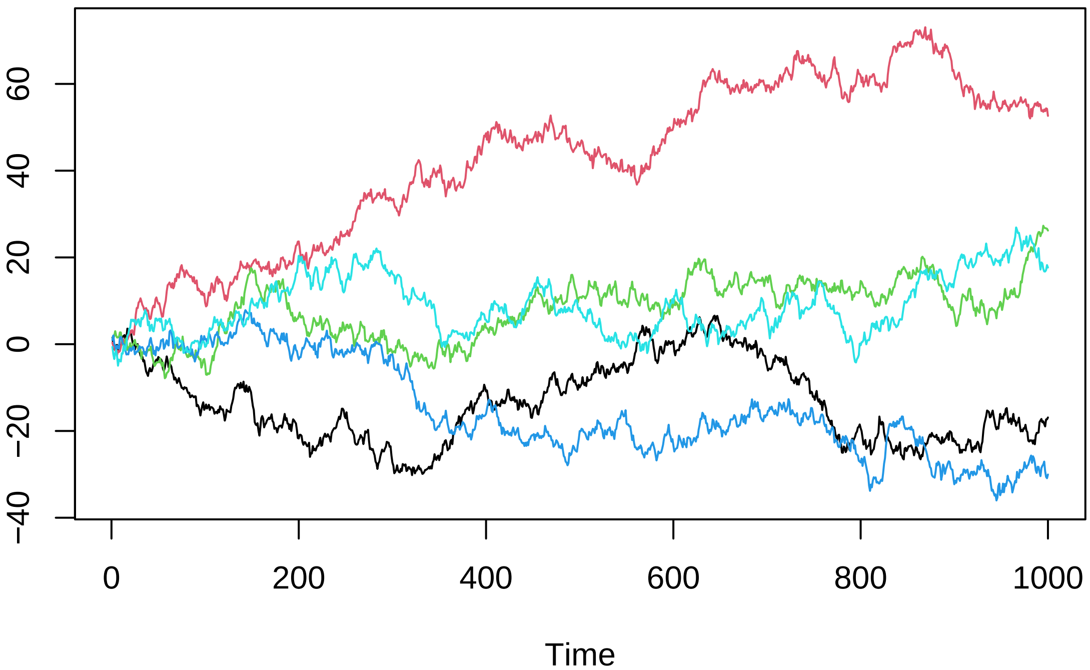
- White noise processes are the simplest stochastic processes.
- Real data does typically not have complete independence between measurements at different time points, so white noise is generally not a good model for real data, but it is a building block for more complicated stochastic processes.

---

3 Example 2: Random walk

- A **random walk** is defined by $X_t = X_{t-1} + W_t$, where W_t is white noise.
- Here are 5 simulations of a random walk:

```r
x = matrix(0,1000,5)
for (i in 1:5) x[,i] = cumsum(rnorm(1000,0,1))
ts.plot(x,col=1:5)
```



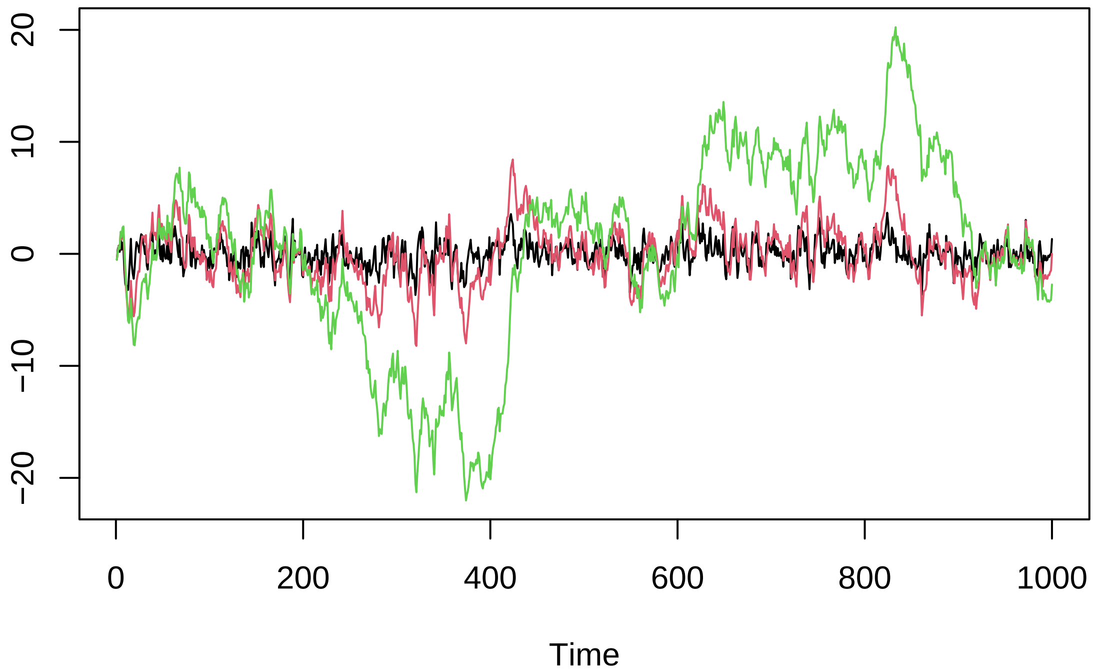
- The random walk may come back to zero after some time, but often it has a tendency to wander of in some random direction.

---

4 Example 3: First order autoregressive process

- A **first order autoregressive process**, AR(1), is defined by $X_t = \alpha X_{t-1} + W_t$, where W_t is white noise and $\alpha \in \mathbb{R}$.
  - Typically $-1 \leq \alpha \leq 1$
  - For $\alpha = 0$ we get white noise
  - For $\alpha = 1$ we get a random walk
- Simulation of 3 AR(1)-processes with different α values:

```
w = ts(rnorm(1000))
x1 = filter(w,0.5,method="recursive")
x2 = filter(w,0.9,method="recursive")
x3 = filter(w,0.99,method="recursive")
ts.plot(x1,x2,x3,col=1:3)
```



- Next time we will consider autoregressive processes in much more detail and higher order, where they become quite flexible models for data.

5 Mean, autocovariance and stationarity

5.1 Mean function

- The **mean function** of a stochastic process is given by

$$\mu_t = \mathbb{E}(X_t)$$

- A process is called first order stationary if $\mu_t = \mu$.
- **Examples:**
  - The white noise process: $\mu_t = 0$ by definition.
  - The random walk:

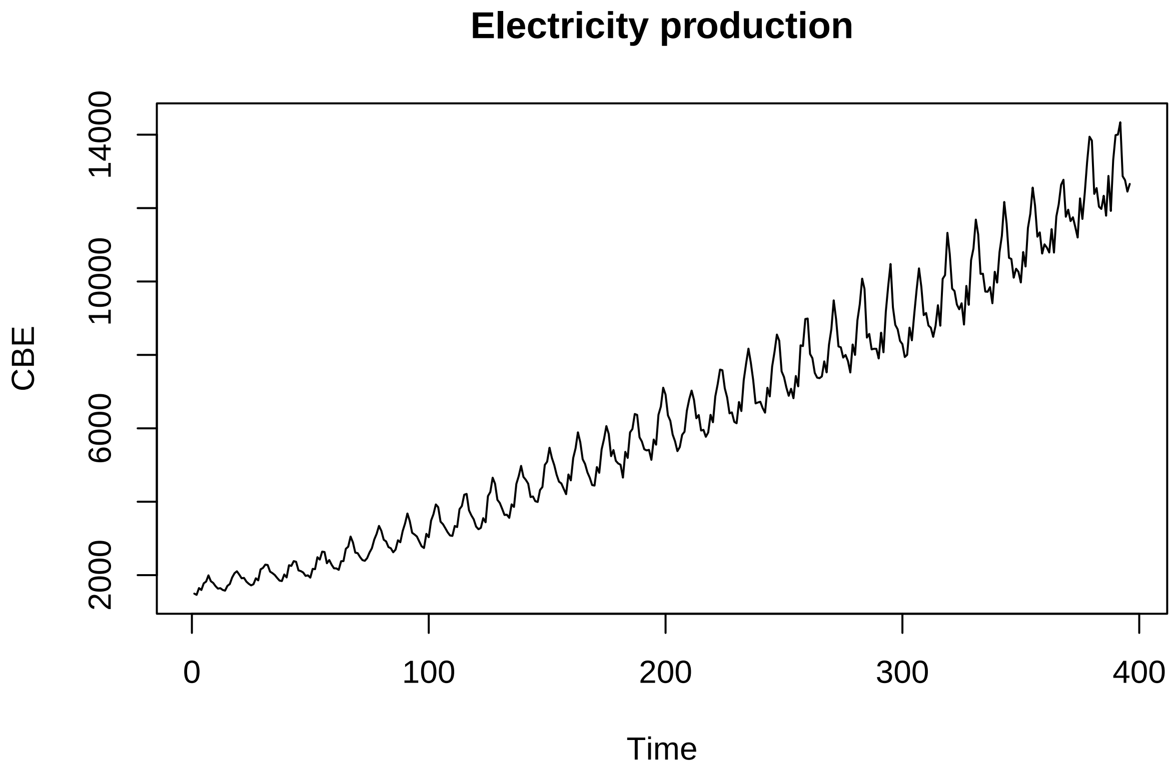
$$\mu_t = \mathbb{E}(X_t) = \mathbb{E}(X_{t-1} + W_t) = \mathbb{E}(X_{t-1}) + \mathbb{E}(W_t) = \mathbb{E}(X_{t-1}) = \mu_{t-1}$$

    So the random walk is first order stationary.
  - Similarly,

$$\mu_t = \mathbb{E}(X_t) = \mathbb{E}(\alpha X_{t-1} + W_t) = \alpha\mathbb{E}(X_{t-1}) + \mathbb{E}(W_t) = \alpha\mathbb{E}(X_{t-1}) = \alpha\mu_{t-1}$$

    The AR(1)-model is first order stationary if $\mu_0 = 0$ or $\alpha = 1$, otherwise not.
  - The electricity production in Australia did not look first order stationary.

```
plot(CBE,main="Electricity production")
```



- The mean function shows the mean behavior of the process, but individual simulations may move far away from this. For example, the random walk has a tendency to move far away from the mean. White noise on the other hand will stay close to the mean.

5.2 Autocovariance/autocorrelation functions

- The **autocovariance** function is given by

$$\gamma(t, t+h) = \mathrm{Cov}(X_t, X_{t+h}) = \mathbb{E}((X_t - \mu_t)(X_{t+h} - \mu_{t+h}))$$

- h is called the **lag**.
- Note that

$$\gamma(t, t) = \mathrm{Var}(X_t) = \sigma_t^2$$

  is the variance at time t.
- The **autocorrelation function (ACF)** is

$$\rho(t, t+h) = \mathrm{Cor}(X_t, X_{t+h}) = \frac{\mathrm{Cov}(X_t, X_{t+h})}{\sigma_t \sigma_{t+h}}$$

- It holds that $\rho(t, t) = 1$, and $\rho(t, t+h)$ is between -1 and 1 for any h.
- The autocorrelation function measures how correlated X_t and X_{t+h} are related:
  - If X_t and X_{t+h} are independent, then $\rho(t, t+h) = 0$
  - If $\rho(t, t+h)$ is close to one, then X_t and X_{t+h} tends to be either high or low at the same time.
  - If $\rho(t, t+h)$ is close to minus one, then when X_t is high X_{t+h} tends to be low and vice versa.

5.3 Stationarity

- We call a stochastic process **second order stationary** if

- the mean is constant, $\mu_t = \mu$
- the variance $\sigma_t^2 = \mathrm{Var}(X_t, X_t)$ is constant.
- the autocorralation function only depends on the lag h:

$$\rho(t, t+h) = \rho(h)$$

- If a process is second order stationary, then also the autocovariance is stationary $\gamma(t, t+h) = \gamma(h)$, i.e. it is a function of only the lag and is easier to work with and plot.
- Intuitively, stationarity means that the process behaves in the same way no matter which time we look at.
- There are other kinds of stationarity, but *in this course, stationarity will always mean second order stationarity.*

5.4 Stationarity and autocorrelation - example

- Consider an AR(1) process $X_t = \alpha X_{t-1} + W_t$. We consider stationarity and autocorrelation for this process.
  - We have already seen that we need $\mu_t = 0$ to have first order stationarity.
- Now consider the variance. Since $X_t = \alpha X_{t-1} + W_t$,

$$\sigma_t^2 = \mathrm{Var}(X_t) = \mathrm{Var}(\alpha X_{t-1} + W_t) = \mathrm{Var}(\alpha X_{t-1}) + \mathrm{Var}(W_t) = \alpha^2 \mathrm{Var}(X_{t-1}) + \mathrm{Var}(W_t) = \alpha^2 \sigma_{t-1}^2 + \sigma^2$$

  - (Here we used that $\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$ when X and Y are independent).
- If the variance is constant, then $\sigma_t^2 = \sigma_{t-1}^2$ and

$$\sigma_t^2 = \alpha^2 \sigma_t^2 + \sigma^2$$

  - We see that the variance can only be constant if $-1 < \alpha < 1$. In this case $\sigma_t^2 = \frac{\sigma^2}{1-\alpha^2}$.
  - For $|\alpha| \geq 1$, the variance will increase over time. The process is cannot be stationary (including random walk).
- To find the autocorrelation, first observe

$$X_{t+h} = \alpha X_{t+h-1} + W_{t+h} = \cdots = \alpha^h X_t + \sum_{i=0}^{h-1} \alpha^i W_{t+h-i}$$

- Then we find the autocovariance:

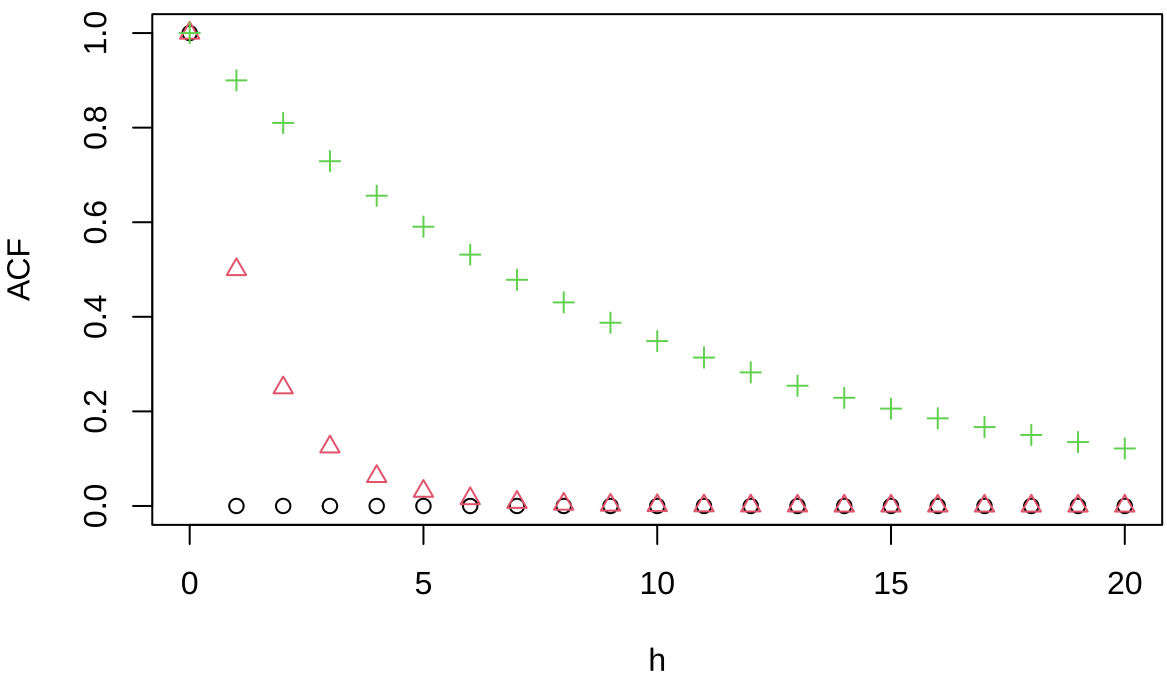
$$\gamma(t, t+h) = \mathrm{Cov}(X_t, X_{t+h}) = \mathrm{Cov}(X_t, \alpha^h X_t + \sum_{i=0}^{h-1} \alpha^i W_{t+h-i}) = \mathrm{Cov}(X_t, \alpha^h X_t) + \mathrm{Cov}(X_t, \sum_{i=0}^{h-1} \alpha^i W_{t+h-i}) = \alpha^h \mathrm{Cov}(X_t, X$$

  - (Here we used the computation rules $\mathrm{Cov}(X, Y+Z) = \mathrm{Cov}(X,Y) + \mathrm{Cov}(X,Z)$ and $\mathrm{Cov}(X, aY) = a\mathrm{Cov}(X,Y)$.)
- If the variance is constant, we can calculate the autocorrelation:

$$\frac{\mathrm{Cov}(X_t, X_{t+h})}{\sigma_t \sigma_{t+h}} = \frac{\alpha^h \sigma^2/(1-\alpha^2)}{\sigma^2/(1-\alpha^2)} = \alpha^h.$$

- So: the AR(1)-model is stationary if $-1 < \alpha < 1$ and $\sigma_t^2 = \sigma^2/(1-\alpha^2)$ - otherwise not.
- The autocorrelation decays exponentially for a stationary AR(1)-model. This is illustrated for 3 different α values:

```
h = 0:20
acf1 = 0^h  # AR(1) with alpha = 0 (or white noise)
acf2 = 0.5^h  # AR(1) with alpha = 0.5
acf3 = 0.9^h  # AR(1) with alpha = 0.9
plot(matrix(rep(h,3),3),cbind(acf1,acf2,acf3),col=rep(1:3,each=length(h)),
     pch=rep(1:3,each = length(h)),xlab="h",ylab="ACF")
```



6 Estimation

6.1 Estimation

- The mean and autocovariance/autocorrelation functions are theoretical constructions defined for stochastic processes, but what about data? Here we have to estimate them.
- We will assume that the process is stationary.
- The (constant) mean can be estimated the usual way:

$$\hat{\mu} = \bar{x} = \frac{1}{n}\sum_{t=1}^{n} x_t$$

- The autocovariance function can be estimated as follows (remember it only depends on h, not on t in the case of stationarity):

$$\hat{\gamma}(h) = \frac{1}{n}\sum_{t=1}^{n-h}(x_t - \bar{x})(x_{t+h} - \bar{x})$$

- The (constant) variance is estimated as $\hat{\sigma}^2 = \hat{\gamma}(0)$.
- An estimate of the autocorrelation function is obtained as

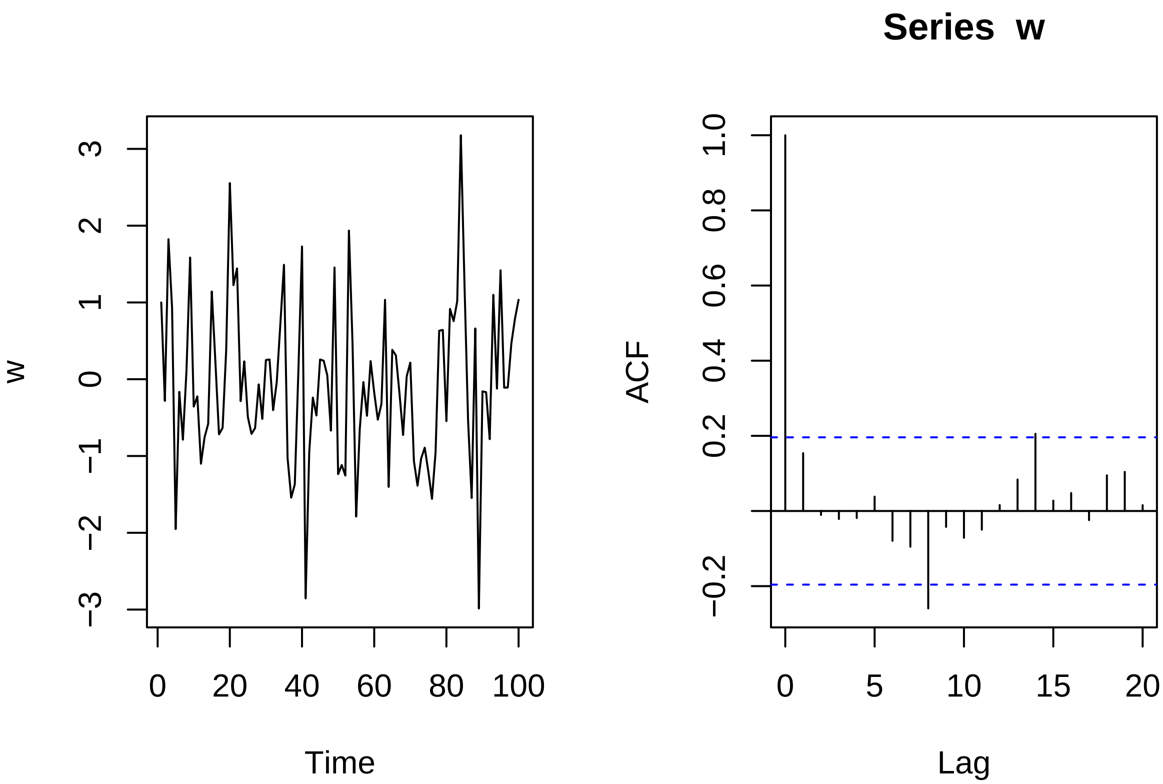
$$\hat{\rho}(h) = \frac{\hat{\gamma}(h)}{\hat{\gamma}(0)}$$

6.2 The correlogram

- A plot of the sample acf as a function of the lag is called a **correlogram**.
- To get an idea of how a correlogram looks, we make simulated data from different models and plot the correlograms below.

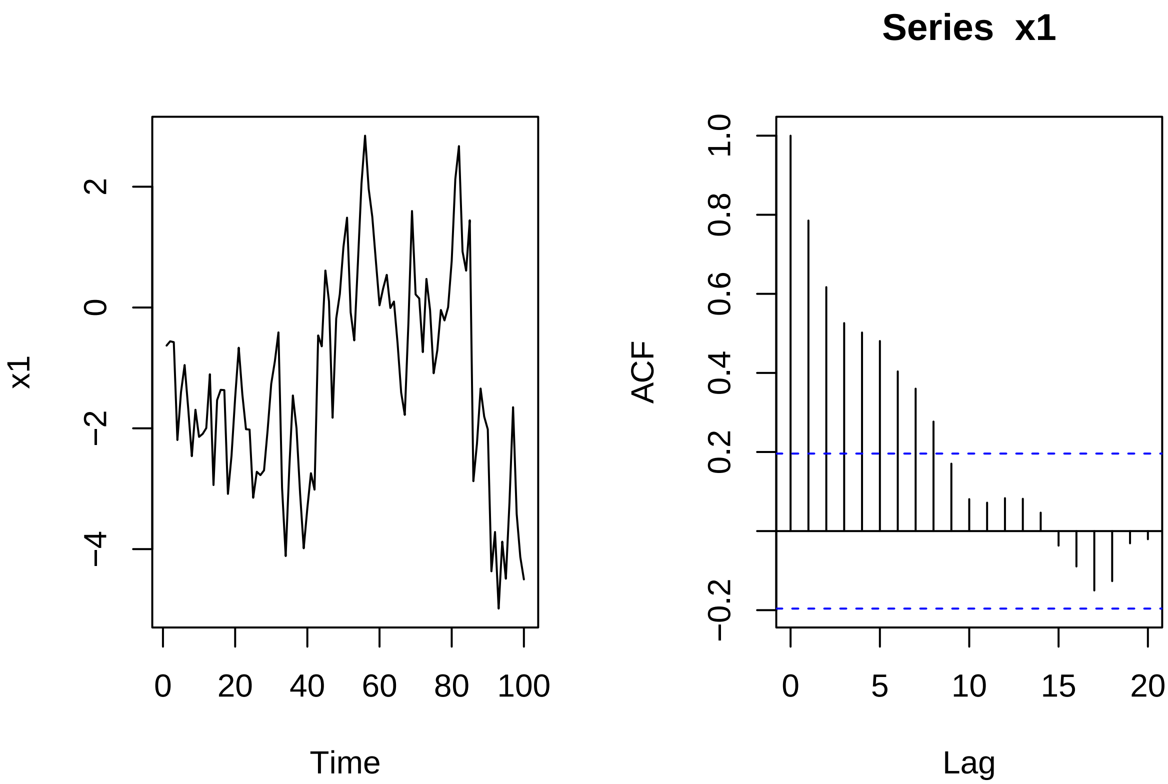
White noise:

```
w = ts(rnorm(100))
par(mfrow=c(1,2))
plot(w)
acf(w)
```



- The correlogram is always 1 at lag 0
- For white noise, the true autocorrelation drops to zero.
- The estimated autocorrelation is never exactly zero - hence we get the small bars.
- The blue lines is a 95% confidence band for a test that the true autocorrelation is zero.
- Remember that there is 5% chance of rejecting a true null hypothesis. Thus, 5% of the bars can be expected to exeed the blue lines.
- AR(1) process with $\alpha = 0.9$:

```
w = ts(rnorm(100))
x1 = filter(w,0.9,method="recursive")
par(mfrow=c(1,2))
plot(x1)
acf(x1)
```



- The true acf decays exponentially.

7 Non-stationary data

7.1 Check for stationarity

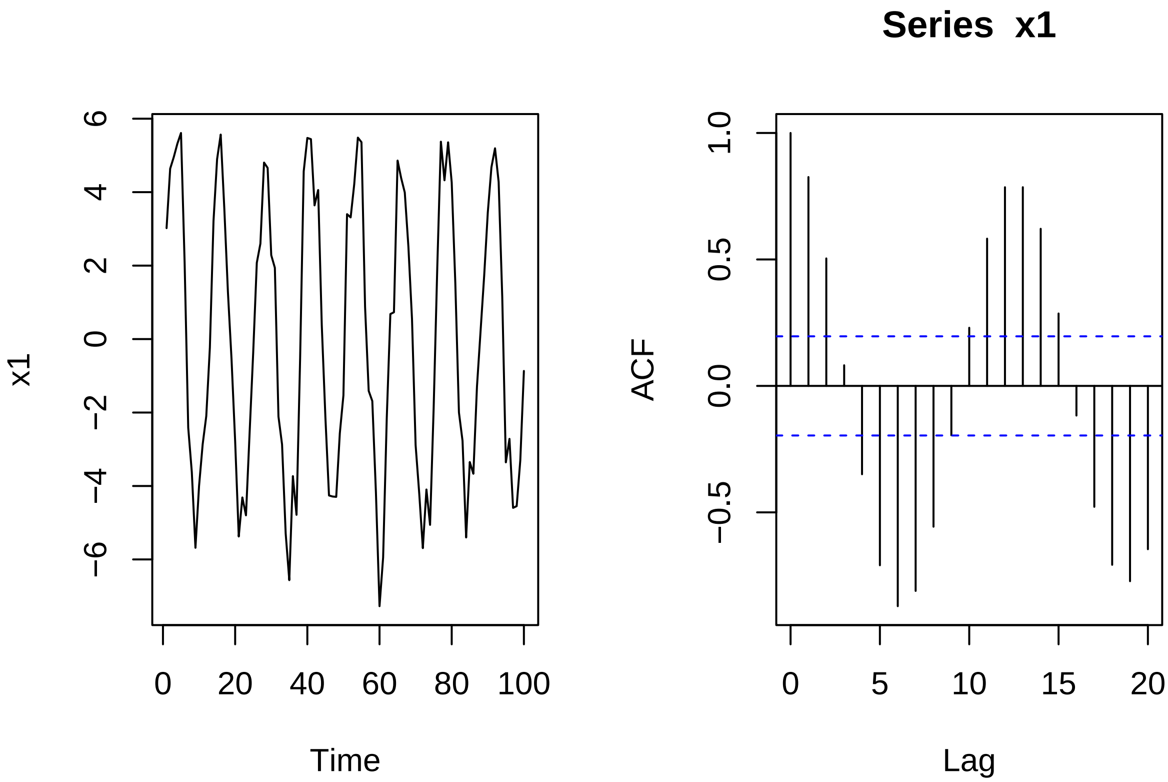
- We will primarily look at stationary processes the next time, but these will not always be good models for data.
- First we need to check whether the assumption of stationarity is okay.
  - One check is visual inspection of a plot of x_t vs t to see whether there is any indication of non-stationarity.
  - Another visual check is a plot of the correlogram. If this tends very slowly to zero, this indicates non-stationarity.
- Note: even though $\rho(h)$ is only well-defined for stationary models, we can plug any data (stationary or not) into the estimation formula. The estimate may help detecting deviations from stationarity.

7.2 Correlograms for non-stationary data

- Sine curve with added white noise:

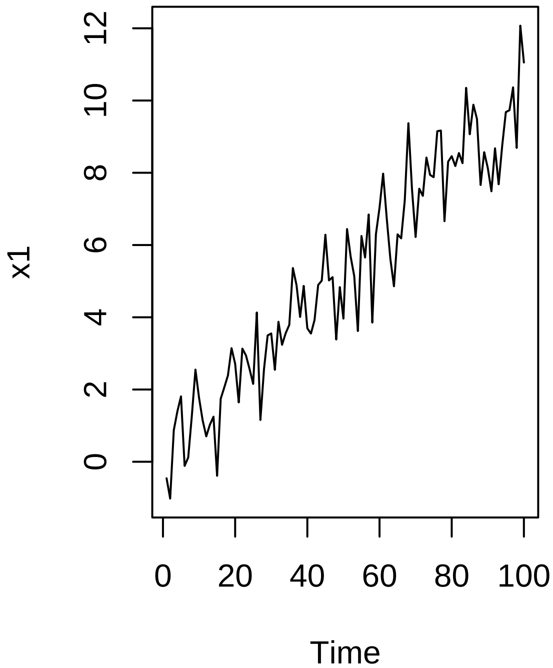
```
w = ts(rnorm(100))
x1 = 5*sin(0.5*(1:100)) + w
par(mfrow=c(1,2))
```

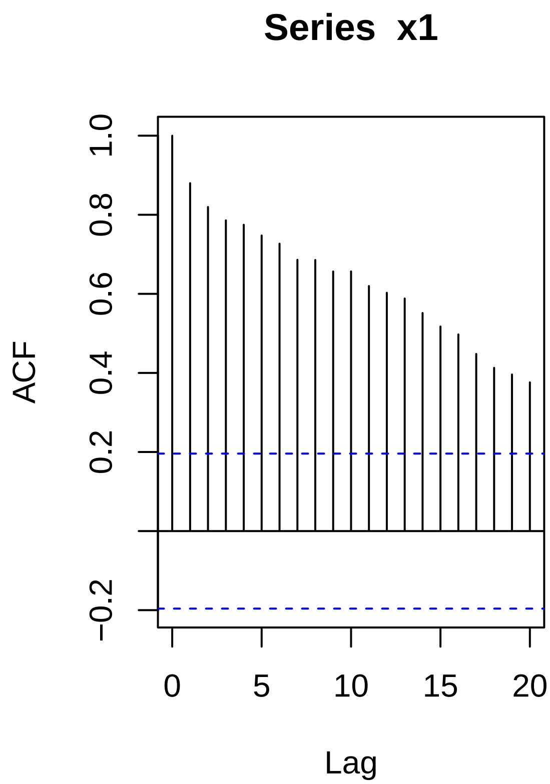
```r
plot(x1)
acf(x1)
```



- The periodic mean of the process results in a periodic behavior of the correlogram.
- A periodic behavior in the correlogram suggests seasonal behavior in the process.
- Straight line with added white noise:

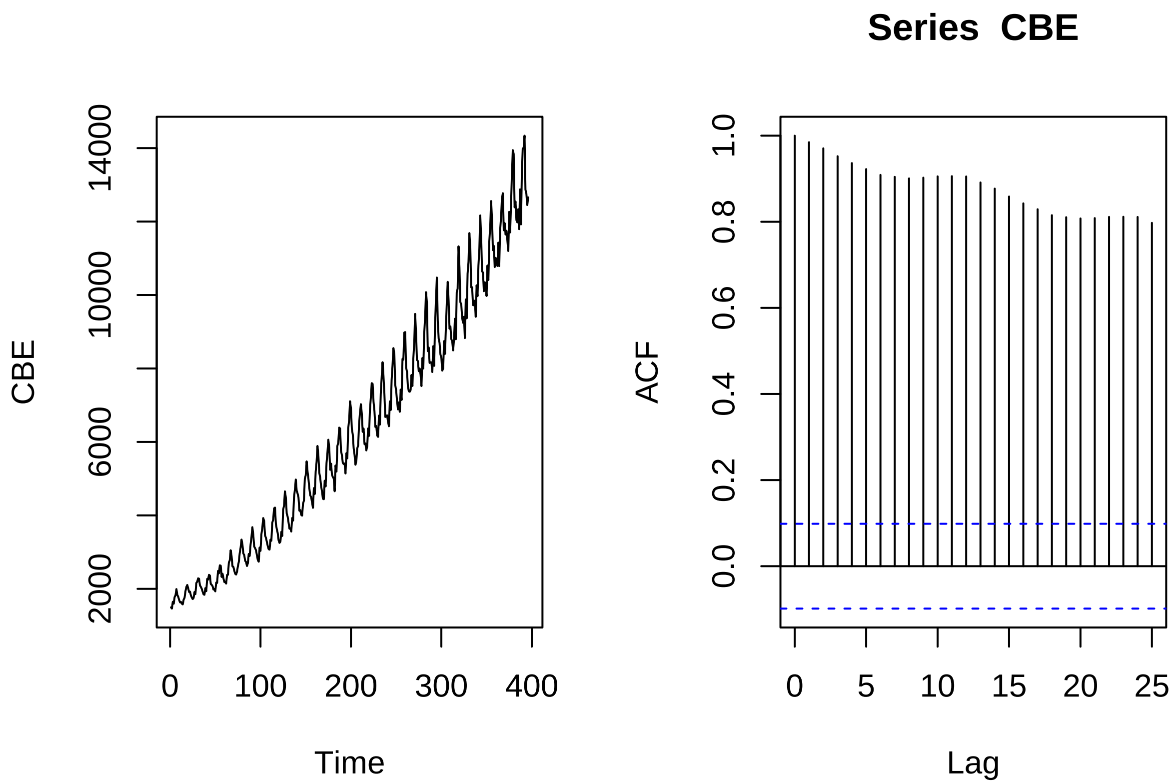
```r
w = ts(rnorm(100))
x1 = 0.1*(1:100) + w
par(mfrow=c(1,2))
plot(x1)
acf(x1)
```





- The linear trend results in a slowly decaying, almost linear correlogram.
- Such a correlogram suggests a trend in the data.
- Data example: Electricity production.

```
par(mfrow=c(1,2))
plot(CBE)
acf(CBE)
```



- There seems to be an increasing trend in the data.
- There is a periodic behavior around the increasing trend.
- It is reasonable to believe that the period is 12 months.
- We have the model

$$X_t = m_t + s_t + Z_t$$

where

  - m_t is the (deterministic) trend
  - s_t is a (deterministic) seasonal term ($s_t = s_{t+12}$)
  - Z_t is a random (hopefully) stationary part

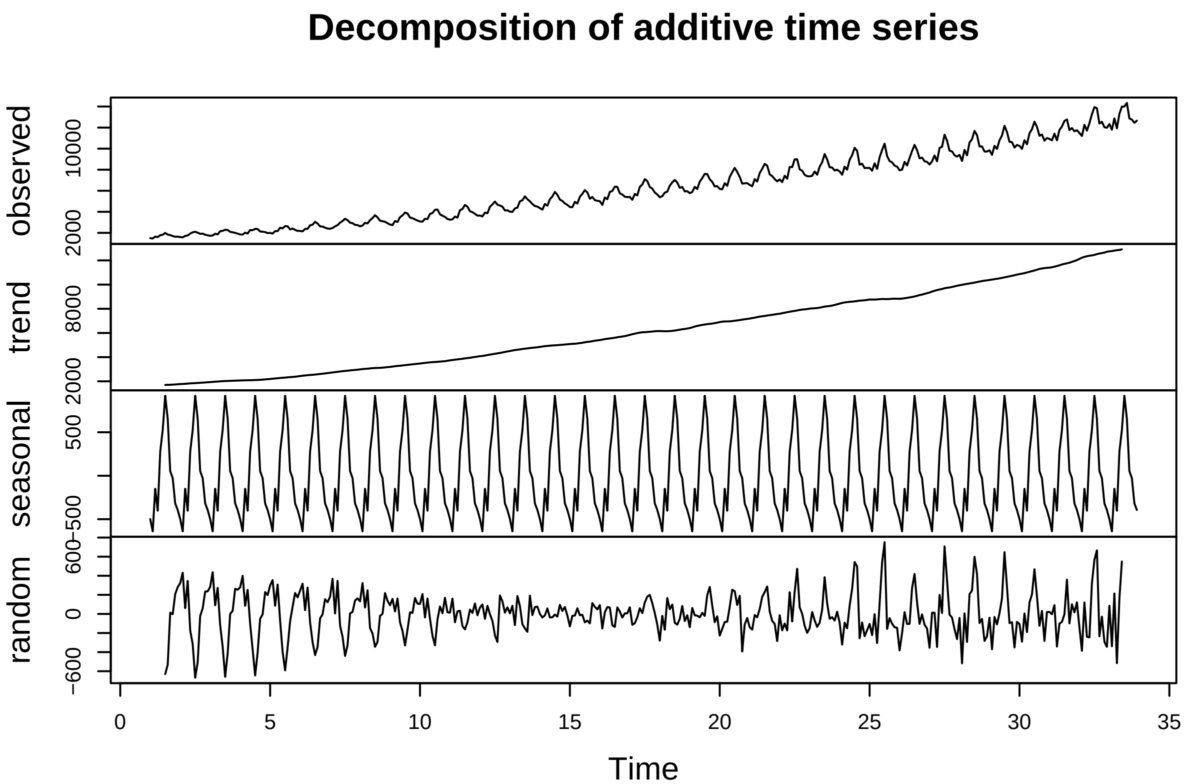
7.3 Detrending data

- The trend m_t in the data can be estimated by a **moving average**.
- In the case of monthly variation,

$$\hat{m}_t = \frac{\frac{1}{2}x_{t-6} + x_{t-5} + \cdots + x_t + \cdots + x_{t+5} + \frac{1}{2}x_{t+6}}{12}$$

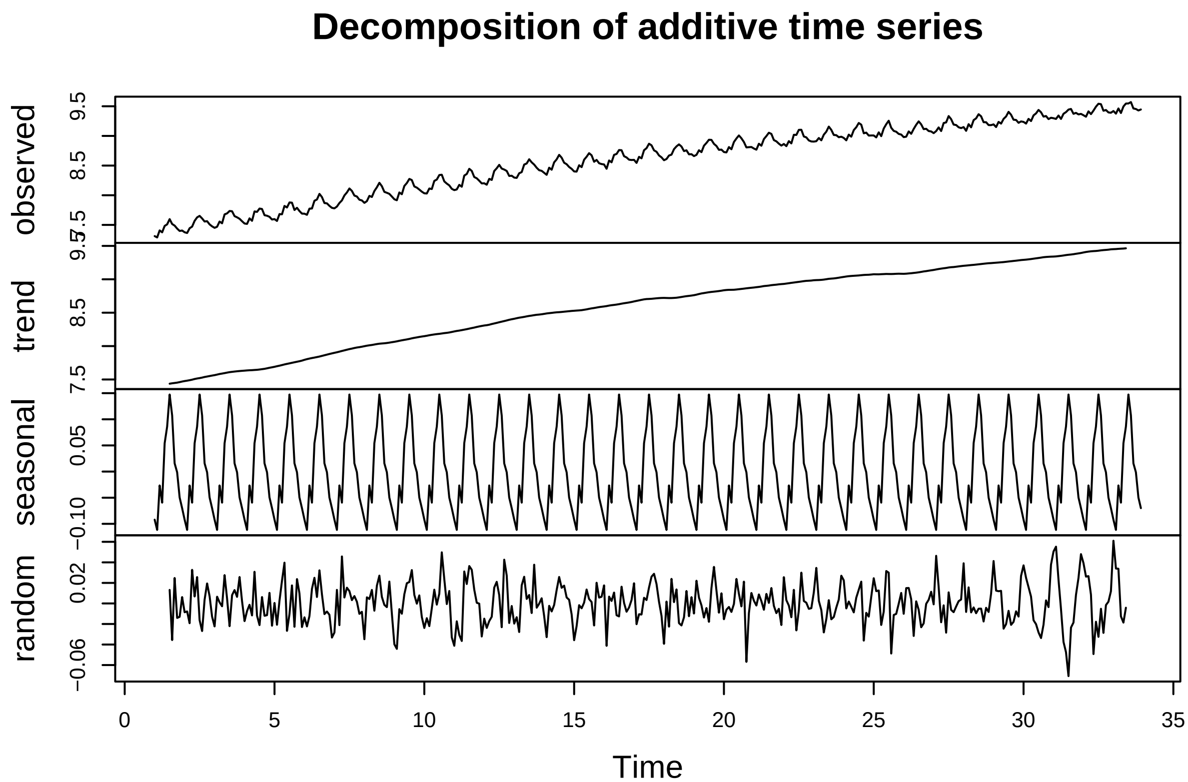
- We remove the trend by considering $x_t - \hat{m}_t$.
- Next we find the seasonal term s_t by averaging $x_t - \hat{m}_t$ over all measurements in the given month.
  - E.g., the value of s_t for January is given by averaging all values from January.
- We are left with the random part $\hat{z}_t = x_t - \hat{m}_t - \hat{s}_t$.
- For the Australian electricity data:

```r
CBE <- ts(CBE,frequency=12)
plot(decompose(CBE))
```

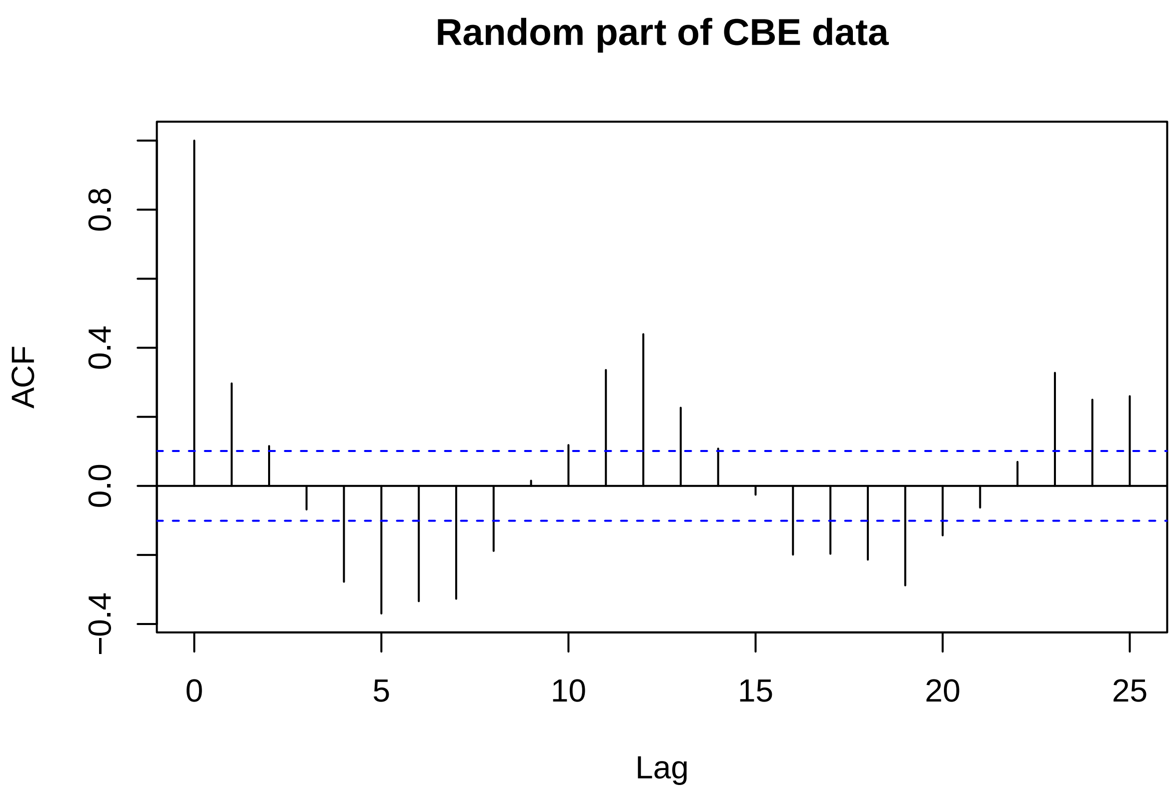


- The random term does not look stationary. The solution is to log-transform the data - see Ch. 1.5.5 in the book.

```r
logCBE <- ts(log(CBEdata[,3]),frequency=12)
plot(decompose(logCBE))
```



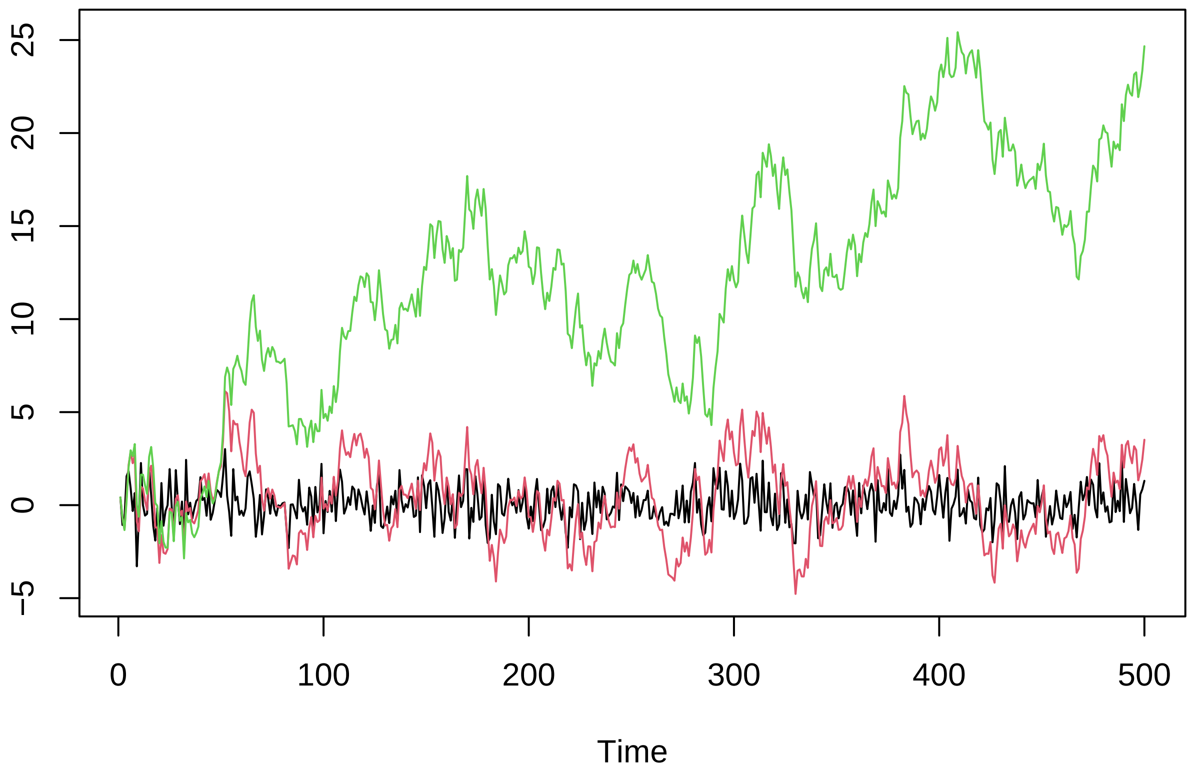

```r
random<-decompose(logCBE)$random[7:382]
acf(random, main="Random part of CBE data")
```



8 Basic stochastic process models

8.1 Stochastic processes

- Last time we introduced stochastic processes



- We saw three basic models:
  - **White noise:** independent random variables - not very interesting by itself, but an essential building block in more complicated/realistic models.
  - **Random walk:** cumulatively adding white noise - non-stationary, tendency to wander off.
  - **Autoregressive process AR(1):** weaker dependence on previous value than random walk, is stationary or not depending on choice of parameters.
- Today we will consider more stochastic process models.
- First we take a recap of the three models above, and add some details.

9 White noise

9.1 White noise

- A time series W_t, $t = 1, \dots, n$ is **white noise** if the variables $W_1, W_2, \dots, W_n$ are
  - independent
  - identically distributed
  - have mean zero and variance σ^2
- From the definition it follows that white noise is a second order stationary process since

  - The mean is constant ($= 0$)
  - The variance function $\sigma^2(t) = \sigma^2$ is constant
  - The autocorrelation $Cor(W_t, W_{t+k}) = 0$ for all $k \neq 0$, which does not depend on t.
- Hence, we have a well-defined mean,

$$\mu = 0,$$

autocovariance function

$$\gamma(k) = Cov(W_t, W_{t+k}) = \begin{cases} \sigma^2 \text{ for } k = 0, \\ 0 \text{ for } k \neq 0, \end{cases}$$

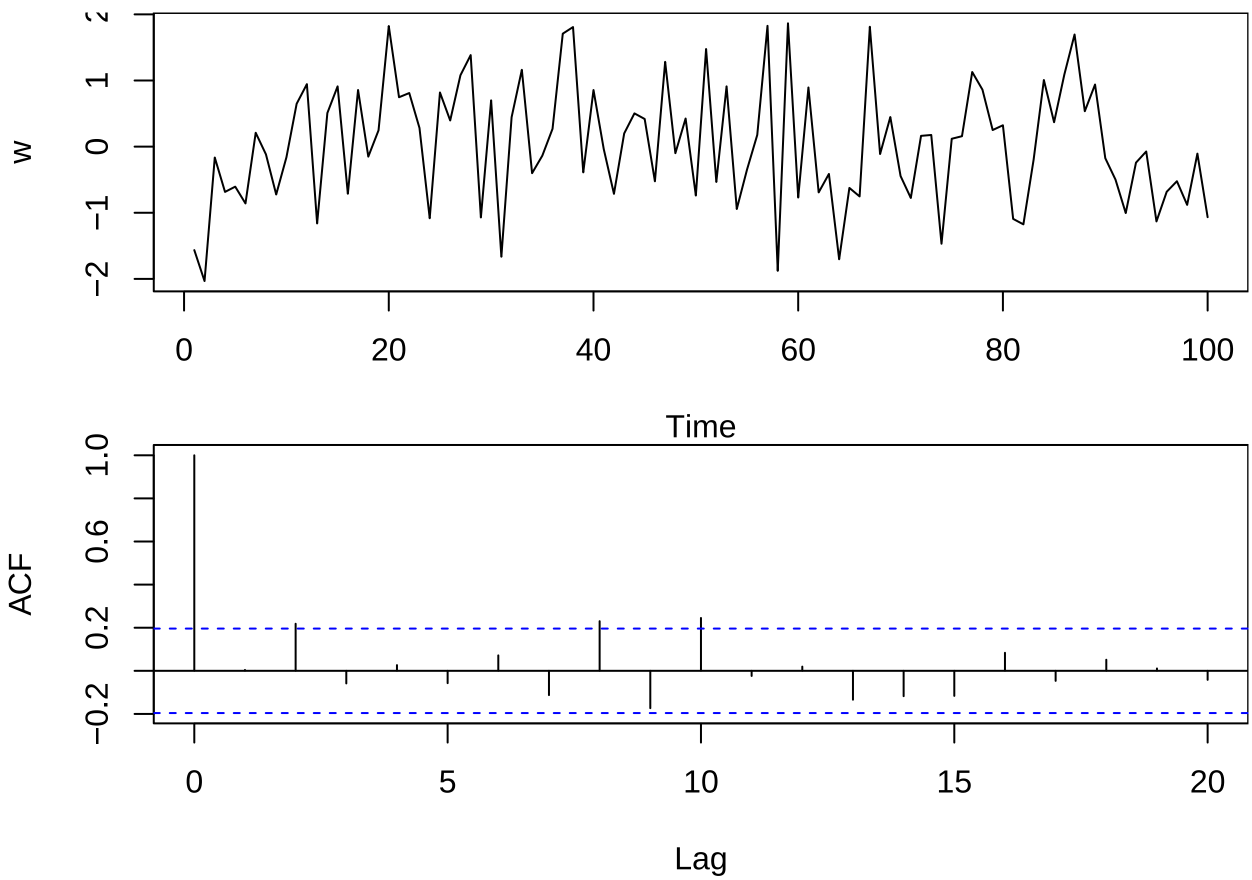
and autocorrelation function

$$\rho(k) = \begin{cases} 1 \text{ for } k = 0, \\ 0 \text{ for } k \neq 0. \end{cases}$$

9.2 Correlogram for white noise

- To understand how white noise behaves we can simulate it with R and plot both the series and the autocorrelation:

```
w <- rnorm(100, mean = 0, sd = 1)
par(mfrow = c(2,1), mar = c(4,4,0,0))
ts.plot(w)
acf(w)
```



- 95% of the estimated autocorrelations at lag $k > 0$ lie between the blue lines.
- It is a good idea to repeat this simulation and plot a few times to appreciate the variability of the results.

10 Random walk

10.1 Random walk

- A time series X_t is called a **random walk** if

$$X_t = X_{t-1} + W_t$$

  where W_t is a white noise series.
- Using $X_{t-1} = X_{t-2} + W_{t-1}$ we get

$$X_t = X_{t-2} + W_{t-1} + W_t$$

- Substituting for X_{t-2} we get

$$X_t = X_{t-3} + W_{t-2} + W_{t-1} + W_t$$

- Continuing this way, assuming we start at $X_0 = 0$,

$$X_t = W_1 + W_2 + \cdots + W_t$$

10.2 Properties of random walk

- A random walk X_t has a constant mean function

$$\mu(t) = 0.$$

- The variance function is given by

$$\sigma^2(t) = Var(X_t) = Var(W_1 + \cdots + W_t) = Var(W_1) + \cdots + Var(W_t) = t \cdot \sigma^2,$$

  which clearly depends on the time t, so the process is not stationary.
- Similarly, one can compute the autocovariance and autocorrelation function (see Cowpertwait and Metcalfe for details)

$$Cov(X_t, X_{t+k}) = t\sigma^2.$$

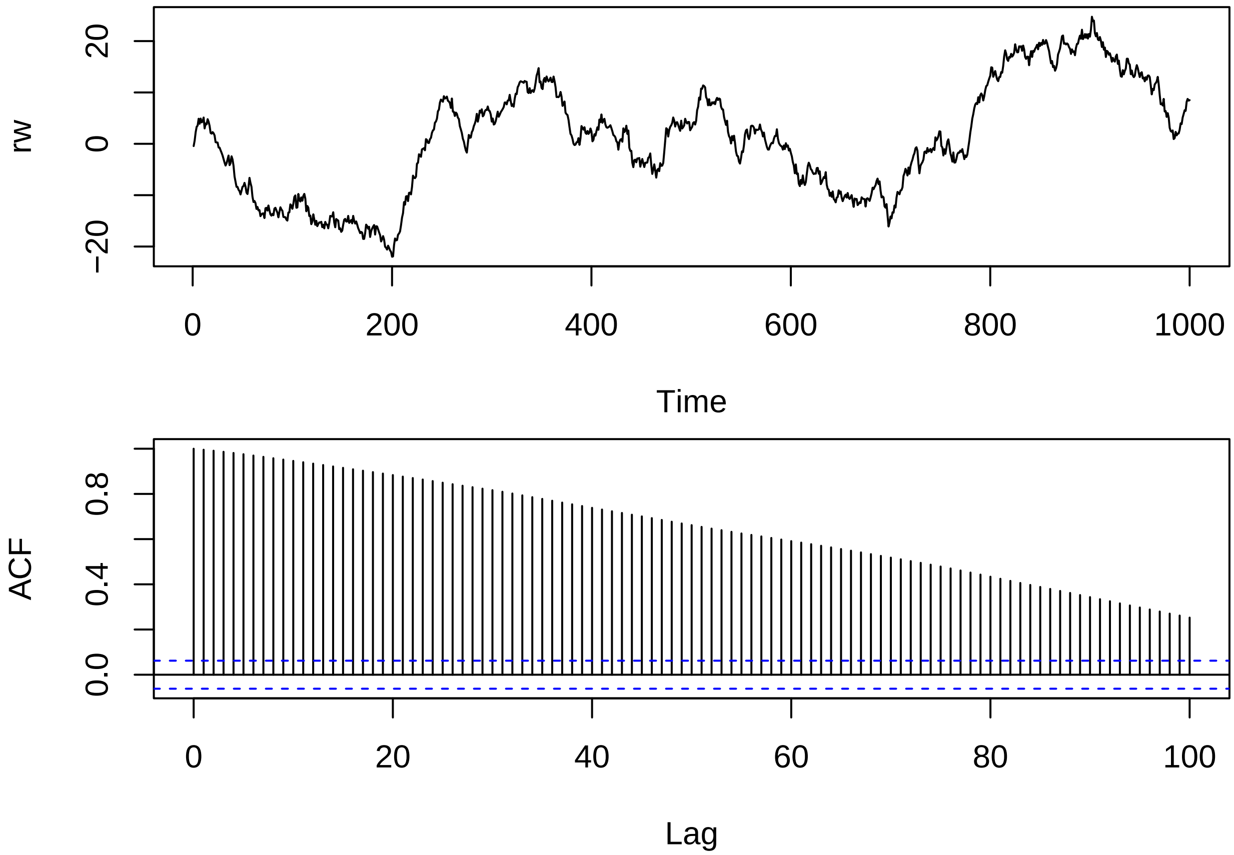
$$Cor(X_t, X_{t+k}) = \frac{1}{\sqrt{1 + k/t}}$$

- The autocorrelation is non-stationary.
- When t is large compared to k we have very high correlation (close to 1)
- We expect the correlogram of a reasonably long random walk to show very slow decay.

10.3 Simulation and correlogram of random walk

- We already know how to simulate Gaussian white noise W_t(with `rnorm`) and the random walk is just generated by repeatedly adding noise terms:

```
w <- rnorm(1000)
rw<-w
for(t in 2:1000) rw[t]<- rw[t-1] + w[t]
par(mfrow = c(2,1), mar = c(4,4,0.5,0.5))
ts.plot(rw)
acf(rw, lag.max = 100)
```



- The slowly decaying acf for random walk is a classical sign of non-stationarity, indicating there may be some kind of trend. In this case there is no real trend, since the theoretical mean is constant zero, but we refer to the apparent trend which seems to change directions unpredictiably as a **stochastic trend**.

10.4 Differencing

- If a non-stationary time series shows a stochastic trend we can try to study the **time series of differences** and see if that is stationary and easier to understand:

$$\nabla X_t = X_t - X_{t-1}.$$

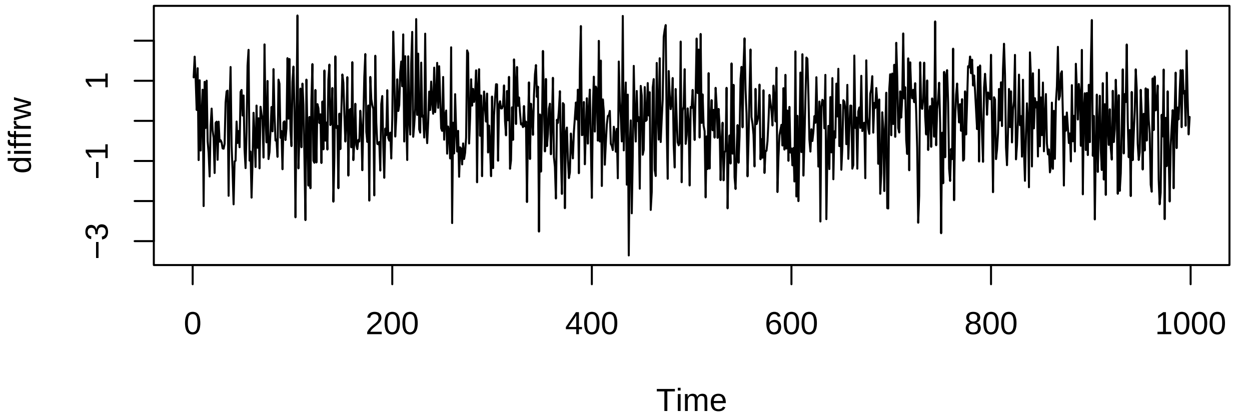
- Since we assume $X_0 = 0$ we get

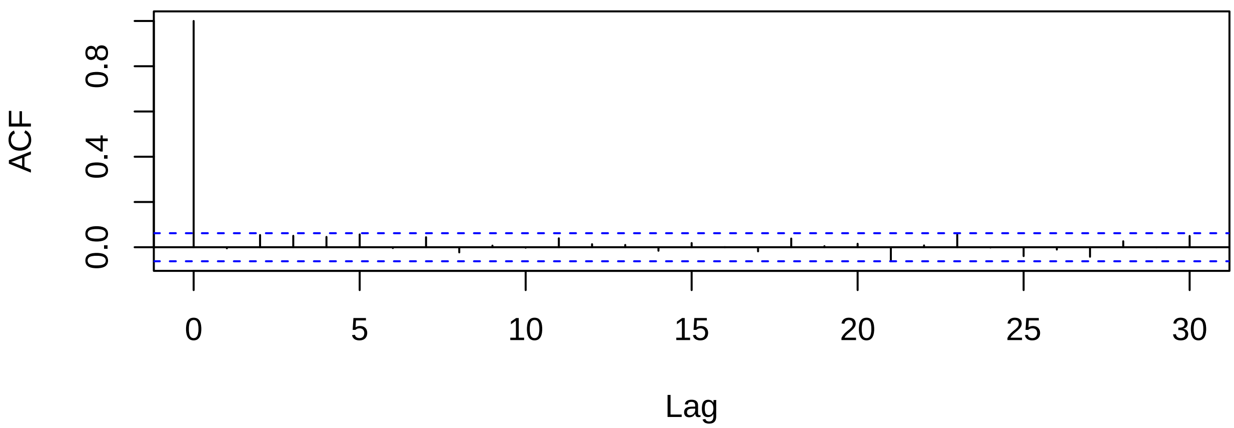
$$\nabla X_1 = X_1.$$

- Specifically when we difference a random walk $X_t = X_{t-1} + W_t$ we recover the white noise series

$$\nabla X_t = W_t$$

```r
diffrw <- diff(rw)
par(mfrow = c(2,1), mar = c(4,4,0.5,0.5))
ts.plot(diffrw)
acf(diffrw, lag.max = 30)
```



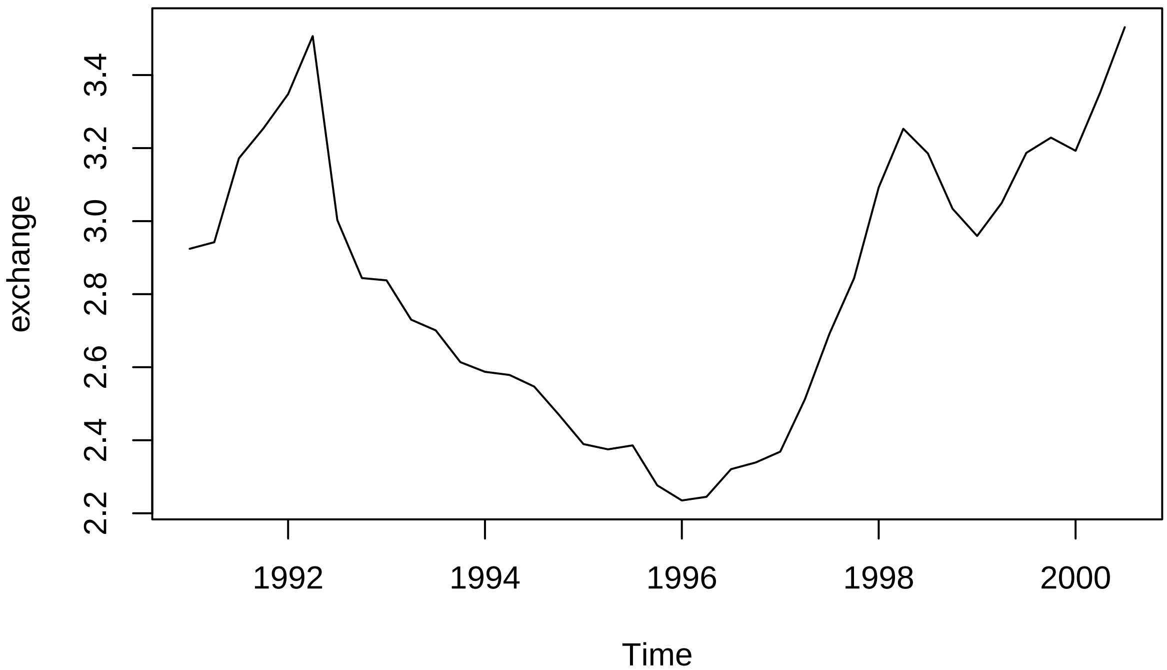


- If a time series needs to be differenced to become stationary, we say that the series is **integrated** (of order 1).

10.5 Example: Exchange rate

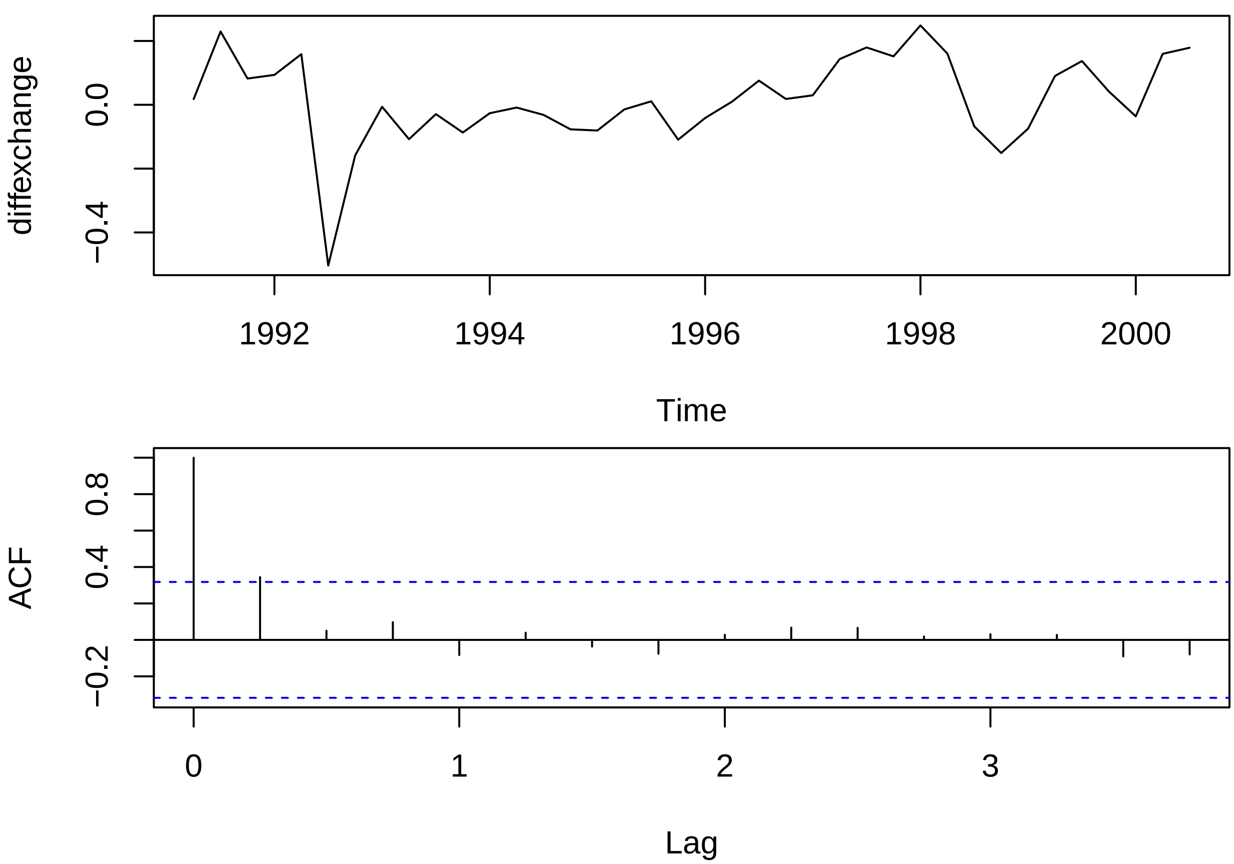
- We consider a data set for the exchange rate from GBP to NZD. We observe what looks like an unpredictable stochastic trend and we would like to see if it could reasonably be described as a random walk.

```r
www <- "https://asta.math.aau.dk/eng/static/datasets?file=pounds_nz.dat"
exchange_data <- read.table(www, header = TRUE)[[1]]
exchange <- ts(exchange_data, start = 1991, freq = 4)
plot(exchange)
```



- We difference the series and see if the differences looks like white noise:

```
diffexchange <- diff(exchange)
par(mfrow = c(2,1), mar = c(4,4,0.5,0.5))
plot(diffexchange)
acf(diffexchange)
```



- The first order difference looks reasonably stationary, so the original exchange rate series could be considered integrated of order 1. However, there is an indication of significant autocorrelation at lag 1, so a random walk might not be a completely satisfactory model for this dataset.

11 First order auto-regressive models AR(1)

11.1 Auto-regressive model of order 1: AR(1)

- If the correlogram shows a significant auto-correlation at lag 1, it means that X_t and X_{t-1} are correlated.
- The simplest way to model this, is the auto-regressive model of order one, AR(1):

$$X_t = \alpha_1 X_{t-1} + W_t$$

where W_t is white noise and the auto-regressive coefficient α_1 is a parameter to be estimated from data.

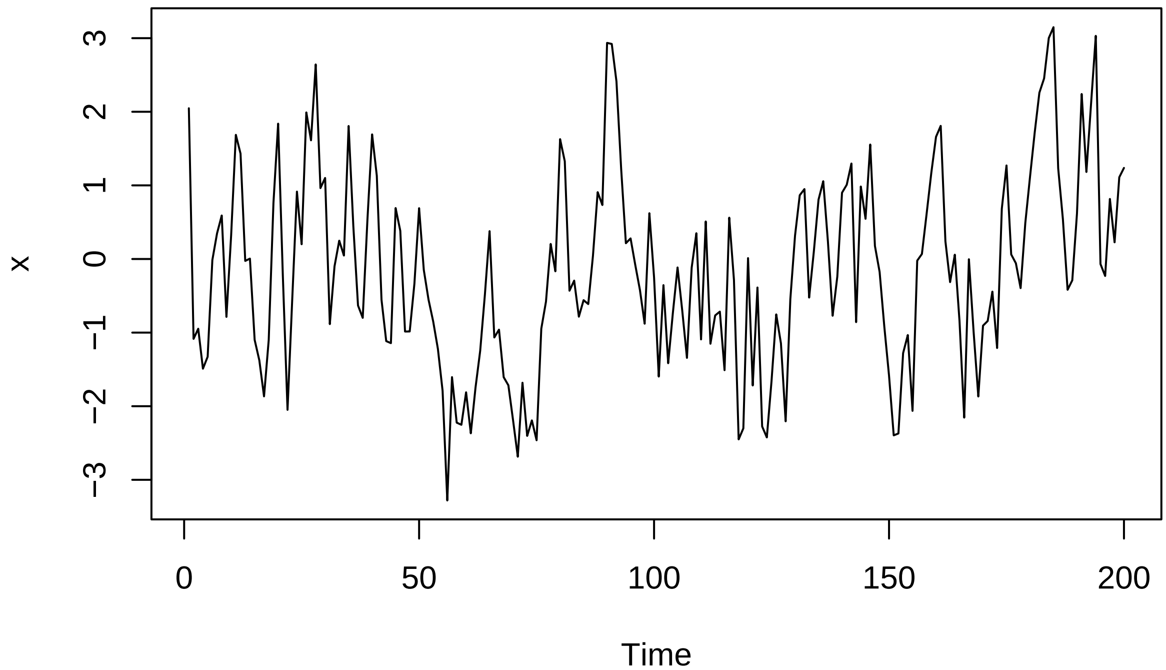
11.2 Properties of AR(1) models

- From last time: The model can only be stationary if $-1 < \alpha_1 < 1$ such that the dependence on the past decreases with time.
- For a stationary AR(1) model with $-1 < \alpha_1 < 1$ we found
  - $\mu(t) = 0$
  - $Var(X_t) = \sigma_t^2 = \sigma^2/(1-\alpha_1^2)$
  - $\gamma(k) = \alpha_1^k \sigma^2/(1-\alpha_1^2)$
  - $\rho(k) = \alpha_1^k$

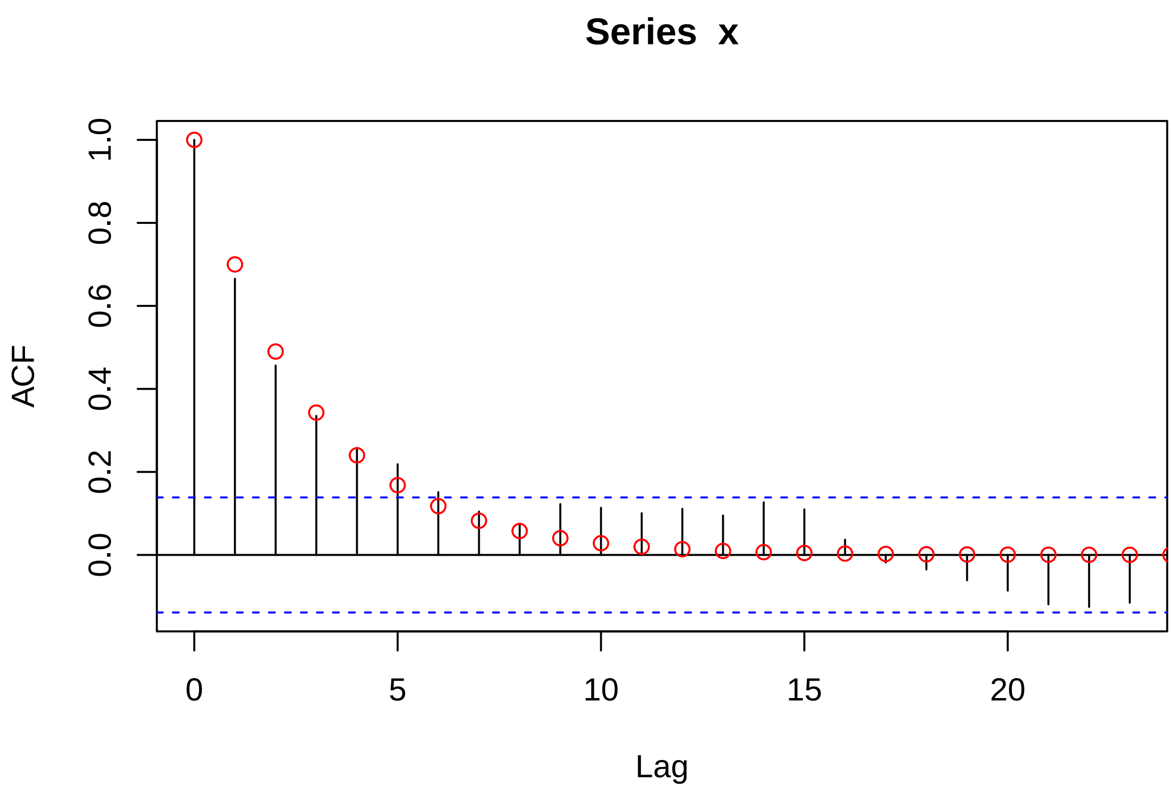
11.3 Simulation of AR(1) models

- R has a built-in function `arima.sim` to simulate AR(1) models (and other more complicated models called ARMA and ARIMA).
- It needs the model (i.e. the autoregressive coefficient α_1) and the desired number of time steps `n`.
- To simulate 200 time steps of AR(1) with $\alpha_1 = 0.7$ we do:

```r
x <- arima.sim(model = list(ar = 0.7), n = 200)
plot(x)
```




```r
acf(x)
k = 0:30
points(k, 0.7^k, col = "red")
```



- Here we have compared the empirical correlogram with the theoretical values of the model.

11.4 Fitted AR(1) models

- There are several ways to estimate the parameters in an AR(1) process. We use the so-called maximum likelihood estimation method (MLE).
- We skip the details of this, and simply use the function `ar`:

```
fit <- ar(x, order.max = 1, method="mle")
```

- The resulting object contains the value of the estimated parameter $\hat{\alpha}_1$ and a bunch of other information.
- In this example, the input data are artificial so we know that $\hat{\alpha}_1$ should ideally be close to 0.7:

```
fit$ar
```

```
## [1] 0.675656
```

- An estimate of the variance (squared standard error) of $\hat{\alpha}_1$ is given in `fit$asy.var.coef`

```
fit$asy.var.coef
```

```
##             [,1]
## [1,] 0.002740389
```

- The estimated std. error is the square root of this:

```
se <- sqrt(fit$asy.var.coef)
se
```

```
##            [,1]
## [1,] 0.05234872
```

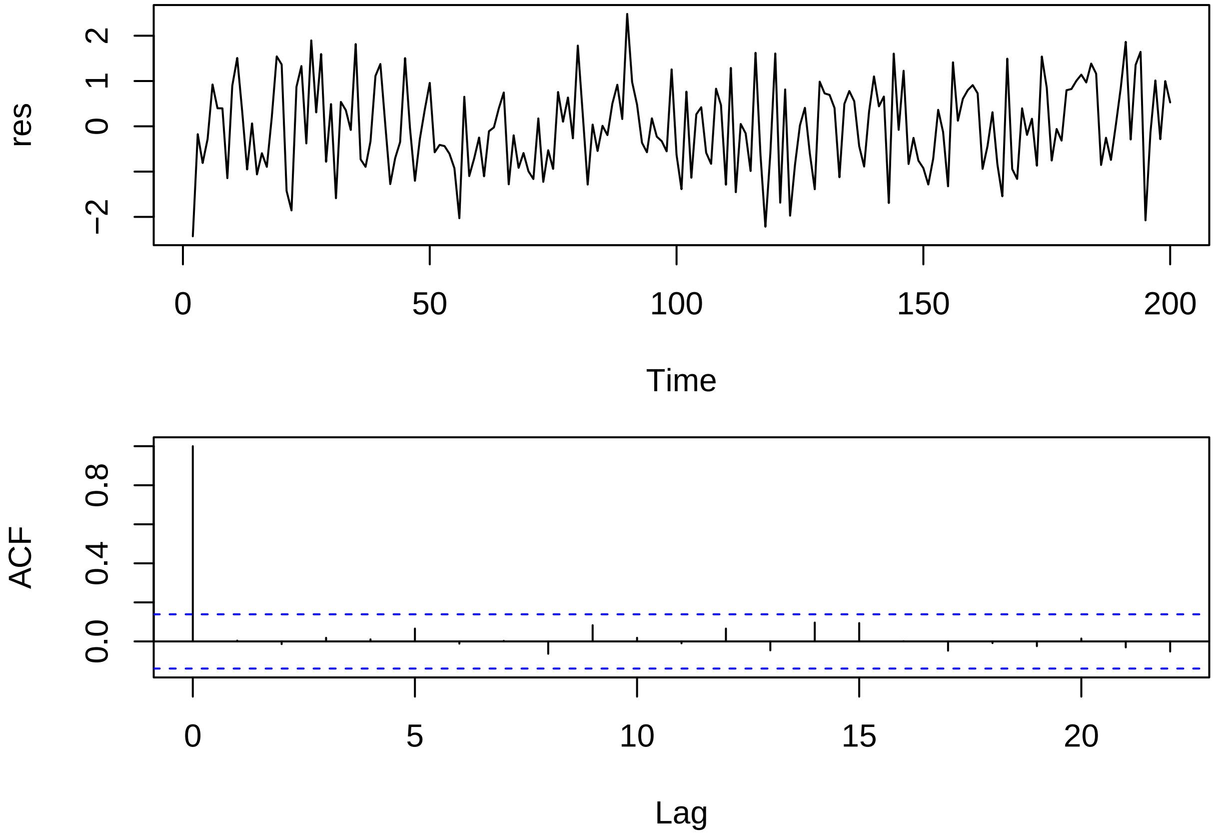
```
ci <- c(fit$ar - 2*se, fit$ar + 2*se)
ci
```

```
## [1] 0.5709586 0.7803535
```

11.5 Residuals for AR(1) models

- The AR(1) model defined earlier has mean 0. However, we cannot expect data to fulfill this.
- This is fixed by subtracting the average $\bar{X}$ of the data before doing anything else, so the model that is fitted is actually:
$$X_t - \bar{X} = \alpha_1 \cdot (X_{t-1} - \bar{X}) + W_t$$
- The fitted model can be used to make predictions. To predict X_t from X_{t-1}, we use that W_t is white noise, so we expect it to be zero on average:
$$\hat{x}_t - \bar{x} = \hat{\alpha}_1 \cdot (x_{t-1} - \bar{x}),$$
so
$$\hat{x}_t = \bar{x} + \hat{\alpha}_1 \cdot (x_{t-1} - \bar{x}), \quad t \geq 2.$$
- We can estimate the white noise terms as usual by the model residuals:
$$\hat{w}_t = x_t - \hat{x}_t, \quad t \geq 2.$$
- If the model is correct, the residuals should look like a sample of white noise. We check this by plotting the acf:

```
res <- na.omit(fit$resid)
par(mfrow = c(2,1), mar = c(4,4,1,1))
plot(res)
acf(res)
```



- It naturally looks good for this artificial dataset.

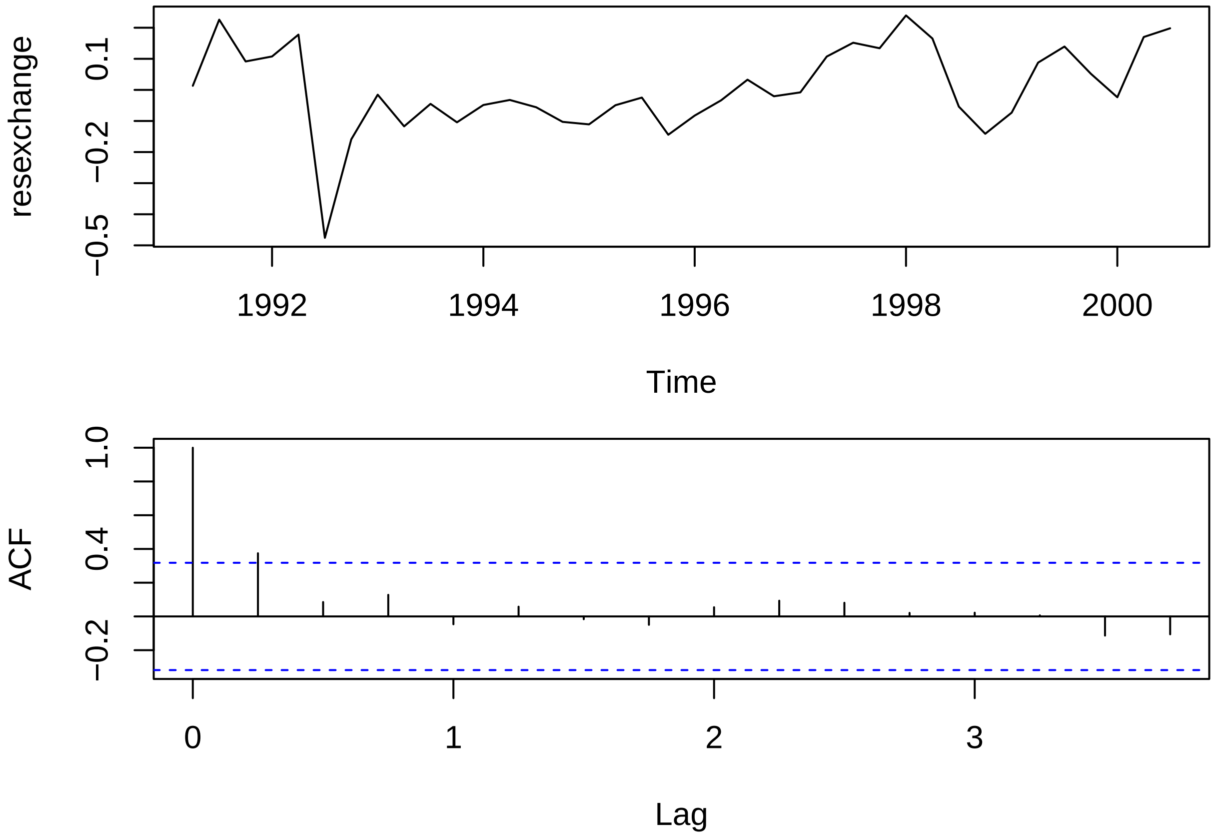
11.6 AR(1) model fitted to exchange rate

- A random walk is an example of an AR(1) model with $\alpha_1 = 1$. It didn't provide an ideal fit for the exchange rate dataset, so we might suggest a stationary AR(1) model with α_1 as a parameter to be estimated from data:

```
fitexchange <- ar(exchange, order.max = 1, method="mle")
fitexchange$ar
```

```
## [1] 0.9437125
```

```
resexchange <- na.omit(fitexchange$resid)
par(mfrow = c(2,1), mar = c(4,4,1,1))
plot(resexchange)
acf(resexchange)
```

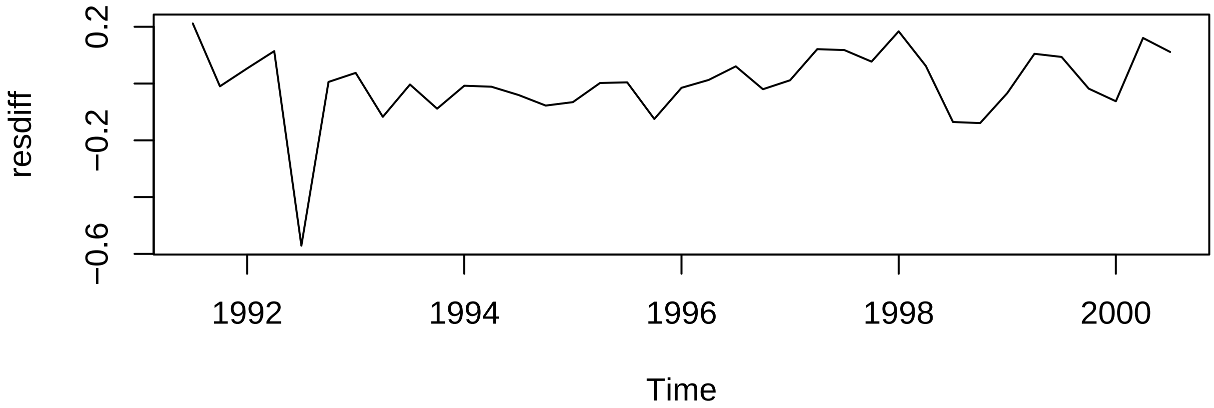


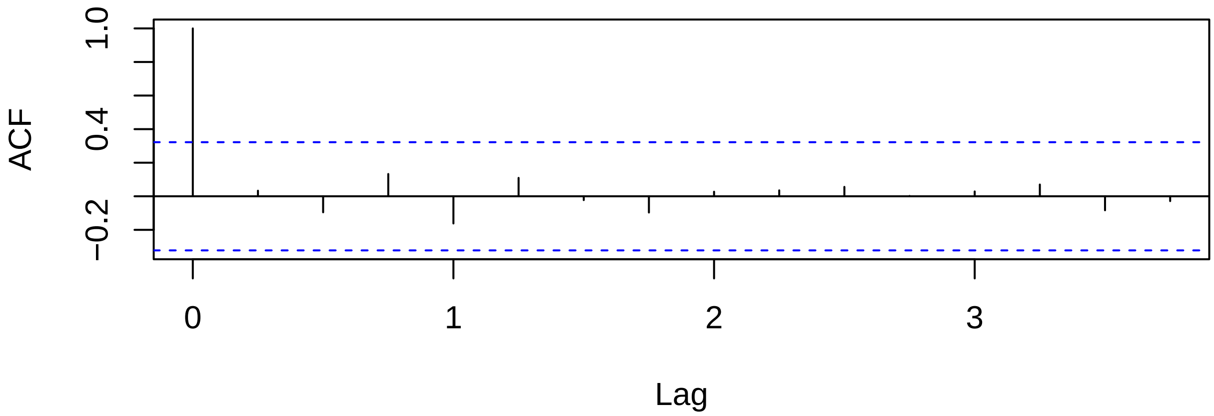
- This does not appear to really provide a better fit than the random walk model proposed earlier.
- An alternative would be to propose an AR(1) model for the differenced time series $\nabla X_t = X_t - X_{t-1}$:

```
dexchange <- diff(exchange)
fitdiff <- ar(dexchange, order.max = 1,method="mle")
fitdiff$ar
```

```
## [1] 0.3495624
```

```
resdiff <- na.omit(fitdiff$resid)
par(mfrow = c(2,1), mar = c(4,4,1,1))
plot(resdiff)
acf(resdiff)
```





11.7 Prediction/forecasting from AR(1) model

- We can use a fitted AR(1) model to predict the next value of an observed time series $x_1, \dots, x_n$.

$$\hat{x}_{n+1} = \bar{x} + \hat{\alpha}_1 \cdot (x_n - \bar{x}).$$

- This can be iterated to predict x_{n+2}

$$\hat{x}_{n+2} = \bar{x} + \hat{\alpha}_1 \cdot (\hat{x}_{n+1} - \bar{x}),$$

  and we can continue this way.

- Prediction is performed by `predict` in R.
- E.g. for the AR(1) model fitted to the exchange rate data, the last observation is in third quarter of 2000. If we want to predict 1 year ahead to third quarter of 2001:

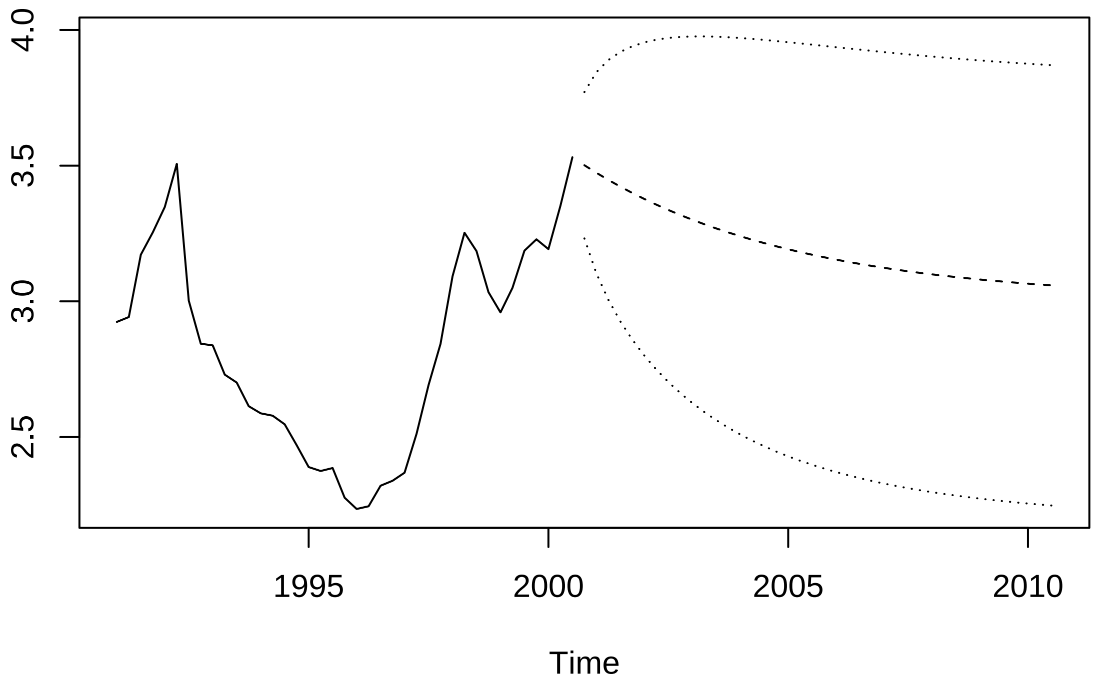
```
pred1 <- predict(fitexchange, n.ahead = 4)
pred1
```

```
## $pred
##          Qtr1     Qtr2     Qtr3     Qtr4
## 2000                            3.501528
## 2001 3.473716 3.447469 3.422699
##
## $se
##           Qtr1      Qtr2      Qtr3      Qtr4
## 2000                               0.1348262
```

```
## 2001 0.1853845 0.2208744 0.2482461
```

- Note how the prediction returns both the predicted value and a standard error for this value.
- We can use this to say that we are 95% confident that the exchange rate in third quarter of 2001 would be within $3.42 \pm 2 \cdot 0.25$.
- We can plot a prediction and approximate 95% pointwise prediction intervals with `ts.plot` (where we use a 10 year prediction – which is a very bad idea – to see how it behaves in the long run):

```
pred10 <- predict(fitexchange, n.ahead = 40)
lower10 <- pred10$pred-2*pred10$se
upper10 <- pred10$pred+2*pred10$se
ts.plot(exchange, pred10$pred, lower10, upper10, lty = c(1,2,3,3))
```



12 Auto-regressive models of higher order (AR(p)-models)

12.1 Auto-regressive models of higher order

- The first order auto-regressive model can be generalised to higher order by adding more lagged terms to explain the current value X_t.
- An **AR(p) process** is
$$X_t = \alpha_1 X_{t-1} + \alpha_2 X_{t-2} + \cdots + \alpha_p X_{t-p} + W_t$$
where W_t is white noise and $\alpha_1, \alpha_2, \ldots, \alpha_p$ are parameters to be estimated from data.
- To simplify the notation, we introduce the **backshift operator** B, that takes the time series one step back, i.e.
$$BX_t = X_{t-1}$$

- This can be used repeatedly, for example $B^2X_t = BBX_t = BX_{t-1} = X_{t-2}$. We can then make a "polynomial" of backshift operators

$$\alpha(B) = 1 - \alpha_1 B - \alpha_2 B^2 - \cdots - \alpha_p B^p$$

and write the AR(p) process as

$$W_t = \alpha(B) X_t.$$

12.2 Stationarity for AR(p) models

- Not all AR(p) models are stationary. To check that a given AR(p) model is stationary we consider the **characteristic equation**

$$1 - \alpha_1 z - \alpha_2 z^2 - \cdots - \alpha_p z^p = 0.$$

- The characteristic equation can be written with the polynomial α from previous slide, but with z inserted instead of B, i.e.

$$\alpha(z) = 0.$$

- The process is stationary if and only if all roots of the characteristic equation have an absolute value that is greater than 1.
- Solving a p-order polynomial is hard for high values of p, so we will let R do this for us.

12.3 Estimation of AR(p) models

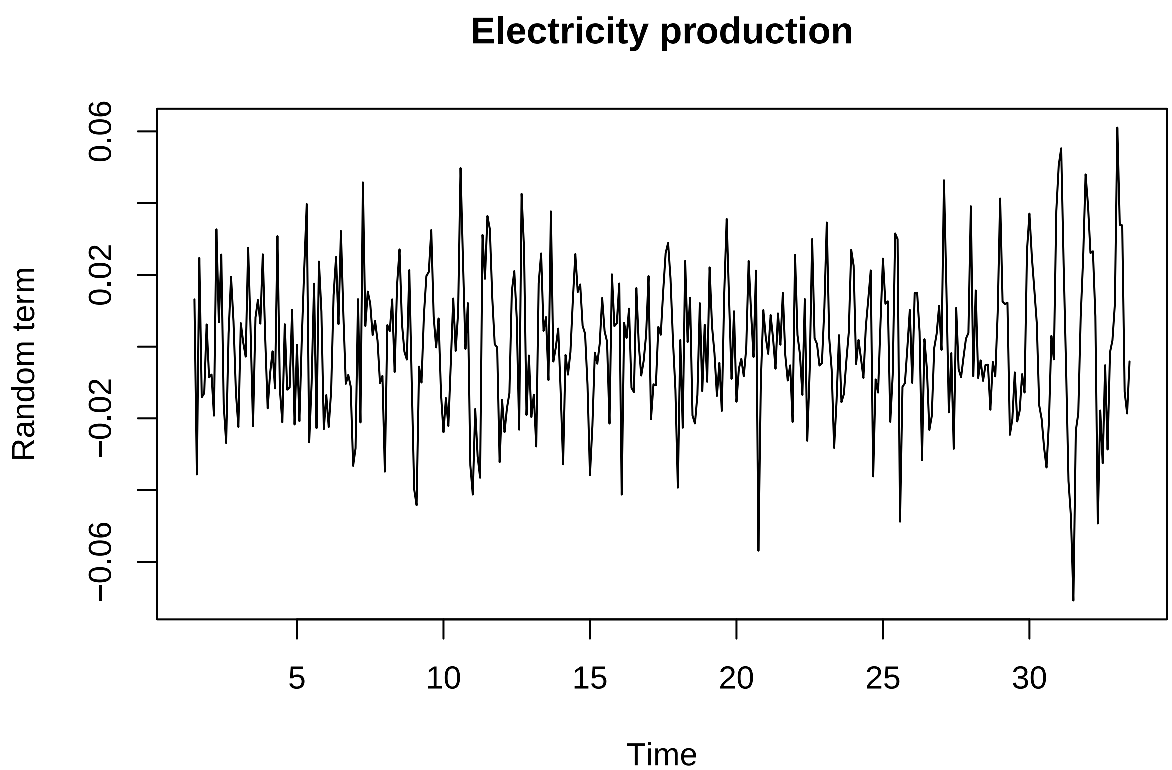
- For an AR(p) model there are typically two things we need to estimate:

1. The maximal non-zero lag p in the model.
2. The autoregressive coefficients/parameters $\alpha_1, \ldots, \alpha_p$.

- To select p we can use **AIC (Akaike's Information Criterion)**.
  - More complex models can always fit a dataset better
  - AIC is essentially a balance between model simplicity and good fit.
  - The AIC results in a single real number, where smaller is better.
  - The ar function in R uses AIC to automatically select the value for p by calculating the AIC for all models with p between 1 and some chosen maximal value, and picking the one with the smallest AIC.
- Once the order p is chosen, the estimates $\hat{\alpha}_1, \ldots, \hat{\alpha}_p$ can be computed by the ar function.
  - The corresponding standard errors can be found as the square root of the diagonal of the matrix stored as `asy.var.coef` in the fitted model object.

12.4 Example of AR(p) model for electricity data

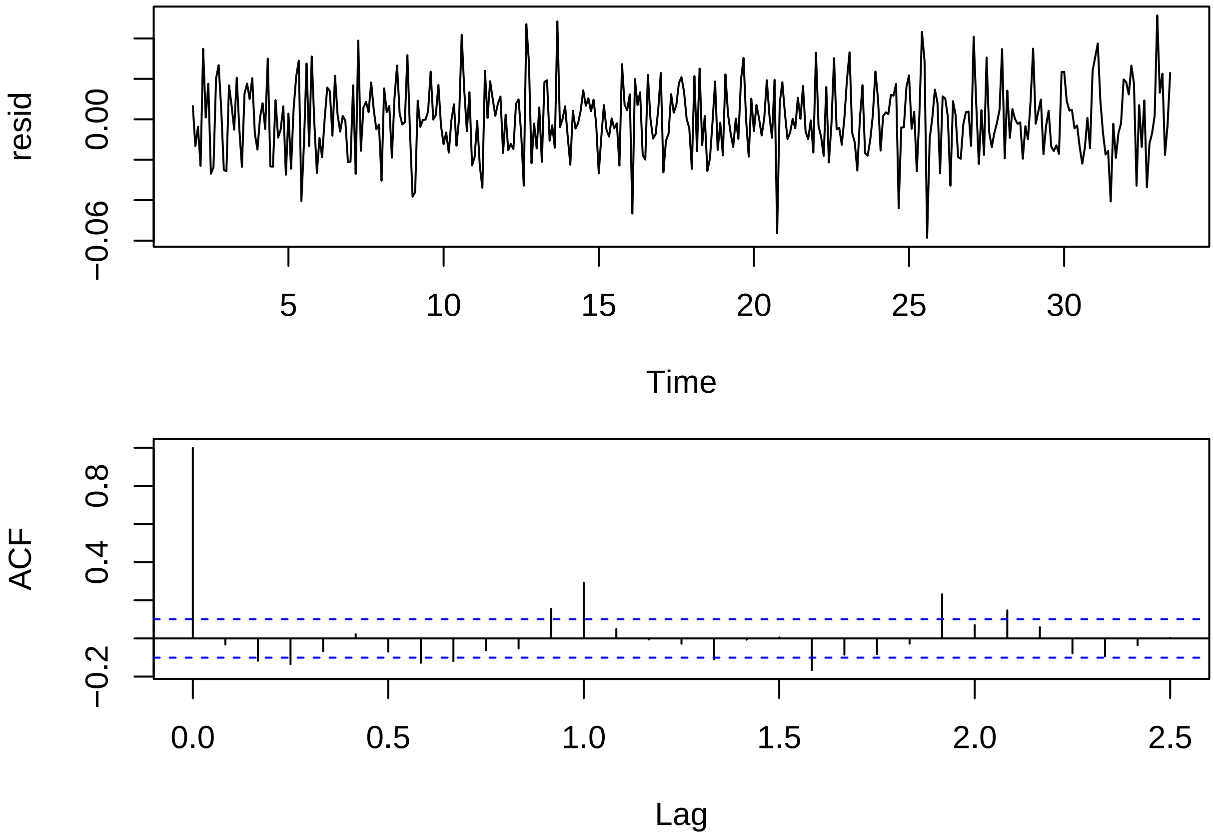
- We try to fit an AR(p) model to the detrended data for the log of the electricity production in Australia:




```
fit <- ar(random, order.max = 5,method="mle")
fit
```

```
##
## Call:
## ar(x = random, order.max = 5, method = "mle")
##
## Coefficients:
##       1        2        3        4        5
##  0.2047   0.0714  -0.0280  -0.1975  -0.2440
##
## Order selected 5  sigma^2 estimated as  0.0003103
```

```
resid <- na.omit(fit$resid)
par(mfrow = c(2,1), mar = c(4,4,1,1))
plot(resid)
acf(resid, lag.max = 30)
```



- There are too many significant autocorrelations. Suggests the model is not a good fit.
- We are not assured that the estimated model will be stationary. To check stationarity we solve $\alpha(z) = 0$:

```
abs(polyroot(c(1,-fit$ar)))
```

```
## [1] 1.148425 1.415987 1.415987 1.148425 1.549525
```

- All absolute values of the roots are greater than 1, indicating stationarity.

13 Moving average models (MA(q)-models)

13.1 The moving average model

- A **moving average process** of order q, MA(q), is defined by

$$X_t = W_t + \beta_1 W_{t-1} + \beta_2 W_{t-2} + \cdots + \beta_q W_{t-q}$$

where W_t is a white noise process with mean 0 and variance σ_w^2 and $\beta_1, \beta_2, \ldots, \beta_q$ are parameters to be estimated.

- The moving average process can also be written using the backshift operator:

$$X_t = \beta(B) W_t$$

where the polynomial $\beta(B)$ is given by

$$\beta(B) = 1 + \beta_1 B + \beta_2 B^2 + \cdots + \beta_q B^q.$$

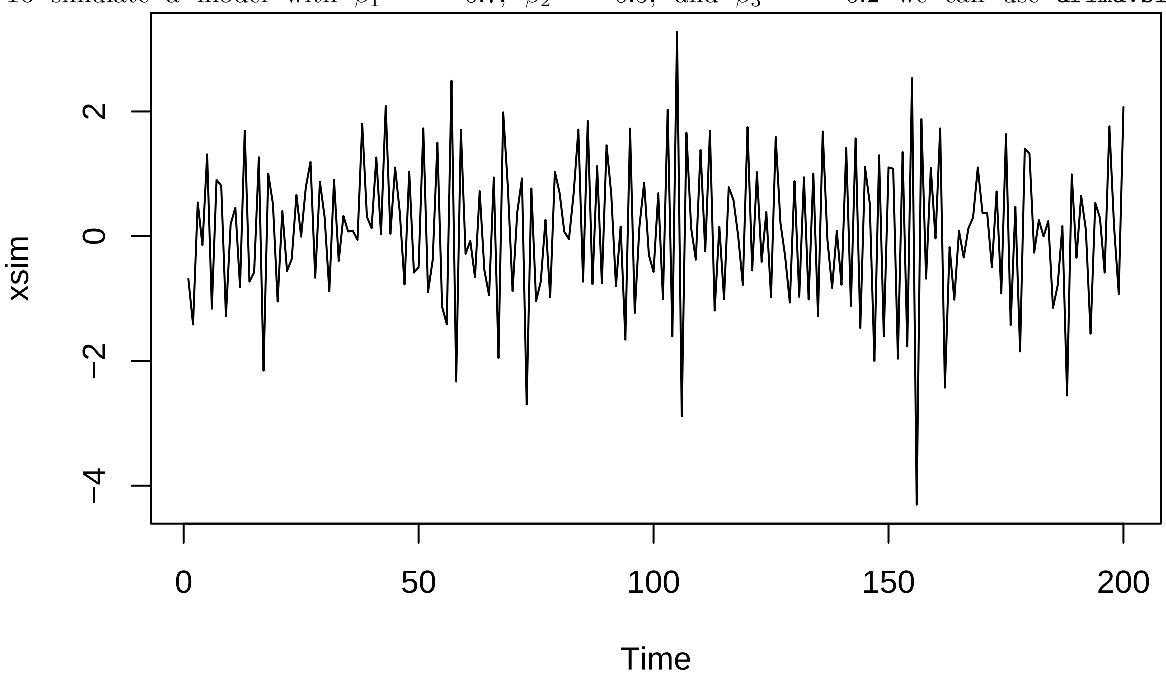
- Since a moving average process is a finite sum of stationary white noise terms it is itself stationary with mean and variance:
  - Mean $\mu(t) = 0$.
  - Variance $\sigma^2(t) = \sigma_w^2(1 + \beta_1^2 + \beta_2^2 + \cdots + \beta_q^2)$.
- The autocorrelation function is

$$\rho(k) = \begin{cases} 1 & k = 0 \\ \sum_{i=0}^{q-k} \beta_i \beta_{i+k} / \sum_{i=0}^{q} \beta_i^2 & k = 1, 2, \ldots, q \\ 0 & k > q \end{cases}$$

  where $\beta_0 = 1$.

13.2 Simulation of MA(q) processes

- An MA(q) process can be simulated by first generating a white noise process W_t and then transforming it using the MA coefficients.
- To simulate a model with $\beta_1 = -0.7$, $\beta_2 = 0.5$, and $\beta_3 = -0.2$ we can use `arima.sim`:



- The theoretical autocorrelations are in this case:

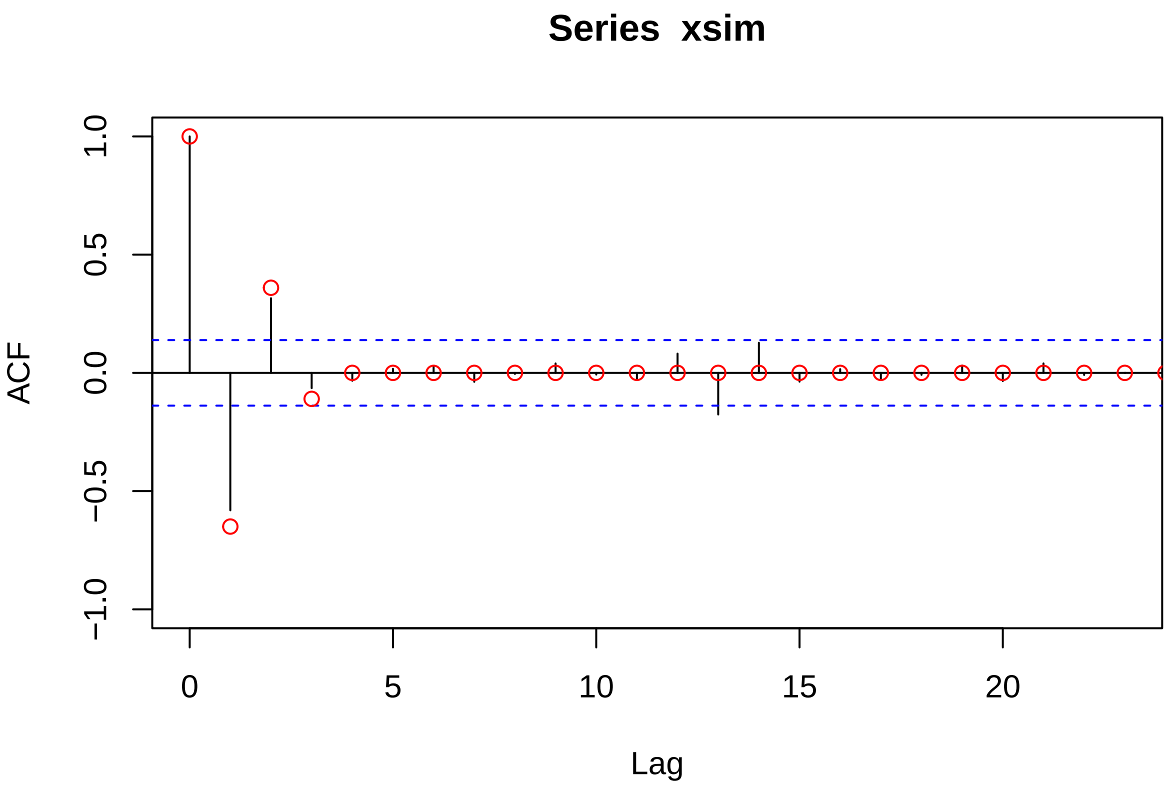
$$\rho(1) = \frac{1 \cdot (-0.7) + (-0.7) \cdot 0.5 + 0.5 \cdot (-0.2)}{1 + (-0.7)^2 + 0.5^2 + (-0.2)^2} = -0.65$$

$$\rho(2) = \frac{1 \cdot 0.5 + (-0.7) \cdot (-0.2)}{1 + (-0.7)^2 + 0.5^2 + (-0.2)^2} = 0.36$$

$$\rho(3) = \frac{1 \cdot (-0.2)}{1 + (-0.7)^2 + 0.5^2 + (-0.2)^2} = -0.11$$

  and $\rho(k) = 0$ for $k > 3$.
- We plot the acf of the simulated series together with the theoretical one.

```r
acf(xsim,ylim=c(-1,1))
points(0:25, c(1,-.65, .36, -.11, rep(0,22)), col = "red")
```



13.3 Estimation of MA(q) models

- To estimate the parameters of an MA(3) model we use `arima`:

```r
xfit <- arima(xsim, order = c(0,0,3))
xfit
```

```
##
## Call:
## arima(x = xsim, order = c(0, 0, 3))
##
## Coefficients:
##           ma1     ma2      ma3  intercept
##       -0.5996  0.4471  -0.1811     0.0711
## s.e.   0.0690  0.0766   0.0784     0.0442
##
## sigma^2 estimated as 0.8772:  log likelihood = -270.96,  aic = 551.91
```

- The function `arima` does not include automatic selection of the order of the model so this has to be chosen beforehand or selected by comparing several proposed models and choosing the model with the minimal AIC.

```r
AIC(xfit)
```

```
## [1] 551.9113
```

14 Auto-regressive moving average models (ARMA)

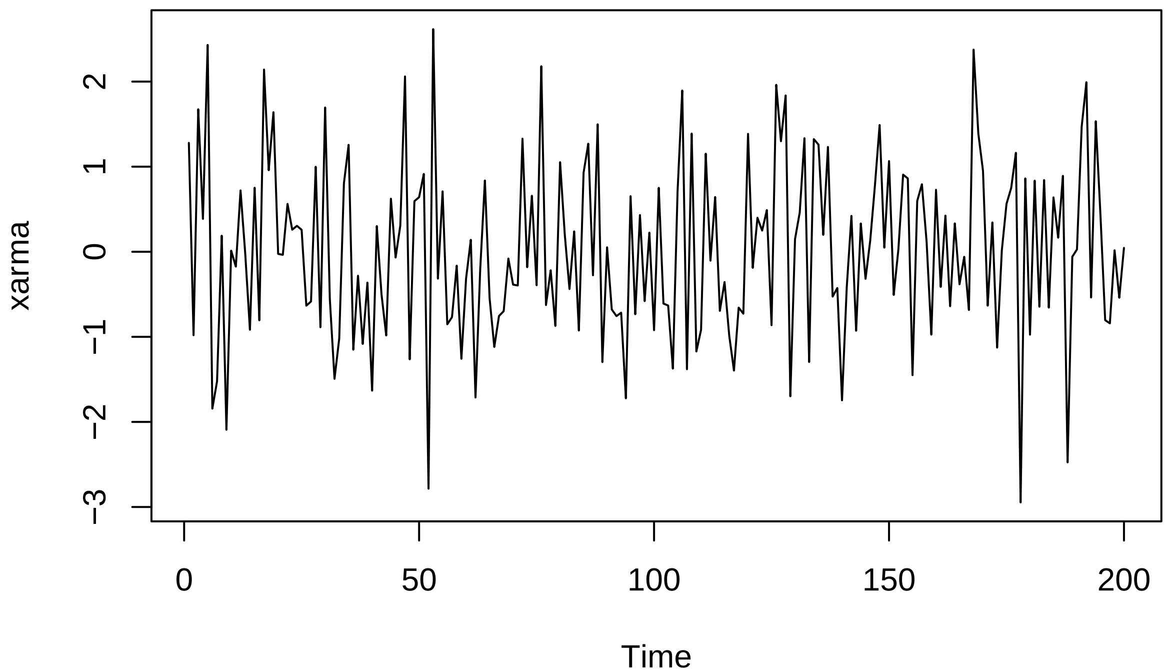
14.1 Mixed models: Auto-regressive moving average models

- A time series X_t follows an auto-regressive moving average (ARMA) process of order (p, q), denoted $ARMA(p, q)$, if

$$X_t = \alpha_1 X_{t-1} + \alpha_2 X_{t-2} + \cdots + \alpha_p X_{t-p} + W_t + \beta_1 W_{t-1} + \beta_2 W_{t-2} + \cdots + \beta_q W_{t-q}$$

  where W_t is a white noise process and $\alpha_1, \alpha_2, \ldots, \alpha_p, \beta_1, \beta_2, \ldots, \beta_q$ are parameters to be estimated.
- We can simulate an ARMA model with `arima.sim`. E.g. an ARMA$(1,1)$ model with $\alpha_1 = -0.6$ and $\beta_1 = 0.5$:

```
xarma <- arima.sim(model = list(ar = -0.6, ma = 0.5), n = 200)
plot(xarma)
```



- Estimation is done with `arima` as before.

14.2 Example with exchange rate data

For the exchange rate data we may e.g. suggest either a AR(1), MA(1) or ARMA$(1,1)$ model. We can compare fitted models using `AIC` (smaller is better):

```
exchange_ar <- arima(exchange, order = c(1,0,0))
AIC(exchange_ar)
```

```
## [1] -37.40417
```

```
exchange_ma <- arima(exchange, order = c(0,0,1))
AIC(exchange_ma)
```

```
## [1] -3.526895
```

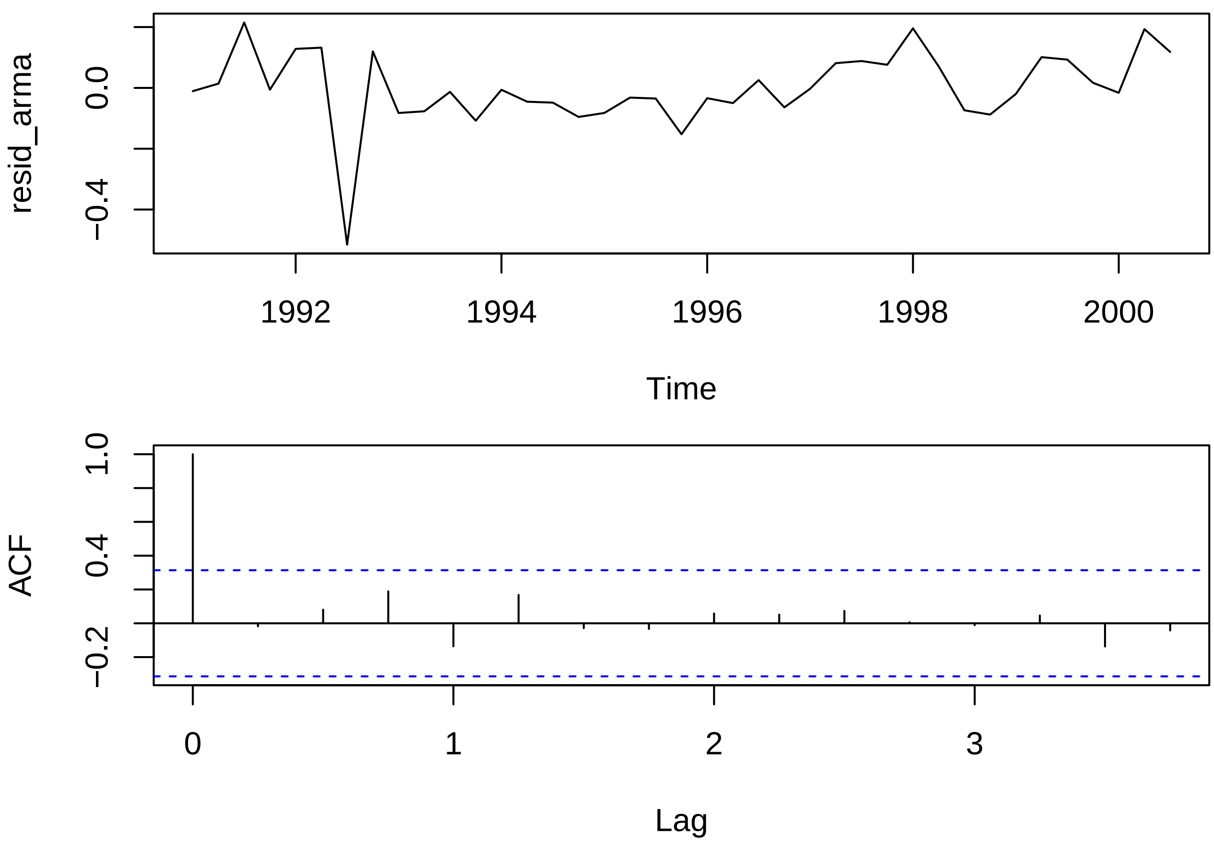
```r
exchange_arma <- arima(exchange, order = c(1,0,1))
AIC(exchange_arma)
```

```
## [1] -42.27357
```

```r
exchange_arma
```

```
##
## Call:
## arima(x = exchange, order = c(1, 0, 1))
##
## Coefficients:
##          ar1     ma1  intercept
##       0.8925  0.5319     2.9597
## s.e.  0.0759  0.2021     0.2435
##
## sigma^2 estimated as 0.01505:  log likelihood = 25.14,  aic = -42.27
```

```r
par(mfrow = c(2,1), mar = c(4,4,1,1))
resid_arma <- na.omit(exchange_arma$residuals)
plot(resid_arma)
acf(resid_arma)
```



15 Models with exogenous variables

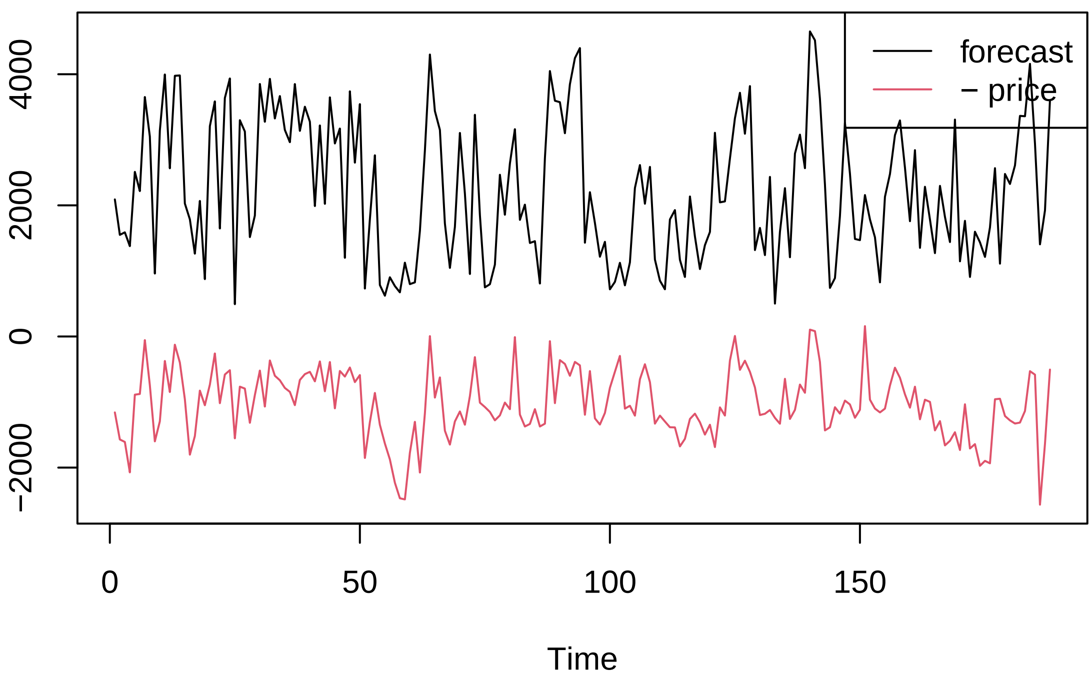
15.1 Exogenous variables

- The ARMA processes are flexible models for a time series Y_t, $t = 1, \dots, n$ evolving randomly over time, but they do not include the possibility that anything is influencing Y_t.
- An exogeneous variable is another variable, say X_t, that influences the behaviour of Y_t
  - Wind power production Y_t is influenced by the wind speed X_t
  - The velocity of a DC motor Y_t is influenced by the input voltage X_t
- Here X_t may be another stochastic process, which we do not model, but only consider as given, or it might be something we can control.

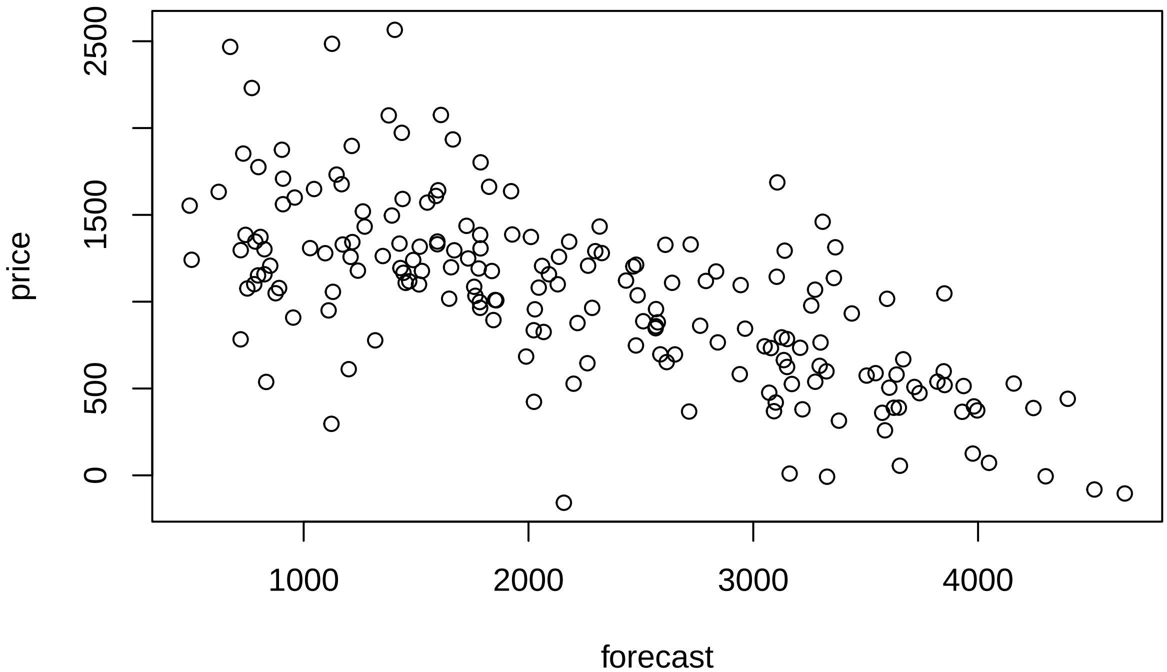
15.2 Data example

- The dataset below contains data from Jan 7 to Jul 13 2022 on two variables
  - forecast: Total day ahead forecasted wind and solar energy production
  - price: Day ahead elspot prices with weekly variation removed

```
elspot<-read.csv("https://asta.math.aau.dk/eng/static/datasets?file=elspot.csv", header = TRUE)
forecast<-elspot[,2]
price<-elspot[,3]
ts.plot(ts(forecast),ts(-price),col=1:2)
legend("topright",legend=c("forecast","- price"),col=1:2,lty=1)
```




```
plot(forecast,price)
```



15.3 Regression models with exogenous variables

- We can combine regression models with ARMA models to obtain a stochastic process which is influenced by exogenous variables.
- Consider a linear regression of Y_t on X_t, but where the noise term is an ARMA process:

$$Y_t = \gamma_0 + \gamma_1 X_t + \epsilon_t, \qquad \alpha(B)\epsilon_t = \beta(B)W_t$$

- If we isolate $\epsilon_t = Y_t - \gamma_0 - \gamma_1 X_t$ and insert into the ARMA expression, we get something that looks more like an ARMA process, but with Y_t adjusted by the exogenous variable:

$$\alpha(B)(Y_t - \gamma_0 - \gamma_1 X_t) = \beta(B)W_t$$

- The purpose of fitting such a model is both to obtain a good model for the evolution of the data and to obtain an understanding of the relation between Y_t and X_t.
- Above, X_t is a single stochastic process, but we can also include multiple stochastic processes by making a multiple regression model with an ARMA model for the errors.

15.4 Example

- As an example consider a simple linear regression combined with an AR(1) process for noise terms:

$$Y_t = \gamma_0 + \gamma_1 X_t + \epsilon_t, \qquad \epsilon_t = \alpha_1 \epsilon_{t-1} + W_t$$

or, since $\epsilon_{t-1} = Y_{t-1} - \gamma_0 - \gamma_1 X_{t-1}$,

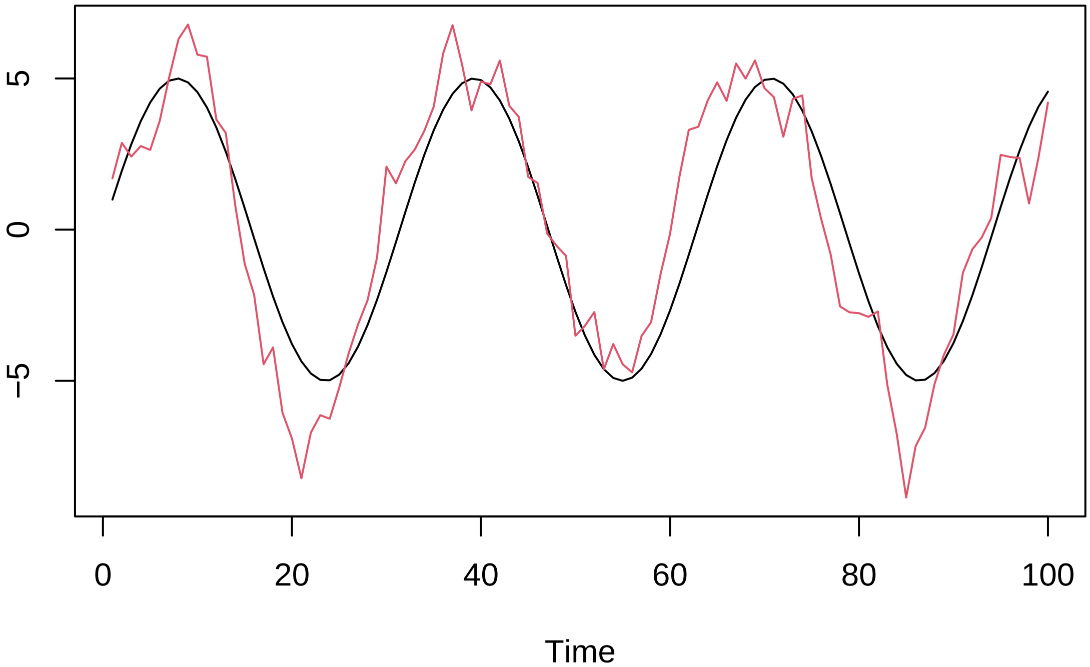
$$Y_t = \alpha_1 Y_{t-1} + (1 - \alpha_1)\gamma_0 + \gamma_1 (X_t - \alpha_1 X_{t-1}) + W_t$$

- Notice that the model behaves like an AR(1) process, but instead of having a constant mean of 0, its mean is constantly adjusted by the exogenous variable.

15.5 Simulation of the example

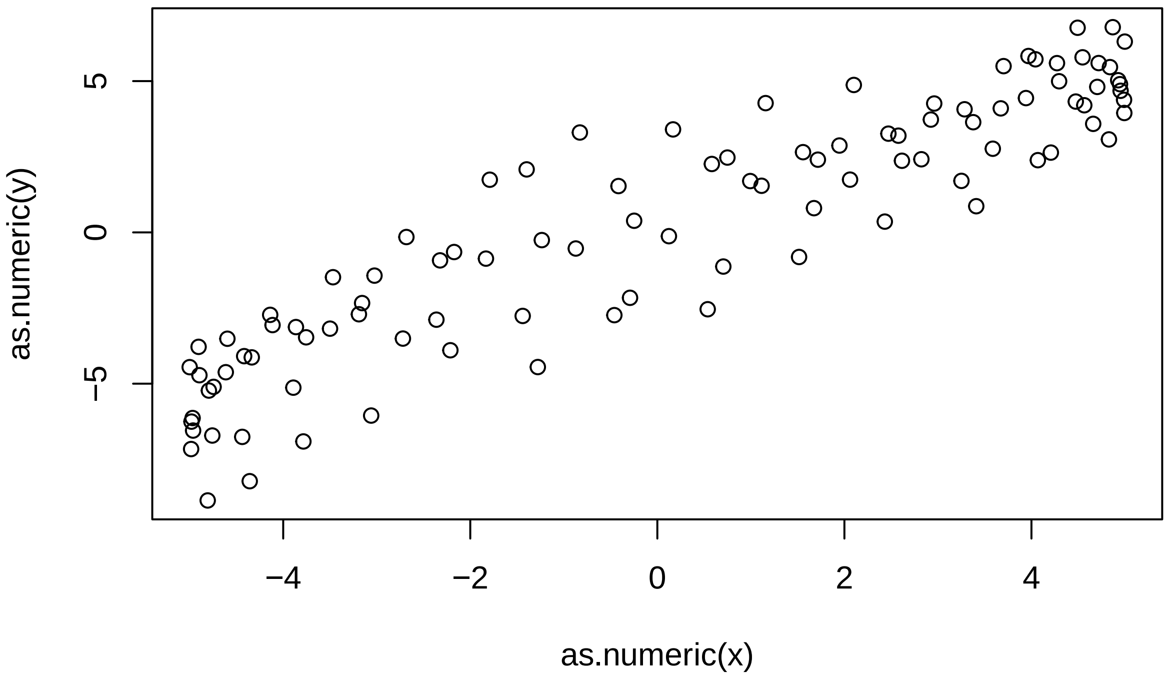
- We simulate some data resembling the example, where we let X_t follow a sine curve:

```
alpha = 0.9; gamma = 1; n = 100
x = as.ts(5*sin(1:n/5))
eps = arima.sim(model=list(ar=alpha),n=n)
y = gamma*x+eps
ts.plot(x,y,col=1:2)
```



- We should think of the red curve as some data we want to model, and the black curve as another variable which we believe may influence the data.
- We can also plot X_t against Y_t to get a view of the relation between the two variables.

```
plot(as.numeric(x),as.numeric(y))
```



15.6 Estimation and model checking

- We can estimate the parameters using the arima function in R.
- We fit a linear regression model with AR(1) noise to the simulated data (i.e. the true model used for simulation):

```r
mod=arima(y,order=c(1,0,0),xreg=x); mod
```

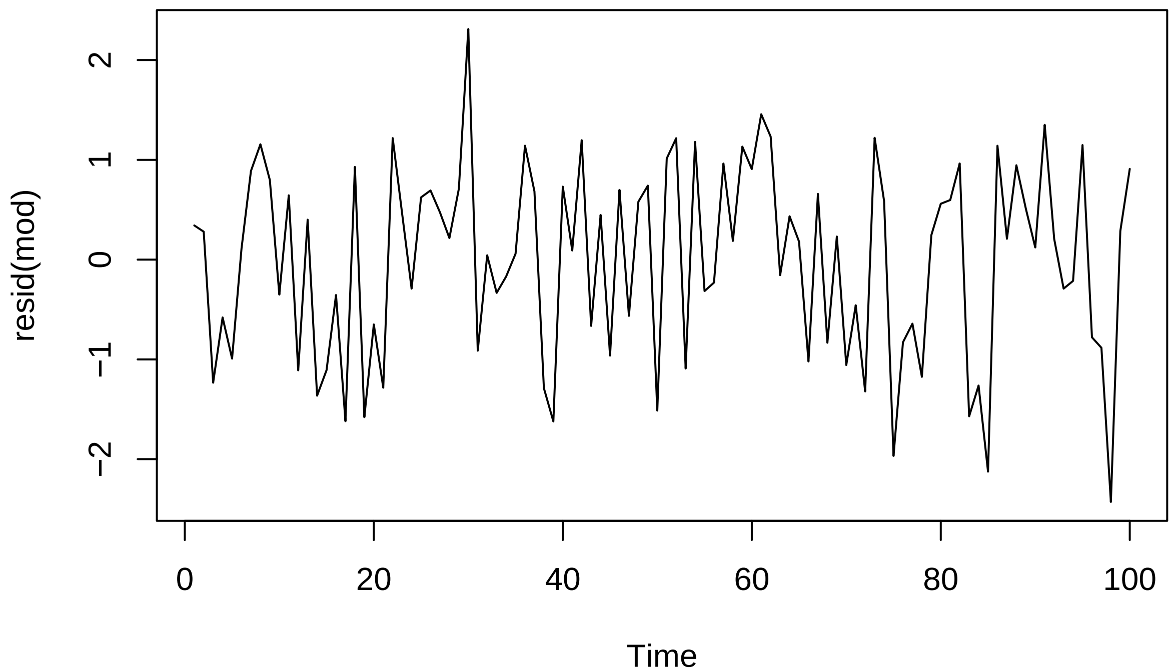
```
##
## Call:
## arima(x = y, order = c(1, 0, 0), xreg = x)
##
## Coefficients:
##          ar1  intercept       x
##       0.8069     0.0679  1.0551
## s.e.  0.0569     0.4795  0.1018
##
## sigma^2 estimated as 0.923:  log likelihood = -138.41,  aic = 284.82
```

- The fitted model becomes

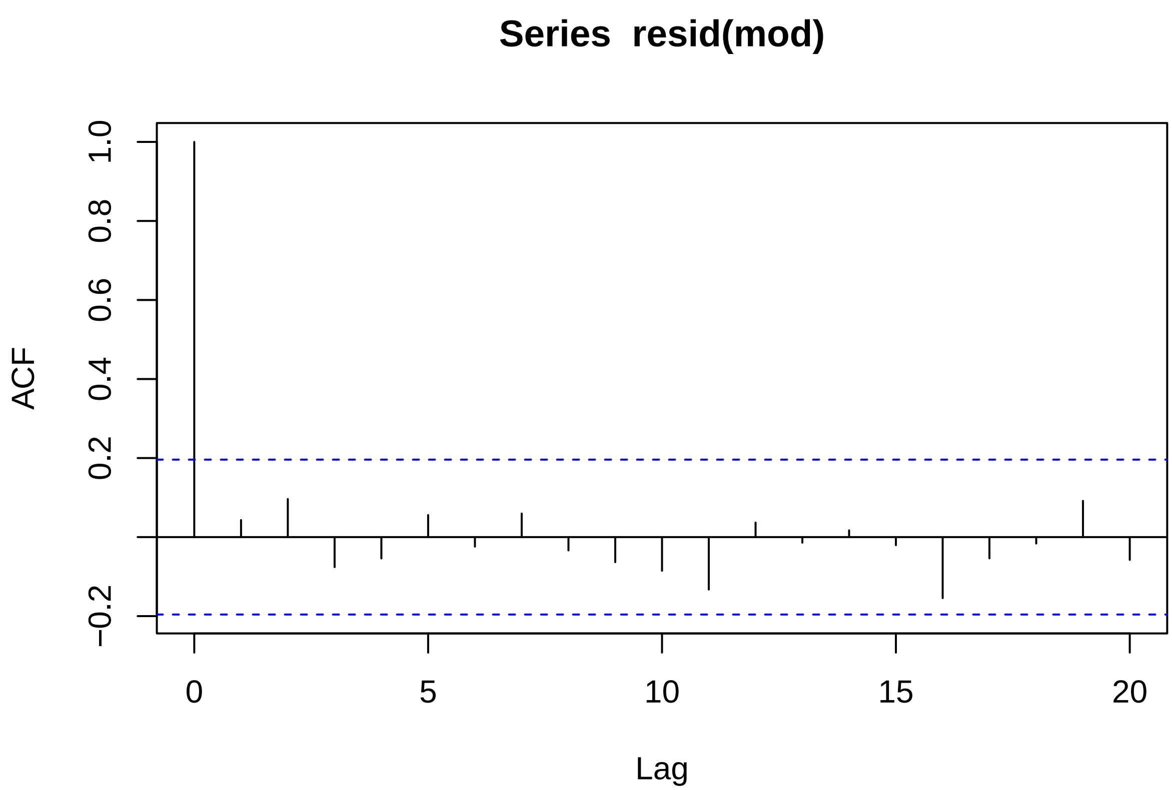
$$Y_t = 0.0679 + 1.0551 \cdot X_t + \epsilon_t, \qquad \epsilon_t = 0.8069 \cdot \epsilon_{t-1} + W_t$$

- The errors $\hat{\epsilon}_t = y_t - 0.0679 + 1.0551 \cdot x_t$ should behave like an AR(1)-model with $\hat{\alpha} = 0.8069$.
  - So the residuals $\epsilon_t - 0.8069 \cdot \epsilon_{t-1}$ should look like white noise.

```
plot(resid(mod))
```




```
acf(resid(mod))
```



15.7 Fitting AR(1) model to data example

- Recall the elspot price dataset

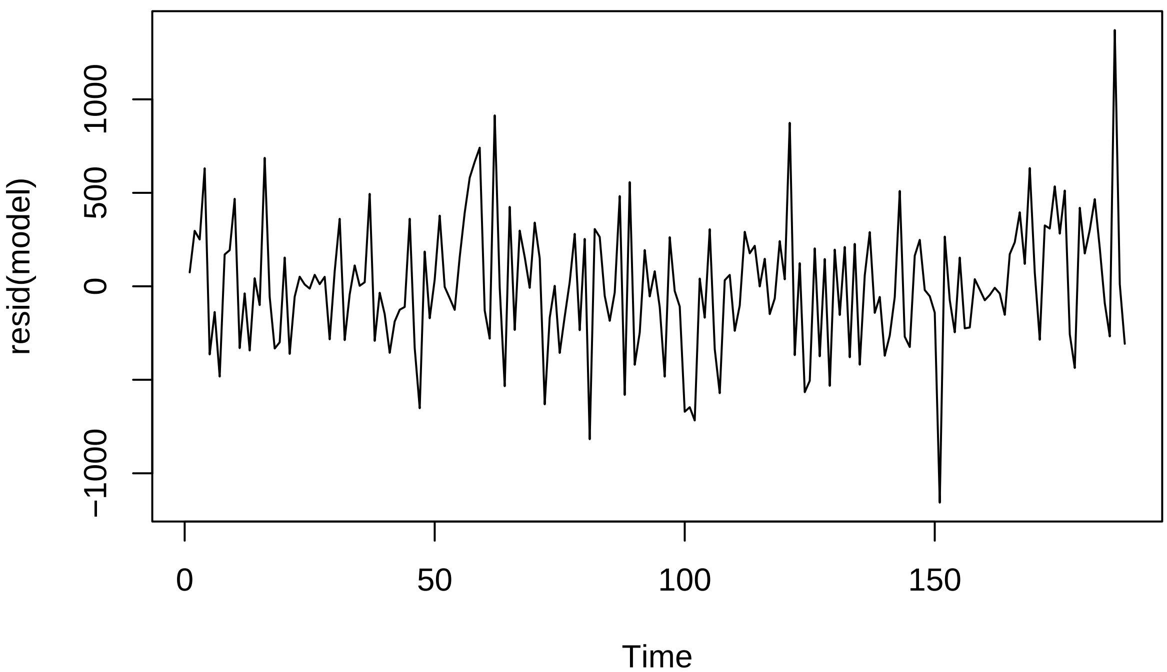
```
forecast<- ts(forecast)
price<-ts(price)
model=arima(price,order=c(1,0,0),xreg=forecast); model
```

```
##
## Call:
## arima(x = price, order = c(1, 0, 0), xreg = forecast)
##
## Coefficients:
##          ar1  intercept  forecast
##       0.3886  1715.8412   -0.3053
## s.e.  0.0680    73.2894    0.0271
##
## sigma^2 estimated as 117486:  log likelihood = -1364.2,  aic = 2736.41
```

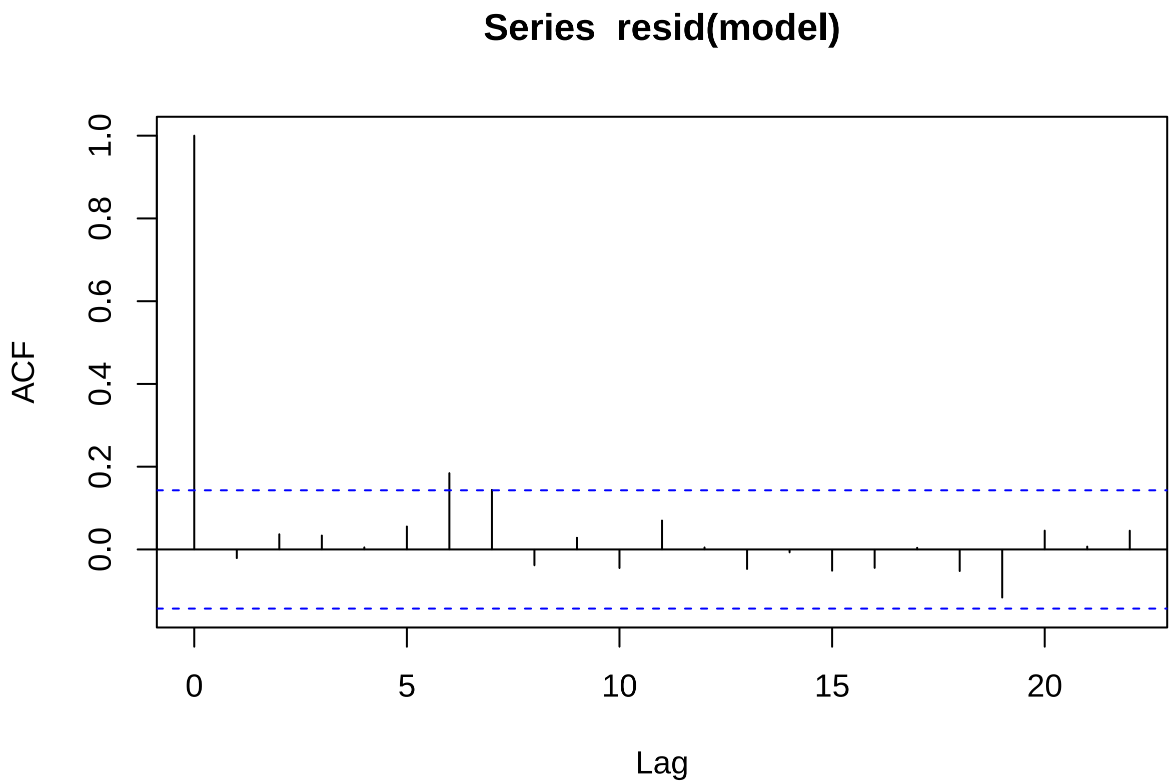
- So we get the model

$$\text{price}_t = 1715.8412 - 0.3053 \cdot \text{forecast}_t + \epsilon_t, \qquad \epsilon_t = 0.3886 \cdot \epsilon_{t-1} + W_t.$$

```
plot(resid(model))
```




```
acf(resid(model))
```

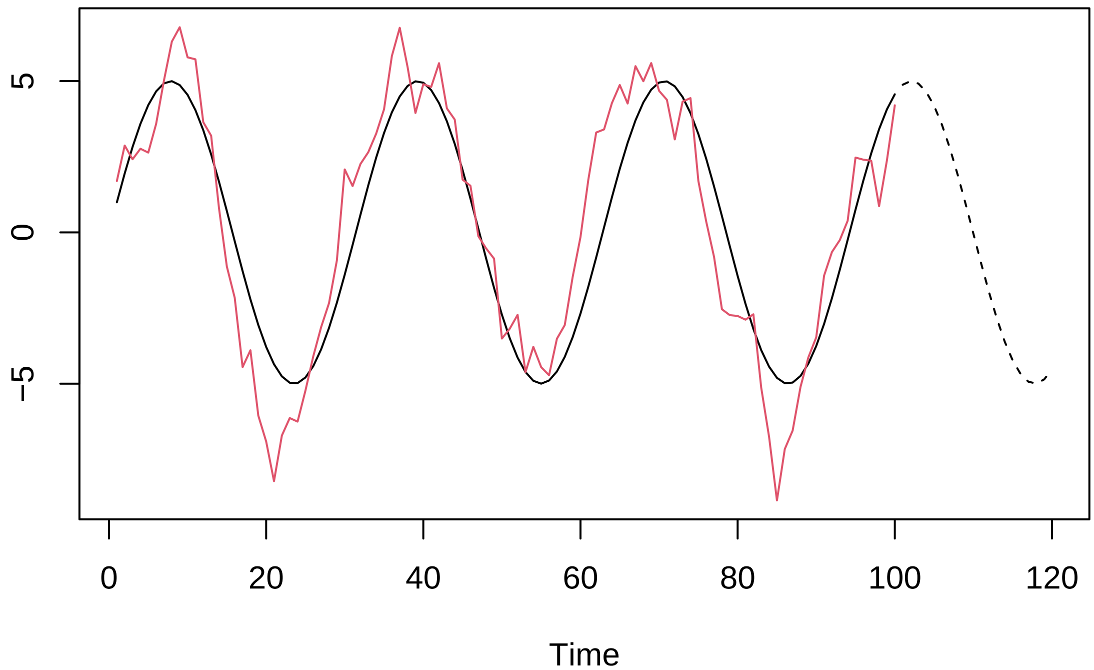


- Residuals indicate that there could be some weekly variation not accounted for.

15.8 Prediction

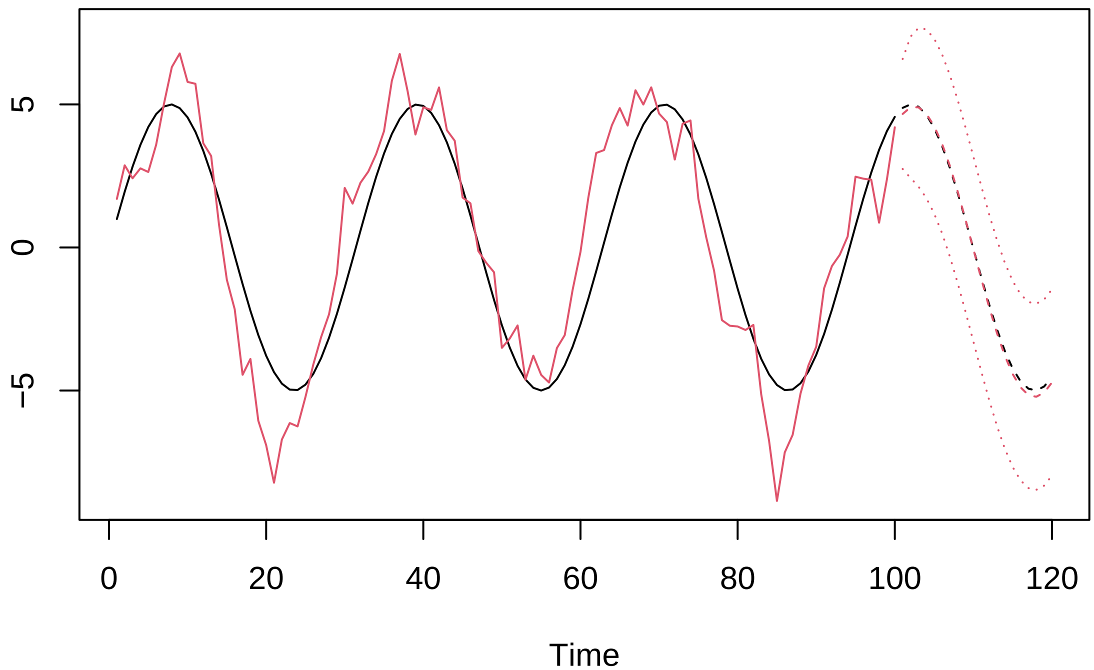
- Prediction can only be performed if we know the behavior of X_t for future time points, for example if we are able to control it.
- For the previous example we assume that the sine curve continues:

```
nnew = 20
xnew = lag(as.ts(5*sin(((n+1):(n+nnew))/5)),-n)
ts.plot(x,y,xnew,col=c(1,2,1),lty=c(1,1,2))
```



- We use the predict function.

```
p = predict(mod,n.ahead=nnew,newxreg=xnew)
ts.plot(x,y,xnew,p$pred,p$pred+2*p$se,p$pred-2*p$se,col=c(1,2,1,2,2,2),lty=c(1,1,2,2,3,3))
```

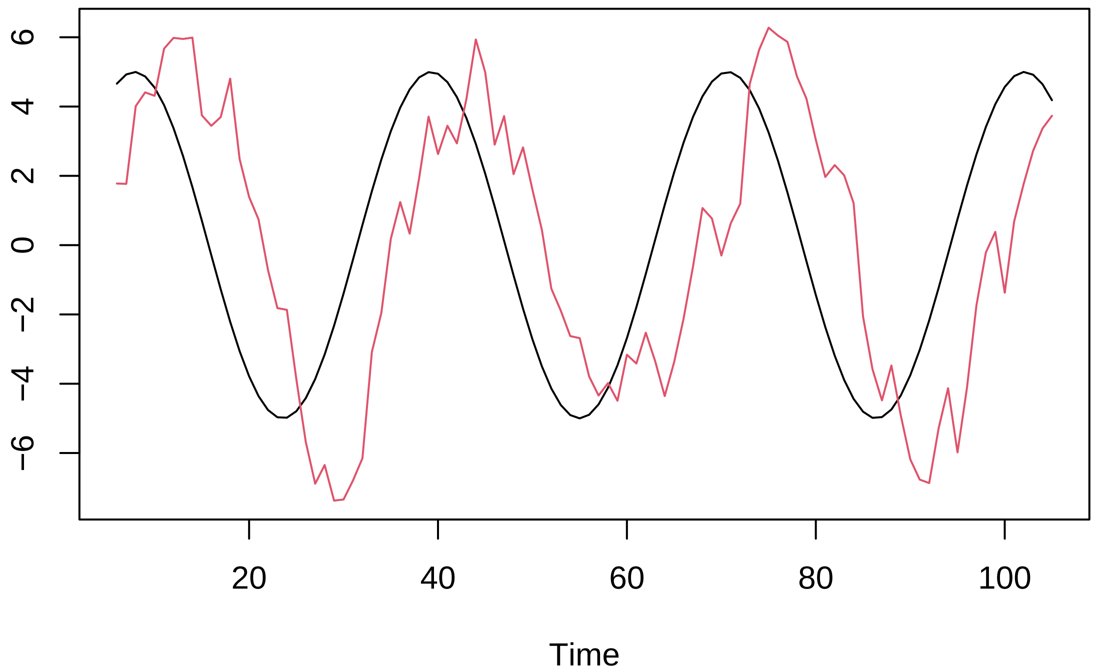


15.9 An example with delay

- If we model the influence of X_t on Y_t, it may take some time before Y_t responds to a change in X_t.
  - Say the delay is k time steps.

- We want to model the effect of X_{t-k} on Y_t.
    - We may not know the delay k, so we may need to estimate it first.
- We simulate a dataset with a built-in delay, and then we model this afterwards.

```
alpha = 0.5; gamma = 1; n = 100; delay = 5
x = as.ts(5*sin(1:(n+delay)/5))
eps = arima.sim(model=list(ar=alpha),n=n+delay)
y = gamma*lag(x,-delay)+eps
dat_lag = ts.intersect(x,y)
ts.plot(dat_lag[,1],dat_lag[,2],col=1:2)
```



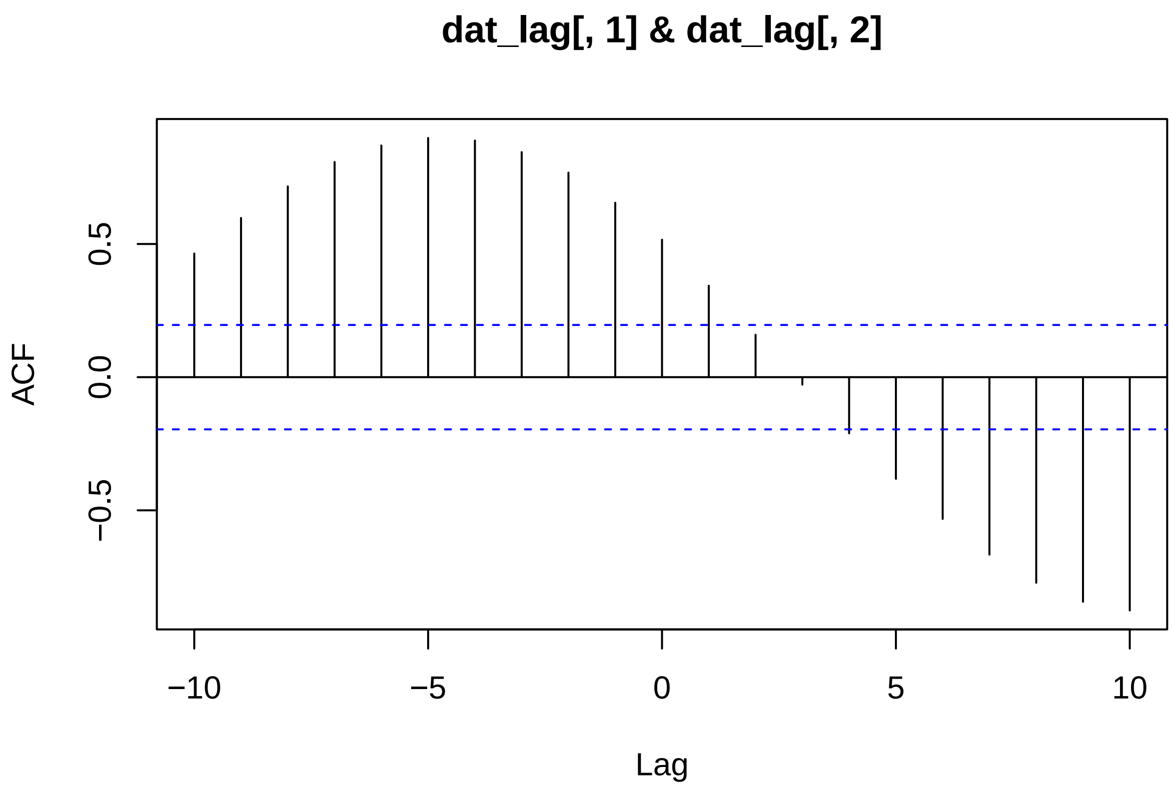
15.10 The cross-correlation function

- The **cross correlation function** is used for checking the relation between two time series at different time points:

$$\rho_{xy}(t+k,t) = \mathrm{Cor}(X_{t+k}, Y_t).$$

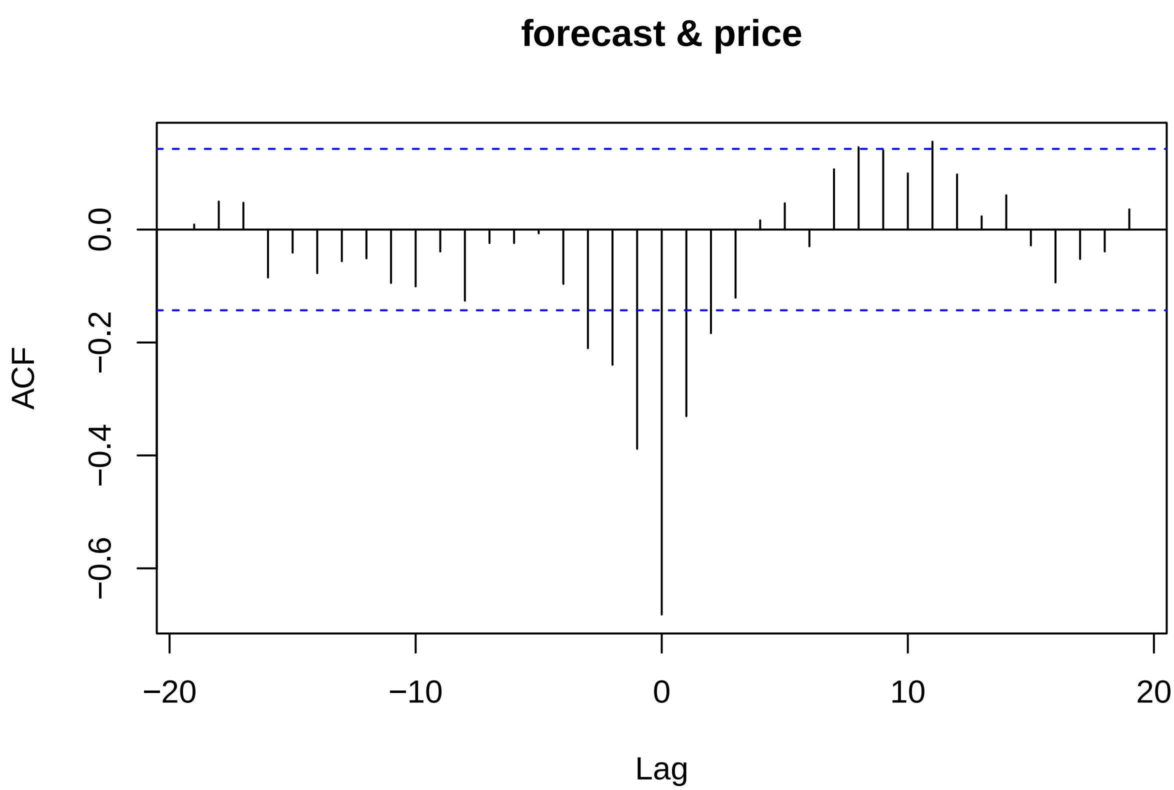
- Values that are close to 1 or -1 indicate that the two time series are closely related if X_t is delayed by k time steps.
- Cross-correlation function for the simulated data

```
cc = ccf(dat_lag[,1],dat_lag[,2],lag.max=10)
```



- Cross-correlation function for the elspot data:

```
ccf(forecast,price)
```



15.11 Fitting models with lag

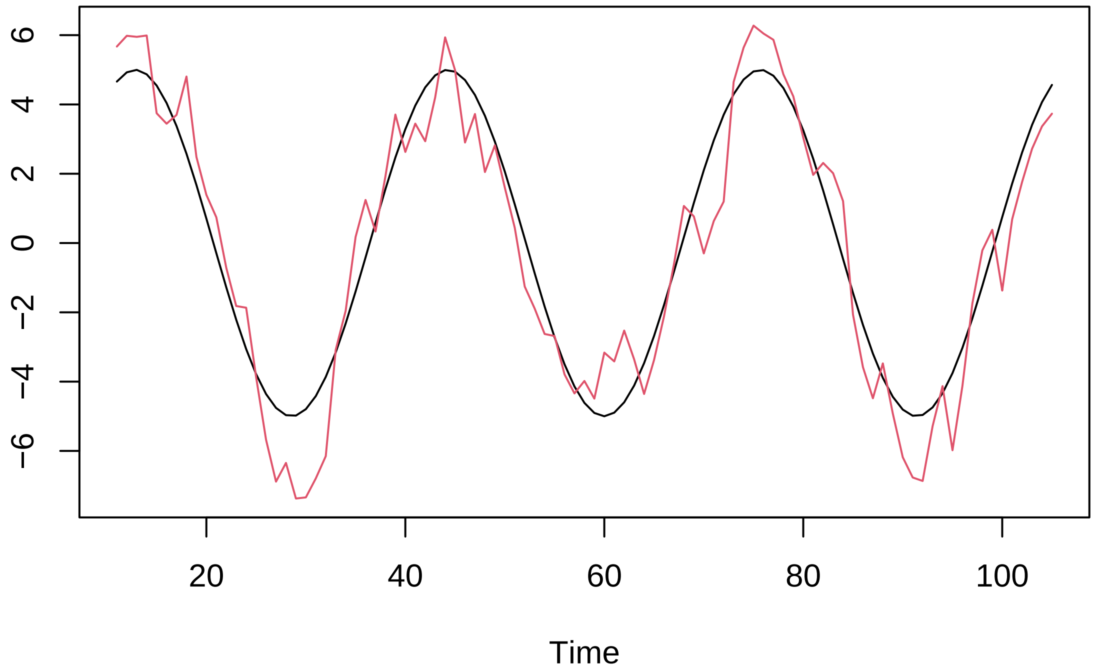
- We estimate the lag to be the one where the cross-correlation function is maximal:

```
estlag = cc$lag[which(cc$acf==max(abs(cc$acf)))]
estlag
```

```
## [1] -5
```

- Plotting the data with this lags can be useful to check the choice:

```
dat_shifted = ts.intersect(lag(as.ts(dat_lag[,1]),estlag),dat_lag[,2] )
ts.plot(dat_shifted[,1],dat_shifted[,2],col=1:2)
```



- We can now fit a model with this lag:

```
mod=arima(dat_shifted[,2],order=c(1,0,0),xreg=dat_shifted[,1]); mod
```

```
##
## Call:
## arima(x = dat_shifted[, 2], order = c(1, 0, 0), xreg = dat_shifted[, 1])
##
## Coefficients:
##          ar1  intercept  dat_shifted[, 1]
##       0.5938    -0.2047            1.0526
## s.e.  0.0820     0.2347            0.0615
##
## sigma^2 estimated as 0.8884:  log likelihood = -129.39,  aic = 266.79
```

15.12 ARMAX models

- An alternative way of including exogenous variables into an ARMA model is an ARMAX model.
- The $ARMAX(p, q, b)$ model is an $ARMA(p, q)$ model including b terms of an exogenous variable, i.e. it

is defined by

$$Y_t = \sum_{i=1}^{p} \alpha_i Y_{t-i} + \sum_{i=1}^{b} \gamma_i X_{t-i} + W_t + \sum_{i=1}^{q} \beta_i W_{t-i}$$

- Using the backshift operator, this can be written as

$$\alpha(B)Y_t = \gamma(B)X_t + \beta(B)W_t$$

  with $\alpha(B) = 1 - \sum_{i=1}^{p} \alpha_i B^i$, $\beta(B) = 1 + \sum_{i=1}^{q} \beta_i B^i$, and $\gamma(B) = \sum_{i=1}^{b} \gamma_i B^i$.
- Compare with the regression with ARMA noise:

$$\alpha(B)(Y_t - \gamma_0 - \gamma_1 X_t) = \beta(B)W_t \quad \Rightarrow \quad \alpha(B)Y_t = \alpha(B)(\gamma_0 + \gamma_1 \gamma X_t) + \beta(B)W_t$$

- The difference is only how the model includes the exogenous variable.
- It is mostly a matter of taste which kind of model you should choose.
- Only the regression with ARMA noise is included into R as standard.

16 Continuous time processes

16.1 Discrete vs. continuous time

There are two fundamentally different model classes for time series data.

- Discrete time stochastic processes
  - Variables given at equally spaced time points
- Continuous time stochastic processes
  - Variables that evolve over a continuous time scale

So far we have only looked at the discrete time case. We will finish todays lecture by looking a bit at the continuous time case, just to give you an idea of this topic.

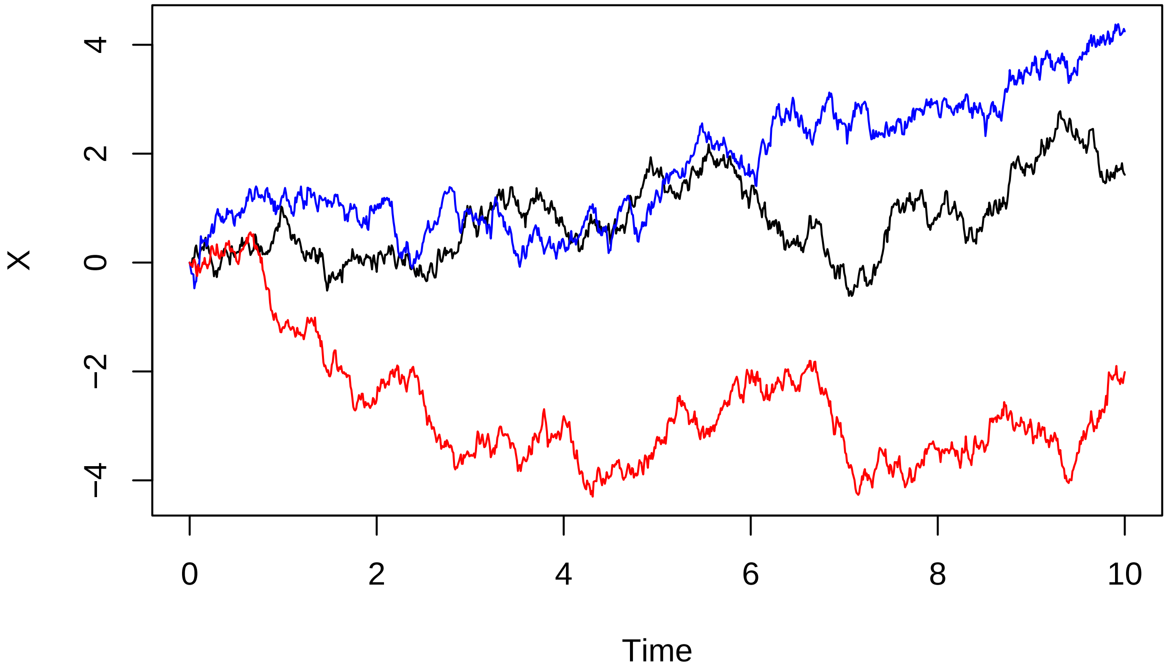
16.2 Continuous time stochastic processes

- In this setup we see the underlying X_t as a continuous function of t for t in some interval $[0, T]$.
- In principle we imagine that there are infinitely many data points, simply because there are infinitely many time points between 0 and T.
- In practice we will always only have finitely many data points.
- But we can imagine that the real data actually contains all the data points. We are just not able to measure them (and to store them in a computer).
- With a model for all datapoints, we are - through simulation - able to describe the behaviour of data. Also between the observations.

16.3 The Wiener process

- A key example of a process in continuous time will be the so–called **Wiener process** or **Brownian motion**.
- Here are three simulated realizations (black, blue and red) of this process: here

```
## Package 'Sim.DiffProc', version 4.9
## browseVignettes('Sim.DiffProc') for more informations.
```



- A Wiener process has the following properties:
  - It starts in 0: $B_0 = 0$.
  - It has independent increments: For $0 < s < t$ it holds that $B_t - B_s$ is independent of everything that has happened up to time s, that is B_u for all $u \leq s$.
  - It has normally distributed increments: For $0 < s < t$ it holds that the increment $B_t - B_s$ is normally distributed with mean zero and variance $t - s$:

$$B_t - B_s \sim \texttt{norm}(0, t - s).$$

- The intuition of this process is that it somehow changes direction all the time: How the process changes after time s will be independent of what has happened before time s. So whether the process should increase or decrease after s will not be affected by how much it was increasing or decreasing before. This gives the very bumpy behaviour over time.

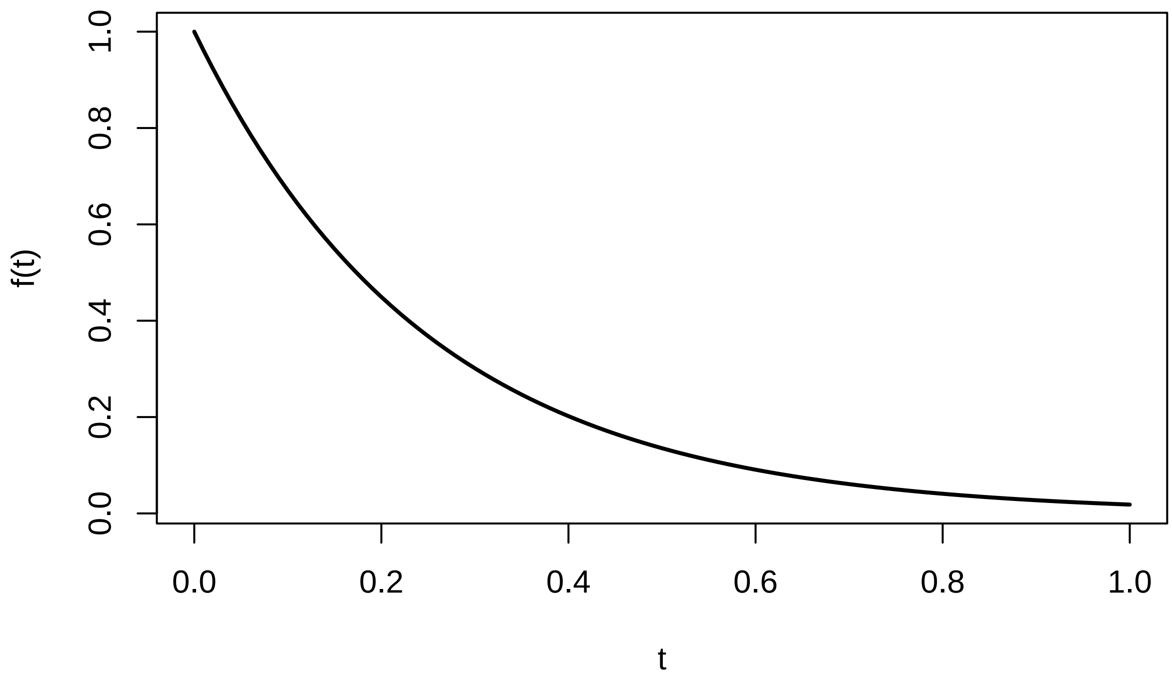
16.4 Stochastic differential equations

- A common way to define a continuous time stochastic process model is through a stochastic differential equation (SDE) which we will turn to shortly, but before doing so we will recall some basic things about ordinary differential equations.
- **Example:** Suppose f is an unknown differentiable function satisfying the differential equation

$$\frac{df(t)}{dt} = -4f(t)$$

with initial condition $f(0) = 1$. This equation has the solution

$$f(t) = \exp(-4t)$$



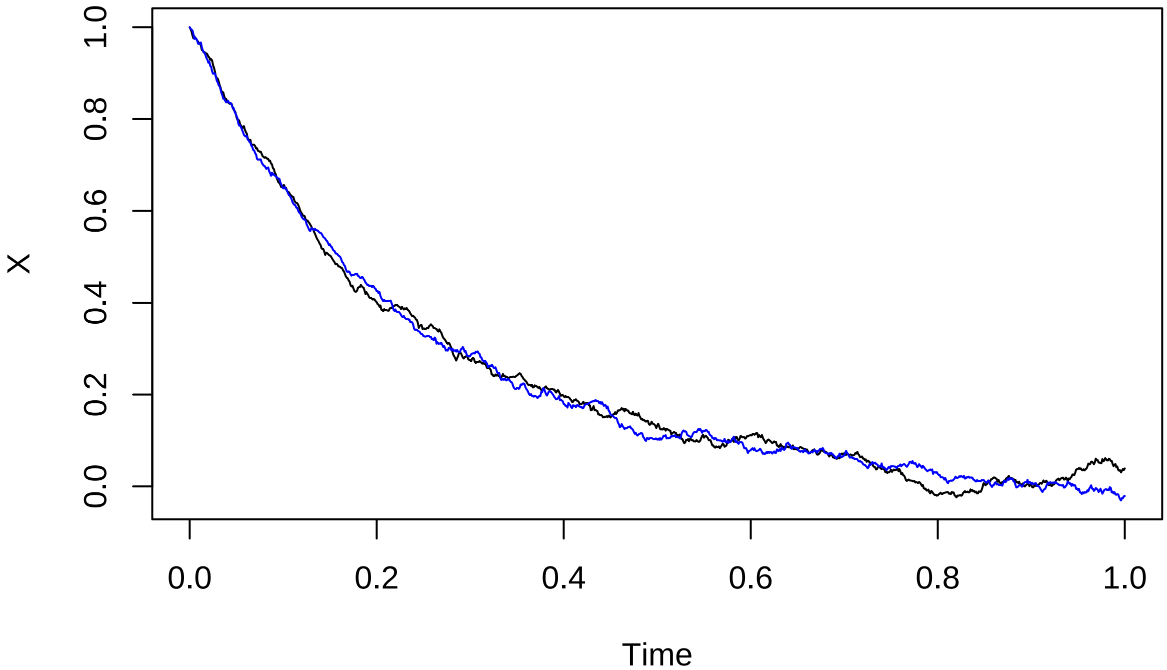
- With a slightly unusual notation we can rewrite this as

$$df(t) = -4 \cdot f(t)dt$$

- This equation has the following (hopefully intuitive) interpretation:
  - When time increases by a small amount dt (from t to $t + dt$) the value of f changes (approximately) by $-4f(t) \cdot dt$.
- So when t is increased, then f is decreased, and the decrease is proportional to the value of $f(t)$. That is why f decreases slower and slower, when t is increased.
- We say that the function has a **drift** towards zero, and this drift is determined by the value of the function.

16.5 Stochastic differential equations

- It will probably never be true that data behaves exactly like the exponentially decreasing curve on the previous slide.
- Instead we will consider a model, where some random noise from a Wiener process has been added to the growth rate. Two different (black/blue) simulated realizations can be seen below



- The type of process that is simulated above is described formally by the equation

$$dX_t = -4X_t dt + 0.1dB_t$$

- This is called a **Stochastic Differential Equation** (SDE), and the processes simulated above are called solutions of the stochastic differential equation.
- The SDE $dX_t = -4X_t dt + 0.1dB_t$ has two terms:
  - $-4X_t dt$ is the **drift term**.
  - $0.1dB_t$ is the **diffusion term**.
- The intuition behind this notation is very similar to the intuition in the equation $df(t) = -4 \cdot f(t)\, dt$ for an ordinary differential equation. When the time is increased by the small amount dt, then the process X_t is increased by $-4X_t\, dt$ AND by how much the process $0.1B_t$ has increased on the time interval $[t, t + dt]$.
- So this process has a **drift** towards zero, but it is also pushed in a random direction (either up or down) by the Wiener process (more precisely, the process $0.1B_t$)